

FUJITA FREENESS IN DIMENSION SIX

JIHAO LIU

ABSTRACT. We prove that $K_X + 7L$ is globally generated for every smooth complex projective sixfold X and every ample Cartier divisor L . The proof constructs, at each point, an effective rational boundary of coefficient strictly less than seven with that point as its minimal log canonical center. It combines bounds for complete cutting chains with uniform local dimension estimates and amplification of complete spaces of sections, while retaining the full boundary and all fixed components. The main result of this paper was obtained using the Danus system.

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1. INTRODUCTION

Let X be a smooth complex projective variety, and let L be an ample Cartier divisor. We write K_X for a canonical divisor. The adjoint line bundle $\mathcal{O}_X(K_X + mL)$ is globally generated if its global sections generate its fiber at every point. Such a linear series defines a morphism to projective space without a base locus. Fujita's freeness conjecture predicts that $K_X + (n + 1)L$ is globally generated when $\dim X = n$; see [Fuj87, AS95]. The coefficient depends only on dimension, although the polarization L itself may have base points.

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The conjecture is known in dimensions at most five. Reid's vector-bundle method proves the surface case [Rei88]. Ein and Lazarsfeld prove the threefold case by cohomological methods from the minimal model program [EL93], and Kawamata proves the fourfold case [Kaw97]. Ye and Zhu prove the fivefold case [YZ20]. In arbitrary dimension, Angehrn and Siu obtain the effective bound

$$m \geq \frac{n^2 + n + 2}{2}$$

for global generation of $K_X + mL$ [AS95, Corollary 0.2]. Helmke develops a method based on the local geometry of log canonical centers [Hel99]. Ghidelli and Lacini estimate the cost of a full sequence of cuts simultaneously and obtain bounds of order $n \log \log n$ in the dimension. In dimension six they prove that $K_X + mL$ is globally generated for every integer $m \geq 8$ [GL24, Theorem 1.2].

The local inputs used here concern both singularities of centers and the number of sections that survive prescribed vanishing. Fujino and Hashizume's adjunction theorem supplies discriminant and moduli divisors on the normalization of a log canonical center [FH23]. We also use local subadjunction of Fujino and Gongyo [FG12], normality and rational singularities of minimal centers as in de Fernex and Hacon [dFH09], and the negativity lemma in the form recorded by Fujino [Fuj11]. Han relates the deficit of a minimal center to normal Hilbert coefficients [Han26]. The Hilbert-function and Rees-algebra methods draw on Huckaba and Marley [HM97], the criteria surveyed by Verma [Ver08], and the normality criterion of Polini, Ulrich, and Vasconcelos [PUV05]. These inputs retain information about the full tangent scheme, to which the small-degree results of Eisenbud, Green, Hulek, and Popescu apply [EGHP04]. We use the precise geometric factorization, irreducibility, and node statements specified in the Stacks Project [Sta26].

Multiplier-ideal restriction and generation, in the forms collected by Lazarsfeld [Laz09], connect local thresholds with actual global sections. Monomial threshold formulas of Howald [How01] and de Fernex, Ein, and Mustařă [dFEM02], together with the threshold estimates of Demailly and Kollár [DK01], underlie the local dimension estimates. The global comparison uses differentiability and concavity of the appropriate root of volume [BFJ09], and the complete-series volume limit of Lazarsfeld and Mustařă [LM09]. Convex fixed-divisor profiles and section counting already occur in the proof of Kawamata's Theorem 2.2, following Helmke and Fujita [Kaw97].

To reach coefficient seven in dimension six, it is necessary to control a complete continuation from every possible center. Separate bounds on each center's dimension and multiplicity leave expensive chains. Moreover, a local ideal with a favorable threshold does not by itself provide a section of the required global line bundle. The proof below addresses these two issues together: it uses restrictions between successive centers of the same pair, and counts the full images of complete spaces of sections.

We show that the conjectured coefficient is sufficient in dimension six. In the following statement, a sixfold has dimension six and global generation has the meaning fixed above.

Theorem 1.1. *For every smooth complex projective sixfold X and every ample Cartier divisor L , the line bundle $\mathcal{O}_X(K_X + 7L)$ is globally generated.*

The hypothesis imposes no global-generation condition on L and no extra condition on the volume or on an auxiliary center. Equivalently, for every closed point $x \in X$, the map

$$H^0(X, \mathcal{O}_X(K_X + 7L)) \longrightarrow \mathcal{O}_X(K_X + 7L) \otimes k(x)$$

is surjective. For $X = \mathbb{P}^6$ and $L = \mathcal{O}_{\mathbb{P}^6}(1)$, the adjoint bundle at coefficient seven is $\mathcal{O}_{\mathbb{P}^6}$. At coefficient six it is $\mathcal{O}_{\mathbb{P}^6}(-1)$, which has no nonzero global sections. Thus seven is the smallest coefficient possible in a statement for every such polarization.

The quantitative part of the argument has three connected features. First, the discriminant calculation retains codimension-one centers of the same current pair, including a parent that

was minimal at an earlier stage. Combined with Hilbert moments, this gives restrictions on successor multiplicities and hence bounds for entire cutting chains. The positive errors in the published cutting theorem are paid only after a strict total margin has been established. Second, we prove bounds for the complete point-initial image of a primary ideal, with an error uniform in the ideal and its threshold. The full tangent scheme and all of its components remain in these estimates. Third, asymptotic multiplier ideals give a comparison uniform over divisorial valuations for the actual complete base ideals. A supporting line to the resulting convex profile controls every point-order layer, including those below the original jet order. Summing the uniform local estimates produces stronger complete spaces of global sections.

The geometric endpoint arguments preserve the full residual boundary. Intrinsic divisors are descended through normalization with a fixed conductor correction and lifted to actual ambient divisors by restriction and Serre vanishing. A finite sequence of stronger complete sources then covers every possible new center at its own threshold. The final reduction leaves only a double fourfold center for a sufficiently late original source; its complete-source estimate supplies a point boundary of coefficient strictly below seven. Tie breaking and Nadel vanishing yield the displayed evaluation map. These quantitative estimates improve the six-dimensional freeness coefficient from eight to seven while using the classical cutting, adjunction, and volume methods just described.

Section 2 fixes the conventions. Section 3 treats approximate cuts and point evaluation, Section 4 develops the local Hilbert and center-flag inequalities, and Section 5 bounds complete cutting chains. Sections 6 and 7 give the local capacities and global complete-source comparisons. Section 8 constructs and lifts intrinsic divisors, and Section 9 settles the threefold endpoints. The stronger-source criterion and the case of volume at least two are proved in Sections 10 and 11. Sections 12–16 then handle, respectively, higher-multiplicity fourfolds, double fivefolds, smooth fivefolds, cubic fourfolds, and double fourfolds of the original source. Section 17 combines these endpoints and proves the theorem.

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2. NOTATION AND CONVENTIONS

Set-up 2.1. Unless another ambient variety is specified, X is a smooth complex projective variety of dimension six, L is an ample Cartier divisor on X , and x is a closed point. A boundary means an effective $\mathbb{Q}$ -divisor. Log canonicity and log canonical centers through x are considered on a neighborhood of x . Global divisors and linear equivalences remain on X .

Definition 2.2. For a boundary B that is log canonical near x , put

$$\mathrm{def}_x(B) = \mathrm{lct}_x((X, B); \mathfrak{m}_x).$$

The threshold is the supremum of the real numbers $t \geq 0$ for which the ideal pair $(X, B; \mathfrak{m}_x^t)$ is log canonical near x . If W is a positive-dimensional integral subvariety through x , write

$$d(W) = \dim W, \quad m(W) = \mathrm{mult}_x W, \quad g(W) = L^{d(W)} \cdot W, \quad r(W) = \left(\frac{m(W)}{g(W)} \right)^{1/d(W)}.$$

The notation s and M , for a specified boundary $B \sim_{\mathbb{Q}} sL$, denotes its entire coefficient s and its entire point multiplicity $M = \text{mult}_x B$. Its deficit is always computed for this same entire boundary.

Definition 2.3. An additive cutting chain is a finite sequence of boundaries $B_{i+1} = B_i + t_i H_i$, where $H_i \sim_{\mathbb{Q}} L$ is effective, $t_i > 0$ is rational, every pair is log canonical near x , and their minimal centers through x strictly decrease. For a chain ending at $\{x\}$, put $b_i = \text{def}_x(B_i)$, $r_i = r(W_i)$, and

$$C = \sum_i (b_i - b_{i+1}) r_i.$$

Here $b_i = 0$ at the final point. A point boundary is a boundary log canonical near x with minimal center $\{x\}$. An additive continuation retains every coefficient of its initial boundary. A separately constructed point boundary need not do so.

Definition 2.4. For an analytically unramified local ring $(R, \mathfrak{n})$ of dimension $d \geq 1$, the first normal Hilbert coefficient is specified by the eventual expansion

$$\text{length}(R/\overline{\mathfrak{n}^k}) = e_0(\mathfrak{n}) \binom{k+d-1}{d} - \bar{e}_1(\mathfrak{n}) \binom{k+d-2}{d-1} + O(k^{d-2}).$$

The ordinary first Hilbert coefficient $e_1(\mathfrak{n})$ uses $\mathfrak{n}^k$ in place of its integral closure. A parameter reduction of the maximal ideal in a Cohen–Macaulay local ring uses a system of parameters forming an R -regular sequence; this does not assert that R is a regular local ring.

Definition 2.5. For rational numbers $a, r > 0$ and positive integers k with $ak, rk \in \mathbb{Z}$, write

$$V_k(a, r) = H^0(X, \mathcal{O}_X(akL) \otimes \mathfrak{m}_x^{rk}).$$

Its base ideal is the image of its evaluation map after tensoring by $\mathcal{O}_X(-akL)$. The space is complete with the specified point-order condition. Its full fixed divisor includes every divisorial component of this base ideal with its full multiplicity. In particular,

$$U_j = V_j(65, 64), \quad \mathfrak{q}_j = \mathfrak{b}(U_j).$$

For any nonzero complete space $V = V_k(a, r)$, a pure threshold average means $\Gamma = (\tau/N) \sum_{i=1}^N H_i$, where $\tau = \text{lct}_x(X; \mathfrak{b}(V))$ and the H_i are a sufficiently large sufficiently general tuple of actual divisors from V . For $V = U_j$, this threshold is $\text{lct}_x(X; \mathfrak{q}_j)$. “Pure” means that no earlier boundary is retained.

Notation 2.6. Let $\pi : Y = \text{Bl}_x X \rightarrow X$ be the blow-up and let E be its exceptional divisor. The symbol $\text{vol}_Y(D)$ denotes the usual volume of a real divisor D ; for rational D it is $\limsup 6! h^0(Y, \mathcal{O}_Y(kD))/k^6$, over positive integers for which kD is integral. Thus the spaces $V_k(a, r)$ are the actual spaces of sections of $k(a\pi^*L - rE)$.

3. CUTTING AND POINT EVALUATION

We retain the conventions of Section 2. We first pass from a point boundary to a section of an adjoint line bundle. We then keep track of the approximation errors in a sequence of cuts. Throughout a sequence, the deficit belongs to the entire current boundary on the same smooth ambient variety. We conclude by realizing the point-ideal threshold as a supremum of orders of actual divisors.

3.1. From a point boundary to evaluation. We use the center-intersection theorem in the smooth ambient setting of this paper; see [GL24, Lemma 2.7].

Theorem 3.1 (Minimal centers). *For an effective pair log canonical near a point, every irreducible component through that point of the intersection of two lc centers is an lc center. Consequently there is a unique minimal lc center through a non-klt point.*

The minimal center is contained in every center through the point. There can still be larger centers through that point. To apply vanishing, we will first remove these larger centers by tie breaking.

The vanishing theorem we need is [GL24, Theorem 2.5].

Theorem 3.2 (Nadel vanishing). *Let Z be a smooth projective variety, let B be an effective $\mathbb{Q}$ -divisor, and let N be an integral divisor such that $N - B$ is big and nef. Then*

$$H^i(Z, \mathcal{O}_Z(K_Z + N) \otimes \mathcal{I}(Z, B)) = 0 \quad (i > 0).$$

Here the positivity condition is global. The boundary need not be log canonical away from the point where we seek a nonvanishing section.

We also use the one-point case of [GL24, Lemma 2.8].

Theorem 3.3 (Tie breaking at a point). *Let Z be klt, let Δ be an effective $\mathbb{Q}$ -Cartier divisor, and suppose (Z, Δ) is log canonical at z with $\{z\}$ a log canonical center. For an ample $\mathbb{Q}$ -Cartier divisor A , there is a rational $a > 0$ such that for every sufficiently small rational $\gamma > 0$ an effective divisor $F \sim_{\mathbb{Q}} aA$ can be chosen so that $(Z, (1 - \gamma)\Delta + \gamma F)$ is log canonical at z and its only log canonical center containing z is $\{z\}$.*

The coefficient a is fixed before γ is chosen. Thus the change in the ample coefficient can be made arbitrarily small.

Proposition 3.4. *Let Z be a smooth complex projective variety, let A be ample Cartier, and let $\Delta \sim_{\mathbb{Q}} cA$, $c \geq 0$, be a point boundary at z . For every positive integer $m > c$, the line bundle $\mathcal{O}_Z(K_Z + mA)$ is generated at z .*

Only a strict inequality between the boundary coefficient and the integer m is needed.

Proof. We isolate the non-klt support at z , choose a section on its finite local component, and lift that section by Nadel vanishing.

Step 1. In this step, we isolate the non-klt locus at z while keeping the coefficient below m . Theorem 3.3 applies because Z is smooth and $\{z\}$ is an lc center. Let $a > 0$ be its rational coefficient. For a sufficiently small rational $0 < \gamma < 1$, choose the divisor F furnished by that theorem and put

$$\Gamma = (1 - \gamma)\Delta + \gamma F, \quad c_\gamma = (1 - \gamma)c + \gamma a.$$

Then $\Gamma \geq 0$, $\Gamma \sim_{\mathbb{Q}} c_\gamma A$, and $c_\gamma - c = \gamma(a - c)$. We can therefore choose γ small enough that $c_\gamma < m$. The pair (Z, Γ) is lc on a neighborhood of z , and its only lc center through z is $\{z\}$.

Step 2. In this step, we identify the open-and-closed Artinian component of the multiplier-ideal quotient supported at z . Put $\mathfrak{j} = \mathcal{I}(Z, \Gamma)$ and let $T \subseteq Z$ be the closed subscheme defined by $\mathfrak{j}$. Since the pair is not klt at z ,

$$\mathfrak{j}_z \subseteq \mathfrak{m}_z \mathcal{O}_{Z,z}.$$

The support of T is the non-klt locus. On an lc neighborhood, its irreducible components are lc centers: on a log resolution this locus is the image of the components of the crepant boundary with coefficient one. Thus no positive-dimensional component of $\text{Supp } T$ contains z . The support has finitely many irreducible components, so $\{z\}$ is disjoint from the union of the other components and is both open and closed in $\text{Supp } T$. Consequently there is a decomposition into open-and-closed subschemes

$$T = T_z \amalg T_{\text{away}}, \quad T_z = \text{Spec}(\mathcal{O}_{Z,z}/\mathfrak{j}_z), \quad z \notin \text{Supp } T_{\text{away}}.$$

The ring of T_z is Artinian and local. This argument uses the support and the inclusion $\mathfrak{j}_z \subseteq \mathfrak{m}_z \mathcal{O}_{Z,z}$; it requires no equality of these ideals.

Step 3. In this step, we lift a section with nonzero residue at z by Nadel vanishing. Write $\mathcal{F} = \mathcal{O}_Z(K_Z + mA)$. Its restriction to the Artinian local scheme T_z is free of rank one. Choose a

generator there and extend it by zero on T_{away} . This gives a section of $\mathcal{F}|_T$ with nonzero image in $\mathcal{F} \otimes k(z)$. Since

$$mA - \Gamma \sim_{\mathbb{Q}} (m - c_\gamma)A$$

is ample, Theorem 3.2 gives $H^1(Z, \mathcal{F} \otimes \mathfrak{j}) = 0$. The exact sequence

$$0 \longrightarrow \mathcal{F} \otimes \mathfrak{j} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}|_T \longrightarrow 0$$

therefore makes $H^0(Z, \mathcal{F}) \longrightarrow H^0(T, \mathcal{F}|_T)$ surjective. A lift of the chosen section has nonzero value at z . Since the fiber of $\mathcal{F}$ is one-dimensional, its evaluation map at z is surjective. $\square$

3.2. Cutting with a fixed error allowance. The cutting result is the one-point specialization of the published [GL24, Proposition 3.2 and Remark 3.3].

Theorem 3.5 (One-point cutting). *Let Z be smooth projective, A ample $\mathbb{Q}$ -Cartier, and B effective rational. Assume either $B = 0$ and $W = Z$, or (Z, B) is log canonical but not klt at z and W is its positive-dimensional minimal center through z . Set*

$$b = \text{lct}_z((Z, B); \mathfrak{m}_z), \quad r = \left(\frac{\text{mult}_z W}{A^{\dim W} \cdot W} \right)^{1/\dim W}.$$

For every $\epsilon > 0$, there are an effective rational $H \sim_{\mathbb{Q}} A$ and a rational $t > 0$ such that $B + tH$ is log canonical at z , its minimal center through z is properly contained in W , and its deficit b' satisfies

$$t \leq br + \epsilon, \quad b' \leq b - t/r + \epsilon.$$

Both estimates concern the same actual divisor H , coefficient t , and new boundary. In the cited proposition, take both finite sets to be $\{z\}$. Its nonempty output subset must then be $\{z\}$, and its separation assumptions have no additional content. A coefficient $t = 0$ could not decrease the minimal center; in the zero-boundary case it would leave the pair klt. Thus the coefficient here is positive.

We explain the normalization of the deficit in this citation. For $n = \dim Z \geq 2$, let $\sigma : \widehat{Z} \rightarrow Z$ be the blow-up at z , with exceptional divisor F_z . Then

$$\mathfrak{m}_z \mathcal{O}_{\widehat{Z}} = \mathcal{O}_{\widehat{Z}}(-F_z), \quad K_{\widehat{Z}} = \sigma^* K_Z + (n - 1)F_z.$$

For a real parameter u , the log pullback of the ideal pair $(Z, B; \mathfrak{m}_z^u)$ has boundary

$$\sigma^* B - (n - 1 - u)F_z.$$

This is the point-blow-up definition used in the cited cutting theorem. Valuations with zero order on $\mathfrak{m}_z$ are unaffected; those with positive order have center z . Thus its local discrepancy is precisely the threshold in Definition 2.2. On a smooth curve the ideal of the point is already invertible, and the same comparison uses the divisor z on the curve itself.

The errors in Theorem 3.5 are retained separately. The first bounds the added coefficient, while the second compares that coefficient with the deficit lost by the actual new boundary. Rearranging the second gives

$$(3.1) \quad t \leq (b - b')r + \epsilon r.$$

In particular, the weighted deficit drop need not pay for the entire cut without an error allowance. The radii of later centers, as well as their deficits, can depend on the errors already chosen.

We record the elementary threshold properties used to sum these drops. For a prime divisor P over a smooth variety Z , write $A_{Z,B}(P)$ for its log discrepancy with respect to (Z, B) and v_P for its order valuation.

Lemma 3.6. *Let Z be a smooth complex variety of dimension $n \geq 1$, let z be a closed point, and let $B \leq B'$ be effective rational divisors such that both pairs are log canonical near z . Then the following hold:*

$$(i) \quad 0 \leq \text{def}_z(B') \leq \text{def}_z(B).$$

- (ii) If $\{z\}$ is an lc center of (Z, B) , then $\text{def}_z(B) = 0$.
 (iii) $\text{def}_z(0) = n$.

Thus additive cuts give nonnegative deficit drops, and a chain ending at the point has terminal deficit zero.

Proof. We use the valuative formula for the first two assertions and the point blow-up for the zero boundary.

Step 1. In this step, we prove monotonicity of the deficit and its vanishing at a point center. The valuative criterion for the ideal pair gives

$$(3.2) \quad \text{def}_z(B) = \inf_{P: v_P(\mathfrak{m}_z) > 0} \frac{A_{Z,B}(P)}{v_P(\mathfrak{m}_z)}.$$

Indeed, its lc condition is $A_{Z,B}(P) - u v_P(\mathfrak{m}_z) \geq 0$ for every prime divisor over a neighborhood of z . The valuations in the display are exactly those centered at z . The other valuations have zero denominator and impose only the assumed lc condition for (Z, B) .

Log canonicity makes each numerator nonnegative. Moreover,

$$A_{Z,B'}(P) = A_{Z,B}(P) - v_P(B' - B) \leq A_{Z,B}(P).$$

Taking infima with the same positive denominators proves (i). If $\{z\}$ is an lc center, one such divisor P has $A_{Z,B}(P) = 0$, so (ii) follows from the same formula.

Step 2. In this step, we compute the deficit of the zero boundary. For (iii), suppose first that $n \geq 2$. The point blow-up principalizes $\mathfrak{m}_z$ and is a log resolution of that ideal. Denote its exceptional divisor by F_z . The crepant boundary for $(Z; \mathfrak{m}_z^u)$ is $(u - n + 1)F_z$. It has simple normal crossing support and is lc exactly when $u - n + 1 \leq 1$, or $u \leq n$. If $n = 1$, the ideal is $\mathcal{O}_Z(-z)$, and the corresponding boundary is $u\{z\}$, which is lc exactly when $u \leq 1$. $\square$

Proposition 3.7. *Let $B_0 \sim_{\mathbb{Q}} s_0 L$ satisfy the hypotheses of Theorem 3.5 on X , and fix $h > 0$. Fix positive numbers $a_0, \dots, a_5$ with sum less than h . At each stage, after its radius r_i is known, choose $0 < \epsilon_i < a_i/(1 + r_i)$ and apply that theorem. The resulting chain ends at a point in at most six steps and its actual final coefficient satisfies*

$$s_N < s_0 + C + h.$$

In particular, if a number K determined by the initial data bounds C for every additive cutting chain from B_0 to the point and $s_0 + K < 7$, then $K_X + 7L$ is generated at x . The margin is fixed before any cutting error is chosen.

The hypothesis on K concerns all output chains. A strict bound for a single chain whose centers change when the errors change would not supply the fixed margin used here.

Proof. We construct the finite additive chain, sum the actual errors, and then apply the uniform bound within a margin chosen in advance.

Step 1. In this step, we construct the additive chain and prove that its centers reach the point after at most six cuts. At a positive-dimensional stage let W_i be the current minimal center, with $W_0 = X$ if $B_0 = 0$, and set

$$b_i = \text{def}_x(B_i), \quad r_i = \left(\frac{\text{mult}_x W_i}{L^{\dim W_i} \cdot W_i} \right)^{1/\dim W_i}.$$

Ampleness and positive multiplicity make r_i finite and positive. It is known before we choose ϵ_i . Applying Theorem 3.5 with the prescribed error gives

$$B_{i+1} = B_i + t_i H_i, \quad H_i \sim_{\mathbb{Q}} L, \quad t_i > 0,$$

and a proper smaller center $W_{i+1} \subsetneq W_i$ through x . Each boundary is effective and rational. Each pair is lc near x , after shrinking the neighborhood when needed. The same hypotheses therefore

hold at every subsequent positive-dimensional stage. Strict containment decreases dimension, so after $N \leq \dim W_0 \leq 6$ cuts the center is $W_N = \{x\}$. Lemma 3.6 gives

$$b_0 \geq b_1 \geq \cdots \geq b_N = 0.$$

When $B_0 = 0$, that lemma also gives $b_0 = 6$.

Step 2. In this step, we bound the total added coefficient by summing the deficit drops and both approximation errors. For every performed cut, the two published estimates remain

$$t_i \leq b_i r_i + \epsilon_i, \quad b_{i+1} \leq b_i - t_i / r_i + \epsilon_i.$$

Our choice of ϵ_i controls the unweighted and weighted errors simultaneously:

$$\epsilon_i + \epsilon_i r_i = \epsilon_i(1 + r_i) < a_i.$$

In particular, the first estimate retains a coefficient error less than a_i . For the sum of the costs we use the second estimate in the form (3.1), with its weighted error still present. Since every addition is retained,

$$\begin{aligned} s_N &= s_0 + \sum_{i=0}^{N-1} t_i \\ &\leq s_0 + \sum_{i=0}^{N-1} (b_i - b_{i+1})r_i + \sum_{i=0}^{N-1} \epsilon_i r_i \\ &\leq s_0 + C + \sum_{i=0}^{N-1} \epsilon_i(1 + r_i) \\ &< s_0 + C + \sum_{i=0}^{N-1} a_i < s_0 + C + h. \end{aligned}$$

Here C is exactly the cost in Definition 2.3 for this constructed chain. All its centers and deficits may depend on the preceding errors; none are held fixed while an error tends to zero.

Step 3. In this step, we choose the cutting allowances inside the uniform margin and obtain generation at x . Suppose now that K has the stated uniform property. Before making any cut, put

$$\eta = 7 - s_0 - K > 0, \quad h = \frac{\eta}{2}, \quad a_i = \frac{\eta}{24} \quad (0 \leq i \leq 5).$$

Then $\sum_{i=0}^5 a_i = \eta/4 < h$. Run the construction with these choices. The resulting chain is additive, its minimal centers strictly decrease, and it ends at the point. Hence the hypothesis gives $C \leq K$ for this very chain, and

$$s_N < s_0 + K + \frac{\eta}{2} = 7 - \frac{\eta}{2} < 7.$$

The effective rational divisor $B_N \sim_{\mathbb{Q}} s_N L$ is a point boundary. Proposition 3.4, with integer $m = 7$, now proves generation at x . The margin pays for the errors in the additive construction before the final use of tie breaking. $\square$

3.3. Divisors realizing the point-ideal threshold. We finally compare the ideal threshold with the order of a divisor that can be added to the boundary. This comparison concerns the ambient pair and preserves every coefficient of its boundary.

Lemma 3.8. *Let Z be a smooth complex projective variety, let z be a closed point, and let B be an effective rational divisor log canonical near z . Then*

$$\text{lct}_z((Z, B); \mathbf{m}_z) = \sup\{\text{ord}_z D : D \text{ effective rational on } Z, (Z, B + D) \text{ lc near } z\}.$$

The divisors on the right are global divisors, although their lc condition is imposed only near z .

Proof. Write $b = \text{lt}_z((Z, B); \mathfrak{m}_z)$. We first bound the order of every admissible divisor by b . We then construct a global divisor of each rational order strictly between zero and b .

Step 1. In this step, we bound the point order of every admissible divisor by the point-ideal threshold. Let $D \geq 0$ be rational with $(Z, B + D)$ lc near z , and put $\alpha = \text{ord}_z D$. Choose a positive integer p such that pD is Cartier near z , and let f be a local equation. Then $p\alpha$ is an integer and $f \in \mathfrak{m}_z^{p\alpha} \mathcal{O}_{Z,z}$. For every prime divisor P over Z centered at z , this gives

$$v_P(D) = \frac{v_P(f)}{p} \geq \alpha v_P(\mathfrak{m}_z).$$

Log canonicity of $(Z, B + D)$ gives $A_{Z,B}(P) \geq v_P(D)$. Thus

$$\alpha \leq \frac{A_{Z,B}(P)}{v_P(\mathfrak{m}_z)}.$$

Taking the infimum as in (3.2) proves $\alpha \leq b$. If $b = 0$, this already proves the assertion, since the zero divisor is admissible.

Step 2. In this step, we construct an admissible global divisor of every positive rational order below b . Fix a rational number $0 < q < b$ and an integer $k > q$. Take a simultaneous log resolution $\mu : \tilde{Z} \rightarrow Z$ of B and $\mathfrak{m}_z$, with

$$\mathfrak{m}_z \mathcal{O}_{\tilde{Z}} = \mathcal{O}_{\tilde{Z}}(-G).$$

The support of G , the exceptional divisors, and the strict transform of B form a simple normal crossing divisor. Fix a very ample Cartier divisor T on Z . For a sufficiently large integer ℓ , the sheaf $\mathfrak{m}_z^k \otimes \mathcal{O}_Z(\ell T)$ is globally generated, and the restriction map

$$H^0(Z, \mathfrak{m}_z^k \otimes \mathcal{O}_Z(\ell T)) \longrightarrow H^0(Z, (\mathfrak{m}_z^k / \mathfrak{m}_z^{k+1}) \otimes \mathcal{O}_Z(\ell T))$$

is surjective. The latter assertion follows by twisting the exact sequence for $\mathfrak{m}_z^{k+1} \subseteq \mathfrak{m}_z^k$ and taking ℓ large enough that $H^1(Z, \mathfrak{m}_z^{k+1} \otimes \mathcal{O}_Z(\ell T)) = 0$.

After pullback and removal of the fixed divisor kG , these global sections generate $\mathcal{O}_{\tilde{Z}}(\mu^*(\ell T) - kG)$. Indeed, their evaluation generates $\mathfrak{m}_z^k$ downstairs, whose image upstairs is exactly $\mathcal{O}_{\tilde{Z}}(-kG)$. Consequently a general member H of the corresponding subspace of $|\ell T|$ satisfies

$$\text{ord}_z H = k, \quad \mu^* H = kG + \tilde{H},$$

where $\tilde{H}$ is reduced, contains none of the fixed components, and meets their simple normal crossing strata transversely. To obtain the first condition, choose a section with nonzero image in the displayed jet quotient; this is a nonempty open condition. The other conditions hold on a nonempty open set by Bertini applied to the generated moving system and its restrictions to the finitely many strata. These open conditions can be imposed simultaneously.

Put $D = (q/k)H$. It is an effective rational divisor on Z , and $\text{ord}_z D = q$. Define the crepant boundary $B_{\tilde{Z}}$ by

$$K_{\tilde{Z}} + B_{\tilde{Z}} = \mu^*(K_Z + B).$$

Then the crepant boundary for $(Z, B + D)$ is

$$B_{\tilde{Z}} + qG + \frac{q}{k} \tilde{H}.$$

For a component P of G , the corresponding log discrepancy is

$$A_{Z,B}(P) - q v_P(\mathfrak{m}_z) > 0,$$

because $q < b$. Components of the old resolved boundary outside G keep their coefficients, which are at most one over an lc neighborhood of z . Each new moving component has coefficient $q/k < 1$. The resulting boundary has simple normal crossing support and all coefficients at most one over that neighborhood, so $(Z, B + D)$ is lc there. No lc assertion away from that neighborhood is needed.

Thus the supremum in the statement is at least every rational q with $0 < q < b$, and hence is at least b . The valuative upper bound gives the reverse inequality. $\square$

4. LOCAL HILBERT COEFFICIENTS AND CENTER FLAGS

We retain the deficit and normal Hilbert coefficient conventions of Section 2. We compare the point threshold with adjunction on the normalized blow-up of a center. Keeping the divisorial part of the discriminant gives a bound for codimension-one subcenters, including those inside a center which is no longer minimal.

4.1. Adjunction and Cartier additions. We use the following form of adjunction [FH23, Theorem 1.2 and Definition 1.3].

Theorem 4.1 (Adjunction to a center). *Let T be a normal complex variety, let $\Delta \geq 0$ be a real divisor with $K_T + \Delta$ real Cartier, and let W be a log canonical center. On its normalization $\nu : Z \rightarrow W$ there are discriminant and moduli b -divisors $\mathcal{B}, \mathcal{M}$ such that*

$$\nu^*((K_T + \Delta)|_W) = K_Z + \mathcal{B}_Z + \mathcal{M}_Z.$$

Here a b -divisor specifies compatible traces on birational models; the closure of a Cartier divisor has its pullback as every higher trace. The base trace $\mathcal{B}_Z$ is effective. On a smooth birational model $p : Z' \rightarrow Z$, the b -divisor $\mathcal{M}$ is the closure of a potentially nef divisor $\mathcal{M}_{Z'}$, and $K + \mathcal{B}$ is the closure of $K_{Z'} + \mathcal{B}_{Z'}$. The support of $\mathcal{B}_{Z'}$ is simple normal crossing. The image of its coefficients greater than one is the non-log-canonical locus on W ; the image of its coefficients at least one is that locus together with the intersections with lc centers not containing W . Potential nefness means extension to a nef divisor on a completion; in particular the trace is nef on a projective such model. For a prime P on a higher model of Z , its discriminant coefficient is one minus the infimum, over ambient lc places dominating W , of the generic sub-lc thresholds of the pullback of P on those places with their induced boundaries. Rational input gives rational discriminant and moduli data. No effectivity of every higher discriminant trace or nefness of every lower moduli trace is asserted.

The coefficient formula permits us to compare the discriminant with the point threshold without choosing an auxiliary boundary on the center. For a Cartier center, the corresponding adjunction statement is [FH23, Corollary 1.4].

Theorem 4.2 (Cartier adjunction). *For an effective pair $(T, W + R)$, with W a prime Cartier divisor not contained in $\text{Supp } R$ and $K_T + W + R$ rational Cartier, adjunction on the normalization of W has zero moduli part and boundary the different. The pair is plt near W if and only if the adjoint pair is klt. In the smooth local situation at a point of W , the different is $R|_W$; lc and smaller-center questions are governed by inversion of adjunction for this full pair.*

The residual divisor in this statement is the entire divisor R . For a center of arbitrary codimension, a Cartier addition has the following exact effect on the adjunction data.

Proposition 4.3. *In Theorem 4.1, let D be an effective rational Cartier divisor not containing W , and let $t \geq 0$. Whenever the same center adjunction is defined for $\Delta + tD$, its discriminant is $\mathcal{B}(t) = \mathcal{B}(0) + t\nu^*(D|_W)$. The moduli class is unchanged; compatible representatives make the moduli b -divisor unchanged. The bar denotes Cartier closure.*

Thus a Cartier addition can be followed on every birational trace of the discriminant.

Proof. Put $D_Z = \nu^*(D|_W)$. The two ambient pairs agree over the generic point of W , because D does not contain W . Their lc places with center W are therefore the same.

Step 1. In this step, we compute the change in every discriminant coefficient under the Cartier addition. Fix a sufficiently high model $p : Z' \rightarrow Z$, a prime divisor P on Z' , and an ambient lc

place S with center W . Resolve so that the induced map $f_S : S \rightarrow Z'$ is a morphism. Let $\Theta_S(t)$ be the induced boundary on S for the ambient boundary $\Delta + tD$, and put

$$a_P = \text{coeff}_P(p^*D_Z).$$

Cartier pullback commutes with restriction to S , so $\Theta_S(t) = \Theta_S(0) + tf_S^*p^*D_Z$. Over the generic point of P , the added divisor is $ta_P f_S^*P$. If $\alpha_{P,S}(t)$ denotes the generic sub-lc threshold of f_S^*P for this induced pair, then

$$\alpha_{P,S}(t) = \alpha_{P,S}(0) - ta_P.$$

The threshold description in Theorem 4.1 now gives

$$\begin{aligned} \text{coeff}_P \mathcal{B}(t)_{Z'} &= 1 - \inf_S \alpha_{P,S}(t) \\ &= \text{coeff}_P \mathcal{B}(0)_{Z'} + ta_P. \end{aligned}$$

This holds for every prime on every sufficiently high model. Compatibility of traces proves the asserted equality of b-divisors.

Step 2. In this step, we prove that compatible representatives leave the moduli b-divisor unchanged. To check the moduli part, choose a divisor representative Λ_0 of the pulled-back adjunction class on Z' , and use $\Lambda_t = \Lambda_0 + tp^*D_Z$ for the augmented pair. Its moduli trace is

$$\begin{aligned} \mathcal{M}(t)_{Z'} &= \Lambda_t - K_{Z'} - \mathcal{B}(t)_{Z'} \\ &= \Lambda_0 - K_{Z'} - \mathcal{B}(0)_{Z'} = \mathcal{M}(0)_{Z'}. \end{aligned}$$

The same calculation on higher models gives equality with compatible representatives. Changing those representatives changes the moduli data by real linear equivalence, so the moduli class is unchanged independently of this choice. $\square$

4.2. Local rings and the normalized point blow-up. We record the affine local subadjunction theorem [FG12, Theorem 7.2].

Theorem 4.4 (Local subadjunction). *Let (T, Δ) be an effective rational lc pair on a normal affine variety with $K_T + \Delta$ rational Cartier. A minimal lc center W admits an effective rational boundary Δ_W such that (W, Δ_W) is klt and $K_W + \Delta_W \sim_{\mathbb{Q}} (K_T + \Delta)|_W$.*

This is a local assertion about the center. The local ring properties we need are supplied by the following result [dFH09, Corollary 7.12, Proposition 7.11, and Corollary 7.7].

Theorem 4.5. *On a smooth complex variety, the minimal log canonical center of an effective rational pair through a point where the pair is log canonical but not klt is normal and has rational singularities near that point. Its local ring is Cohen–Macaulay.*

Such a local ring is excellent and normal, hence analytically unramified. Its residue field is $\mathbb{C}$. We can therefore use the following Hilbert coefficient comparison [Han26, Lemmas 4.2 and 4.4].

Theorem 4.6. *Let $(R, \mathfrak{n})$ be an analytically unramified Cohen–Macaulay local ring of dimension d with infinite residue field. Let h_i be the Hilbert function of a general Artinian minimal reduction of $\mathfrak{n}$, and put $m = e_0(\mathfrak{n})$. Then*

$$h_0 = 1, \quad h_1 = \text{edim } R - d, \quad \sum_i h_i = m, \quad \sum_i ih_i \leq e_1(\mathfrak{n}) \leq \bar{e}_1(\mathfrak{n}).$$

If $h_1 > 0$, then $h_i \leq \binom{h_1+i-1}{i}$. If $h_1 = 0$, the Artinian reduction is the residue field.

Here the Artinian ring is R/J , where J is generated by a general system of parameters forming an R -regular sequence. The sequence (h_i) is the Hilbert function of its maximal-ideal filtration. In particular its first moment is a lower bound for the normal Hilbert coefficient.

The geometric calculation below needs normality, but does not require the normalized blow-up to be Cohen–Macaulay. Intersections of a Weil divisor with Cartier divisors are understood as intersections with its Weil cycle; no Cartier assumption on the canonical divisor is implicit.

Lemma 4.7. *Let Z be a normal complex projective variety of dimension $d \geq 2$, let $x \in Z$ be a closed point, and let $f : T \rightarrow Z$ be the normalized blow-up of its ideal sheaf. Write*

$$\mathfrak{m}_x \mathcal{O}_T = \mathcal{O}_T(-F), \quad H = -F, \quad F = \sum_i r_i F_i, \quad w_i = F_i \cdot H^{d-1},$$

where the F_i are the prime components of F . Put $m = \text{mult}_x Z$ and $\mathfrak{n} = \mathfrak{m}_{Z,x}$. Then

$$r_i, w_i \in \mathbb{Z}_{>0}, \quad m = -H^d = \sum_i r_i w_i, \quad K_T \cdot H^{d-1} = (d-1)m - 2\bar{e}_1(\mathfrak{n}).$$

For every prime divisor P on Z , its strict transform P^{str} on T satisfies

$$P^{\text{str}} \cdot H^{d-1} = \begin{cases} -\text{mult}_x P, & x \in P, \\ 0, & x \notin P. \end{cases}$$

The first normal Hilbert coefficient is thus determined by a canonical intersection, and each divisor through the point contributes its full multiplicity with a negative sign.

Proof. We first recover the two leading Hilbert coefficients on the normalized blow-up, and then compute the contribution of a strict transform.

Step 1. In this step, we prove positivity and integrality of the exceptional multiplicities and intersection weights. The normalization map from T to $\text{Bl}_x Z$ is finite. The line bundle $\mathcal{O}_T(H)$ is the pullback of the tautological $\mathcal{O}(1)$, so H is f -ample. Every F_i is an integral projective variety of dimension $d-1$ over x . Consequently

$$w_i = \deg((H|_{F_i})^{d-1}) \in \mathbb{Z}_{>0}.$$

Also $r_i = \text{ord}_{F_i}(\mathfrak{n})$ is a positive integer.

Step 2. In this step, we recover the multiplicity and first normal Hilbert coefficient from intersection numbers on the normalized blow-up. Put $R = \mathcal{O}_{Z,x}$. The normalized Rees algebra identifies

$$f_* \mathcal{O}_T(kH) = \overline{\mathfrak{m}_x^k}.$$

Relative Serre vanishing gives $R^q f_* \mathcal{O}_T(kH) = 0$ for $q > 0$ and sufficiently large k . The quotient of $\mathcal{O}_Z$ by this integral-closure ideal is supported at x . Its length is $\text{length}(R/\overline{\mathfrak{n}^k})$. Thus

$$(4.1) \quad \chi(T, \mathcal{O}_T(kH)) = \chi(Z, \mathcal{O}_Z) - \text{length}(R/\overline{\mathfrak{n}^k}) \quad (k \gg 0).$$

Choose a projective resolution $\mu : \tilde{T} \rightarrow T$ which is an isomorphism over the regular locus. Normality gives $\mu_* \mathcal{O}_{\tilde{T}} = \mathcal{O}_T$, and each $R^q \mu_* \mathcal{O}_{\tilde{T}}$, for $q > 0$, is supported in dimension at most $d-2$. The Euler characteristic of such a coherent sheaf after tensoring with $\mathcal{O}_T(kH)$ is a polynomial in k of degree at most $d-2$. The Leray spectral sequence and the projection formula therefore give

$$\chi(T, \mathcal{O}_T(kH)) = \chi(\tilde{T}, \mathcal{O}_{\tilde{T}}(k\mu^*H)) + O(k^{d-2}).$$

Riemann–Roch on the smooth projective variety $\tilde{T}$ now yields

$$(4.2) \quad \chi(T, \mathcal{O}_T(kH)) = \frac{H^d}{d!} k^d - \frac{K_T \cdot H^{d-1}}{2(d-1)!} k^{d-1} + O(k^{d-2}).$$

Indeed $\mu_* K_{\tilde{T}} = K_T$ as Weil cycles, so projection against the Cartier divisor H gives the displayed canonical coefficient. This argument accounts for the higher direct images instead of assuming that they vanish.

Expanding the binomial coefficients in Definition 2.4 gives

$$\text{length}(R/\overline{\mathfrak{n}^k}) = \frac{m}{d!} k^d + \frac{(d-1)m - 2\bar{e}_1(\mathfrak{n})}{2(d-1)!} k^{d-1} + O(k^{d-2}).$$

Comparison with (4.1) and (4.2) proves

$$-H^d = m, \quad K_T \cdot H^{d-1} = (d-1)m - 2\bar{e}_1(\mathfrak{n}).$$

Since $F = -H$, it also gives $\sum_i r_i w_i = m$.

Step 3. In this step, we compute the intersection of each strict transform with H^{d-1} . If $x \notin P$, the strict transform is disjoint from F , so its intersection with H^{d-1} is zero. Suppose $x \in P$. Let $\hat{P} = \text{Bl}_x P$ and write its exceptional Cartier divisor as F_P , with $H_P = -F_P$. The point ideal becomes the invertible ideal $\mathcal{O}_{P^{\text{str}}}(H)$ on P^{str} . The universal property of the blow-up gives a projective birational morphism $P^{\text{str}} \rightarrow \hat{P}$ pulling H_P back to $H|_{P^{\text{str}}}$. Hence

$$P^{\text{str}} \cdot H^{d-1} = H_P^{d-1}.$$

For $k \gg 0$, the ordinary Rees algebra and relative Serre vanishing on $\hat{P}$ give

$$\chi(\hat{P}, \mathcal{O}_{\hat{P}}(kH_P)) = \chi(P, \mathcal{O}_P) - \text{length}(\mathcal{O}_{P,x}/\mathfrak{m}_{P,x}^k).$$

The leading coefficients of the two sides are respectively $H_P^{d-1}/(d-1)!$ and $-\text{mult}_x P/(d-1)!$. This proves $H_P^{d-1} = -\text{mult}_x P$, including $d = 2$. $\square$

Lemma 4.8. *Let $\mathcal{X}$ be a smooth complex projective variety, let $x \in \mathcal{X}$ be a closed point, and let B be an effective rational divisor on $\mathcal{X}$ such that $(\mathcal{X}, B)$ is log canonical near x . Let W be a positive-dimensional lc center through x , normal near x , and put $b = \text{def}_x(B)$. Let $\nu : Z \rightarrow W$ be the normalization and $\mathcal{B}$ its adjunction discriminant. For every prime divisor P over Z centered at x , one has*

$$1 - \text{coeff}_P \mathcal{B} \geq b \text{ord}_P(\mathfrak{m}_{Z,x}).$$

Minimality of W is not a hypothesis of this threshold comparison.

Proof. The normalization is an isomorphism near x , so we use x for its unique inverse image. Choose a smooth model $Z' \rightarrow Z$ which extracts P and principalizes the point ideal, and put $a_P = \text{ord}_P(\mathfrak{m}_{Z,x})$.

Step 1. In this step, we construct a sub-lc adjoint pair on each ambient lc place dominating W . Fix any ambient lc place S with center W . On a sufficiently high log resolution $\rho : \tilde{\mathcal{X}} \rightarrow \mathcal{X}$ of the ambient pair and point ideal, we may realize S as a smooth divisor and arrange that its induced map $f_S : S \rightarrow Z'$ is a morphism. Write

$$K_{\tilde{\mathcal{X}}} + \tilde{B} = \rho^*(K_{\mathcal{X}} + B), \quad \mathfrak{m}_x \mathcal{O}_{\tilde{\mathcal{X}}} = \mathcal{O}_{\tilde{\mathcal{X}}}(-D_x).$$

The coefficient of S in $\tilde{B}$ is one. Since W has positive dimension, $\text{ord}_S(\mathfrak{m}_x) = 0$, so S is not a component of D_x .

The threshold b is attained: on a log resolution it is the minimum of the finitely many discrepancy-to-order ratios with positive point-ideal order. Thus $(\tilde{\mathcal{X}}, \tilde{B} + bD_x)$ is sub-lc over a neighborhood of x . Put $\Theta_S = (\tilde{B} - S)|_S$. Adjunction to its coefficient-one component gives a sub-lc pair

$$(S, \Theta_S + bD_x|_S).$$

Step 2. In this step, we compare the generic thresholds over P with the point deficit. The restriction of the ambient point ideal to S is the pullback of $\mathfrak{m}_{Z,x}$. Over the generic point of P , this implies

$$D_x|_S \geq a_P f_S^* P.$$

Decreasing the effective added divisor preserves sub-log-canonicity. The generic threshold of $f_S^* P$ for (S, Θ_S) is therefore at least ba_P . Taking the infimum over all these lc places, as in Theorem 4.1, proves the assertion. $\square$

Lemma 4.9. *Let B be an effective rational divisor on a smooth complex projective variety, log canonical near x , and let W be an lc center through x of dimension $d \geq 2$, normal near x . Let $Z \rightarrow W$ be its normalization, let $f : T \rightarrow Z$ be the normalized point blow-up, and write $\mathfrak{m}_x \mathcal{O}_T = \mathcal{O}_T(-F)$ and $H = -F$. Let $\mathcal{B}, \mathcal{M}$ be the adjunction b -divisors, and let $\mathcal{B}^{\text{str}}$ be the strict*

transform of $\mathcal{B}_Z$. There is a smooth projective common model $g : V \rightarrow T$ on which $\mathcal{M}_V$ is nef. With $H_V = g^*H$, one has

$$(K_T + \mathcal{B}_T) \cdot H^{d-1} = -\mathcal{M}_V \cdot H_V^{d-1} \geq 0, \quad -\mathcal{B}^{\text{str}} \cdot H^{d-1} \geq 0.$$

The two signs use different models: the moduli trace on V and the effective discriminant trace on Z .

Proof. Choose V to dominate both T and the smooth model of descent in Theorem 4.1. These models can be chosen projective because Z is projective. Potential nefness on the model of descent, followed by pullback, makes $\mathcal{M}_V$ nef on this projective model. We use this trace throughout the argument.

Step 1. In this step, we express the canonical-discriminant intersection as the negative moduli intersection. The adjunction formula on V identifies $K_V + \mathcal{B}_V + \mathcal{M}_V$ with the pullback of the adjunction class on Z . At least one factor of H_V^{d-1} is represented by a divisor supported over x . The projection formula consequently gives

$$(K_V + \mathcal{B}_V + \mathcal{M}_V) \cdot H_V^{d-1} = 0.$$

Pushing the first two Weil cycles to T yields

$$(K_T + \mathcal{B}_T) \cdot H^{d-1} = -\mathcal{M}_V \cdot H_V^{d-1}.$$

Step 2. In this step, we prove the required sign of the moduli intersection on V . Write $g^*F = \sum_j s_j G_j$, where $s_j > 0$ and the G_j are prime divisors over x . On the normalization of each G_j , the restriction of H_V is nef: it is the pullback of H restricted to the projective fiber of f , where H is ample. The restriction of $\mathcal{M}_V$ is also nef. Since $H_V = -\sum_j s_j G_j$, we obtain

$$\mathcal{M}_V \cdot H_V^{d-1} = -\sum_j s_j (\mathcal{M}_V|_{G_j}) \cdot (H_V|_{G_j})^{d-2} \leq 0.$$

The products can be computed on the normalizations of the G_j . Their nonnegativity follows by approximating the nef restrictions by ample classes on these proper varieties.

Step 3. In this step, we compute the strict-transform discriminant contribution using its effective base trace. Finally write $\mathcal{B}_Z = \sum_P c_P P$, with $c_P \geq 0$, using base effectivity from Theorem 4.1. Lemma 4.7 gives the exact identity

$$(4.3) \quad -\mathcal{B}^{\text{str}} \cdot H^{d-1} = \sum_{P \ni x} c_P \text{mult}_x P \geq 0.$$

No sign for the exceptional coefficients of $\mathcal{B}_T$ or $\mathcal{B}_V$ is used. □

4.3. Deficits and divisorial subcenters. The following inequality is Han's normal Hilbert coefficient bound [Han26, Theorem 1.4]. We include the proof in a form which keeps the separate intersection terms available.

Proposition 4.10. *Let B be an effective rational divisor on a smooth complex projective variety and let W be its positive-dimensional minimal log canonical center through x . Suppose the pair is log canonical near x . With $d = \dim W$, $b = \text{def}_x(B)$, and $m = \text{mult}_x W$, one has*

$$b > 0, \quad 2\bar{e}_1(\mathfrak{m}_{W,x}) \leq (d - b)m.$$

This turns the available point deficit into an upper bound for the normal Hilbert coefficient of the actual center.

Proof. We first verify positivity of the deficit, then combine the exceptional threshold bounds with the intersection identities.

Step 1. In this step, we prove that a positive-dimensional minimal center has positive deficit. By Theorem 4.5, W is normal near x . Its global normalization $Z \rightarrow W$ is therefore an isomorphism near x , and the local rings and point ideals there agree. Choose its adjunction b-divisors $\mathcal{B}, \mathcal{M}$

by Theorem 4.1. Write $A_B(G)$ for the log discrepancy of a divisorial valuation G for the entire pair with boundary B . On a log resolution of the ambient boundary and point ideal, the deficit is a minimum of finitely many ratios

$$\frac{A_B(G)}{\text{ord}_G(\mathfrak{m}_x)} \quad \text{with} \quad \text{ord}_G(\mathfrak{m}_x) > 0.$$

Every such divisor has center $\{x\}$. If the minimum were zero, $\{x\}$ would be an lc center, contrary to the positive dimension of the minimal center W . Thus $b > 0$.

Step 2. In this step, we derive the normal Hilbert coefficient bound from the discriminant thresholds and intersection identities. Suppose $d \geq 2$, and use the normalized blow-up notation of Lemma 4.7 for Z . Set $\beta_i = \text{coeff}_{F_i} \mathcal{B}_T$. By Lemma 4.8,

$$br_i \leq 1 - \beta_i.$$

Choose V, H_V and $\mathcal{B}^{\text{str}}$ as in Lemma 4.9. The decomposition $\mathcal{B}_T = \mathcal{B}^{\text{str}} + \sum_i \beta_i F_i$ and Lemma 4.9 give

$$K_T \cdot H^{d-1} + \sum_i \beta_i w_i = -\mathcal{M}_V \cdot H_V^{d-1} - \mathcal{B}^{\text{str}} \cdot H^{d-1} \geq 0.$$

Multiply the threshold inequalities by $w_i > 0$ and sum. Since $r_i \geq 1$ and $\sum_i r_i w_i = m$, it follows that

$$\begin{aligned} bm &\leq \sum_i (1 - \beta_i) w_i \\ &\leq m + K_T \cdot H^{d-1} \\ &= dm - 2\bar{e}_1(\mathfrak{m}_{W,x}). \end{aligned}$$

This is the desired inequality in dimension at least two.

If $d = 1$, the normal local ring $\mathcal{O}_{W,x}$ is a discrete valuation ring. Its multiplicity is one and $\text{length}(\mathcal{O}_{W,x}/\mathfrak{m}_{W,x}^k) = k$, with all powers integrally closed. Hence its first normal Hilbert coefficient is zero. Applying Lemma 4.8 to the prime divisor x on Z , and using $\mathcal{B}_Z \geq 0$, gives

$$b \leq 1 - \text{coeff}_x \mathcal{B}_Z \leq 1.$$

Thus the asserted inequality also holds for $d = 1$. $\square$

The strict-transform contribution used above also measures divisorial lc subcenters. Retaining it gives the following strengthening.

Proposition 4.11. *Let B be effective rational on a smooth complex projective variety, log canonical near x , and let W be a log canonical center through x of dimension $d \geq 2$, normal near x . Suppose $b = \text{def}_x(B) > 0$. For any distinct log canonical centers $T_1, \dots, T_r$ through x of codimension one in W ,*

$$\sum_i \text{mult}_x T_i \leq (d - b) \text{mult}_x W - 2\bar{e}_1(\mathfrak{m}_{W,x}).$$

Here W need not be minimal for the current pair.

The deficit in this inequality belongs to the current boundary, even when the normal parent was minimal only at an earlier stage.

Proof. We identify each T_i in the base discriminant and retain its intersection contribution in the normal Hilbert calculation.

Step 1. In this step, we identify every divisorial subcenter T_i with a coefficient-one prime of the base discriminant. Let $\nu : Z \rightarrow W$ be the normalization. It is an isomorphism near x , so the closure in Z of $\nu^{-1}(T_i)$ near that point is a prime divisor P_i , with $\text{mult}_x P_i = \text{mult}_x T_i$. Choose the adjunction b-divisors $\mathcal{B}, \mathcal{M}$ in Theorem 4.1. Its description of the coefficient-at-least-one locus gives a divisor on a high model dominating P_i , since T_i is an lc center contained properly

in W . A proper birational morphism to the normal variety Z is an isomorphism at the generic point of P_i . Thus the discriminant coefficient of P_i on Z is at least one. The ambient pair is lc near this generic point, so the coefficient-greater-than-one assertion in the same theorem bounds it above by one. Consequently

$$\text{coeff}_{P_i} \mathcal{B}_Z = 1 \quad \text{for each } i.$$

Step 2. In this step, we retain the multiplicities of the T_i in the normal Hilbert intersection bound. Let $f : T \rightarrow Z$ be the normalized point blow-up. Write

$$\mathfrak{m}_x \mathcal{O}_T = \mathcal{O}_T(-F), \quad H = -F, \quad F = \sum_j a_j F_j,$$

and put

$$u_j = F_j \cdot H^{d-1}, \quad \beta_j = \text{coeff}_{F_j} \mathcal{B}_T, \quad m = \text{mult}_x W.$$

All a_j and u_j are positive integers. Use the strict transform $\mathcal{B}^{\text{str}}$ and the common model V , with divisor H_V , of Lemma 4.9. By (4.3) and base effectivity,

$$-\mathcal{B}^{\text{str}} \cdot H^{d-1} \geq \sum_i \text{mult}_x T_i.$$

The two intersection signs in Lemma 4.9 now imply

$$\begin{aligned} \sum_i \text{mult}_x T_i &\leq -\mathcal{M}_V \cdot H_V^{d-1} - \mathcal{B}^{\text{str}} \cdot H^{d-1} \\ &= K_T \cdot H^{d-1} + \sum_j \beta_j u_j. \end{aligned}$$

By Lemma 4.8, $\beta_j \leq 1 - ba_j$. The identities of Lemma 4.7 therefore give

$$\begin{aligned} K_T \cdot H^{d-1} + \sum_j \beta_j u_j &\leq K_T \cdot H^{d-1} + \sum_j u_j - bm \\ &\leq K_T \cdot H^{d-1} + (1 - b)m \\ &= (d - b)m - 2\bar{e}_1(\mathfrak{m}_{W,x}). \end{aligned}$$

These calculations use normality near x , not minimality or a Cohen–Macaulay assertion for the current parent. They prove the claimed inequality. $\square$

For use along an additive chain, we spell out why an old center persists. Suppose $B' \geq B$ and both pairs are lc near x . If a divisor G is an lc place of B with center W through x , write $A_B(G)$ and $A_{B'}(G)$ for its log discrepancies for the two pairs. Then

$$0 \leq A_{B'}(G) = A_B(G) - \text{ord}_G(B' - B) = -\text{ord}_G(B' - B) \leq 0.$$

Thus G is still an lc place for B' . If W was minimal for B , its normality near x follows from Theorem 4.5 and is unchanged by the addition. When $\text{def}_x(B') > 0$, Proposition 4.11 applies to this same parent with that later deficit. The later deficit is no larger than the earlier one, since every ideal pair which is lc after the addition is lc before it.

Proposition 4.12. *Let W be a minimal center of dimension $d \geq 2$ for an effective rational log canonical boundary B on a smooth projective variety, at the closed point x . Put $b = \text{def}_x(B)$ and $m = \text{mult}_x W$. On the normalized blow-up T of the point on the normalization $\bar{W}$, write*

$$\mathfrak{m}_x \mathcal{O}_T = \mathcal{O}_T(-F), \quad H = -F, \quad F = \sum_i r_i F_i, \quad w_i = F_i \cdot H^{d-1}.$$

Let $\mathcal{B}$ and $\mathcal{M}$ be the adjunction discriminant and moduli b -divisors. Write $\beta_i = \text{coeff}_{F_i} \mathcal{B}_T$, let $\mathcal{B}^{\text{str}}$ be the strict transform of $\mathcal{B}_{\bar{W}}$, and choose a smooth common model V with nef moduli trace

$\mathcal{M}_V$ and pulled-back divisor H_V . Then

$$(d-b)m - 2\bar{e}_1(\mathfrak{m}_{W,x}) = \sum_i (r_i - 1)w_i + \sum_i (1 - \beta_i - br_i)w_i \\ - \mathcal{M}_V \cdot H_V^{d-1} - \mathcal{B}^{\text{str}} \cdot H^{d-1}.$$

Each term on the right is nonnegative, and every w_i is a positive integer. In particular, if the left side is less than one, the exceptional divisor F is reduced.

This separates the four nonnegative contributions which were combined in Proposition 4.10.

Proof. All the intersections are unchanged on a further common model by the projection formula, so we may take V smooth and projective.

Step 1. In this step, we decompose the Hilbert defect into the four displayed contributions. Lemma 4.7 gives

$$m = \sum_i r_i w_i, \quad (d-b)m - 2\bar{e}_1(\mathfrak{m}_{W,x}) = K_T \cdot H^{d-1} + (1-b)m.$$

By Lemma 4.9 and the decomposition of $\mathcal{B}_T$,

$$K_T \cdot H^{d-1} = -\mathcal{M}_V \cdot H_V^{d-1} - \mathcal{B}^{\text{str}} \cdot H^{d-1} - \sum_i \beta_i w_i.$$

Substitute this identity into the preceding one. For each i ,

$$(1-b)r_i - \beta_i = (r_i - 1) + (1 - \beta_i - br_i),$$

which proves the exact decomposition.

Step 2. In this step, we prove nonnegativity of all four contributions and deduce reducedness when the defect is less than one. The first sum is nonnegative because $r_i \geq 1$ and $w_i \in \mathbb{Z}_{>0}$. Each summand of the second sum is nonnegative by Lemma 4.8. The last two terms are nonnegative by Lemma 4.9, with the moduli term computed on V and base effectivity used on $\bar{W}$.

If some $r_i \geq 2$, the first sum is at least $w_i \geq 1$, so the whole defect is at least one. A defect less than one therefore forces every $r_i = 1$. This makes F generically reduced. Since T is normal, it satisfies S_2 , and its effective Cartier divisor F satisfies S_1 . In particular F has no embedded associated primes. Generic reducedness then implies that F is reduced as a scheme. $\square$

Corollary 4.13. *A surface minimal center on X has multiplicity $m \leq 5$ and deficit at most $2/m$. A curve minimal center is smooth at x and has deficit at most one.*

For a surface, positivity of the deficit forces the Artinian reduction to have no terms in degree at least two.

Proof. Let W be a surface minimal center for the boundary B , let $R = \mathcal{O}_{W,x}$, and put $b = \text{def}_x(B) > 0$. By Theorem 4.5, R is Cohen–Macaulay, normal, and analytically unramified. Choose a general minimal reduction J of its maximal ideal, generated by a system of parameters forming an R -regular sequence, and let h_i be the Hilbert function of R/J . Since R is a quotient of the local ring of the smooth sixfold, $\text{edim } R \leq 6$. Theorem 4.6 yields

$$h_0 = 1, \quad h_1 = \text{edim } R - 2 \leq 4, \quad m = \sum_i h_i.$$

Its first moment can be written as

$$\sum_i i h_i = m - 1 + \sum_{i \geq 2} (i-1) h_i.$$

Combining the moment inequality with Proposition 4.10 gives

$$0 < bm \leq 2 - 2 \sum_{i \geq 2} (i-1) h_i.$$

The sum is a nonnegative integer and must be zero. Thus $h_i = 0$ for $i \geq 2$, whence $m = 1 + h_1 \leq 5$ and $b \leq 2/m$.

A curve minimal center is normal near x by Theorem 4.5. A normal complex curve is smooth there. Its multiplicity is one and its first normal Hilbert coefficient is zero, so Proposition 4.10 gives $b \leq 1$. $\square$

4.4. Associated graded rings and normal powers. We finish by recording the local algebra criteria used with these Hilbert bounds. The first statement combines Sally's minimal-multiplicity theorem and the Rees criterion in [Ver08, Theorem 2.7 and Corollary 6.6].

Theorem 4.14 (Associated graded and Rees criteria). *Let $(R, \mathfrak{n})$ be a Cohen–Macaulay local ring of dimension $d \geq 2$ with infinite residue field. If $e_0(\mathfrak{n}) = \text{edim } R - d + 1$, then for a minimal reduction J , one has $\mathfrak{n}^2 = J\mathfrak{n}$ and $\text{gr}_{\mathfrak{n}} R$ is Cohen–Macaulay. More generally, for an $\mathfrak{n}$ -primary ideal I and a minimal reduction J , the Rees algebra $R[It]$ is Cohen–Macaulay if and only if*

$$e_1(I) = \sum_{n=1}^{d-1} \text{length}(I^n / (J \cap I^n)).$$

Thus the Rees conclusion requires the displayed equality of reduction lengths. For a specified parameter sequence, regularity of its initial forms is characterized by the following intersection equalities [Ver08, Theorem 5.16].

Theorem 4.15 (Regular initial forms). *Let I be an ideal in a Noetherian local ring R , and let $x_1, \dots, x_s \in I \setminus I^2$ be an R -regular sequence. Their initial forms are a regular sequence on $\text{gr}_I R$ if and only if $(x_1, \dots, x_s) \cap I^n = (x_1, \dots, x_s)I^{n-1}$ for every $n \geq 1$.*

These equalities control passage from the local ring to its associated graded ring. Normality of the powers is governed by the next criterion [PUV05, Corollary 2.5]. Here $\mathfrak{n}$ denotes the maximal ideal of the local ring under consideration.

Theorem 4.16 (Normality of powers). *Let R be an analytically unramified Cohen–Macaulay local ring of dimension at least two, with a canonical module and infinite residue field. An $\mathfrak{n}$ -primary ideal I has every power integrally closed if and only if $R[It]$ satisfies S_2 and $e_1(I) = \bar{e}_1(I)$.*

The condition on the Rees algebra and the equality of Hilbert coefficients are both part of this criterion.

5. BOUNDS FOR ENTIRE CUTTING CHAINS

We retain the notation of Section 2. We bound the scalar cost of an entire additive chain, with its actual nested centers. The inequalities for successive centers are used before maximizing over their deficits. At the end, Proposition 3.7 converts these bounds into actual point boundaries.

5.1. Deficit drops and persistent centers. Fix an additive chain as in Definition 2.3, and write $b = b_0$. Effective addition decreases every log discrepancy, so the definition of the point-ideal threshold gives

$$b_0 \geq b_1 \geq \dots \geq b_N = 0.$$

In particular, for any real number a ,

$$(5.1) \quad C = ab + \sum_{i=0}^{N-1} (b_i - b_{i+1})(r_i - a).$$

Thus replacing any radius by an upper bound increases the estimated cost. If every radius is at most a , the sum telescopes to $C \leq ab$. More generally, if the initial portion of a chain ends at deficit z and all its radii are at most a , that portion costs at most $a(b - z)$. This applies even when dimensions are skipped or two consecutive deficits agree.

Every center that was minimal at an earlier stage remains an lc center at every later stage. Indeed, let F be an lc place with center V for the earlier boundary B_i . For $j \geq i$,

$$0 \leq A_{X, B_j}(F) = -\operatorname{ord}_F(B_j - B_i) \leq 0.$$

Consequently the added boundary has order zero at F , and F remains an lc place. The variety V is unchanged and is normal near x by Theorem 4.5. If a later minimal center $U \subset V$ has codimension one in V , Proposition 4.11, applied to the later boundary and its current deficit z , therefore gives

$$(5.2) \quad m(U) \leq (\dim V - z)m(V) - 2\bar{e}_1(\mathfrak{m}_{V,x}).$$

The parent in this inequality need not remain minimal.

All local rings of minimal centers used below are normal, Cohen–Macaulay, and essentially of finite type over $\mathbb{C}$. They are excellent and analytically unramified, and their residue field is infinite. Thus Theorem 4.6 applies. For an Artinian minimal reduction at a center of dimension d and multiplicity m , write h_i for its Hilbert function and $T = \sum_i ih_i$. Together with Proposition 4.10, this gives

$$(5.3) \quad \begin{aligned} h_0 &= 1, & h_1 &= \operatorname{edim}_x W - d \leq 6 - d, & \sum_i h_i &= m, \\ T &\leq \bar{e}_1(\mathfrak{m}_{W,x}), & 2T &\leq (d - b)m. \end{aligned}$$

Every contained center inherits an embedding-dimension bound on its parent, since the surjection of local rings induces a surjection of cotangent spaces. Also $g(W)$ is a positive integer, so $r(W) \leq m(W)^{1/d(W)}$.

Lemma 5.1. *Let a threefold minimal center on X have multiplicity m and deficit $z > 0$. Then $m \leq 9$. For $m = 8$, $z \leq 1/4$, every positive-deficit surface successor in an additive cutting chain is smooth, and the entire chain has $C \leq 2z$. For $m = 9$, $z \leq 1/9$, no positive-deficit surface successor occurs, and every entire chain has $C \leq (9/4)z$. Both assertions include direct jumps to a curve or to the point.*

These are the last two possible multiplicities of a threefold with positive deficit. Their surface successors are more restricted than the independent surface bounds would suggest.

Proof. We first bound the multiplicity, and then apply the flag inequality to the same threefold at a later surface stage.

Step 1. In this step, we bound the threefold multiplicity and its initial deficit. Since $h_1 \leq 3$,

$$T \geq h_1 + 2 \sum_{i \geq 2} h_i = 2m - 2 - h_1 \geq 2m - 5.$$

Consequently (5.3) gives

$$(5.4) \quad 2(2m - 5) \leq (3 - z)m, \quad z \leq \frac{10}{m} - 1, \quad m \leq 9.$$

The last conclusion uses $z > 0$ and the integrality of m .

Step 2. In this step, we bound surface successors and the complete tail at multiplicity eight. Suppose a surface of multiplicity t and deficit $w > 0$ occurs later in the chain. By (5.2),

$$(5.5) \quad t \leq (3 - w)m - 2\bar{e}_1(\mathfrak{m}_{W,x}) \leq 10 - m - mw.$$

At $m = 8$, the initial deficit is at most $1/4$, the normal Hilbert coefficient is at least 11, and $t \leq 2 - 8w < 2$. Hence $t = 1$. Its Artinian reduction has length one, so its embedding dimension equals its dimension; this surface is smooth at x . The threefold radius is at most $8^{1/3} = 2$, and every later surface or curve radius is at most one, using Corollary 4.13 for curves. Equation (5.1) yields

$$(5.6) \quad z \leq \frac{1}{4}, \quad C \leq 2z.$$

Step 3. In this step, we exclude positive-deficit surface successors and bound the complete tail at multiplicity nine. At $m = 9$, the initial deficit is at most $1/9$, the normal Hilbert coefficient is at least 13, and $t \leq 1 - 9w < 1$. No positive-deficit surface can occur. The initial radius is less than $9/4$, since $9 \cdot 4^3 = 576 < 729 = 9^3$. A subsequent curve has radius at most one. Therefore

$$(5.7) \quad z \leq \frac{1}{9}, \quad C \leq \frac{9}{4}z.$$

The same telescoping argument applies to a direct curve or point successor, so both bounds concern the complete chain. $\square$

5.2. The local tables. We record the remaining numerical data before using them in the chain estimates. The elementary moment bounds are

$$T \geq m - 1, \quad T \geq 2m - 2 - h_1, \quad T \geq 3m - 3 - 2h_1 - h_2.$$

For a fourfold, $h_1 \leq 2$ and $h_2 \leq 3$, so the last two bounds give $T \geq 2m - 4$ and $T \geq 3m - 10$. The latter and positive deficit imply $6m - 20 < 4m$, hence $m \leq 9$. Combining all three moment bounds with (5.3) gives the following deficit caps:

$$(5.8) \quad \begin{array}{c|ccccccccc} m & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline \beta_m (d=3) & 3 & 2 & 5/3 & 3/2 & 1 & 2/3 & 3/7 & 1/4 & 1/9 \\ \theta_m (d=4) & 4 & 3 & 8/3 & 2 & 8/5 & 4/3 & 6/7 & 1/2 & 2/9 \end{array}$$

Here β_m and θ_m denote only the numerical upper bounds in this table. The threefold range is exhaustive by Lemma 5.1.

For a fivefold, $h_1 \leq 1$. If $h_1 = 0$, its reduction is the residue field. Otherwise its associated graded Artinian algebra is a quotient of $\mathbb{C}[t]$, so its Hilbert function consists of m consecutive ones. In both cases

$$(5.9) \quad \bar{e}_1(\mathfrak{m}_{W,x}) \geq \frac{m(m-1)}{2}, \quad b \leq 6 - m, \quad m \leq 5.$$

In any dimension, multiplicity one forces $h_1 = 0$ and hence smoothness at x .

For later use, the surface moment inequality also gives the embedding dimension precisely. Set $J = \sum_{i \geq 2} (i-1)h_i$. Then

$$0 < bm \leq 2 - 2J.$$

Since J is a nonnegative integer, $J = 0$. Thus a surface center satisfies $m = 1 + h_1 = \text{edim}_x W - 1$ and $b \leq 2/m$. This refines Corollary 4.13 when an embedding dimension bound is inherited.

Lemma 5.2. *Let B be an effective rational boundary on X , log canonical near x , with a positive-dimensional minimal center of deficit b . Consider an additive cutting chain from B to the point. If $1 \leq a \leq 9/4$ and every center in the chain of dimension at least three has radius at most a , then*

$$C \leq ab + \frac{9}{10} - \frac{2}{5}a.$$

If every center has embedding dimension at most four and $1 \leq a \leq 7/4$, under the same radius hypothesis one has

$$C \leq ab + \frac{2}{3} \left(\frac{7}{4} - a \right).$$

Only a surface can contribute an excess over the reference radius in these inequalities. Its complete curve continuation is included.

Proof. We first bound a complete surface tail and then compare its excess over the reference radius in the two embedding-dimension ranges.

Step 1. In this step, we bound a complete surface tail by a rational majorant of its radius. For surface multiplicities $t = 1, \dots, 5$, use

$$q_t = 1, \frac{17}{12}, \frac{7}{4}, 2, \frac{9}{4}, \quad \text{respectively.}$$

These dominate $\sqrt{t}$, since $17^2 = 289 > 288 = 2 \cdot 12^2$, $7^2 = 49 > 48 = 3 \cdot 4^2$, and $9^2 = 81 > 80 = 5 \cdot 4^2$. If a surface begins with deficit z and a later curve has deficit y , Corollary 4.13 and telescoping give

$$(5.10) \quad C_{\text{surface}} \leq q_t(z - y) + y \leq q_t z.$$

The formula includes a direct point step by taking $y = 0$.

Step 2. In this step, we bound the surface surplus without an embedding-dimension restriction. For $1 \leq a \leq 9/4$, we claim

$$(5.11) \quad \max_{1 \leq t \leq 5} \frac{2}{t} (q_t - a)_+ \leq \frac{9}{10} - \frac{2}{5} a.$$

The right side is nonnegative. The $t = 1$ term is zero, and $t = 5$ gives equality. For $t = 2, 3, 4$, subtracting the untruncated expression from the right side gives, respectively,

$$\frac{3}{5}a - \frac{31}{60}, \quad \frac{4}{15}(a - 1), \quad \frac{1}{10}(a - 1),$$

all nonnegative. If the expression is truncated to zero, the nonnegativity of the right side proves the same comparison.

A strictly decreasing chain contains at most one surface. If it has deficit z and multiplicity t , the earlier portion costs at most $a(b - z)$, and its remaining tail costs at most $q_t z$. Its excess over ab is at most $(q_t - a)_+ z \leq (2/t)(q_t - a)_+$. If there is no surface, all radii are at most a . This proves the first assertion.

Step 3. In this step, we sharpen the surface surplus when the embedding dimension is at most four. With embedding dimension at most four, surface multiplicities are at most three. For $1 \leq a \leq 7/4$, their analogous comparison is

$$(5.12) \quad \max_{1 \leq t \leq 3} \frac{2}{t} (q_t - a)_+ \leq \frac{2}{3} \left(\frac{7}{4} - a \right).$$

The $t = 3$ term gives equality. For $t = 2$, the difference from the untruncated expression is $a/3 - 1/4 \geq 1/12$; the $t = 1$ term is zero. The preceding chain argument proves the second assertion as well. $\square$

Lemma 5.3. *Let an effective rational boundary on X be log canonical near x , with a positive-dimensional minimal center W of dimension at most three and deficit $b > 0$. Every additive cutting chain to the point satisfies $C \leq b + 87/80$. If W is a threefold of multiplicity m , the same chain has $b \leq \beta_m$ and $C \leq c_m b + d_m$, for the following data:*

$$(5.13) \quad \begin{array}{c|c|c|c} m & \beta_m & c_m & d_m \\ \hline 1 & 3 & 1 & 1/2 \\ 2 & 2 & 19/15 & 29/120 \\ 3 & 5/3 & 13/9 & 11/54 \\ 4 & 3/2 & 127/80 & 33/160 \\ 5 & 1 & 7/4 & 1/20 \\ 6 & 2/3 & 2 & 0 \\ 7 & 3/7 & 2 & 0 \\ 8 & 1/4 & 2 & 0 \\ 9 & 1/9 & 9/4 & 0 \end{array}$$

The intercepts in the table measure the possible increase of the radius at a surface stage. They use the deficit allowed by the actual parent.

Proof. We bound the parent radius, use its flag to control every possible surface, and then maximize the excess over the initial deficit.

Step 1. In this step, we bound the parent radius and express the chain cost through a possible surface surplus. The deficit caps were proved in (5.8). The parent radius is at most c_m : the nontrivial cube comparisons are

$$\begin{aligned} 19^3 &= 6859 > 6750 = 2 \cdot 15^3, & 13^3 &= 2197 > 2187 = 3 \cdot 9^3, \\ 127^3 &= 2048383 > 2048000 = 4 \cdot 80^3, & 7^3 &= 343 > 320 = 5 \cdot 4^3, \\ 2^3 &= 8, & 9^3 &= 729 > 576 = 9 \cdot 4^3. \end{aligned}$$

Use the surface majorants q_t in the proof of Lemma 5.2. For a surface successor of multiplicity t and deficit w , the whole chain is bounded by

$$(5.14) \quad C \leq c_m(b - w) + q_t w = c_m b + (q_t - c_m)w.$$

If a surface is skipped, its excess is zero.

Step 2. In this step, we compute the threefold-tail intercepts for multiplicities one through four. For $m = 1$, the first surface comparison with $a = 1$ gives $d_1 = 1/2$. For $m = 2$, the inequality $1 + h_1 \leq m$ gives embedding dimension at most four, so $t \leq 3$. The flag inequality and $\bar{e}_1 \geq 1$ give $t \leq 4 - 2w$. Thus the positive excesses for $t = 2, 3$ are at most

$$\frac{17}{12} - \frac{19}{15} = \frac{3}{20}, \quad \frac{1}{2} \left(\frac{7}{4} - \frac{19}{15} \right) = \frac{29}{120}.$$

For $m = 3$, the embedding dimension is at most five, so $t \leq 4$. Since $\bar{e}_1 \geq 2$, the flag gives $t \leq 5 - 3w$. The positive excesses, at $t = 3, 4$, are

$$\frac{2}{3} \left(\frac{7}{4} - \frac{13}{9} \right) = \frac{11}{54}, \quad \frac{1}{3} \left(2 - \frac{13}{9} \right) = \frac{5}{27}.$$

For $m = 4$, use $\bar{e}_1 \geq 3$ to obtain $t \leq 6 - 4w$. For $t = 3, 4, 5$, the allowed deficits are at most $2/3, 1/2, 1/4$, respectively, and the excesses are

$$\frac{13}{120}, \quad \frac{33}{160}, \quad \frac{53}{320}.$$

Smaller surface multiplicities have radius below c_4 . The maxima of these lists give the first four intercepts.

Step 3. In this step, we compute the complete-tail bounds for threefold multiplicities five through nine. For $m \geq 5$, (5.5) gives $t \leq 10 - m - mw$. At $m = 5$, only $t \leq 4$ occurs, and $t = 4$ requires $w \leq 1/5$. The sole positive excess over $7b/4$ is at most $(2 - 7/4)/5 = 1/20$. At $m = 6, 7$, surface multiplicities are at most three and two, respectively; all radii are at most two. The final two rows are exactly (5.6) and (5.7). Curves and direct point steps have already been included in every comparison.

Step 4. In this step, we obtain the uniform bound for the surplus over b . Subtracting b and using the deficit caps gives, in increasing multiplicity order,

$$\frac{1}{2}, \quad \frac{31}{40}, \quad \frac{17}{18}, \quad \frac{87}{80}, \quad \frac{4}{5}, \quad \frac{2}{3}, \quad \frac{3}{7}, \quad \frac{1}{4}, \quad \frac{5}{36}.$$

Their maximum is $87/80$. Curves and surfaces have excess at most $1/2$ by Lemma 5.2 with $a = 1$. $\square$

Lemma 5.4. *Let an effective rational boundary on X be log canonical near x , with a fourfold minimal center of multiplicity $n \in \{2, 3, 4, 5, 6\}$ and deficit $b > 0$. Every additive cutting chain*

to the point satisfies $C \leq a_n b + P_n$, with

$$(5.15) \quad \begin{array}{c|c|c|c|c} n & \theta_n & a_n & P_n & D_n \\ \hline 2 & 3 & 6/5 & 19/32 & 6 \\ 3 & 8/3 & 79/60 & 817/1440 & 8 \\ 4 & 2 & 17/12 & 181/480 & 8 \\ 5 & 8/5 & 3/2 & 3/10 & 8 \\ 6 & 4/3 & 8/5 & 13/50 & 8 \end{array}$$

Here θ_n is the deficit cap in (5.8); the constants D_n bound a threefold successor of multiplicity m and deficit $z > 0$ by $m \leq D_n - nz$.

The successor restriction in the last column is imposed before taking the maximum that defines the intercept.

Proof. We derive the allowed successor deficits and then maximize the surpluses of their complete tails.

Step 1. In this step, we bound the fourfold radii and the deficits of their threefold successors. The moment bounds give lower bounds 1, 2, 4, 6, 8 for $\bar{e}_1$ in these five rows. Subtracting twice these numbers from $4n$ gives D_n , by (5.2). Thus a positive-deficit threefold successor must satisfy

$$(5.16) \quad m < D_n, \quad 0 < z \leq \min \left\{ \beta_m, \frac{D_n - m}{n} \right\}.$$

The parent radii are at most a_n , as follows from

$$\begin{aligned} 6^4 &= 1296 > 1250 = 2 \cdot 5^4, & 79^4 &= 38950081 > 38880000 = 3 \cdot 60^4, \\ 17^4 &= 83521 > 82944 = 4 \cdot 12^4, & 3^4 &= 81 > 80 = 5 \cdot 2^4, & 8^4 &= 4096 > 3750 = 6 \cdot 5^4. \end{aligned}$$

Step 2. In this step, we maximize the complete successor surpluses to obtain the fourfold intercepts. If $c_m > a_n$, the whole threefold tail from Lemma 5.3, together with the initial deficit drop, bounds the excess over $a_n b$ by

$$(5.17) \quad (c_m - a_n) \min \left\{ \beta_m, \frac{D_n - m}{n} \right\} + d_m.$$

If $c_m \leq a_n$, all stages of dimension at least three have radius at most a_n , so Lemma 5.2 gives instead $9/10 - 2a_n/5$. This also covers a direct surface, curve, or point successor.

The following are all the candidates. Each row lists the surface comparison first, followed by (5.17) for the indicated threefold multiplicities:

$$(5.18) \quad \begin{array}{c|c|c|c} n & 9/10 - 2a_n/5 & m \text{ with } c_m > a_n & \text{successor excesses} \\ \hline 2 & 21/50 & 2, 3, 4, 5 & 3/8, 77/135, 19/32, 13/40 \\ 3 & 28/75 & 3, 4, 5, 6, 7 & 5/12, 817/1440, 29/60, 41/90, 41/180 \\ 4 & 1/3 & 3, 4, 5, 6, 7 & 103/432, 181/480, 3/10, 7/24, 7/48 \\ 5 & 3/10 & 4, 5, 6, 7 & 221/800, 1/5, 1/5, 1/10 \\ 6 & 13/50 & 5, 6, 7 & 1/8, 2/15, 1/15 \end{array}$$

For example, the largest candidate in the second row is

$$\left(\frac{127}{80} - \frac{79}{60} \right) \frac{4}{3} + \frac{33}{160} = \frac{817}{1440}.$$

The maxima of the five rows are exactly P_n . Every omitted multiplicity either has $c_m \leq a_n$ or violates the positive deficit condition in (5.16). In particular $m = 8, 9$, although possible threefold multiplicities, cannot occur in these rows because $D_n \leq 8$. This proves the asserted bounds for the entire chain. $\square$

5.3. Uniform bounds for the entire chain.

Proposition 5.5. *For any positive-dimensional minimal center W of an effective rational boundary on X , let b be its deficit. Every additive cutting chain from this boundary to the point satisfies the following bounds, under the indicated hypotheses:*

<i>hypothesis on W</i>	<i>upper bound for C</i>
<i>none</i>	$\frac{5}{3}b + \frac{113}{480}$
$\text{edim}_x W \leq 4$	$b + \frac{77}{90}$
$\dim W \leq 4$	$b + \frac{2033}{1440}$
$\dim W \leq 4, \text{edim}_x W \leq 5$	$b + \frac{3887}{3360}$
$\dim W = 5, W \text{ smooth at } x$	$b + \frac{3887}{3360}$
$\dim W = 5, \text{mult}_x W = 2$	$\frac{23}{20}b + \frac{1417}{1440}$
$\dim W = 5, \text{mult}_x W = 3$	$\frac{5}{4}b + \frac{57}{80}$

Every bound concerns the whole scalar cost of the same actual chain, including chains that omit an intermediate dimension.

The first bound is independent of the initial center. The other bounds retain its dimension, embedding dimension, or multiplicity throughout the comparison with its actual descendants.

Proof. We prove the universal estimate first, then the estimates with an inherited embedding-dimension bound, and finally the two fivefold estimates. All smaller tails below are portions of the fixed chain. The deficit of its first proper center is denoted by z , with $z = 0$ when that center is the point.

Step 1. In this step, we prove the universal affine bound for the entire cutting chain. Put $A = 5/3$ and $\eta = 113/480$. All radii in any chain are at most $9/4$. For surfaces and threefolds this follows from their multiplicity bounds. For fourfolds use $9^{1/4} = \sqrt{3} < 7/4$, and for fivefolds use $5^{1/5} < 3/2$, since $243/32 > 5$. Hence

$$(5.19) \quad C \leq \frac{9}{4}b.$$

Every chain beginning in dimension at most three satisfies $C \leq Ab + 7/30$. For curves, surfaces, and threefolds of multiplicity at most four, this is Lemma 5.2 with $a = A$, since $4^{1/3} < 8/5 < A$ and $8^3 > 4 \cdot 5^3$. For multiplicity five, the threefold table gives

$$C - Ab \leq \left(\frac{7}{4} - \frac{5}{3}\right)b + \frac{1}{20} \leq \frac{2}{15} < \frac{7}{30}.$$

For multiplicities six, seven, and eight, the complete bound $C \leq 2b$ and $b \leq 2/3$ give excess at most $2/9 < 7/30$. For multiplicity nine, Lemma 5.1 gives excess at most $(9/4 - 5/3)/9 = 7/108 < 7/30$.

For a fourfold of multiplicity at most seven, its radius is less than A , because $A^4 = 625/81 > 7$. Its initial deficit drop and its entire smaller tail therefore give $C \leq Ab + 7/30$. For a fourfold of multiplicity eight, use $b \leq 1/2$ and $\bar{e}_1 \geq 3 \cdot 8 - 10 = 14$. Any threefold successor of deficit $z > 0$ has

$$m_{\text{child}} \leq (4 - z)8 - 28 = 4 - 8z < 4.$$

Its radius is consequently less than $8/5$. The initial radius is at most $\alpha_8 = 27/16$, since $27^4 = 531441 > 524288 = 8 \cdot 16^4$. Thus every stage of dimension at least three in this chain has radius at most α_8 , including a chain that skips the threefold. The surface comparison gives

$$C - Ab \leq (\alpha_8 - A)b + \frac{9}{10} - \frac{2}{5}\alpha_8 \leq \frac{1}{96} + \frac{9}{40} = \frac{113}{480}.$$

For multiplicity nine, $b \leq 2/9$ and (5.19) give excess at most $(9/4 - 5/3)(2/9) = 7/54 < 113/480$. Since $7/30 = 112/480 < \eta$, the first bound is proved in dimension at most four. A fivefold has

radius less than $3/2 < A$, and its first proper center has dimension at most four. Absorbing the initial deficit drop at radius A proves

$$(5.20) \quad C \leq \frac{5}{3}b + \frac{113}{480}$$

in every possible dimension.

Step 2. In this step, we bound complete chains whose centers inherit embedding dimension at most four. This bound is inherited by every center in the chain. A fourfold is smooth. For a threefold its Artinian embedding dimension is at most one; the consecutive-ones calculation gives $b \leq 4 - m$, hence $m = 1, 2, 3$, with deficit caps 3, 2, 1. Surfaces have multiplicity at most three, and curves are smooth. In particular,

$$(5.21) \quad r_i \leq \frac{7}{4} \quad \text{for every stage,} \quad C \leq \frac{7}{4}b.$$

For the threefold rows, use the respective reference radii 1, 19/15, 13/9 in the second assertion of Lemma 5.2. Their excesses over b , after applying the caps 3, 2, 1, are

$$\frac{1}{2}, \quad 2\left(\frac{19}{15} - 1\right) + \frac{2}{3}\left(\frac{7}{4} - \frac{19}{15}\right) = \frac{77}{90}, \quad \frac{4}{9} + \frac{2}{3}\left(\frac{7}{4} - \frac{13}{9}\right) = \frac{35}{54}.$$

Curves and surfaces have excess at most $1/2$. A smooth fourfold has radius at most one, so adjoining its initial deficit drop preserves the bound $C \leq b + 77/90$.

Step 3. In this step, we bound the surplus for every initial center of dimension at most four. Dimension at most three was treated in Lemma 5.3. Smooth fourfolds are covered by the preceding embedding-dimension bound. For fourfold multiplicities $2, \dots, 6$, (5.15) and the deficit caps give the following respective bounds for $C - b$:

$$\frac{191}{160}, \quad \frac{2033}{1440}, \quad \frac{581}{480}, \quad \frac{11}{10}, \quad \frac{53}{50}.$$

For multiplicities seven, eight, and nine, use $b \leq 6/7$ and (5.20) to get $C - b \leq 4/7 + 113/480 < 1$. The largest of all these bounds is $2033/1440$.

Step 4. In this step, we prove the embedding-dimension-five bound and its consequence for a smooth fivefold. Write $\delta = 3887/3360$. In dimension at most three, the bound $87/80 < \delta$ suffices. For a fourfold of multiplicity n , its Artinian embedding dimension is at most one, so

$$\bar{e}_1 \geq \frac{n(n-1)}{2}, \quad b \leq 5 - n, \quad n \leq 4.$$

The smooth case was already treated. At $n = 4$, the universal estimate and $b \leq 1$ give $C - b \leq 433/480 < \delta$.

Consider $n = 2$. Use the reference radius $a_2 = 6/5$, with $b \leq 3$. A threefold successor of multiplicity m and deficit $z > 0$ satisfies $m \leq 6 - 2z$, and hence $m \leq 5$. We need a sharper comparison for its cubic row. For any threefold minimal center of multiplicity three and deficit z , its embedding dimension is at most five, its normal Hilbert coefficient is at least two, and a surface successor of multiplicity t and deficit w satisfies

$$t \leq 4, \quad t \leq 5 - 3w.$$

Thus the surface deficits for $t = 1, 2, 3, 4$ are at most 2, 1, $2/3$, $1/3$. The sharper radius majorants $\hat{c} = 577/400$ for the threefold and $\hat{q} = 97/56$ for a multiplicity-three surface are valid:

$$577^3 = 192100033 > 192000000 = 3 \cdot 400^3, \quad 97^2 = 9409 > 9408 = 3 \cdot 56^2.$$

Surface multiplicities at most two have radius below $\hat{c}$. The two possible positive excesses over $\hat{c}z$ are

$$\frac{2}{3}(\hat{q} - \hat{c}) = \frac{811}{4200}, \quad \frac{1}{3}(2 - \hat{c}) = \frac{223}{1200}, \quad \frac{811}{4200} - \frac{223}{1200} = \frac{61}{8400} > 0.$$

Using (5.10) and the initial deficit drop, this proves for every such cubic threefold tail that

$$(5.22) \quad C_3 \leq \frac{577}{400}z + \frac{811}{4200}.$$

This argument used only that threefold and its actual descendants.

Return to the double fourfold. If $m = 3$, the parent flag gives $z \leq 3/2$; subtracting a_2z from (5.22) gives at most

$$\frac{97}{400} \cdot \frac{3}{2} + \frac{811}{4200} = \frac{1871}{3360}.$$

The direct-surface comparison, also covering $m = 1$, is $21/50$, and the $m = 2$ comparison in (5.18) is $3/8$. For $m = 4$, inherited embedding dimension at most five gives $\bar{e}_1 \geq 2m - 4 = 4$, so $z \leq 1$. A surface successor satisfies $t \leq 4 - 4w < 4$; at $t = 3$ one has $w \leq 1/4$. Comparing the threefold radius with $8/5$ and surface radii with $1, 17/12, 7/4$ gives

$$C_3 \leq \frac{8}{5}z + \frac{3}{80}, \quad C_3 - \frac{6}{5}z \leq \frac{7}{16}.$$

For $m = 5$, the inherited embedding dimension gives $\bar{e}_1 \geq 6$, and its surface flag is $t \leq 3 - 5w < 3$. All radii in this threefold tail are at most $7/4$. Since its double-fourfold parent gives $z \leq 1/2$, its excess over $6z/5$ is at most $11/40$. The largest of the five bounds

$$\frac{21}{50}, \quad \frac{3}{8}, \quad \frac{1871}{3360}, \quad \frac{7}{16}, \quad \frac{11}{40}$$

is $1871/3360$. Adding $(a_2 - 1)b \leq 3/5$ proves $C - b \leq 3887/3360$ in the double-fourfold case.

For $n = 3$, one has $b \leq 2$ and $\bar{e}_1 \geq 3$. A threefold successor satisfies $m \leq 6 - 3z$. Use $a_3 = 79/60$ and (5.13), with $z \leq \min\{\beta_m, (6 - m)/3\}$. The excesses over the initial b , for the surface comparison and then $m = 3, 4, 5$, are respectively

$$\frac{151}{150}, \quad \frac{521}{540}, \quad \frac{1469}{1440}, \quad \frac{149}{180}.$$

For instance, the $m = 4$ expression is $19/30 + (13/48)(2/3) + 33/160 = 1469/1440$. Rows $m = 1, 2$ have radius below a_3 and are covered by the surface comparison; $m \geq 6$ is impossible at positive deficit. All four bounds are less than δ . This proves the fourth row of the proposition. For a smooth initial fivefold, its radius is at most one and every proper center has embedding dimension at most five. The same estimate on its entire proper tail therefore proves the fifth row.

Step 5. In this step, we bound every complete continuation from a fivefold of multiplicity two. The initial radius is at most $\alpha = 23/20$, since $23^5 = 6436343 > 6400000 = 2 \cdot 20^5$. Its normal Hilbert coefficient is at least one. If the first proper center is a fourfold of multiplicity n and deficit $z > 0$, the flag at this same fivefold gives

$$(5.23) \quad n \leq 8 - 2z, \quad z \leq \min \left\{ b, \theta_n, \frac{8 - n}{2} \right\}, \quad n \leq 7.$$

For $n = 1$, use the proved bound $C_4 \leq z + 77/90$. For $n = 2, \dots, 6$, use the whole-tail fourfold table; for $n = 7$, use (5.20). This gives

n	upper bound used for z	$C_4 - (23/20)z$ is at most
1	4	77/90
2	3	119/160
3	5/2	1417/1440
4	2	437/480
5	3/2	33/40
6	1	71/100
7	1/2	79/160

(5.24)

For example, the cubic fourfold contributes $(79/60 - 23/20)(5/2) + 817/1440 = 1417/1440$. The smooth row discards the nonpositive term $(1 - \alpha)z$. The maximum of the seven rows is $1417/1440$.

If the first proper center is a threefold, the bounds from (5.13) give the following surpluses over $23z/20$, in multiplicity order:

$$(5.25) \quad \frac{1}{2}, \frac{19}{40}, \frac{25}{36}, \frac{69}{80}, \frac{13}{20}, \frac{17}{30}, \frac{51}{140}, \frac{17}{80}, \frac{11}{90}.$$

Their maximum is $69/80 < 1417/1440$, and the last two entries use Lemma 5.1. A direct surface contributes at most $9/10 - (2/5)(23/20) = 11/25$. A direct curve or point has no positive surplus. Adding the initial cost at most $\alpha(b - z)$ proves the sixth row of the proposition.

Step 6. In this step, we bound every complete continuation from a fivefold of multiplicity three. Its deficit is at most three, its normal Hilbert coefficient is at least three, and its radius is less than $5/4$, since $5^5 = 3125 > 3072 = 3 \cdot 4^5$. A fourfold successor satisfies $n \leq 9 - 3z$, so $n \leq 8$. For a smooth fourfold, the low-embedding estimates already proved give

$$C_4 \leq \min \left\{ \frac{7}{4}z, z + \frac{77}{90} \right\}.$$

After subtracting $5z/4$, the two lines meet at $z = 154/135$, where their common value is $77/135$. This is their maximum, since the two slopes have opposite signs. For the remaining fourfolds, impose $z \leq \min\{\theta_n, (9 - n)/3\}$ before using the fourfold table and, at $n = 7, 8$, the universal estimate. The full list is

$$(5.26) \quad \begin{array}{c|c} n & C_4 - (5/4)z \text{ is at most} \\ \hline 1 & 77/135 \\ 2 & 19/32 \\ 3 & 1009/1440 \\ 4 & 943/1440 \\ 5 & 19/30 \\ 6 & 61/100 \\ 7 & 739/1440 \\ 8 & 539/1440 \end{array}$$

For $n = 2$ the fourfold slope is below $5/4$, so its intercept $19/32$ suffices. Every entry is at most $57/80$. For a direct threefold successor, the nine surpluses are

$$\frac{1}{2}, \frac{11}{40}, \frac{19}{36}, \frac{57}{80}, \frac{11}{20}, \frac{1}{2}, \frac{9}{28}, \frac{3}{16}, \frac{1}{9}.$$

Again the last two are the completed large-threefold cases. The maximum is $57/80$. A direct surface contributes at most $9/10 - (2/5)(5/4) = 2/5$, and curves or points contribute no positive excess. The initial deficit drop is absorbed at radius $5/4$, proving the final row. Every possible first successor and its complete continuation has now been included. $\square$

Proposition 5.6. *Let an effective rational boundary on X be log canonical near x , with a threefold minimal center W and deficit $b > 0$. Every additive cutting chain to the point satisfies the following bounds:*

- (i) *If $\text{mult}_x W = 3$, then $b \leq 5/3$ and $C \leq (577/400)b + 811/4200$.*
- (ii) *If $\text{mult}_x W = 4$ and $\text{edim}_x W = 5$, then $b \leq 1$ and $C \leq (127/80)b + 7/192$.*

The first improvement uses the cubic parent's restriction on multiplicity-four surfaces. The second uses the Artinian Hilbert function of the specified quartic threefold.

Proof. For (i), the cap is $\beta_3 = 5/3$, and the full scalar bound is (5.22). That argument applies to any cubic threefold minimal center, as its only inputs were the inherited embedding dimension, the flag $t \leq 5 - 3w$, and the two displayed rational radius comparisons.

For (ii), the Artinian reduction has length four and Hilbert function $(1, 2, 1)$. Indeed, its degree-zero and degree-one values are one and two; the remaining value must be in degree two because a standard graded Hilbert function has no gaps before it vanishes. Hence $\bar{e}_1 \geq 2 + 2 = 4$, and

$$8 \leq (3 - b)4, \quad b \leq 1.$$

At any surface stage, the same parent flag gives $t \leq 4 - 4w < 4$. Thus $t \leq 3$, and $t = 3$ requires $w \leq 1/4$. The initial radius is at most $127/80$, as checked in the threefold table. Surfaces of multiplicity at most two have radius below that number, and $\sqrt{3} < 26/15$, since $26^2 = 676 > 675 = 3 \cdot 15^2$. There is at most one surface, so

$$C \leq \frac{127}{80}b + \frac{1}{4} \left(\frac{26}{15} - \frac{127}{80} \right) = \frac{127}{80}b + \frac{7}{192}.$$

If the chain jumps to a curve or to the point, the additional $7/192$ is unnecessary. This proves (ii). $\square$

5.4. Point boundaries with fixed margins.

Corollary 5.7. *Let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be log canonical near x with a positive-dimensional minimal center, point multiplicity M , and deficit b . Suppose $s \leq (65/64)M + e$ with $e < 1/320$. If $b \leq 1$, its entire boundary admits an additive point continuation of total coefficient less than $7 - 1/100$. If its minimal center has embedding dimension at most four, the total coefficient can be taken less than $7 - 1/32$.*

The initial coefficient and multiplicity in this statement refer to the entire boundary. The error allowance is chosen only after its uniform scalar margin has been established.

Proof. The exceptional divisor of the blow-up of x in the smooth sixfold has log discrepancy six, point-ideal order one, and boundary order M . Testing the deficit against it gives

$$(5.27) \quad M + b \leq 6.$$

Step 1. In this step, we construct the additive point continuation when the deficit is at most one. If $b \leq 1$, Proposition 5.5 therefore gives

$$\begin{aligned} s + C &\leq \frac{65}{64}(6 - b) + e + \frac{5}{3}b + \frac{113}{480} \\ &\leq \frac{6701}{960} + e = 7 - \frac{19}{960} + e. \end{aligned}$$

Choose total cutting allowance $h = 1/640$ in Proposition 3.7. Its actual point boundary has coefficient strictly less than

$$7 - \frac{19}{960} + \frac{1}{320} + \frac{1}{640} = 7 - \frac{29}{1920} < 7 - \frac{1}{100}.$$

Step 2. In this step, we obtain the stronger margin under the embedding-dimension-four hypothesis. If the initial embedding dimension is at most four, the corresponding scalar bound gives instead

$$s + C \leq \frac{195}{32} - \frac{b}{64} + \frac{77}{90} + e < 7 - \frac{137}{2880}.$$

Take $h = 1/100$. Since $137/2880 - 1/32 = 47/2880 > 1/100$, Proposition 3.7 gives coefficient less than $7 - 1/32$. In both cases its hypotheses concern every chain from the fixed boundary, so the estimates apply to the chain produced by the adaptive error choices without changing B . $\square$

Proposition 5.8. *If an actual effective rational divisor $D \sim_{\mathbb{Q}} qL$, $q > 0$, satisfies $\text{mult}_x D/q \geq 140/99$, there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/630$.*

This criterion starts with a global effective divisor. Its whole threshold divisor is included in the coefficient being estimated.

Proof. We threshold the given divisor, bound every possible center and its whole tail, and then choose the cutting errors inside the resulting margin. Put

$$\gamma = \frac{q}{\text{mult}_x D} \leq \frac{99}{140}, \quad \kappa = \frac{2039}{2880}, \quad D_0 = \frac{6}{\text{mult}_x D} D \sim_{\mathbb{Q}} 6\gamma L.$$

Step 1. In this step, we threshold the given divisor and relate its coefficient to the resulting deficit. Take the actual threshold $\tau = \text{lt}_x(X; D_0)$, and put $B_0 = \tau D_0$, $s_0 = 6\gamma\tau$. On a log resolution the threshold is a positive rational number, and B_0 is lc near x but not klt at x . The point blow-up gives $\tau \leq 1$. If the minimal center is the point, the coefficient $0 < s_0 \leq 6\gamma < 9/2$ already suffices.

Otherwise write $b > 0$ for its deficit. The same point blow-up gives

$$(5.28) \quad b \leq 6 - 6\tau, \quad s_0 \leq \gamma(6 - b).$$

Step 2. In this step, we bound the total coefficient for centers of dimension at most four and for smooth or double fivefolds. We have $\gamma < \kappa < 3/4$. Write $A = 5/3$, $\eta = 113/480$, and $K = 113/80$. Proposition 5.5 implies the convenient relaxed bounds

$$(5.29) \quad \begin{aligned} C &\leq \min\{Ab + \eta, b + K\} \quad (\dim W \leq 4), \\ C &\leq b + \frac{6}{5} \quad (W \text{ a smooth fivefold}). \end{aligned}$$

Here $2033/1440 < 113/80$ and $3887/3360 < 6/5$. If the initial center has dimension at most four, then $b \leq 4$, so (5.28) gives

$$s_0 + C \leq 4 + 2\gamma + K < \frac{553}{80} < \frac{279}{40}.$$

For a smooth fivefold, $b \leq 5$, and the bound is

$$s_0 + C \leq 5 + \gamma + \frac{6}{5} < \frac{139}{20} < \frac{279}{40}.$$

For a double fivefold, $b \leq 4$, and its full scalar estimate gives

$$s_0 + C \leq 2\gamma + \frac{23}{5} + \frac{1417}{1440} = 7 - 2(\kappa - \gamma).$$

Step 3. In this step, we bound the complete tails of fivefolds of multiplicities three, four, and five. It remains to include fivefold multiplicities three, four, and five. Use the respective radius majorants $a = 5/4, 4/3, 7/5$. Their fifth-power comparisons are

$$3125 > 3072 = 3 \cdot 4^5, \quad 1024 > 972 = 4 \cdot 3^5, \quad 16807 > 15625 = 5 \cdot 5^5.$$

The first proper center has dimension at most four. For any $1 < a < A$, its deficit $z \geq 0$ satisfies

$$(5.30) \quad \min\{Az + \eta, z + K\} - az \leq (A - a) \frac{113}{64} + \eta.$$

Indeed, the two affine expressions meet at $z = (K - \eta)/(A - 1) = 113/64$; the first increases and the second decreases as functions of z . For the three values of a , the maxima are $1243/1280, 791/960, 113/160$. Combining the initial cost at most $a(b - z)$, $b \leq 6 - m$, and (5.28) gives

m	$s_0 + C$ is at most	strict upper bound using $\gamma < 3/4$
3	$3\gamma + 15/4 + 1243/1280$	$8923/1280$
4	$4\gamma + 8/3 + 791/960$	$6231/960$
5	$5\gamma + 7/5 + 113/160$	$937/160$

Each is less than $279/40$; in the closest comparison, $279/40 = 8928/1280$. These cases exhaust the initial center by (5.9). The complete tails in (5.29) already include both large-threefold rows and every skipped dimension.

Step 4. In this step, we choose the cutting errors inside the uniform coefficient margin. Thus every chain from this fixed B_0 has

$$s_0 + C \leq K_\gamma := \max \left\{ \frac{279}{40}, 7 - 2(\kappa - \gamma) \right\}.$$

The exact difference

$$\kappa - \gamma \geq \frac{2039}{2880} - \frac{99}{140} = \frac{17}{20160}$$

shows that $K_\gamma \leq 7 - 17/10080$, since $279/40 = 7 - 1/40 < 7 - 17/10080$. Choose $h = 1/10080$ in Proposition 3.7. Its actual coefficient satisfies

$$0 < v < s_0 + C + h \leq 7 - \frac{17}{10080} + \frac{1}{10080} = 7 - \frac{1}{630}.$$

The initial divisor and every added coefficient are retained in this estimate. $\square$

Proposition 5.9. *Let $\Gamma \sim_{\mathbb{Q}} sL$, $s \geq 0$, be log canonical near x with a fivefold minimal center of multiplicity two. If $b = \text{def}_x(\Gamma) \leq 2$, every additive cutting chain to the point satisfies*

$$C \leq \frac{23}{20}b + \frac{437}{480}.$$

If also $s \leq (65/69) \text{mult}_x \Gamma + e$ with $e < 1/128$, the entire Γ admits an additive point continuation of total coefficient less than $7 - 1/4096$.

Keeping the restriction on the incoming deficit lowers the largest fourfold surplus in the double-fivefold comparison.

Proof. We impose the incoming deficit cap on every successor and then pay the actual cuts within the resulting coefficient margin.

Step 1. In this step, we bound every successor tail under the incoming deficit cap $b \leq 2$. Use $a = 23/20$ as the initial radius bound. If the first proper center is a fourfold of multiplicity n and deficit $z > 0$, (5.23) still holds, and now $z \leq b \leq 2$. Repeating the same complete-tail comparison with this cap gives

(5.31)	n	upper bound used for z	$C_4 - (23/20)z$ is at most
	1	2	77/90
	2	2	111/160
	3	2	1297/1440
	4	2	437/480
	5	3/2	33/40
	6	1	71/100
	7	1/2	79/160

For $n = 2$, the value is $(6/5 - 23/20)2 + 19/32 = 111/160$. For $n = 3$, it is $(79/60 - 23/20)2 + 817/1440 = 1297/1440$. The remaining comparisons are unchanged. The maximum is $437/480 = 1311/1440$. The flag excludes $n \geq 8$ at positive deficit.

For a direct threefold successor, (5.25) bounds the surplus by $69/80 < 437/480$. That complete list includes $m = 8, 9$ through Lemma 5.1. A direct surface has surplus at most $11/25$, and a curve or point has no positive surplus. Adding the initial deficit drop at radius a proves the scalar assertion for every chain.

Step 2. In this step, we convert the scalar bound into an additive point continuation with the asserted margin. Set $q = 65/69$ for the coefficient comparison. The point blow-up inequality

(5.27), applied to the entire Γ , gives

$$\begin{aligned} s + C &\leq 6q + \left(\frac{23}{20} - q\right)b + \frac{437}{480} + e \\ &\leq 4q + \frac{23}{10} + \frac{437}{480} + e = 7 - \frac{79}{3680} + e. \end{aligned}$$

The scalar margin after allowing $e < 1/128$ exceeds $1/100$, since

$$\frac{79}{3680} - \frac{1}{128} = \frac{201}{14720} > \frac{1}{100}.$$

Choose total allowance $h = 1/65536$ in Proposition 3.7. Since $1/100 > 1/4096 + 1/65536$, the resulting additive point continuation has coefficient less than $7 - 1/4096$. $\square$

5.5. Two further linear envelopes. The following comparisons will be useful when a new divisor has already paid for part of the deficit drop. They include every threefold multiplicity permitted by positive deficit.

Lemma 5.10. *Let an effective rational boundary on X be log canonical near x , with a positive-dimensional minimal center W of dimension at most four and deficit $z > 0$. Set $\gamma = 2301/2000$. Every additive cutting chain to the point satisfies*

$$C - \frac{9}{10}z \leq \frac{100663}{60000}.$$

If W is not a fourfold of multiplicity three or four, then the same chain also satisfies

$$C - \gamma z \leq \frac{3447}{4000}.$$

The first estimate includes the two fourfold types excluded in the second. The constants are obtained from complete tails in both cases.

Proof. We slightly sharpen the double- and cubic-fourfold reference radii, then compare the finite lists in each dimension.

Step 1. In this step, we sharpen the complete-tail bounds for double and cubic fourfolds. For a double fourfold, replace $6/5$ by $119/100$. This remains a radius bound because $119^4 = 200533921 > 200000000 = 2 \cdot 100^4$. The parent flag remains $m \leq 6 - 2w$ for its threefold successor, and its deficit cap remains $z \leq 3$. Formula (5.17), with this new reference radius, has candidates

$$\frac{53}{125}, \quad \frac{79}{200}, \quad \frac{3161}{5400}, \quad \frac{483}{800}, \quad \frac{33}{100},$$

where the first is the surface comparison and the remaining entries correspond to $m = 2, 3, 4, 5$. Their maximum is $483/800$. Consequently the entire double-fourfold tail satisfies

$$(5.32) \quad C \leq \frac{119}{100}z + \frac{483}{800}, \quad z \leq 3.$$

For a cubic fourfold, put $a = 13161/10000$ and $p = 34087/60000$. Here $a^2 > 17321/10000$ and $17321^2 = 300017041 > 300000000$, so $a > 3^{1/4}$. The first inequality follows from $13161^2 = 173211921 > 173210000$. The parent flag is $m \leq 8 - 3w$, and $z \leq 8/3$. The surface comparison is $9/10 - 2a/5$. The remaining candidates, for $m = 3, 4, 5, 6, 7$, are

$$\left(\frac{13}{9} - a\right)\frac{5}{3} + \frac{11}{54}, \quad \left(\frac{127}{80} - a\right)\frac{4}{3} + \frac{33}{160} = p, \quad \frac{9}{5} - a, \quad \frac{2}{3}(2 - a), \quad \frac{1}{3}(2 - a).$$

Every candidate except p is less than $1/2$, by $a > 13/10$, while $p > 1/2$. As in Lemma 5.4, threefold radii below the reference radius and all skipped dimensions are covered by the surface comparison. Thus

$$(5.33) \quad C \leq \frac{13161}{10000}z + \frac{34087}{60000}, \quad z \leq \frac{8}{3}.$$

For these two parent types, $m = 8, 9$ cannot be threefold successors because their positive-deficit flags require $m < 6$ and $m < 8$, respectively.

Step 2. In this step, we bound both affine surpluses for every possible threefold multiplicity. For a slope λ , (5.13) bounds the surplus by $(c_m - \lambda)_+ \beta_m + d_m$. The complete list for the two slopes is

$$(5.34) \quad \begin{array}{c|cc} m & C - \gamma z \text{ is at most} & C - (9/10)z \text{ is at most} \\ \hline 1 & 1/2 & 4/5 \\ 2 & 237/500 & 39/40 \\ 3 & 2497/3600 & 10/9 \\ 4 & 3447/4000 & 99/80 \\ 5 & 1299/2000 & 9/10 \\ 6 & 1699/3000 & 11/15 \\ 7 & 5097/14000 & 33/70 \\ 8 & 1699/8000 & 11/40 \\ 9 & 733/6000 & 3/20 \end{array}$$

For example, the fourth row in the first column of surpluses is $(127/80 - 2301/2000)(3/2) + 33/160 = 3447/4000$. All other entries in that column are less than $7/10$. The maximum in the last column is $99/80$.

The last two rows follow from (5.6)–(5.7); their comparisons with the asserted envelope constants are explicitly

$$(5.35) \quad \begin{aligned} m = 8: \quad C - \gamma z &\leq \left(2 - \frac{2301}{2000}\right) \frac{1}{4} = \frac{1699}{8000} < \frac{3447}{4000}, \\ m = 9: \quad C - \gamma z &\leq \left(\frac{9}{4} - \frac{2301}{2000}\right) \frac{1}{9} = \frac{733}{6000} < \frac{3447}{4000}. \end{aligned}$$

The integer comparisons are $1699 < 6894$ and $1466 < 10341$, on denominators 8000 and 12000, respectively. For the other slope they are

$$(5.36) \quad \begin{aligned} m = 8: \quad C - \frac{9}{10}z &\leq \left(2 - \frac{9}{10}\right) \frac{1}{4} = \frac{11}{40} < \frac{100663}{60000}, \\ m = 9: \quad C - \frac{9}{10}z &\leq \left(\frac{9}{4} - \frac{9}{10}\right) \frac{1}{9} = \frac{3}{20} < \frac{100663}{60000}. \end{aligned}$$

Here the numerators on denominator 60000 are 16500 and 9000, both less than 100663. Since Lemma 5.1 excludes $m \geq 10$, the displayed table exhausts the positive-deficit threefolds, including those that jump directly to a curve or to the point.

Step 3. In this step, we prove both affine envelopes for fourfolds, surfaces, and curves. For fourfolds, use the embedding-dimension-four estimate in the smooth case, (5.32) and (5.33) in multiplicities two and three, and (5.15) in multiplicities four through six. For multiplicities seven through nine use (5.20) and $z \leq 6/7$. The results are

$$(5.37) \quad \begin{array}{c|cc} n & C - \gamma z \text{ is at most} & C - (9/10)z \text{ is at most} \\ \hline 1 & 77/90 & 113/90 \\ 2 & 2889/4000 & 1179/800 \\ 3 & \text{excluded} & 100663/60000 \\ 4 & \text{excluded} & 677/480 \\ 5 & 537/625 & 63/50 \\ 6 & 1289/1500 & 179/150 \\ 7, 8, 9 & \text{less than } 7/10 & \text{less than } 1 \end{array}$$

The cubic value in the last column is

$$\left(\frac{13161}{10000} - \frac{9}{10}\right) \frac{8}{3} + \frac{34087}{60000} = \frac{100663}{60000}.$$

For the last row, the first surplus is at most $(5/3 - 2301/2000)(6/7) + 113/480 < 7/10$; the second is at most $23/35 + 113/480 < 1$. The closest comparisons in the first column have positive gaps

$$\frac{3447}{4000} - \frac{77}{90} = \frac{223}{36000}, \quad \frac{3447}{4000} - \frac{537}{625} = \frac{51}{20000}, \quad \frac{3447}{4000} - \frac{1289}{1500} = \frac{29}{12000}.$$

Every entry in the last column except the cubic value is less than $3/2 < 100663/60000$.

Finally, the surface comparison at $a = \gamma$ gives surplus at most $9/10 - 2\gamma/5 = 2199/5000$. At slope $9/10$, the five surface surpluses are

$$\frac{1}{5}, \quad \frac{31}{60}, \quad \frac{17}{30}, \quad \frac{11}{20}, \quad \frac{27}{50},$$

whose maximum is $17/30$. A curve contributes at most zero for slope γ , and at most $1/10$ for slope $9/10$. The threefold table, fourfold table, and these surface and curve comparisons prove both assertions. The point case, if appended with deficit and tail cost both zero, also satisfies both bounds. $\square$

6. CAPACITIES OF FULL LEADING IDEALS

We use the threshold notation of Section 2. We bound the degree pieces of a full point initial ideal by passing its generic threshold to monomial ideals. One component gives a bound without any tangent hypothesis. A reduced Cohen–Macaulay tangent with no linear equations supplies three simultaneous constraints.

We first record the external results used below. The monomial formula is [How01, Main Theorem]; its generic-point interpretation is the formula preceding Lemma 1.3 and the coefficient-field argument in the proof of Theorem 1.1 of [dFEM02].

Theorem 6.1 (Monomial thresholds). *For a monomial ideal I on a smooth affine space and $c > 0$, the multiplier ideal $\mathcal{J}(I^c)$ is generated by those monomials x^u for which $u + (1, \dots, 1)$ lies in the interior of $c\text{Newt}(I)$. In particular its log canonical threshold is the reciprocal of the first intersection parameter of the positive diagonal with the Newton polyhedron. The threshold formula is unchanged by the characteristic-zero coefficient-field extension and completion used at a generic center.*

Thus a supporting face at the diagonal point imposes a linear inequality on every monomial of the ideal. For the cubic tangent we use the following classification, in the form of [EGHP04, Theorems 0.1 and 0.4, Proposition 2.4].

Theorem 6.2 (Reduced schemes of minimal degree). *An integral nondegenerate projective variety of degree equal to its codimension plus one is a linear space, a quadric hypersurface, a rational normal scroll, or a cone over the Veronese surface. A reduced projective scheme is 2-regular if and only if its irreducible components are varieties of minimal degree which admit a linearly joined ordering. In such an ordering, the scheme-theoretic intersection of a component with the preceding union is the intersection of their linear spans. Linear degeneracy is removed by working in the span.*

The scheme-theoretic intersection assertion will determine the equations of each reducible cubic. Finally, we use [Laz09, Theorem 3.2 and the ideal version after Remark 3.7].

Theorem 6.3 (Restriction of multiplier ideals). *Let T be a smooth divisor in a smooth complex variety Z , and let $\mathfrak{a}$ be an ideal with nonzero restriction to T . For every real $c > 0$,*

$$\mathcal{J}(T, (\mathfrak{a}\mathcal{O}_T)^c) \subseteq \mathcal{J}(Z, \mathfrak{a}^c)\mathcal{O}_T.$$

The inclusion is preserved by localization.

Only the ambient variety and its parameter divisor need to be smooth in this theorem. The support of the ideal can have nilpotents.

6.1. Threshold specialization and a uniform count.

Lemma 6.4. *Let $\mathcal{R}$ be a regular local ring essentially of finite type over $\mathbb{C}$, let t be a nonzero element of its maximal ideal, and assume that $\mathcal{R}/(t)$ is regular. Let $\mathfrak{a}$ be an ideal whose image $\mathfrak{a}_0$ in $\mathcal{R}/(t)$ is nonzero and proper. For every prime $\mathfrak{q}$ of $\mathcal{R}$ with $t \notin \mathfrak{q}$, one has*

$$\mathrm{lct}_{\mathcal{R}/(t)}(\mathfrak{a}_0) \leq \mathrm{lct}_{\mathcal{R}}(\mathfrak{a}) \leq \mathrm{lct}_{\mathcal{R}_{\mathfrak{q}}}(\mathfrak{a}\mathcal{R}_{\mathfrak{q}}).$$

This gives the direction of threshold comparison at every central component used in a degeneration.

Proof. For an exponent below the first threshold, the multiplier ideal on $\mathcal{R}/(t)$ is the unit ideal. By Theorem 6.3, the ambient multiplier ideal becomes the unit ideal modulo t . It is therefore the unit ideal in the local ring $\mathcal{R}$. This proves the first inequality. Localization preserves a unit multiplier ideal and proves the second. Regularity gives the smooth neighborhoods to which the restriction theorem applies, since the ground field has characteristic zero. Adjoining a transcendental coefficient field does not change any threshold in this comparison: a log resolution base changes, with the same ideal orders and discrepancies. $\square$

Lemma 6.5. *Let $(S, \mathfrak{m})$ be the local ring at a closed point of a smooth complex sixfold, let P be a height-two prime, and let I be P -primary. Put*

$$G = \mathrm{gr}_{\mathfrak{m}} S, \quad Q = \mathrm{in}_{\mathfrak{m}} P, \quad J = \mathrm{in}_{\mathfrak{m}} I, \quad \lambda = \mathrm{lct}_{S_P}(IS_P).$$

There is an integer $e \geq 1$ with $Q^e \subseteq J \subseteq Q$. For every height-two minimal prime $\mathfrak{p}$ of Q , the ideal $JG_{\mathfrak{p}}$ is nonzero and $\mathfrak{p}G_{\mathfrak{p}}$ -primary, and

$$\mathrm{lct}_{G_{\mathfrak{p}}}(JG_{\mathfrak{p}}) \leq \lambda.$$

No condition on the tangent support is needed for this specialization.

Proof. Choose e with $P^e \subseteq I \subseteq P$.

Step 1. In this step, we construct the simultaneous point degeneration and preserve the full-ideal inclusions. In the extended point Rees algebra

$$\mathcal{R} = \sum_{k \in \mathbb{Z}} \mathfrak{m}^k t^{-k} \subseteq S[t, t^{-1}], \quad \mathfrak{m}^k = S \quad (k \leq 0),$$

contract the full ideals $IS[t, t^{-1}]$ and $PS[t, t^{-1}]$ to $\mathcal{I}$ and $\mathcal{P}$. Both quotients are t -torsion-free. Their parameter-zero fibers are J and Q , and $P^e \subseteq \mathcal{I} \subseteq \mathcal{P}$. Specialization gives the asserted sandwich. In particular the support of J is the full support of Q .

Step 2. In this step, we specialize the generic threshold at each height-two tangent component. The quotient $\mathcal{R}/\mathcal{P}$ is the extended Rees algebra of the induced filtration on the domain S/P . It is a domain, so $\mathcal{P}$ is the horizontal support prime. Also $\dim G/Q = \dim S/P = 4$, so height-two minimal primes exist. Let $\tilde{\mathfrak{p}}$ be the inverse image of such a prime under $\mathcal{R} \rightarrow G = \mathcal{R}/(t)$. The element t is regular in $\mathcal{R}_{\tilde{\mathfrak{p}}}$, and its quotient is the regular two-dimensional local ring $G_{\mathfrak{p}}$. Two parameter lifts together with t generate the maximal ideal. The total local ring has dimension three and is regular. Its parameter-zero divisor is regular as well.

The sandwich gives

$$\sqrt{JG_{\mathfrak{p}}} = \sqrt{QG_{\mathfrak{p}}} = \mathfrak{p}G_{\mathfrak{p}},$$

where the last equality uses the minimality of $\mathfrak{p}$. Thus $JG_{\mathfrak{p}}$ is a nonzero $\mathfrak{p}G_{\mathfrak{p}}$ -primary ideal of finite colength. Since $\mathcal{P} \subseteq \tilde{\mathfrak{p}}$ and $t \notin \mathcal{P}$, Lemma 6.4 applies. The horizontal localization is S_P after adjoining the transcendental parameter to its coefficient field. Its threshold is λ , giving the conclusion. This argument uses regularity of the ambient local ring; it makes no reducedness assertion about $QG_{\mathfrak{p}}$. $\square$

Lemma 6.6. *Let G be a polynomial ring in six variables over $\mathbb{C}$, and let $J \subseteq Q$ be homogeneous ideals with $Q^e \subseteq J$ for some $e \geq 1$ and $\dim G/Q = 4$. Assume that*

$$\text{lct}_{G_{\mathfrak{q}}}(JG_{\mathfrak{q}}) \leq \lambda$$

for every height-two minimal prime $\mathfrak{q}$ of Q , where $\lambda > 0$ is fixed. For an integral weight, let K and H be the weight initial ideals of J and Q . Then

$$H^e \subseteq K \subseteq H, \quad \dim_{\mathbb{C}} K_n = \dim_{\mathbb{C}} J_n, \quad \dim_{\mathbb{C}} H_n = \dim_{\mathbb{C}} Q_n \quad (n \geq 0).$$

At every height-two minimal prime $\mathfrak{p}$ of H , $KG_{\mathfrak{p}}$ is nonzero and $\mathfrak{p}G_{\mathfrak{p}}$ -primary, and its threshold is at most λ .

The assertion concerns both full ideals, even when the special support has several components or a nonreduced structure.

Proof. We use the simultaneous weight family to compare its generic and central ideals.

Step 1. In this step, we preserve the ideal inclusions and both graded Hilbert functions under the weight degeneration. Homogenization with the weight gives simultaneous ideals $\mathcal{J}, \mathcal{Q}$ in $G[t]$, contracted from their diagonally transformed generic fibers. The quotients are t -torsion-free, and each ordinary graded Hilbert function is preserved. Initial forms of products multiply, which proves the sandwich. Adding a common integer to the weights does not affect initial forms of homogeneous polynomials, so negative weights cause no change in this construction.

Step 2. In this step, we prove the threshold bound at each central component and retain it under simultaneous monomialization. Localize $G[t]$ at $(t, \mathfrak{p})$. This is a regular local ring of dimension three with regular parameter-zero quotient $G_{\mathfrak{p}}$. Choose a minimal prime $\mathfrak{q}_h$ of $\mathcal{Q}$ contained in $(t, \mathfrak{p})$. Torsion-freeness excludes t from every such minimal prime. Its height is at least two because its generic fiber is a component of a four-dimensional support in affine six-space. Its strict containment in the height-three prime $(t, \mathfrak{p})$ gives height at most two. Thus it has height two and generizes to a height-two minimal component of the transformed Q . The generic threshold is at most λ by hypothesis. The sandwich and the minimality of $\mathfrak{p}$ give

$$\sqrt{KG_{\mathfrak{p}}} = \sqrt{HG_{\mathfrak{p}}} = \mathfrak{p}G_{\mathfrak{p}}.$$

The central ideal is therefore nonzero and $\mathfrak{p}G_{\mathfrak{p}}$ -primary, with finite colength. Lemma 6.4 now proves its threshold bound.

In particular, a single weight can monomialize both ideals: choose a monomial term order, take finite Groebner bases for both ideals, and choose an integral weight realizing all their leading-term inequalities. The degreewise dimension equalities are exact and do not depend on the size of this weight. $\square$

Lemma 6.7. *Let $G = \mathbb{C}[t_1, \dots, t_6]$, let $K \subseteq G$ be a monomial ideal, and put $\mathfrak{p} = (t_1, t_2)$. Let $\lambda > 0$, and suppose $KG_{\mathfrak{p}}$ is a nonzero $\mathfrak{p}G_{\mathfrak{p}}$ -primary ideal with $\text{lct}_{G_{\mathfrak{p}}}(KG_{\mathfrak{p}}) \leq \lambda$. There is a real number $0 < p < 1$ such that every monomial $t_1^{a_1} \cdots t_6^{a_6}$ of K satisfies*

$$pa_1 + (1-p)a_2 \geq \frac{1}{\lambda}.$$

The other four exponents remain unrestricted in this inequality.

Proof. At $\mathfrak{p}$ the other coordinates are units. The $\mathfrak{p}G_{\mathfrak{p}}$ -primary hypothesis gives $(\mathfrak{p}G_{\mathfrak{p}})^e \subseteq KG_{\mathfrak{p}}$ for some integer $e \geq 1$. In particular the ideal contains t_1^e and t_2^e . Over the coefficient field of the other coordinates it is a monomial ideal of finite colength in two variables. Its Newton polygon meets the positive diagonal first at (b, b) , where $b = 1/\text{lct}(KG_{\mathfrak{p}}) \geq 1/\lambda$, by Theorem 6.1. A compact supporting edge through this point has positive normal weights (r, s) . Such an edge exists also when the diagonal point is a vertex: finite colength gives pure powers of both variables, and the unbounded axial edges do not meet the positive diagonal boundary. Normalize by $p = r/(r+s)$. The supporting inequality is $pa_1 + (1-p)a_2 \geq b$. Every monomial of the original ideal remains in the localization, so it satisfies this same inequality. $\square$

Lemma 6.8. *For an integer $n \geq 1$, let Ω be the intersection of*

$$\{(t_1, \dots, t_5) \in \mathbb{R}_{\geq 0}^5 : t_1 + \dots + t_5 \leq n\}$$

with at most three closed halfspaces. There is an absolute constant C_{lat} such that

$$\left| \#(\Omega \cap \mathbb{Z}^5) - \text{vol}_{\mathbb{R}^5}(\Omega) \right| \leq C_{\text{lat}}(n+1)^4.$$

The constant is independent of the defining coefficients and intercepts of the halfspaces.

This uniformity allows both the ideal and the supporting weights to vary with the degree being counted.

Proof. Associate to each lattice point its half-open unit cube. The difference between the union of cubes anchored at points of Ω and Ω itself is supported on cubes meeting its boundary. There are at most nine bounding hyperplanes, including the simplex faces. For one nontrivial hyperplane choose a nonzero normal coefficient of largest absolute value, and solve for that coordinate. Each of the other four slopes has absolute value at most one. Over a unit cube in those four coordinates, the graph varies by at most four and hence meets a bounded number of five-dimensional unit cubes. The relevant four-dimensional box has $O((n+1)^4)$ unit cubes, with an absolute constant. Summing over the at most nine faces proves the assertion. A halfspace with zero normal either has no effect or makes the region empty. A region of smaller dimension lies in its bounding hyperplanes and satisfies the same estimate. No bound on a normal coefficient is used. $\square$

For later summations, if at most Dk degrees satisfy $n \leq Dk$ for a fixed $D > 0$, their lattice errors sum to $O_D(k^5)$. Indeed the sum is at most $DkC_{\text{lat}}(Dk+1)^4$. The ideals, components and weights can be chosen independently at every degree in this assertion.

6.2. The general height-two profile.

Definition 6.9. For $d \geq 2$, let Δ_{d-1} be the simplex of nonnegative coordinates $t_1, \dots, t_d$ with sum one. We use the measure obtained by eliminating t_d and multiplying Lebesgue measure by $(d-1)!$; thus Δ_{d-1} has measure one. For $0 \leq p \leq 1$ and $c \geq 0$, let $H_d(p, c)$ be the measure of the region

$$pt_1 + (1-p)t_2 \geq c$$

in this simplex.

Lemma 6.10. *For every $0 \leq p \leq 1$ and $c \geq 0$, one has*

$$H_6(p, c) \leq \max\{(1-c)_+^5, (1-2c)_+^4(1+8c)\}.$$

For $c \geq 2/9$, the maximum on the right is $(1-c)_+^5$.

The largest permissible image occurs at the balanced weights or at an endpoint of the weight interval.

Proof. We compute the volume, locate its maxima on the small parameter intervals, and then propagate the endpoint bound to larger c .

Step 1. In this step, we compute the one-pair volume and its balanced and endpoint values. Put $q = 1 - p$ and $f_c(t) = t(1 - c/t)_+^5$ for $t > 0$, with $f_c(0) = 0$. For $p > q$, integration with $s = t_1 + t_2$ gives

$$H_6(p, c) = \frac{20}{p-q} \int_0^1 (1-s)^3((ps-c)_+ - (qs-c)_+) ds = \frac{f_c(p) - f_c(q)}{p-q}.$$

In fact the integral of $(ts-c)_+(1-s)^3$ is $t(1-c/t)_+^5/20$. Symmetry and continuity give the other weights, including

$$(6.1) \quad H_6(1, c) = (1-c)_+^5, \quad H_6(1/2, c) = (1-2c)_+^4(1+8c).$$

The cases $c = 0$ and $c \geq 1$ follow directly, so we consider $0 < c < 1$ and $p \geq 1/2$.

Step 2. In this step, we locate the maximum over the weights for $0 < c \leq 2/9$. When $p \leq 1 - c$, set $z = p(1 - p)$. Expanding the divided difference gives

$$(6.2) \quad H_6(p, c) = 1 - \frac{10c^2}{z} + \frac{10c^3 + 5c^4}{z^2} - \frac{5c^4 + 2c^5}{z^3} + \frac{c^5}{z^4}.$$

Its derivative with respect to z is $c^2 P_c(z)/z^5$, where

$$P_c(z) = 10z^3 - (20c + 10c^2)z^2 + (15c^2 + 6c^3)z - 4c^3.$$

On the remaining interval $p \geq 1 - c$, only $f_c(p)$ is nonzero. Where the capacity is positive, its derivative with respect to p has the sign of

$$(6.3) \quad 10c - 1 - \frac{4c}{p}.$$

This expression is increasing in p , so this interval has no interior maximum.

For $0 < c \leq 1/5$, the discriminant of P'_c is

$$c^2(-200 + 880c + 400c^2) < 0.$$

Its leading coefficient is positive, so $P'_c > 0$. At the left endpoint of the z -interval,

$$P_c(c(1 - c)) = -c^3(2c - 1)^2(5c - 1) \geq 0.$$

Thus (6.2) increases with z , and its maximum in this interval is the balanced value. Together with (6.3), this leaves only the two values in (6.1). The logarithmic derivative of their ratio, balanced over endpoint, has the sign of $1 - 10c$. The ratio is one at zero and $1053/1024 > 1$ at $c = 1/5$. It first increases and then decreases, so the balanced value dominates throughout this range.

For $1/5 \leq c \leq 2/9$, one has

$$P''_c(z) = 60z - 40c - 20c^2 \geq 20c(1 - 4c) > 0 \quad (c(1 - c) \leq z \leq 1/4).$$

Now P_c is nonpositive at the left endpoint. Its convexity allows at most one change from negative to positive on this interval. Thus the capacity has no interior maximum as a function of z . On the one-active interval, (6.3) is nonnegative already at $p = 1 - c$, because there its value is $(5c - 1)(1 - 2c)/(1 - c)$. It then increases. The overall maximum is again one of the two values in (6.1).

Step 3. In this step, we extend the endpoint bound from $c = 2/9$ to every larger threshold. At $c = 2/9$ the balanced and endpoint values are respectively

$$\frac{15625}{59049}, \quad \frac{16807}{59049}.$$

Fix $p > q > 0$. For $c < q$, put

$$U = \frac{p - c}{p(1 - c)}, \quad V = \frac{q - c}{q(1 - c)}.$$

The derivative of $H_6(p, c)/(1 - c)^5$ has the sign of $pV^4 - qU^4$. Since $pq \leq 1/4$,

$$\frac{pq}{1 + \sqrt{pq}} \leq \frac{1}{6}, \quad \frac{V}{U} \leq \sqrt{\frac{q}{p}} \iff c \geq \frac{pq}{1 + \sqrt{pq}}.$$

For $c \geq 1/6$ it follows that $pV^4 \leq q^2U^4/p \leq qU^4$. Hence the normalized capacity is nonincreasing. If $q \leq c < p$, its second term vanishes and its first normalized term decreases directly; if $c \geq p$, the capacity is zero. The balanced and endpoint weights follow by continuity. The endpoint bound at $2/9$ therefore holds at every larger c . This proves all the claims. $\square$

Theorem 6.11. *Let $(S, \mathfrak{m})$ be the local ring at a closed point of a smooth complex sixfold. Let P be any height-two prime and I any P -primary ideal. Set $\lambda = \text{lct}_{S_P}(IS_P) > 0$, and let $J = \text{in}_{\mathfrak{m}} I \subseteq \text{gr}_{\mathfrak{m}} S$. There is an absolute constant C_0 such that for every $n \geq 1$,*

$$\dim_{\mathbb{C}} J_n \leq \frac{n^5}{120} H_* \left(\frac{1}{n\lambda} \right) + C_0(n+1)^4,$$

where

$$H_*(c) = \max\{(1-c)_+^5, (1-2c)_+^4(1+8c)\} \quad (c \geq 0).$$

In particular, if $n\lambda \leq 9/2$, then

$$\dim_{\mathbb{C}} J_n \leq \frac{(n-1/\lambda)_+^5}{120} + C_0(n+1)^4.$$

No hypothesis is imposed on the tangent scheme of S/P ; linear equations in P are permitted.

Only one generic component is needed for this bound on the full leading image.

Proof. Put $G = \text{gr}_{\mathfrak{m}} S$ and $Q = \text{in}_{\mathfrak{m}} P$. Lemma 6.5 bounds the generic threshold of J at every height-two component of Q . Apply one simultaneous monomial degeneration to J, Q , obtaining K, H as in Lemma 6.6. Since $\dim G/H = 4$, choose a height-two minimal prime $\mathfrak{p}$ of H . It is generated by two coordinates because H is monomial. For the exponent $e \geq 1$ in that lemma, the localized inclusions $H^e G_{\mathfrak{p}} \subseteq K G_{\mathfrak{p}} \subseteq H G_{\mathfrak{p}}$ show that $K G_{\mathfrak{p}}$ is a nonzero $\mathfrak{p} G_{\mathfrak{p}}$ -primary ideal of finite colength, as proved in Lemma 6.6. Its threshold is at most λ .

Lemma 6.7 puts every degree- n monomial of K in a halfspace of the degree- n simplex. Its volume is

$$\frac{n^5}{120} H_6 \left(p, \frac{1}{n\lambda} \right)$$

for some $0 < p < 1$. Lemmas 6.10 and 6.8 bound its lattice count by the first asserted expression, with an absolute constant. Hilbert preservation identifies $\dim K_n$ with $\dim J_n$. If $n\lambda \leq 9/2$, the normalized threshold is at least $2/9$, where the endpoint profile applies. This gives the second inequality. No ideal has been replaced by its localization in the dimension count; the localization supplies only the constraint. $\square$

6.3. Three simultaneous normal pairs.

Lemma 6.12. *Let $(S, \mathfrak{m})$ be the local ring at a closed point of a smooth complex sixfold. Let $P \subseteq \mathfrak{m}^2$ be a height-two prime, suppose $\text{gr}_{\mathfrak{m}}(S/P)$ is reduced and Cohen–Macaulay, and let I be P -primary. Put*

$$G = \text{gr}_{\mathfrak{m}} S, \quad Q = \text{in}_{\mathfrak{m}} P, \quad J = \text{in}_{\mathfrak{m}} I, \quad \lambda = \text{lct}_{S_P}(IS_P).$$

There are coordinates u, v, w, z_1, z_2, z_3 and simultaneous homogeneous flat degenerations of J, Q to monomial ideals K, H preserving both ordinary graded Hilbert functions, such that, for some integer $N \geq 1$,

$$\mathfrak{q}_{\Delta} = (uv, uw, vw), \quad \mathfrak{q}_{\Delta}^N \subseteq K \subseteq H \subseteq \mathfrak{q}_{\Delta}, \quad \sqrt{K} = \sqrt{H} = \mathfrak{q}_{\Delta}.$$

For each $\mathfrak{p} \in \{(u, v), (u, w), (v, w)\}$, the ideal $K G_{\mathfrak{p}}$ is nonzero and $\mathfrak{p} G_{\mathfrak{p}}$ -primary and has threshold at most λ .

The degree of G/H can exceed three. The inclusions retain the scheme structure created when its components collide.

Proof. We take three regular hyperplane sections by degenerations, retain three noncollinear points during a collision, and then monomialize. At every stage we specialize the two full ideals together.

Step 1. In this step, we bound the generic threshold on every component of the full tangent scheme. Choose e with $P^e \subseteq I \subseteq P$. Lemma 6.5 gives $Q^e \subseteq J \subseteq Q$ and the threshold bound at

every minimal prime of Q . All these primes have height two, since the reduced Cohen–Macaulay ring G/Q is pure of dimension four.

Step 2. In this step, we choose three reduced regular hyperplane sections containing three noncollinear points. The projective scheme $\mathcal{T} = \text{Proj}(G/Q)$ is reduced, pure of dimension three, and spans $\mathbb{P}^5$. We choose three general linear forms z_1, z_2, z_3 such that each successive section ring is reduced and Cohen–Macaulay. Here is the incidence argument for this choice. For a smooth d -dimensional locally closed piece in $\mathbb{P}^M$, the incidence of a point and a hyperplane containing its tangent space has dimension at most

$$d + (M - d - 1) = M - 1.$$

Its image in the M -dimensional dual projective space has proper closure. A general hyperplane is therefore transverse on the smooth locus of every component. The complementary bad locus has dimension at most $d - 1$; a general hyperplane meets it in dimension at most $d - 2$, or avoids it for $d = 1$. The section is consequently generically reduced. Choose the hyperplane also to contain no component. Its equation is regular on the Cohen–Macaulay ring, and its quotient is Cohen–Macaulay of dimension at least one, with no embedded associated primes. A nonzero nilradical in that quotient would have an associated prime which is also an associated prime of the ring. This would be a minimal prime, contradicting generic reducedness. The whole section ring is reduced, including at its graded vertex.

After three choices we obtain the homogeneous coordinate ring of a finite reduced set $Z \subseteq \mathbb{P}^2$. It is saturated: depth one excludes elements killed by a power of the irrelevant ideal. Its degree-one piece has dimension $6 - 3 = 3$, because Q has no linear equations. Thus Z is not contained in a line and contains three noncollinear points.

Step 3. In this step, we realize the hyperplane sections as simultaneous degenerations of the two full ideals. Suppose B is the current homogeneous support ideal and z is a regular section coordinate on G/B . Give z weight zero and every other coordinate weight one, using largest weight for an initial form. For a homogeneous $f \in B$, write $f = z^a g$ with a maximal. Since $(B : z) = B$, one has $g \in B$, and

$$\text{in}_{\text{wt}}(f) = z^a g|_{z=0}.$$

Conversely every homogeneous element of the section ideal is the specialization of an element of B . Thus the new support ideal is exactly the extension of the section ideal, with z free. Apply this same degeneration to the full current image ideal. Lemma 6.6 preserves its Hilbert function, its sandwich by powers of the support, and all generic threshold bounds. Repeat for z_1, z_2, z_3 ; previously freed variables remain free. The resulting ideals J_*, B_* satisfy

$$B_* = (I_Z \subseteq \mathbb{C}[u, v, w])G, \quad B_*^e \subseteq J_* \subseteq B_*.$$

Every component threshold of J_* is at most λ . Make a linear change of u, v, w taking the three selected points to the coordinate vertices.

Step 4. In this step, we collide the points while retaining three fixed branches and their threshold bounds. Let $r = \#Z$. Move the points by $\text{diag}(1, t, t^2)$ and take the induced flat weight families of both ideals. For a point $[c_0 : c_1 : c_2]$, let j be its first nonzero coordinate, and write $(x_0, x_1, x_2) = (u, v, w)$. The closure of its moving linear cone has ideal generated by

$$x_i - (c_i/c_j)t^{i-j}x_j \quad (i \neq j, c_i \neq 0), \quad x_i \quad (c_i = 0).$$

All exponents of t are nonnegative. Its quotient is $\mathbb{C}[t, x_j, z_1, z_2, z_3]$, and its central ideal is the coordinate prime $\mathfrak{p}_j$ generated by the other two active variables. The three selected vertex branches are fixed horizontal primes throughout this family.

The full support family is the intersection of these r component ideals: this equality holds after inverting t , and contraction commutes with finite intersections. Its product of component

ideals is contained in this intersection. If B_0, J_0 are the full central ideals, specialization gives

$$\mathfrak{q}_\Delta^r \subseteq \prod_{p \in Z} \mathfrak{p}_{j(p)} \subseteq B_0 \subseteq \bigcap_{p \in Z} \mathfrak{p}_{j(p)} = \mathfrak{q}_\Delta.$$

Together with $B_0^e \subseteq J_0 \subseteq B_0$, this yields

$$(6.4) \quad \mathfrak{q}_\Delta^{re} \subseteq J_0 \subseteq B_0 \subseteq \mathfrak{q}_\Delta.$$

These are inclusions of actual ideals. In particular the degree of G/B_0 remains r .

For each $\mathfrak{p}_j$, localize the smooth ambient family at $(t, \mathfrak{p}_j)$. Its central ring is the regular local surface $G_{\mathfrak{p}_j}$, and (6.4) makes the restricted image ideal nonzero and $\mathfrak{p}_j G_{\mathfrak{p}_j}$ -primary, since $\mathfrak{q}_\Delta G_{\mathfrak{p}_j} = \mathfrak{p}_j G_{\mathfrak{p}_j}$. The fixed vertex branch is a horizontal generization whose threshold is at most λ . Lemma 6.4 gives the same bound for $J_0 G_{\mathfrak{p}_j}$, for all three choices of j . Other colliding points do not affect the smoothness of this ambient ring or the nonzero restriction.

Step 5. In this step, we monomialize both ideals while preserving the triangle inclusions and all three generic threshold bounds. Use a single integral weight realizing a monomial term order for both J_0 and B_0 . The triangle ideal, its powers, and its three coordinate primes stay fixed. Hence the final ideals K, H satisfy the asserted sandwich with $N = re$. For each of the three coordinate primes $\mathfrak{p}$, it localizes to

$$(\mathfrak{p}G_{\mathfrak{p}})^N \subseteq KG_{\mathfrak{p}} \subseteq \mathfrak{p}G_{\mathfrak{p}},$$

so each localization is nonzero and has finite colength. Lemma 6.6 preserves the three threshold bounds and both full Hilbert functions. This proves the assertion without removing any component or nilpotent structure. $\square$

Lemma 6.13. *Let $A, B, D > 0$, let $h \geq 1/6$, and let $V(A, B, D; h)$ be the normalized measure in Δ_5 , with active coordinates u, v, w , of the three inequalities*

$$Au + v \geq (A + 1)h, \quad Bu + w \geq (B + 1)h, \quad Dv + w \geq (D + 1)h.$$

For $1/6 \leq h < 1/3$, one has

$$V(A, B, D; h) \leq (1 - h)^5 H_* \left(\frac{h}{1 - h} \right).$$

For $h \geq 1/3$, one has $V(A, B, D; h) \leq (1 - 2h)_+^5$. Consequently,

$$V(A, B, D; 1/6) \leq \frac{13}{96}, \quad V(A, B, D; h) \leq (1 - 2h)_+^5 \quad (h \geq 2/11).$$

The three constraints use the same three active coordinates. Their weights may be chosen independently.

Proof. We split the simplex into the central region and three exterior regions, then reduce the optimization to the one-pair profile.

Step 1. In this step, we bound the feasible volume for $h \geq 1/3$. Each pair inequality forces at least one of its two coordinates to be at least h . Thus at least two of u, v, w are at least h . If $h > 1/3$, the three alternatives specifying which two are at least h are disjoint closed subsets of the simplex. The feasible region is convex, hence connected, and lies in one alternative. Its measure is at most $(1 - 2h)_+^5$. At $h = 1/3$ the same bound follows by continuity: for fixed positive weights the moving bounding hyperplanes have measure zero, so their indicator functions converge almost everywhere. The assertion includes $h \geq 1/2$, where the feasible volume is zero.

Step 2. In this step, we express the triangle volume as the sum of its central and exterior integrals. Now let $1/6 \leq h < 1/3$, and put $w_0 = 1/h - 3$, so $0 < w_0 \leq 3$. Translate u, v, w by h . At most one translated coordinate is negative. The region where all are nonnegative has measure $(1 - 3h)^5 = h^5 w_0^5$. In the region $u - h = -s < 0$, the first two inequalities say $v - h \geq As$, $w - h \geq Bs$, and the third follows from these. The remaining five nonnegative coordinates have

sum $1 - 3h - (A + B - 1)s$. Integrating their fourth-power slice measure, and doing the other two exterior regions, gives exactly

$$(6.5) \quad V(A, B, D; h) = h^5 \left(w_0^5 + \sum_{i=1}^3 F_{w_0}(c_i) \right), \quad F_{w_0}(c) = 5 \int_0^1 (w_0 - cz)_+^4 dz,$$

where

$$c_1 = A + B - 1, \quad c_2 = A^{-1} + D - 1, \quad c_3 = B^{-1} + D^{-1} - 1.$$

Here $c_i > -1$. The factor five in the integral is $5!/4!$, from the normalization in Definition 6.9.

Step 3. In this step, we reduce the triangle-volume bound to the one-pair profile. Set $x_i = 1/(c_i + 2)$. Bringing these three fractions to a common denominator gives

$$(6.6) \quad 1 - \sum_{i=1}^3 x_i = \frac{(B - AD)^2}{(1 + A + B)(1 + A + AD)(B + D + BD)} \geq 0.$$

Define $\phi(x) = F_{w_0}(1/x - 2)$ for $0 < x \leq 1$, with $\phi(0) = 0$. This is increasing, since F_{w_0} is decreasing. We claim that $\phi(x)/x$ is nondecreasing on $(0, 1/2]$, or equivalently that $(c + 2)F_{w_0}(c)$ is nonincreasing for $c \geq 0$.

First take $w_0 = 3$. For $c \geq 3$, $(c + 2)F_3(c) = 243 + 486/c$. For $0 \leq c \leq 3$,

$$F_3(c) = 405 - 270c + 90c^2 - 15c^3 + c^4,$$

and the derivative of $(c + 2)F_3(c)$ is $-135 - 180c + 180c^2 - 52c^3 + 5c^4$. Substituting $c = 3t$ expresses it as

$$-135(1 - t)^4 - 1080t(1 - t)^3 - 810t^2(1 - t)^2 - 324t^3(1 - t) - 54t^4 < 0 \quad (0 \leq t \leq 1).$$

For general $0 < w_0 \leq 3$, put $a = w_0/3$. Then

$$F_{w_0}(c) = a^4 F_3(c/a), \quad (c + 2)F_{w_0}(c) = a^5 (t + 2/a)F_3(t), \quad t = c/a.$$

The derivative of the latter expression has the sign of $((t + 2/a)F_3(t))' + (2/a - 2)F_3'(t)$, which is nonpositive. This proves the claim uniformly in w_0 .

Increase the three x_i to $y_i \geq x_i$ with sum one, using (6.6). If all $y_i \leq 1/2$, the ratio monotonicity gives $\sum \phi(y_i) \leq 2\phi(1/2)$. Otherwise let $y_1 = p > 1/2$. Since $y_2 + y_3 = 1 - p < 1/2$, the same monotonicity gives

$$\phi(y_2) + \phi(y_3) \leq \phi(y_2 + y_3) = \phi(1 - p).$$

Both cases bound the bracket in (6.5) by

$$\max_{1/2 \leq p \leq 1} (w_0^5 + \phi(p) + \phi(1 - p)).$$

Using $F_{w_0}(c) = (w_0^5 - (w_0 - c)_+^5)/c$, with its continuous value at zero, the divided-difference formula gives

$$(6.7) \quad h^5 (w_0^5 + \phi(p) + \phi(1 - p)) = (1 - h)^5 H_6 \left(p, \frac{h}{1 - h} \right).$$

This identity also holds at $p = 1/2, 1$ by continuity. Lemma 6.10 proves the first bound. At $h = 1/6$, the balanced profile at $c = 1/5$ is $1053/3125$ and dominates the endpoint profile. Hence

$$V(A, B, D; 1/6) \leq \left(\frac{5}{6} \right)^5 \frac{1053}{3125} = \frac{1053}{7776} = \frac{13}{96}.$$

For $2/11 \leq h < 1/3$, one has $h/(1 - h) \geq 2/9$. The endpoint profile in (6.7) is then $(1 - 2h)^5$. The large-threshold argument covers the rest. $\square$

Theorem 6.14. *In the local setting of Theorem 6.11, suppose $P \subseteq \mathfrak{m}^2$ and $\text{gr}_{\mathfrak{m}}(S/P)$ is reduced and Cohen–Macaulay. There is an absolute constant C_1 such that*

$$\text{lct}_{S_P}(IS_P) \leq \frac{6}{n} \implies \dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}} I)_n \leq \frac{13}{96} \frac{n^5}{120} + C_1(n+1)^4.$$

The estimate uses the full initial ideal and is uniform in the primary exponent, multiplicity and degeneration weights.

The absence of linear tangent equations supplies the three noncollinear branches needed for this stronger estimate.

Proof. Use Lemma 6.12 to obtain the full monomial ideal K in $G = \text{gr}_{\mathfrak{m}} S$. Its localized power inclusions make $KG_{\mathfrak{p}}$ a nonzero $\mathfrak{p}G_{\mathfrak{p}}$ -primary ideal of finite colength at each of its three coordinate primes $\mathfrak{p}$. Hence Lemma 6.7 gives a pair inequality with right side at least $1/\lambda \geq n/6$. Enlarge each halfspace by replacing its right side with $n/6$. After division by n , the resulting region is the one in Lemma 6.13 for $h = 1/6$, with some positive A, B, D . Its normalized volume is at most $13/96$. Its actual five-dimensional volume is therefore at most $(13/96)n^5/120$. Lemma 6.8 adds at most $C_{\text{lat}}(n+1)^4$ to its lattice count. The Hilbert function of K equals that of the original initial ideal. All degeneration steps preserve these dimensions exactly, so no error depends on the number of colliding points, a primary exponent, or a choice of weights. $\square$

6.4. Cubic tangents and a two-unit shift.

Lemma 6.15. *Let $(S, \mathfrak{m})$ be the local ring at a closed point of a smooth complex sixfold. Let $P \subseteq \mathfrak{m}^2$ be a height-two prime, and suppose that $\text{gr}_{\mathfrak{m}}(S/P)$ is reduced and Cohen–Macaulay with Hilbert series $(1+2z)/(1-z)^4$. Let I be P -primary, and put $G = \text{gr}_{\mathfrak{m}} S$ and $\lambda = \text{lct}_{S_P}(IS_P)$. After a linear coordinate change there is a simultaneous monomial degeneration of $J = \text{in}_{\mathfrak{m}} I$ and $Q = \text{in}_{\mathfrak{m}} P$ to K, Q_0 , preserving both Hilbert functions, with*

$$Q_0 = (ac, ad, bd) \quad \text{or} \quad Q_0 = (ab, ac, bc)$$

in six coordinates, the undisplayed coordinates being free. For some integer $e \geq 1$, one has $Q_0^e \subseteq K \subseteq Q_0$. At every minimal prime $\mathfrak{p}$ of Q_0 , the ideal $KG_{\mathfrak{p}}$ is nonzero and $\mathfrak{p}G_{\mathfrak{p}}$ -primary and has threshold at most λ .

Here the two displayed support ideals are squarefree. This is a description of actual initial ideals, with the image ideal still intact.

Proof. We establish 2-regularity from the Hilbert series, list the cubic equations, and then apply the full-ideal specialization lemmas.

Step 1. In this step, we prove that the full cubic tangent scheme is 2-regular. Write $A = G/Q$. Four general linear forms are an A -regular sequence. The quotient has Hilbert series $1+2z$, and hence is $\mathbb{C}[\xi, \eta]/(\xi, \eta)^2$. Its minimal graded resolution over $\mathbb{C}[\xi, \eta]$ has free terms of ranks 1, 3, 2 and shifts 0, 2, 3. A minimal graded resolution of A over G remains exact and minimal after these regular linear forms are factored out: the higher tensor homology vanishes by their Koszul complex, and every differential still has positive-degree entries. Consequently the same free terms resolve A , and sheafification gives

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-3)^2 \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-2)^3 \longrightarrow \mathcal{I}_{\text{Proj } A} \longrightarrow 0.$$

The cohomology of these line bundles gives $H^i(\mathcal{I}_{\text{Proj } A}(2-i)) = 0$ for $i > 0$. Thus $\text{Proj } A$ is 2-regular. It is reduced, pure of dimension three, spans $\mathbb{P}^5$, and has degree three. We may apply Theorem 6.2. The component degrees partition three.

Step 2. In this step, we determine the full initial ideals of the integral cubic scrolls. The integral possibilities are the scrolls $S(1, 1, 1), S(0, 1, 2), S(0, 0, 3)$: a three-dimensional scroll has three nonnegative parameters whose sum is its degree, with zero parameters accounting for its

vertex. A Veronese cone has degree four. The respective scroll parametrizations, with vertex coordinates left free, are

$$\begin{aligned}(a, b, c, d, e, f) &= (s\alpha, s\beta, s\gamma, t\alpha, t\beta, t\gamma), \\ (a, b, c, d, e) &= (s\alpha, s^2\beta, st\beta, t\alpha, t^2\beta), \\ (a, b, c, d) &= (s^3, s^2t, st^2, t^3).\end{aligned}$$

Thus their ideals contain the two-by-two minors of the respective matrices

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ d & c & e \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ b & c & d \end{pmatrix}.$$

Their leading ideals can be chosen as

$$(ae, af, bf), \quad (ac, ae, be), \quad (ac, ad, bd),$$

using respectively the weights $(2, 1, 0, 0, 1, 2)$, $(2, 0, 0, 0, 2)$, and $(2, 0, 0, 2)$, and then adding a common positive constant. These are all chain ideals after renaming four variables. To verify that they are the full leading ideals, for the first matrix denote its minors by $f_1 = ae - bd$, $f_2 = af - cd$, $f_3 = bf - ce$. The nontrivial S-polynomials reduce by

$$ff_1 - ef_2 = -df_3, \quad bf_2 - af_3 = cf_1.$$

The remaining leading terms are relatively prime. Substitution of the second and third matrices in these same identities gives the corresponding reductions in those cases. Thus any term-order refinement preserving the displayed strict choices has exactly the indicated initial ideal for the ideal of minors. Inclusion-exclusion on the three generators of a chain ideal gives the quotient Hilbert series

$$\frac{1 - 3z^2 + 2z^3}{(1 - z)^6} = \frac{1 + 2z}{(1 - z)^4}.$$

This is the Hilbert series of the full tangent quotient. The ideal of minors is contained in its ideal and has the same Hilbert function, so the two ideals agree in every degree. This also checks that the displayed degenerations concern the entire scroll ideals.

Step 3. In this step, we determine the full initial ideal when the tangent consists of a quadric and a linear component. Their spans are $\mathbb{P}^4$ and $\mathbb{P}^3$, and they span $\mathbb{P}^5$. Their intersection of spans is $\mathbb{P}^2$. By the linearly joined property, this plane lies on the irreducible quadric. The quadric has rank three or four: its rank is at least three by irreducibility, and the three-dimensional isotropic vector subspace of its five-dimensional space excludes rank five. Linear coordinates preserving that plane give a quadric equation $ac - b^2$ or $ac - bd$, with the quadric's span $e = 0$ and the other component $a = b = 0$. Their full union ideal is respectively

$$(e, ac - b^2) \cap (a, b) = (ac - b^2, ae, be), \quad (e, ac - bd) \cap (a, b) = (ac - bd, ae, be).$$

Choose ac as the first leading monomial. Its S-polynomial with ae is $-b^2e$ or $-bde$, divisible by be . The other noncoprime pair has relation $b(ae) - a(be) = 0$. Thus both full initial ideals are (ac, ae, be) , a chain.

Step 4. In this step, we determine the chain and triangle ideals for three linearly joined linear components. Take a linearly joined ordering of the three $\mathbb{P}^3$'s. The first two span $\mathbb{P}^4$. Indeed, if they already spanned $\mathbb{P}^5$, the scheme-theoretic joining condition would put the third component in their union, contradicting its being a distinct irreducible component. They meet in a $\mathbb{P}^2$. The third meets their span in another $\mathbb{P}^2$, contained in one of the first two components. Call that component the middle component. If its two intersection planes agree, all three share that plane; quotienting by its vector space leaves three independent coordinate points in $\mathbb{P}^2$. The ideal is (ab, ac, bc) , with three free coordinates. If the two planes differ, they meet in a $\mathbb{P}^1$, which is precisely the intersection of the outer components. Quotienting by its vector space leaves three lines forming a chain in $\mathbb{P}^3$. The two intersection directions and the remaining direction on each

outer line give coordinates in which the ideal is (ac, ad, bd) , with two free coordinates. This covers every component-degree partition.

Step 5. In this step, we specialize the full image ideal while preserving its Hilbert function and generic threshold bounds. Lemma 6.5 bounds every generic threshold on Q . For the term order just found, choose one integral weight realizing Groebner bases of both Q and J . Such a weight retains the strict leading choices for Q . Lemma 6.6 then gives the full monomial ideal K , the sandwich by powers of Q_0 , and the threshold inequality at every minimal prime of Q_0 . No dimension is estimated during this degeneration. $\square$

Lemma 6.16. *For $0 \leq p \leq 1$ and $c \geq 3/8$, one has*

$$H_4(p, c) \leq (1 - c)_+^3.$$

This bound is used on the four coordinates remaining after one normal pair is fixed.

Proof. The same integration as for H_6 , now with the factor $6(1 - s)$, gives

$$H_4(p, c) = \frac{g_c(p) - g_c(1 - p)}{2p - 1}, \quad g_c(t) = t(1 - c/t)_+^3,$$

with the continuous midpoint value. By symmetry assume $p \geq 1/2$. For $3/8 \leq c < 1/2$ and $p \leq 1 - c$, put $z = p(1 - p)$. Expansion gives

$$H_4(p, c) = 1 - \frac{3c^2}{z} + \frac{c^3}{z^2}, \quad \frac{\partial H_4}{\partial z} = \frac{c^2(3z - 2c)}{z^3} \leq 0.$$

Thus the capacity increases with p . For $p \geq 1 - c$, where it is positive, its derivative has the sign of $6c - 1 - 2c/p$. Multiplication by p gives an increasing linear expression whose value at $1 - c$ is $(3c - 1)(1 - 2c) \geq 0$. If $c \geq 1/2$, only this one-active branch can be positive; its derivative sign is at least $6c - 3 \geq 0$, since $p > c$. Values for $c \geq 1$ are zero. In every case the maximum occurs at $p = 1$, proving the asserted axis bound. $\square$

Theorem 6.17. *In the same local setting, suppose $P \subseteq \mathfrak{m}^2$ and $\text{gr}_{\mathfrak{m}}(S/P)$ is reduced and Cohen–Macaulay with Hilbert series $(1 + 2z)/(1 - z)^4$. There is a constant C_2 , independent of I, n, Λ , such that for $0 < \Lambda \leq 11/18$,*

$$\text{lct}_{S_P}(IS_P) \leq \frac{6\Lambda}{n} \implies \dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}} I)_n \leq \frac{2(1 - 1/(3\Lambda))_+^5}{120} n^5 + C_2(n + 1)^4.$$

The chain and triangle configurations admit direct volume bounds; the chain bound has no factor two.

Proof. We obtain the simultaneous pair constraints, estimate the chain and triangle volumes separately, and apply one uniform lattice count. Put $G = \text{gr}_{\mathfrak{m}} S$, $J = \text{in}_{\mathfrak{m}} I$, $\lambda = \text{lct}_{S_P}(IS_P)$, and $h = 1/\lambda$. The hypothesis gives

$$\frac{h}{n} \geq \frac{1}{6\Lambda} \geq \frac{3}{11}.$$

Step 1. In this step, we obtain simultaneous normal-pair constraints on every monomial of the full image ideal. Take the full monomial degeneration K, Q_0 from Lemma 6.15. For its exponent $e \geq 1$ and every minimal prime $\mathfrak{p}$ of Q_0 , one has $Q_0 G_{\mathfrak{p}} = \mathfrak{p} G_{\mathfrak{p}}$ and

$$(\mathfrak{p} G_{\mathfrak{p}})^e \subseteq K G_{\mathfrak{p}} \subseteq \mathfrak{p} G_{\mathfrak{p}}.$$

Thus these are nonzero $\mathfrak{p} G_{\mathfrak{p}}$ -primary ideals of finite colength. Lemma 6.7 constrains every monomial of K by a positive weighted pair with right side h . These inequalities define a polytope in the degree- n simplex.

Step 2. In this step, we bound the chain volume using its two disjoint normal pairs. For $Q_0 = (ac, ad, bd)$, two minimal primes are (a, b) and (c, d) . Keep these two pair inequalities; omitting the third only enlarges the comparison region. They use disjoint coordinates. Their

pair sums are each at least h , so if $2h \geq n$ the continuous volume is zero. Suppose $2h < n$. Fix the coordinates c, d and put $M_0 = n - c - d$. On the second pair constraint,

$$\frac{h}{M_0} \geq \frac{h}{n-h} \geq \frac{3}{8}.$$

For the four remaining coordinates, the first pair constraint therefore has no more volume than $a \geq h$, by Lemma 6.16. This replacement is made slice by slice and then integrated in c, d . Translate a by h . The remaining region is a six-coordinate simplex of total degree $n - h$ with just the second pair inequality. Its normalized threshold is at least $3/8 > 2/9$, so Lemma 6.10 bounds it by an axis constraint. We conclude that the original chain volume is at most

$$(6.8) \quad \frac{(n-2h)_+^5}{120}.$$

Step 3. In this step, we bound the triangle volume by integrating its three-coordinate slices. For $Q_0 = (ab, ac, bc)$, fix $S_0 = a + b + c$. The three pair inequalities cut out a convex polygon, and every point in it has at least two of a, b, c at least h . It is empty for $S_0 < 2h$. For $2h \leq S_0 < 3h$, the three triangles specifying which two coordinates are at least h are disjoint. Connectedness puts the feasible polygon in one of them. Its area in the coordinates a, b , with $c = S_0 - a - b$, is at most $(S_0 - 2h)^2/2$.

For $S_0 \geq 3h$, put

$$u = a - h, \quad v = b - h, \quad w = c - h, \quad t_0 = S_0 - 3h.$$

The central triangle $u, v, w \geq 0$, with sum t_0 , has area $t_0^2/2$ and satisfies all three constraints. At most one translated coordinate is negative. In the region $u = -z < 0$, the two constraints involving u give $v \geq \alpha z$, $w \geq \beta z$ for positive α, β ; the third is automatic. Its cross-section has length

$$[t_0 - (\alpha + \beta - 1)z]_+, \quad 0 \leq z \leq h.$$

Among the three exterior slopes at most one is negative. Indeed, if the two ratios at one coordinate sum to less than one, each is less than one, and the reciprocal ratio appearing at either other coordinate is greater than one. A negative slope is greater than -1 . Thus at most one exterior area is bounded by $t_0 h + h^2/2$, and each of the other two is bounded by $t_0 h$. The whole area is at most

$$\frac{t_0^2}{2} + 3t_0 h + \frac{h^2}{2} \leq (t_0 + h)^2 = (S_0 - 2h)^2,$$

since the difference is $(t_0 - h)^2/2$. In both ranges of S_0 , this is at most twice the area obtained by requiring two specified normal coordinates to be at least h . Integrating in S_0 against the density $(n - S_0)_+^2/2$ from the other three coordinates gives the triangle volume bound

$$(6.9) \quad \frac{2(n-2h)_+^5}{120}.$$

Step 4. In this step, we convert the volume bounds into a uniform count for the full initial ideal. The actual comparison polytope has the simplex faces and at most three additional faces. Apply Lemma 6.8 to this polytope. The slice integrals above estimate its continuous volume and introduce no lattice errors of their own. Hilbert preservation, (6.8), and (6.9) yield

$$\dim_{\mathbb{C}} J_n \leq \frac{2(n-2h)_+^5}{120} + C_{\text{lat}}(n+1)^4.$$

Finally $h \geq n/(6\Lambda)$ gives the asserted expression. The fixed number of faces makes its error independent of Λ , the primary exponent, and the threshold weights. $\square$

Proposition 6.18. *Under the local hypotheses of Theorem 6.14, there is an absolute constant C_4 such that*

$$n \operatorname{lct}_{S_P}(IS_P) \leq \frac{11}{2} \implies \dim_{\mathbb{C}}(\operatorname{in}_{\mathfrak{m}} I)_n \leq \frac{(n - 2/\operatorname{lct}_{S_P}(IS_P))_+^5}{120} + C_4(n+1)^4.$$

The smaller threshold range yields the full shift by twice the reciprocal generic threshold.

Proof. Set $\lambda = \operatorname{lct}_{S_P}(IS_P)$ and $h = 1/(n\lambda) \geq 2/11$. Lemma 6.12 gives the simultaneous full-ideal degeneration under exactly the tangent hypotheses of Theorem 6.14. Its structural assertion has no numerical restriction on $n\lambda$. For its full monomial ideal K in $G = \operatorname{gr}_{\mathfrak{m}} S$, the localized power inclusions make $KG_{\mathfrak{p}}$ a nonzero $\mathfrak{p}G_{\mathfrak{p}}$ -primary ideal of finite colength at each of the three coordinate primes $\mathfrak{p}$. Lemma 6.7 therefore gives three simultaneous constraints on every degree- n monomial. After normalization their common right side is h . By Lemma 6.13, their normalized feasible volume is at most $(1 - 2h)_+^5$. Hilbert preservation and Lemma 6.8 therefore give

$$\dim_{\mathbb{C}}(\operatorname{in}_{\mathfrak{m}} I)_n \leq \frac{n^5 (1 - 2h)_+^5}{120} + C_{\text{lat}}(n+1)^4,$$

which is the required estimate. The proof uses the local three-pair construction only, with no hypothesis about the global origin of the primary ideal. $\square$

7. COMPLETE SOURCES AND ASYMPTOTIC COMPARISON

We retain the notation of Section 2, in particular the point blow-up $\pi : Y \rightarrow X$ and its exceptional divisor E . We first establish the limits and fixed-divisor estimates for complete point-jet spaces. We then compare their base ideals uniformly over divisorial valuations. A convex profile for all point orders turns this comparison into a lower bound for the complete source at order $75k$.

7.1. Complete limits and fixed divisors. We use the following forms of the asymptotic multiplier-ideal theorem, volume concavity, and the complete-series volume theorem. Their source statements are, respectively, [Laz09, Theorems 2.11 and 4.7, Exercise 4.10], [BFJ09, Theorem A, Proposition 2.12, Theorems 2.16, 3.1 and 3.2], and [LM09, Theorem 2.3, equation (2.6)].

Theorem 7.1 (Asymptotic multiplier ideals). *Let D be a big Cartier divisor on a smooth projective variety. For every positive integer m for which its complete system is nonempty, its base ideal is contained in $\mathcal{J}(\|mD\|)$. For positive integers m, q ,*

$$\mathcal{J}(\|qmD\|) \subseteq \mathcal{J}(\|mD\|)^q.$$

There is a fixed sufficiently positive Cartier divisor A such that $\mathcal{O}(mD + A) \otimes \mathcal{J}(\|mD\|)$ is globally generated for every $m \geq 1$.

The twist in the last assertion is independent of the multiple. This will give one error bound for all valuations.

Theorem 7.2 (Volume and concavity). *On a projective variety of dimension d , volume extends continuously to real divisor classes, and its d -th root is concave on the pseudoeffective cone. On the big cone volume is differentiable, with derivative in direction E equal to $d\langle D^{d-1} \rangle \cdot E$. At a big nef class D this is the usual intersection $dD^{d-1} \cdot E$.*

In particular, limits of real fixed divisors may be used in volume inequalities, and concavity bounds the d -th root of volume by its tangent at an ample class.

Theorem 7.3. *If D is a big Cartier divisor on an irreducible projective variety Z of dimension d , then*

$$\lim_{k \rightarrow \infty} \frac{d! h^0(Z, \mathcal{O}_Z(kD))}{k^d} = \operatorname{vol}_Z(D).$$

The assertion concerns every sufficiently large complete multiple. We will also need fixed twists, to pass from divisible to rounded degrees.

Lemma 7.4. *Let Z be a smooth projective sixfold and let D, T be Cartier divisors on Z . There is a constant $c_{D,T}$ such that, for all sufficiently large positive integers m ,*

$$\left| h^0(Z, \mathcal{O}_Z(mD + T)) - h^0(Z, \mathcal{O}_Z(mD)) \right| \leq c_{D,T} m^5.$$

Thus a fixed twist does not change the leading sixth-degree term, even when D is not big.

Proof. Choose a smooth very ample divisor H and a positive integer c such that $cH + T$ and $cH - T$ are effective. Multiplication by their sections gives

$$h^0(mD - cH) \leq h^0(mD + T) \leq h^0(mD + cH).$$

The same outer bounds contain $h^0(mD)$. The restriction sequences for the successive copies of H give

$$h^0(mD + cH) - h^0(mD - cH) \leq \sum_{i=-c+1}^c h^0(H, \mathcal{O}_H((mD + iH)|_H)).$$

Choose an ample Cartier divisor B such that $B - D$ has an effective member not containing H . Its restricted section, raised to the m -th power, injects each summand into $H^0(H, \mathcal{O}_H((mB + iH)|_H))$. The latter dimension is $O(m^5)$ by the ample Hilbert polynomial. There are only $2c$ summands, which proves the claim. $\square$

Proposition 7.5. *For every fixed positive rational r , put*

$$D_r = 65\pi^*L - rE, \quad V_r(k) = H^0(X, \mathcal{O}_X(65kL) \otimes \mathfrak{m}_x^{\lceil rk \rceil}).$$

Then $720 \dim V_r(k)/k^6$ converges to $\text{vol}_Y(D_r)$. If this volume is positive, the full fixed divisors $F_r(k)$ of all sufficiently large $V_r(k)$ have a common finite support. There is an effective real divisor N_r , supported there, such that

$$F_r(k) \geq kN_r, \quad F_r(k)/k \longrightarrow N_r$$

coefficientwise, in intersection degree and in point multiplicity. All assertions refer to this fixed complete family.

Both limits belong to the chosen complete family. They provide cutoffs before a member or a threshold center is selected.

Proof. We first pass from rounded divisors to complete Cartier multiples. We then use products of actual sections to control every fixed coefficient.

Step 1. In this step, we prove the volume limit in every residue class of rounded degrees. The identity $\pi_* \mathcal{O}_Y(-nE) = \mathfrak{m}_x^n$, for $n \geq 0$, and the projection formula identify

$$V_r(k) = H^0(Y, \mathcal{O}_Y(\lfloor kD_r \rfloor)).$$

Choose a positive integer q with $qr \in \mathbb{Z}$, and write $k = qm + i$, where $0 \leq i < q$. Then

$$\lfloor kD_r \rfloor = m(qD_r) + T_i, \quad T_i = 65i\pi^*L - \lceil ri \rceil E.$$

Lemma 7.4 applies to this finite list of Cartier twists. If qD_r is big, Theorem 7.3 therefore gives, in every residue class,

$$\lim_{m \rightarrow \infty} \frac{720h^0(Y, \mathcal{O}_Y(m(qD_r) + T_i))}{(qm + i)^6} = q^{-6} \text{vol}_Y(qD_r) = \text{vol}_Y(D_r).$$

If qD_r is not big, its volume is zero. The nonnegative sequence $720h^0(Y, m(qD_r))/m^6$ has limsup zero by the definition of volume, and hence converges to zero. The same fixed-twist estimate proves the displayed limit in this case as well. This proves convergence over all positive integers k .

Step 2. In this step, we place all sufficiently late full fixed divisors in a common finite support. Suppose now that the volume is positive, and choose k_0 such that $V_r(k) \neq 0$ for every $k \geq k_0$. Rounding is compatible with multiplication, since

$$\lceil ri \rceil + \lceil rj \rceil \geq \lceil r(i+j) \rceil, \quad V_r(i)V_r(j) \subseteq V_r(i+j).$$

Fix $p \geq k_0$. Choose a nonzero section $u \in V_r(p)$, and a nonzero section $u_d \in V_r(d)$ for each $k_0 \leq d < k_0 + p$. Every sufficiently large k has the form $k = np + d$, with $n \geq 0$ and d in this interval. The product $u^n u_d$ is an actual member of $V_r(k)$. Thus the support of $F_r(k)$ is contained in the finite union $\mathcal{S}$ of the prime components of the divisors of these chosen sections.

Step 3. In this step, we prove convergence and the infimum formula for every fixed coefficient. For a prime $P \in \mathcal{S}$ and a nonzero space $V_r(k)$, set

$$f_P(k) = \min_{0 \neq h \in V_r(k)} \text{ord}_P(h).$$

Products give $f_P(i+j) \leq f_P(i) + f_P(j)$. Fix any index ℓ_0 with $V_r(\ell_0) \neq 0$, including an index below k_0 , and take large representatives d for every residue modulo ℓ_0 . The inequality

$$f_P(n\ell_0 + d) \leq nf_P(\ell_0) + f_P(d)$$

implies

$$\limsup_{k \rightarrow \infty} \frac{f_P(k)}{k} \leq \frac{f_P(\ell_0)}{\ell_0}.$$

Taking the infimum over all such ℓ_0 , and comparing with the corresponding lower bound for the liminf, proves

$$n_P := \lim_{k \rightarrow \infty} \frac{f_P(k)}{k} = \inf_{V_r(k) \neq 0} \frac{f_P(k)}{k} \geq 0.$$

Put $N_r = \sum_{P \in \mathcal{S}} n_P P$. The infimum formula gives $F_r(k) \geq kN_r$, and the finite support gives coefficientwise convergence. In particular,

$$\frac{L^5 \cdot F_r(k)}{k} \longrightarrow L^5 \cdot N_r, \quad \frac{\text{mult}_x F_r(k)}{k} \longrightarrow \sum_{P \in \mathcal{S}} n_P \text{mult}_x P = \text{mult}_x N_r.$$

Here a prime missing x has point multiplicity zero. All components of the full fixed divisor enter these sums. $\square$

For later use, fix any tolerance $\epsilon > 0$ and any finite list of rational orders r . Proposition 7.5 gives one integer J , depending only on that list and X, L, x, ϵ , such that

$$\left| \frac{720 \dim V_r(j)}{j^6} - \text{vol}_Y(D_r) \right| < \epsilon \quad (j \geq J)$$

for every order in the list. This choice precedes any occurrence at a particular level j . If its normalized dimension then exceeds a number c , the volume exceeds $c - \epsilon$, and the normalized dimensions at every sufficiently large level exceed $c - 2\epsilon$. The conclusion is valid even when bigness was initially unknown.

7.2. Actual averages and padding.

Proposition 7.6. *The spaces U_j are nonzero for all sufficiently large j . For such j , $\tau_j = \text{lt}_x(X; \mathbf{q}_j)$ is positive rational. A sufficiently large sufficiently general tuple of actual members gives a pure threshold average Γ_j , log canonical near x but not klt at x , with*

$$\Gamma_j \sim_{\mathbb{Q}} s_j L, \quad s_j = 65j\tau_j, \quad M_j \geq 64j\tau_j, \quad s_j \leq \frac{65}{64}M_j \leq \frac{195}{32}.$$

Its minimal center has dimension at most five. If that center is positive-dimensional, its deficit b_j satisfies $0 < b_j \leq 6 - M_j$.

The average uses members of the complete space with all their fixed components. Its coefficient and point multiplicity refer to that entire average.

Proof. We first show that the spaces exist, then realize their ideal thresholds by actual divisors.

Step 1. In this step, we prove eventual nonemptiness of the complete spaces U_j . Ampleness and integrality give $L^6 \geq 1$. The kernel of point-jet evaluation satisfies

$$\begin{aligned} \dim U_j &\geq h^0(X, \mathcal{O}_X(65jL)) - \text{length}(\mathcal{O}_{X,x}/\mathfrak{m}_x^{64j}) \\ &= \frac{65^6 L^6 - 64^6}{720} j^6 + O(j^5). \end{aligned}$$

The coefficient is positive, so every j beyond a fixed cutoff has $U_j \neq 0$. This inequality only uses the dimension of the jet target.

Step 2. In this step, we compute the positive rational ideal threshold and its point-order bound. For such j , take a log resolution $f : Z \rightarrow X$ of $\mathfrak{q}_j$, with $\mathfrak{q}_j \mathcal{O}_Z = \mathcal{O}_Z(-F)$. Shrink the base around x when discussing log canonicity. If E_i runs through the prime components on this resolution whose images contain x and whose coefficients f_i in F are positive, then

$$\tau_j = \min_i \frac{A_X(E_i)}{f_i}.$$

The simple-normal-crossing criterion computes all other places from this resolution. Smoothness makes the finitely many numerators positive; the formula proves positivity, rationality, and attainment. The point blow-up gives the additional bound

$$0 < \tau_j \leq \frac{6}{\text{ord}_x \mathfrak{q}_j} \leq \frac{6}{64j}.$$

Step 3. In this step, we realize the threshold by an actual average and establish its center and coefficient bounds. The pullback of the actual system, after subtracting F , is basepoint-free. Choose $N > \tau_j$ sufficiently large, and choose $H_1, \dots, H_N \in U_j$ sufficiently generally. Write $f^*H_i = F + H'_i$. Their moving parts and the resolution divisors have simple normal crossings. The crepant boundary for the ideal pair at τ_j is $\tau_j F - K_{Z/X}$; every coefficient is at most one, and some coefficient is one above a center containing x . Passing to $\Gamma_j = (\tau_j/N) \sum_i H_i$ adds the moving components $(\tau_j/N) \sum_i H'_i$, each with coefficient less than one. Consequently the new pair is lc near x , and the coefficient-one fixed strata still give non-klt centers through x . The moving components produce no additional coefficient-one strata. In particular the average realizes the relevant lc centers of the ideal pair.

Theorem 3.1 gives its unique minimal center; it is proper and hence has dimension at most five. The actual classes and point orders give

$$s_j = 65j\tau_j, \quad M_j \geq 64j\tau_j.$$

Log canonicity at the point blow-up gives $M_j \leq 6$, proving all coefficient inequalities. If the minimal center is positive-dimensional, Proposition 4.10 gives $b_j > 0$. The same blow-up has order one on $\mathfrak{m}_x$ and discrepancy $6 - M_j$ for (X, Γ_j) , so the definition of deficit gives $b_j \leq 6 - M_j$. $\square$

Proposition 7.7. *Fix an ample Cartier divisor A , a positive rational r , and the complete source $H^0(X, kA \otimes \mathfrak{m}_x^{rk})$ in admissible degrees. Suppose these spaces are nonzero in all sufficiently large admissible degrees. Let $N = \sum_P n_P P$ be their limiting full fixed divisor. If an actual section of kA has point order $\ell \geq rk$, then $\text{ord}_P(h) \geq \ell n_P / r$ for every fixed prime P . Put $\eta = \text{mult}_x N$. For $\eta \leq r$, the entire degree- ℓ initial image has dimension at most*

$$\binom{\lfloor (1 - \eta/r)\ell \rfloor + 5}{5}.$$

All repeated tangent factors of every fixed prime are included in this bound. On a smooth five-dimensional ambient variety the same assertion holds with the final two occurrences of five replaced by four.

The factor orders increase with the actual point order of a section. Division by these factors gives a local bound on the dimension of the initial image.

Proof. We obtain the larger fixed orders by padding powers of an actual section with global sections, and then count all the resulting initial factors. The finite-support and subadditivity proof of Proposition 7.5 uses only multiplication and eventual nonemptiness. Apply it to the admissible degrees, after clearing the denominator of r . Write $V_k = H^0(X, kA \otimes \mathfrak{m}_x^{rk})$, and let F_k be its full fixed divisor when $V_k \neq 0$. With coefficients normalized by the original degree k , it gives

$$n_P = \inf_{V_k \neq 0} \frac{\text{ord}_P(V_k)}{k}, \quad F_k \geq kN,$$

and a finite set $\mathcal{S}$ containing all eventual fixed supports. The infimum includes earlier nonzero admissible degrees, by the same residue argument.

Step 1. In this step, we amplify each actual section to bound its orders along all limiting fixed primes. Let h have exact point order $\ell \geq rk$. If $\ell = rk$, the infimum formula already gives the claimed bound. Suppose $\ell > rk$. Choose arbitrarily large integers v such that

$$k' = v\ell/r \in \mathbb{Z}, \quad rk' = v\ell \in \mathbb{Z}.$$

The positive integer $k' - vk$ tends to infinity with v . Ampleness makes $(k' - vk)A$ globally generated for large v . Choose a section z_v nonzero at x and at the generic point of every prime in $\mathcal{S}$. Each avoidance requirement excludes a proper linear subspace, and there are only finitely many. The product $h^v z_v$ is an actual global member of $V_{k'}$, with point order $v\ell = rk'$ and P -order $v \text{ord}_P(h)$. Hence

$$v \text{ord}_P(h) \geq \text{ord}_P(V_{k'}) \geq k' n_P = v\ell n_P / r.$$

Division by v proves the first assertion for every fixed prime.

Step 2. In this step, we count the full initial image after retaining every fixed tangent factor. Choose local equations f_P for the primes in $\mathcal{S}$ that contain x , and put $m_P = \text{ord}_x(f_P)$. The local ring is regular and factorial. Thus every actual section of point order at least ℓ , with $\ell \geq rk$, is divisible by

$$G_\ell = \prod_{P \ni x} f_P^{\lceil \ell n_P / r \rceil}.$$

This follows by applying the first assertion to its exact point order, which is at least ℓ . In the polynomial associated graded ring,

$$\text{in}_{\mathfrak{m}_x}(G_\ell) = \prod_{P \ni x} \text{in}_{\mathfrak{m}_x}(f_P)^{\lceil \ell n_P / r \rceil}, \quad d_\ell := \text{ord}_x(G_\ell) = \sum_{P \ni x} m_P \lceil \ell n_P / r \rceil \geq \ell \eta / r.$$

The equality for initial forms uses that this graded ring is a domain. Repeated factors within one initial equation and factors shared by different equations retain all their exponents in this product.

If $d_\ell > \ell$, the degree- ℓ image is zero. Otherwise it lies in the product of this fixed nonzero polynomial with the homogeneous polynomials of degree $\ell - d_\ell$ in six variables. Multiplication is injective, so its dimension is at most

$$\binom{\ell - d_\ell + 5}{5} \leq \binom{\lfloor (1 - \eta/r)\ell \rfloor + 5}{5}.$$

In fact applying the divisibility assertion to any nonzero source section of positive order ℓ gives $\ell \geq \ell \eta / r$, and hence $\eta \leq r$. On a smooth fivefold the same argument has a polynomial graded ring in five variables, whose degree- n part has dimension $\binom{n+4}{4}$. This proves the final assertion. $\square$

7.3. Uniform comparison and the convex profile.

Proposition 7.8. *For a divisorial valuation w whose center on X contains x , let $A_X(w)$ denote its log discrepancy. The limit*

$$\alpha(w) = \lim_{k \rightarrow \infty} \frac{w(\mathfrak{q}_k)}{k} = \inf_{U_k \neq 0} \frac{w(\mathfrak{q}_k)}{k}$$

exists. There are $C_3 \geq 0$ and an integer J , both depending only on X, L, x , such that for all $k \geq J$ and all these valuations simultaneously,

$$k\alpha(w) \leq w(\mathfrak{q}_k) \leq k\alpha(w) + C_3 A_X(w).$$

The constant and the cutoff are uniform over valuations. In particular, the valuation may be chosen by a threshold at a later level.

Proof. We work with the complete divisor on the point blow-up. Products give the asymptotic infimum, and a fixed multiplier-ideal twist bounds the error uniformly.

Step 1. In this step, we identify the complete series on Y and prove the asymptotic valuation infimum. Put $D = 65\pi^*L - 64E$, and let $\mathfrak{a}_k$ be the base ideal of $|kD|$ whenever this system is nonempty. The kernel estimate in the proof of Proposition 7.6 gives

$$h^0(Y, kD) \geq \frac{65^6 L^6 - 64^6}{720} k^6 + O(k^5), \quad 65^6 - 64^6 = 6699413889 > 0.$$

Thus D is big and these complete systems are nonempty in all large degrees. The evaluation maps of the same actual sections give the identity of full ideals

$$(7.1) \quad \mathfrak{q}_k \mathcal{O}_Y = \mathcal{O}_Y(-64kE) \mathfrak{a}_k.$$

For any divisorial valuation w over Y , multiplication gives $w(\mathfrak{a}_{i+j}) \leq w(\mathfrak{a}_i) + w(\mathfrak{a}_j)$. Fix any nonempty degree ℓ_0 , and choose large nonempty degrees d representing every residue modulo ℓ_0 . Then

$$w(\mathfrak{a}_{n\ell_0+d}) \leq nw(\mathfrak{a}_{\ell_0}) + w(\mathfrak{a}_d).$$

Division by $n\ell_0 + d$, followed by the infimum over ℓ_0 , proves

$$(7.2) \quad \theta(w) := \lim_{k \rightarrow \infty} \frac{w(\mathfrak{a}_k)}{k} = \inf_{|kD| \neq \emptyset} \frac{w(\mathfrak{a}_k)}{k} \geq 0.$$

This argument proves convergence separately for each valuation. We next obtain a uniform bound for the error.

Step 2. In this step, we obtain one valuation error bound from a fixed multiplier-ideal twist. By Theorem 7.1, for every positive integer m and every sufficiently large positive integer q ,

$$\mathfrak{a}_{qm} \subseteq \mathcal{J}(\|qmD\|) \subseteq \mathcal{J}(\|mD\|)^q.$$

Taking orders reverses these inclusions. After division by q and passage to the limit in (7.2), we obtain

$$(7.3) \quad w(\mathcal{J}(\|mD\|)) \leq m\theta(w).$$

The global-generation assertion of the same theorem supplies a fixed Cartier divisor B such that

$$\mathcal{O}_Y(mD + B) \otimes \mathcal{J}(\|mD\|)$$

is globally generated for every $m \geq 1$. Choose a positive integer p and an effective divisor $T \in |pD - B|$. Such a choice exists: the positive sixth-degree growth of $h^0(Y, pD)$, together with Lemma 7.4, makes $h^0(Y, pD - B) > 0$ for large p . Multiplication by the section of T sends the globally generating sections above to actual sections of $|(m+p)D|$. Their evaluation therefore gives

$$\mathfrak{a}_{m+p} \supseteq \mathcal{O}_Y(-T) \mathcal{J}(\|mD\|).$$

Using (7.3), we conclude that

$$(7.4) \quad w(\mathfrak{a}_{m+p}) \leq m\theta(w) + w(T).$$

Choose $C_3 > 0$ so large that $(Y, T/C_3)$ is klt. This can be checked on one log resolution of the fixed divisor T : take C_3 larger than every positive multiplicity divided by its positive ambient log discrepancy. The simple-normal-crossing criterion then gives

$$w(T) \leq C_3 A_Y(w)$$

for every divisorial valuation simultaneously. If $T = 0$, any positive constant works. For $k = m + p \geq p + 1$, (7.2) and (7.4), together with $\theta(w) \geq 0$, give

$$k\theta(w) \leq w(\mathfrak{a}_k) \leq (k - p)\theta(w) + C_3 A_Y(w) \leq k\theta(w) + C_3 A_Y(w).$$

Step 3. In this step, we restore the point factor and transfer the uniform bound to the full ideals $\mathfrak{q}_k$. Equation (7.1) gives

$$w(\mathfrak{q}_k) = 64kw(E) + w(\mathfrak{a}_k), \quad \alpha(w) = 64w(E) + \theta(w).$$

It also gives the stated infimum formula for $\alpha(w)$. Since $K_Y = \pi^* K_X + 5E$,

$$A_X(w) = A_Y(w) + 5w(E) \geq A_Y(w).$$

The preceding bounds now prove the assertion. Choose J to be at least $p + 1$ and the fixed nonemptiness cutoff. The choices of B, p, T, C_3, J were made from X, L, x , before selecting any valuation or any later source occurrence. $\square$

More explicitly, for a prescribed $\epsilon > 0$, first choose $J_\epsilon \geq J$ with $C_3/J_\epsilon < \epsilon$. Then

$$0 \leq \frac{w(\mathfrak{q}_j)}{jA_X(w)} - \frac{\alpha(w)}{A_X(w)} < \epsilon \quad (j \geq J_\epsilon)$$

holds for all relevant valuations at once. This is the order of choices needed when a valuation is supplied by one sufficiently late threshold.

Proposition 7.9. *Fix a divisorial valuation w centered on a subvariety containing x , and trivialize $\mathcal{O}_X(65L)$ near x . For rational $t \geq 0$, set*

$$\phi_w(t) = \inf \left\{ \frac{w(h)}{kA_X(w)} : k \geq 1, \quad 0 \neq h \in H^0(X, 65kL), \quad \text{ord}_x h \geq kt \right\},$$

with value $+\infty$ for an empty set. Then ϕ_w is nonnegative, nondecreasing and convex, $\phi_w(0) = 0$, and it is finite for $t < 65$. Moreover $\phi_w(64) = \alpha(w)/A_X(w)$. If $\phi_w(64) > h_0 > 0$, there is a finite supporting slope $a > h_0/64$ such that every actual section of degree $65k$ and point order at least ℓ satisfies

$$w(h) \geq kA_X(w) \max\{0, h_0 + a(\ell/k - 64)\}.$$

The inequality applies on both sides of the order $64k$.

The profile keeps the complete spaces at every point order. Its supporting line therefore constrains lower-order layers as well as higher-order layers.

Proof. Any unit in a neighborhood of x is a unit at the generic center of w . Changing the trivialization therefore does not change $w(h)$. Nonnegativity follows from effectivity of section germs, and monotonicity follows by inclusion of the point-order conditions. Sufficiently ample multiples have a section nonvanishing at x ; its valuation is zero. Thus $\phi_w(0) = 0$.

Step 1. We verify convexity by multiplying actual sections. Choose rational orders t_1, t_2 where the profile is finite, and sections h_i of degrees $65k_i$ whose normalized valuation orders approximate $\phi_w(t_i)$. Given $\lambda = p/q$ with $0 < p < q$, the product

$$h_1^{pk_2} h_2^{(q-p)k_1}$$

has degree $65qk_1k_2$, point order at least $qk_1k_2(\lambda t_1 + (1 - \lambda)t_2)$, and normalized valuation equal to the corresponding convex combination. Taking the infimum and letting the approximation errors tend to zero gives

$$\phi_w(\lambda t_1 + (1 - \lambda)t_2) \leq \lambda \phi_w(t_1) + (1 - \lambda)\phi_w(t_2).$$

The endpoint weights are immediate, and a value $+\infty$ on the right imposes no additional inequality.

Step 2. In this step, we prove finiteness below order 65 and identify the profile value at order 64. For rational $t < 65$, the same complete jet-kernel estimate as before has leading coefficient $(65^6 L^6 - t^6)/720 > 0$. It gives nonzero sections at every sufficiently large degree satisfying the rounded point-order condition. Thus the profile is finite on $[0, 65)$. At $t = 64$, minimizing valuation over the actual sections in each degree gives

$$\phi_w(64) = \inf_{U_k \neq 0} \frac{w(\mathfrak{q}_k)}{kA_X(w)} = \frac{\alpha(w)}{A_X(w)}$$

by Proposition 7.8.

Step 3. In this step, we obtain a supporting line that bounds every actual point-order layer. A finite convex function on the rational points of an open interval extends continuously to a real convex function there. Apply this near 64, and choose a finite supporting slope a at 64. Monotonicity of secant slopes shows that its supporting inequality holds at every rational point of the original domain, including points at least 65 where actual sections exist. Evaluation of that inequality at zero gives

$$0 = \phi_w(0) \geq \phi_w(64) - 64a, \quad a \geq \frac{\phi_w(64)}{64} > \frac{h_0}{64}.$$

For an actual section satisfying the order condition at ℓ , the definition of $\phi_w(\ell/k)$ now gives

$$\frac{w(h)}{kA_X(w)} \geq \phi_w(\ell/k) \geq \phi_w(64) + a(\ell/k - 64) \geq h_0 + a(\ell/k - 64).$$

Combining this with nonnegativity proves the asserted bound on the whole domain. $\square$

7.4. From the full profile to volume.

Proposition 7.10. *Assume $L^6 = 1$. If a divisorial valuation w with height-two center through x satisfies*

$$\frac{\alpha(w)}{A_X(w)} > 15,$$

then

$$\text{vol}_Y(65\pi^*L - 75E) > 1418890625.$$

The estimate uses the full leading images of every actual point-order layer below $75k$, including layers below $64k$.

The strict numerical gain comes from summing the constraints on the entire profile. In particular, the layers below the original source order contribute to the estimate.

Proof. We put each actual layer inside a full primary ideal and apply Theorem 6.11. We then bound the resulting integral for every possible supporting slope and subtract this loss from the complete Hilbert growth.

Step 1. In this step, we place every constrained actual layer in a full primary ideal with a controlled generic threshold. Write $S = \mathcal{O}_{X,x}$, $\mathfrak{m} = \mathfrak{m}_x S$, and let $P \subset S$ be the height-two prime that is the center of w . Set $A_w = A_X(w) > 0$. Proposition 7.9, with $h_0 = 15$, supplies a finite slope $a > 15/64$. Put

$$T_a(t) = 15 + a(t - 64), \quad t_a = 64 - 15/a \in (0, 64), \quad W_{k,\ell} = H^0(X, 65kL \otimes \mathfrak{m}_x^\ell).$$

The order- ℓ layer has dimension $d_{k,\ell} = \dim(W_{k,\ell}/W_{k,\ell+1})$. For $T_a(\ell/k) > 0$, define the ideal

$$I_{k,\ell} = \{f \in S : w(f) \geq kA_w T_a(\ell/k)\}.$$

The valuation is positive on P and zero on $S \setminus P$. The finitely many generators of P have a positive minimum value, so a sufficiently large power of P lies in $I_{k,\ell}$, and $I_{k,\ell} \subseteq P$. If a factor outside P multiplies an element into this ideal, its valuation zero shows that the element already belongs to the ideal. Thus $I_{k,\ell}$ is P -primary.

Its generic threshold $\lambda_{k,\ell}$ is positive. The same valuation on the regular local surface S_P has discrepancy A_w , so

$$(7.5) \quad \lambda_{k,\ell} := \text{lct}_{S_P}(I_{k,\ell}S_P) \leq \frac{A_w}{w(I_{k,\ell})} \leq \frac{1}{kT_a(\ell/k)}.$$

Every actual germ in $W_{k,\ell}$ belongs to $I_{k,\ell}$ by the supporting-line inequality. Taking its degree- ℓ initial form has kernel exactly $W_{k,\ell+1}$, and hence gives an injection

$$(7.6) \quad W_{k,\ell}/W_{k,\ell+1} \hookrightarrow (\text{in}_{\mathfrak{m}} I_{k,\ell})_{\ell}.$$

These are the full germs of the actual global sections, including all their fixed components.

Both terms of H_* in Theorem 6.11 are nonincreasing on $[0, \infty)$. For its second term, the derivative on $(0, 1/2)$ is $-80c(1-2c)^3$; beyond $1/2$ it vanishes. The theorem and (7.5) therefore give, for $\ell \geq 1$ with $T_a(\ell/k) > 0$,

$$(7.7) \quad d_{k,\ell} \leq \frac{\ell^5}{120} H_*\left(\frac{kT_a(\ell/k)}{\ell}\right) + C_0(\ell+1)^4.$$

The constant is independent of k, ℓ , of the primary ideal, and of its threshold. If $T_a(\ell/k) \leq 0$, we use instead the full polynomial bound $d_{k,\ell} \leq \binom{\ell+5}{5}$. The layer $\ell = 0$ has dimension at most one.

Step 2. In this step, we bound every layer below $64k$ by a polynomial majorant of the capacity profile. For $t_a \leq t \leq 64$, the ratio

$$\frac{T_a(t)}{t} = a - \frac{64a-15}{t}$$

increases from zero to $15/64 < 1/4$. On $0 \leq c \leq 1/4$ we claim

$$(7.8) \quad H_*(c) \leq 2(1-c)^5 - (1-2c)^5.$$

For the first term this follows from $(1-c)^5 \geq (1-2c)^5$. Subtracting the second term from the right side gives

$$2c^2(10-50c+85c^2-49c^3).$$

The cubic decreases on $[0, 1/4]$: its derivative is $-50+170c-147c^2 < 0$ there. Its value at $1/4$ is $131/64 > 0$, proving the claim. Define

$$F_a(t) = \begin{cases} t^5, & 0 \leq t \leq t_a, \\ 2(t - T_a(t))^5 - (t - 2T_a(t))^5, & t_a \leq t \leq 64. \end{cases}$$

The bounds just proved give

$$d_{k,\ell} \leq \frac{k^5}{120} F_a(\ell/k) + C(\ell+1)^4 \quad (0 \leq \ell < 64k),$$

after increasing an absolute constant C to include the polynomial and order-zero bounds. Every layer below the original source order is present in this estimate.

Step 3. In this step, we bound the layers from $64k$ to $75k$ using the endpoint capacity profile. For $64 \leq t \leq 75$, the same ratio is at least $15/64$. Consequently (7.5) implies

$$\ell \lambda_{k,\ell} \leq \frac{\ell}{kT_a(\ell/k)} \leq \frac{64}{15} < \frac{9}{2}.$$

The final assertion of Theorem 6.11 applies, and yields

$$d_{k,\ell} \leq \frac{k^5}{120} G_a(\ell/k) + C(\ell+1)^4, \quad G_a(t) = (t - T_a(t))_+^5, \quad 64k \leq \ell < 75k.$$

This also covers the range where the positive part is zero.

Step 4. In this step, we bound the integrated loss uniformly over every possible supporting slope. Put

$$\mathcal{U}(a) = 6 \int_0^{64} F_a(t) dt, \quad \mathcal{V}(a) = 6 \int_{64}^{75} G_a(t) dt.$$

We prove

$$(7.9) \quad \mathcal{U}(a) + \mathcal{V}(a) < 74000000000 \quad \left(a \geq \frac{15}{64} \right).$$

We include the endpoint $a = 15/64$ by taking $t_a = 0$. For a fixed $t < 64$, increasing a decreases $T_a(t)$. Where $0 \leq T \leq t/2$, the derivative in T of $2(t - T)^5 - (t - 2T)^5$ is

$$-10(t - T)^4 + 10(t - 2T)^4 \leq 0.$$

At $T = 0$ this expression for the function agrees with t^5 . It follows that $\mathcal{U}$ is nondecreasing and $\mathcal{U}(a) \leq 64^6$. For $t > 64$, increasing a increases $T_a(t)$, so $\mathcal{V}$ is nonincreasing.

Set $Q = 49^6$ and $R = 34^6$. Integration of the fifth powers gives

$$(7.10) \quad \mathcal{U}(a) = t_a^6 + \frac{2(Q - t_a^6)}{1 - a} - \frac{R - t_a^6}{1 - 2a},$$

$$(7.11) \quad \mathcal{V}(a) = \frac{(60 - 11a)_+^6 - Q}{1 - a}.$$

At $a = 1/2$ or $a = 1$ the integral gives the continuous value. For example, $t - T_a(t) = 49$ when $a = 1$, and $t - 2T_a(t) = 34$ when $a = 1/2$.

The following are strict rational upper bounds, in units of 10^9 . In each finite interval the first bound is for $\mathcal{U}$ at its right endpoint and the second for $\mathcal{V}$ at its left endpoint. Monotonicity makes their sum an upper bound throughout that interval.

Slope interval	$\mathcal{U}/10^9$	$\mathcal{V}/10^9$
$[15/64, 1/2]$	213/5	2981/100
$[1/2, 3/4]$	243/5	249/10
$[3/4, 1]$	262/5	43/2
$[1, 5/4]$	551/10	187/10
$[5/4, 3/2]$	57	163/10
$[3/2, 2]$	597/10	71/5
$[2, 3]$	312/5	109/10
$[3, 4]$	641/10	7
$[4, \infty)$	1718/25	2307/500

We give the arithmetic certificate for both columns. The values needed for the first column are

$$\begin{aligned} \mathcal{U}(1/2) &= 4Q - 282 \cdot 34^5 = 42552359236, \\ \mathcal{U}(3/4) &= 8Q + 2R - 9 \cdot 44^6 = 48513081736, \\ \mathcal{U}(1) &= 180 \cdot 49^5 + R = 52390349236, \\ \mathcal{U}(5/4) &= \frac{25}{3} 52^6 - 8Q + \frac{2}{3} R = 55054652536, \\ \mathcal{U}(3/2) &= \frac{9}{2} 54^6 - 4Q + \frac{1}{2} R = 56984354236, \\ \mathcal{U}(2) &= \frac{8}{3} \left(\frac{113}{2} \right)^6 - 2Q + \frac{1}{3} R < 59616787070, \end{aligned}$$

$$\begin{aligned}\mathcal{U}(3) &= \frac{9}{5}59^6 - Q + \frac{1}{5}R = 62392634236, \\ \mathcal{U}(4) &= \frac{32}{21}\left(\frac{241}{4}\right)^6 - \frac{2}{3}Q + \frac{1}{7}R < 64100000000.\end{aligned}$$

For the sixth line use $(113/2)^3 < 180400$. For the last line use $(241/4)^3 < 219000$; then its upper bound is

$$\frac{32}{21}47961000000 - \frac{2}{3}Q + \frac{1}{7}R < 64100000000.$$

All the displayed equalities use the following integer powers:

$$\begin{array}{cc|cc} 49^5 & 282475249 & 49^6 & 13841287201 \\ 34^5 & 45435424 & 34^6 & 1544804416 \\ 44^6 & 7256313856 & 52^6 & 19770609664 \\ 54^6 & 24794911296 & 59^6 & 42180533641 \end{array}$$

For the last interval we use $\mathcal{U}(a) \leq 64^6 = 68719476736 < 68720000000$.

For the second column, the exact formula (7.11) gives

$$\begin{aligned}\mathcal{V}(15/64) &= \left(\frac{49}{64}\right)^5 (75^6 - 64^6) < 29810000000, \\ \mathcal{V}(1/2) &= 2 \left(\left(\frac{109}{2}\right)^6 - Q \right) < 2(162000^2 - Q) = 24805425598, \\ \mathcal{V}(3/4) &= 4 \left(\left(\frac{207}{4}\right)^6 - Q \right) < 4(138600^2 - Q) = 21474691196, \\ \mathcal{V}(1) &= 66 \cdot 49^5 = 18643366434, \\ \mathcal{V}(5/4) &= 4 \left(Q - \left(\frac{185}{4}\right)^6 \right) < 4(Q - 98900^2) = 16240308804, \\ \mathcal{V}(3/2) &= 2 \left(Q - \left(\frac{87}{2}\right)^6 \right) < 2(Q - 82300^2) = 14135994402, \\ \mathcal{V}(2) &= Q - 38^6 < Q - 3000000000 < 10900000000, \\ \mathcal{V}(3) &< Q/2 < 7000000000, \\ \mathcal{V}(4) &< Q/3 < 4614000000.\end{aligned}$$

Here the cubed comparisons are

$$(109/2)^3 < 162000, \quad (207/4)^3 < 138600, \quad (185/4)^3 > 98900, \quad (87/2)^3 > 82300.$$

For the first line,

$$\frac{49}{64} < \frac{77}{100}, \quad \left(\frac{77}{100}\right)^5 < \frac{271}{1000}, \quad 75^6 - 64^6 = 109259038889 < 110000000000,$$

so the upper bound is $(271/1000)110000000000 = 29810000000$. For the seventh line, $38^6 = 3010936384 > 3000000000$. This verifies every entry of (7.12) by rational comparisons. The nine row sums are, respectively,

$$\frac{7241}{100}, \quad \frac{147}{2}, \quad \frac{739}{10}, \quad \frac{369}{5}, \quad \frac{733}{10}, \quad \frac{739}{10}, \quad \frac{733}{10}, \quad \frac{711}{10}, \quad \frac{36667}{500}.$$

Every one is less than 74. This proves (7.9) for every finite slope, including arbitrarily large slopes.

Step 5. In this step, we subtract the integrated loss from the complete Hilbert growth to obtain the strict volume gain. For the fixed valuation and slope, F_a and G_a are continuous on their respective integration intervals and piecewise polynomial. Their Riemann sums converge. The

uniform error over all $0 \leq \ell < 75k$ has sum $O(k^5)$. Telescoping the actual complete spaces from order zero therefore gives

$$\begin{aligned} \limsup_{k \rightarrow \infty} \frac{720}{k^6} [h^0(X, 65kL) - \dim W_{k,75k}] \\ \leq 6 \int_0^{64} F_a(t) dt + 6 \int_{64}^{75} G_a(t) dt = \mathcal{U}(a) + \mathcal{V}(a) < 74000000000. \end{aligned}$$

Since $L^6 = 1$, the ample Hilbert polynomial gives

$$\lim_{k \rightarrow \infty} \frac{720h^0(X, 65kL)}{k^6} = 65^6 = 75418890625.$$

The space $W_{k,75k}$ is exactly $H^0(Y, k(65\pi^*L - 75E))$. Proposition 7.5 identifies its normalized limit with the volume, whether or not bigness was known in advance. Consequently

$$\text{vol}_Y(65\pi^*L - 75E) \geq 65^6 - \mathcal{U}(a) - \mathcal{V}(a) > 75418890625 - 74000000000 = 1418890625.$$

The primary-image argument accounts for all layers and the slope certificate gives the strict margin. This proves the assertion. $\square$

8. INTRINSIC DIVISORS AND LIFTING

We retain the ambient notation of Section 2. We first transfer divisors from a resolution of a Cartier center. We then construct point boundaries on a smooth fivefold, first with zero initial boundary and then with an arbitrary effective klt boundary. The relative construction will give an additive endpoint for a fivefold center smooth at the specified point.

Lemma 8.1. *Let W be an integral Cartier divisor in X , smooth at x , and let $f : \widetilde{W} \rightarrow W$ be a smooth projective resolution of its normalization, isomorphic near x . Put $P = f^*(L|_W)$ and let y be the point above x . There are an effective Cartier divisor F avoiding y and a positive rational ϵ_0 such that $P - \epsilon_0 F$ is ample. For small positive rational δ, θ with $(1 + \delta)(1 - \theta) > 1$, the divisor*

$$A = (1 + \delta)(P - \theta\epsilon_0 F)$$

is ample and satisfies $A^d \cdot T \geq L^d \cdot f(T) \geq 1$ for every positive-dimensional integral T through y . These A may be chosen arbitrarily close to P . Suppose $\Gamma = W + R$ is an effective rational boundary, klt after adjunction to W near x , and put $D = f^(R|_W)$. If an effective rational $H \sim_{\mathbb{Q}} vA$, $0 < v < 7$, makes $D + H$ lc at y with point minimal center, then for every $\varepsilon > 0$ there is an actual effective rational $Q \sim_{\mathbb{Q}} uL$, not containing W , such that $\Gamma + Q$ has that point minimal center and $u < v(1 + \delta) + \varepsilon$. All of R is retained. The assertion concerns the restriction germ of Q ; it asserts no equality of intrinsic and ambient point orders.*

The two increases in the coefficient are $v\delta$ and the cost of a fixed twist divided by an arbitrarily large power.

Proof. We construct the ample perturbation on the given resolution, descend an actual section through the normalization, and lift it from W .

Step 1. In this step, we construct the ample perturbation using an effective correction away from y . Write $A_0 = L|_W$, and choose an ample Cartier divisor $B_{\widetilde{W}}$ on $\widetilde{W}$. The coherent sheaf

$$f_* \mathcal{O}_{\widetilde{W}}(-B_{\widetilde{W}}) \otimes \mathcal{O}_W(m_0 A_0)$$

is globally generated for all sufficiently large integers m_0 . Since f is an isomorphism near x , its stalk there is invertible. Choose a global section whose value at x is nonzero. The projection formula gives a section of $\mathcal{O}_{\widetilde{W}}(m_0 P - B_{\widetilde{W}})$ nonvanishing at y . Its zero divisor F is effective Cartier, avoids y , and satisfies

$$P - \frac{1}{m_0} F \sim_{\mathbb{Q}} \frac{1}{m_0} B_{\widetilde{W}}.$$

Thus $\epsilon_0 = 1/m_0$ has the required property. In particular, this construction makes no assumption on the exceptional support of a resolution.

The divisor P is nef. For $0 < \theta < 1$, we have

$$P - \theta\epsilon_0 F = (1 - \theta)P + \theta(P - \epsilon_0 F),$$

a positive sum of a nef and an ample class. Let T be integral through y , and put $d = \dim T$. The open set on which f is an isomorphism meets T , so $T \rightarrow f(T)$ is birational. The projection formula and ampleness of L give

$$P^d \cdot T = L^d \cdot f(T) \in \mathbb{Z}_{>0}.$$

Every mixed intersection of P and $P - \epsilon_0 F$ on T is nonnegative. Consequently

$$A^d \cdot T \geq ((1 + \delta)(1 - \theta))^d P^d \cdot T \geq L^d \cdot f(T) \geq 1.$$

This proves the bound simultaneously for every such T , including $T = \widetilde{W}$. Taking $\delta \rightarrow 0$ and $0 < \theta < \delta/(1 + \delta)$ makes A approach P .

Step 2. In this step, we descend powers of the intrinsic section through normalization using a fixed conductor twist. The effective divisor

$$\widehat{H} = H + v(1 + \delta)\theta\epsilon_0 F \sim_{\mathbb{Q}} v(1 + \delta)P$$

has the same germ as H at y . Choose positive integers n_0, M_0 such that $M_0/n_0 = v(1 + \delta)$ and $n_0\widehat{H}$ is the zero divisor of a section

$$a \in H^0(\widetilde{W}, \mathcal{O}_{\widetilde{W}}(M_0 P)).$$

Factor f as $\widetilde{W} \xrightarrow{h} W^\nu \xrightarrow{\nu} W$, where ν is the normalization. Since W^ν is normal, $h_*\mathcal{O}_{\widetilde{W}} = \mathcal{O}_{W^\nu}$; the projection formula therefore descends a to a section $\bar{a}$ on W^ν .

Consider the coherent conductor-Hom sheaf

$$\mathcal{C} = \mathcal{H}om_{\mathcal{O}_W}(\nu_*\mathcal{O}_{W^\nu}, \mathcal{O}_W).$$

Its stalk at x is $\mathcal{O}_{W,x}$, because ν is an isomorphism there. For a fixed sufficiently large positive integer c , the sheaf $\mathcal{C} \otimes \mathcal{O}_W(cA_0)$ is globally generated. Choose a homomorphism

$$\phi : \nu_*\mathcal{O}_{W^\nu} \longrightarrow \mathcal{O}_W(cA_0)$$

whose local multiplier at x is a unit. Tensoring ϕ by $\mathcal{O}_W(rM_0A_0)$ and applying it to $\bar{a}^r$ gives an actual section

$$a_r \in H^0(W, \mathcal{O}_W((rM_0 + c)A_0)).$$

Near x , this section is a unit times $\bar{a}^r$. Thus its local zero divisor, divided by rn_0 , is the germ of H .

Step 3. In this step, we lift the descended section to an actual ambient divisor while retaining the full residual boundary. Since W is Cartier, its restriction sequence is

$$0 \longrightarrow \mathcal{O}_X((rM_0 + c)L - W) \longrightarrow \mathcal{O}_X((rM_0 + c)L) \longrightarrow \mathcal{O}_W((rM_0 + c)A_0) \longrightarrow 0.$$

Serre vanishing kills the first H^1 for all sufficiently large r . Lift a_r to a section on X , and divide its zero divisor by rn_0 . The resulting effective rational divisor Q does not contain W , since $a_r \neq 0$, and satisfies

$$Q \sim_{\mathbb{Q}} uL, \quad u = v(1 + \delta) + \frac{c}{rn_0} < v(1 + \delta) + \varepsilon$$

after increasing r . Its restriction to W agrees with H near the point. The different of $\Gamma + Q = W + R + Q$ is therefore $R|_W + H$ there. Theorem 4.2 gives log canonicity and the point minimal center for $\Gamma + Q$. The entire residual divisor R occurs in this application. $\square$

8.1. Cutting on a smooth fivefold. For a smooth projective fivefold Z , a point z , and an ample rational Cartier divisor A , write

$$r_A(V) = \left(\frac{\text{mult}_z V}{A^{\dim V} \cdot V} \right)^{1/\dim V}.$$

For an additive chain on Z with deficits ξ_i and minimal centers V_i , ending at z , write

$$C_A = \sum_i (\xi_i - \xi_{i+1}) r_A(V_i), \quad \xi_N = 0.$$

This is the scalar cost for the polarization A .

Lemma 8.2. *Let Z be a smooth complex projective fivefold, let $z \in Z$, and let A be ample rational Cartier with $A^{\dim T} \cdot T \geq 1$ for every positive-dimensional integral T through z . Let Δ be effective rational, lc but not klt near z , with positive-dimensional minimal center V . Put $d = \dim V$, $m = \text{mult}_z V$, and $\xi = \text{lct}_z((Z, \Delta); \mathbf{m}_z)$. Then every additive cutting chain from Δ to the point satisfies the following bounds.*

(i) *A curve is smooth and has $C_A \leq \xi \leq 1$. For a surface, $m \leq 4$, $\xi \leq 2/m$, and*

$$C_A \leq q_m \xi, \quad (q_1, q_2, q_3, q_4) = \left(1, \frac{17}{12}, \frac{97}{56}, 2 \right).$$

(ii) *For a threefold, the complete list is*

m	upper bound for ξ	upper bound for C_A
1	3	$\xi + 5/12$
2	2	$(19/15)\xi + 29/120$
3	$5/3$	$(577/400)\xi + 811/4200$
4	1	$(8/5)\xi + 3/80$
5	$3/5$	$(7/4)\xi$
6	$1/3$	2ξ

(iii) *For a fourfold, the complete list is*

m	upper bound for ξ	upper bound for C_A
1	4	$\xi + 4/5$
2	3	$(6/5)\xi + 1871/3360$
3	2	$(4/3)\xi + 3/5$
4	1	$(17/12)\xi + 7/24$

If $\text{edim}_z V \leq 4$, then $C_A \leq \xi + 4/5$.

(iv) *Put $\gamma = 299/295$. Unless V is a fourfold of multiplicity two or three,*

$$C_A - \gamma \xi \leq \frac{91}{100}.$$

For every V ,

$$C_A - \frac{5}{6}\xi \leq \frac{5567}{3360}.$$

These bounds apply to the entire current boundary. In particular, they do not depend on whether the initial boundary was obtained by an absolute or a relative threshold.

Proof. We first bound the Hilbert data, then retain the restrictions from each parent center when estimating its successors. This also accounts for chains that skip a dimension.

Step 1. In this step, we bound the multiplicities and deficits of all proper centers on the smooth fivefold. Theorem 4.5 makes each minimal center normal and Cohen–Macaulay near z . Its local

ring is excellent and analytically unramified. Apply Theorem 4.6 to a general Artinian minimal reduction, with Hilbert function h_i . Writing $\mu_1 = \sum_i i h_i$, we have

$$\begin{aligned} h_0 &= 1, & h_1 &= \text{edim}_z V - d, & m &= \sum_i h_i, \\ \mu_1 &\leq \bar{e}_1(\mathfrak{m}_{V,z}), & 2\bar{e}_1(\mathfrak{m}_{V,z}) &\leq (d - \xi)m. \end{aligned}$$

The last inequality and $\xi > 0$ are Proposition 4.10. In particular $\mu_1 \geq m - 1$. Multiplicity one gives $h_1 = 0$, so V is regular at z .

A normal curve is regular, and the same inequality gives $\xi \leq 1$. For a surface, $\mu_1 < m$, so integrality gives $\mu_1 = m - 1$. Since

$$\mu_1 - (m - 1) = \sum_{i \geq 2} (i - 1) h_i,$$

all h_i for $i \geq 2$ vanish. Thus $m = 1 + h_1 \leq 4$ and $\xi \leq 2/m$. More generally, a surface of embedding dimension at most e has $m \leq e - 1$.

For a threefold, $h_1 \leq 2$ and $h_2 \leq 3$. Hence

$$\mu_1 \geq m - 1, \quad \mu_1 \geq 2m - 4, \quad \mu_1 \geq 3m - 10.$$

Together with $2\mu_1 < 3m$, the last bound gives $m \leq 6$. Substitution of $m = 1, \dots, 6$ gives the deficit caps in (ii). For a fourfold, the Artinian reduction has at most one generator. If it is not a field, its Hilbert function is a string of m ones. Consequently

$$\bar{e}_1(\mathfrak{m}_{V,z}) \geq \frac{m(m-1)}{2}, \quad \xi \leq 5 - m, \quad m \leq 4.$$

The same argument for a threefold of embedding dimension at most four gives $m \leq 3$ and $\xi \leq 4 - m$.

Step 2. In this step, we retain the parent-center restrictions and express complete costs through successor surpluses. An lc place of an earlier boundary remains an lc place after an effective addition preserving log canonicity: its discrepancy was zero, so its order on the addition must be zero. Thus an earlier normal center remains available in Proposition 4.11. If V' is a codimension-one successor of V and w is the deficit of the boundary at that successor, then

$$(8.1) \quad \text{mult}_z V' \leq (d - w)m - 2\bar{e}_1(\mathfrak{m}_{V,z}).$$

This uses the new deficit w . Embedding dimension cannot increase for a closed subvariety of the germ of V .

Every radius is at most $m^{1/d}$. We use the elementary majorants

$$\begin{aligned} \sqrt{2} &< 17/12, & \sqrt{3} &< 97/56 < 7/4, \\ \sqrt[3]{2} &< 19/15, & \sqrt[3]{3} &< 577/400 < 13/9, & \sqrt[3]{4} &< 8/5, & \sqrt[3]{5} &< 7/4, & \sqrt[3]{6} &< 2. \end{aligned}$$

The two close comparisons follow from $97^2 = 9409 > 3 \cdot 56^2 = 9408$ and $577^3 = 192100033 > 3 \cdot 400^3 = 192000000$; the other comparisons follow by taking the indicated powers. A surface tail has only curves and the point after it, so its cost is at most $q_m \xi$. This proves (i).

For clarity, if the parent radius is at most a , and a successor of deficit w has remaining cost C'_A , then

$$(8.2) \quad C_A \leq a(\xi - w) + C'_A = a\xi + (C'_A - aw).$$

We bound the last term over every possible first successor.

Step 3. In this step, we compute all threefold tails and the bound in embedding dimension at most four. For a smooth threefold, a surface successor has multiplicity at most two. Its excess over w is at most $(17/12 - 1) \cdot 1 = 5/12$. For $m = 2$, the embedding dimension is at most four

and (8.1) gives surface multiplicity $j \leq 4 - 2w$. At $j = 2$ the excess over $19w/15$ is at most $3/20$. At $j = 3$, we have $w \leq 1/2$, and the excess is at most

$$\left(\frac{7}{4} - \frac{19}{15}\right) \frac{1}{2} = \frac{29}{120}.$$

For $m = 3$, we have $j \leq 5 - 3w$. At $j = 3, 4$, respectively, this gives $w \leq 2/3, 1/3$. Their excesses over $577w/400$ are bounded by

$$\frac{2}{3} \left(\frac{97}{56} - \frac{577}{400}\right) = \frac{811}{4200}, \quad \frac{1}{3} \left(2 - \frac{577}{400}\right) < \frac{811}{4200}.$$

For $j \leq 2$, the surface radius is smaller than $577/400$. For $m = 4, 5, 6$, the lower bound $\bar{e}_1 \geq 2m - 4$ gives

$$j \leq 8 - m - mw.$$

Thus $j \leq 3, 2, 1$, respectively. At $m = 4, j = 3$, we have $w \leq 1/4$, giving the excess $(7/4 - 8/5)/4 = 3/80$. For $m = 5, 6$, the displayed parent majorant dominates all subsequent radii. Direct curves and points have no positive excess. This proves (ii).

If a threefold has embedding dimension at most four, its multiplicities are at most three. For $m = 2$, the excess over ξ is at most

$$\left(\frac{19}{15} - 1\right) 2 + \frac{29}{120} = \frac{31}{40} < \frac{4}{5}.$$

For $m = 3$, its first Hilbert moment is at least three, so $j \leq 3 - 3w$ and only $j \leq 2$ is possible. All radii are then at most $13/9$, and $\xi \leq 1$ gives excess at most $4/9$. A surface in embedding dimension at most four has $m \leq 3$, giving excess at most $1/2$. The smooth-threefold case has excess at most $5/12$. Finally, a fourfold of embedding dimension at most four is smooth. Its first radius is at most one, and all its proper centers satisfy these bounds. Equation (8.2) proves the embedding-dimension assertion and the smooth-fourfold row.

Step 4. In this step, we bound the complete tails of singular fourfold centers. For multiplicity two or three, the respective threefold successor restrictions are

$$j \leq 6 - 2w, \quad j \leq 6 - 3w.$$

Only $j = 1, \dots, 5$ can occur. For multiplicity two, subtract $6w/5$ from the bounds in (ii) and use $w \leq \min\{\text{the deficit cap in (ii)}, (6 - j)/2\}$. The resulting bounds, including a direct surface, are

j	1	2	3	4	5	surface
$C'_A - (6/5)w$	5/12	3/8	1871/3360	7/16	11/40	2/5.

For the third entry, the value is

$$\left(\frac{577}{400} - \frac{6}{5}\right) \frac{3}{2} + \frac{811}{4200} = \frac{1871}{3360}.$$

The direct-surface entry follows by maximizing $(q_j - 6/5)(2/j)$ over the positive slopes. This third entry is the largest one.

For multiplicity three, use the weaker cubic-threefold bound $13w/9 + 11/54$; both coefficients exceed those in (ii). Subtracting $4w/3$ gives

j	1	2	3	4	5	surface
$C'_A - (4/3)w$	5/12	29/120	17/54	31/144	5/36	1/3.

For each j , we used $w \leq (6 - j)/3$ and its intrinsic deficit cap; a nonpositive slope contributes at most its constant term. All these entries are at most $3/5$. The parent radii satisfy $\sqrt[4]{2} < 6/5$ and $\sqrt[4]{3} < 4/3$.

For multiplicity four, a threefold successor has $j \leq 4 - 4w$, hence $j \leq 3$. At $j = 2$, $w \leq 1/2$ and every remaining radius is at most $7/4$, giving excess at most $1/6$ over $17w/12$. At $j = 3$, $w \leq 1/4$ and every remaining radius is at most two, giving excess at most $7/48$. At $j = 1$, all remaining radii are at most $17/12$. A direct surface contributes at most $7/24$, by (i). Since

$\sqrt[4]{4} < 17/12$, this proves the last row of (iii). Direct curves and points are bounded by all the parent majorants.

Step 5. In this step, we derive the two affine bounds from the complete center tables. For the first inequality in (iv), centers of embedding dimension at most four contribute at most $4/5$, since $\gamma > 1$. For threefold multiplicities 3, 4, 5, 6, the respective upper bounds for $C_A - \gamma\xi$ are

$$\frac{150001}{165200}, \quad \frac{131}{80} - \gamma, \quad \frac{21}{20} - \frac{3}{5}\gamma, \quad \frac{2}{3} - \frac{1}{3}\gamma.$$

The first is $(577/400 - \gamma)(5/3) + 811/4200$, and is less than $91/100$. The other three are smaller. A multiplicity-four fourfold gives $41/24 - \gamma < 91/100$. Surfaces give at most $4/5$ when their embedding dimension is at most four; the remaining surface has multiplicity four and gives $(2 - \gamma)/2 < 91/100$. This exhausts the first inequality.

For the second inequality, the fourfold rows give, in order,

$$\frac{22}{15}, \quad \frac{11}{10} + \frac{1871}{3360} = \frac{5567}{3360}, \quad \frac{8}{5}, \quad \frac{7}{8}.$$

The first two threefold rows are bounded by $13/10$, using the embedding-dimension estimate and $\xi \leq 3$. The cubic row gives

$$\left(\frac{577}{400} - \frac{5}{6}\right) \frac{5}{3} + \frac{811}{4200} < \frac{5}{4}.$$

The remaining threefold rows give $193/240, 11/20, 7/18$. The surface rows give at most $2/3$, and curves give at most $1/6$. All are at most $5567/3360$, as required. $\square$

Lemma 8.3. *Let Z be a smooth complex projective fivefold, let $z \in Z$, and let D be an effective rational divisor klt near z . Put $b = \text{lct}_z((Z, D); \mathfrak{m}_z)$. Let A be ample rational Cartier, let $a > 0$ be rational with aA Cartier, and let*

$$0 \neq V \subseteq H^0(Z, \mathcal{O}_Z(aA) \otimes \mathfrak{m}_z^\ell), \quad \ell \in \mathbb{Z}_{>0}.$$

Let $\mathfrak{a}$ be its base ideal and $\tau = \text{lct}_z((Z, D); \mathfrak{a})$. Then τ is positive rational. Sufficiently many sufficiently general members of V give an effective rational divisor $G \sim_{\mathbb{Q}} a\tau A$ such that $D + G$ is lc and not klt at z , and, with $\xi = \text{lct}_z((Z, D + G); \mathfrak{m}_z)$,

$$a\tau \leq \frac{a}{\ell}(b - \xi).$$

For an effective rational divisor $G_0 \sim_{\mathbb{Q}} cA$ with $\mu = \text{mult}_z G_0 > 0$, its actual threshold t against D likewise satisfies

$$tc \leq \frac{c}{\mu}(b - \xi), \quad \xi = \text{lct}_z((Z, D + tG_0); \mathfrak{m}_z).$$

The estimates charge only the divisor being added; D remains fixed.

Proof. On a simultaneous log resolution of D , $\mathfrak{a}$, and $\mathfrak{m}_z$, the relative threshold is a minimum of positive rational discrepancy-to-order ratios over the relevant divisors. It is finite because $\mathfrak{a} \subseteq \mathfrak{m}_z^\ell$ and hence $\tau \leq b/\ell$.

Step 1. In this step, we realize the relative ideal threshold by an actual divisor while retaining D . Pull the system back to its fixed divisor and its basepoint-free moving part. Choose an integer N with $\tau/N < 1$, and then choose N general members so that their moving parts have simple normal crossings with one another and with the resolved boundary. Their average with coefficient τ/N defines G . The discrepancy calculation on this resolution gives the same lc centers as the ideal threshold pair.

Step 2. In this step, we bound the added coefficient by the decrease of the point deficit. Denote by $A_{Z,D}(w)$ the log discrepancy of a divisorial valuation w for the pair (Z, D) . For $w(\mathfrak{m}_z) > 0$, the discrepancy of $D + G$ is at most

$$A_{Z,D}(w) - \tau w(\mathfrak{a}) \leq A_{Z,D}(w) - \tau \ell w(\mathfrak{m}_z).$$

Divide by $w(\mathfrak{m}_z)$ and take the infimum to obtain $\xi \leq b - \tau\ell$. For the last assertion, take a sufficiently divisible multiple of G_0 and its one-dimensional space of sections. The same argument gives $\xi \leq b - t\mu$. $\square$

After any such threshold, a prescribed error allowance $h > 0$ can be maintained through the subsequent cuts. There are at most four cuts, because the first center is proper in Z . Choose positive numbers h_i with $\sum h_i < h$. At a stage of radius $r_A(V_i)$, use Theorem 3.5 with $\epsilon_i < h_i/(1 + r_A(V_i))$. Its deficit inequality gives

$$t_i \leq (\xi_i - \xi_{i+1})r_A(V_i) + \epsilon_i r_A(V_i).$$

The actual total added coefficient is therefore less than $C_A + h$. In particular every strict bound for a complete scalar tail gives an actual rational point boundary after choosing the errors inside that bound.

8.2. The complete intrinsic systems.

Lemma 8.4. *Let Z be a smooth complex projective fivefold, let $z \in Z$, and let A be ample rational Cartier with $A^{\dim T} \cdot T \geq 1$ for every positive-dimensional integral subvariety T through z , including Z . Put*

$$q = \frac{299}{50}, \quad \beta = \frac{59}{10}, \quad q_1 = \frac{6}{5}q = \frac{897}{125}.$$

Choose a positive integer d_0 clearing the denominators of qA and β , and for $d_0 \mid k$ set

$$\mathcal{E}_k = H^0(Z, \mathcal{O}_Z(kqA) \otimes \mathfrak{m}_z^{\beta k}).$$

Then $\mathcal{E}_k \neq 0$ for all sufficiently large permitted k . Their full fixed divisors have a common finite support in these degrees, and there is a finite effective real divisor $N = \sum_T n_T T$ such that

$$n_T = \lim_{k \rightarrow \infty, d_0 \mid k} \frac{\text{ord}_T(\mathcal{E}_k)}{k} = \inf_{\mathcal{E}_k \neq 0} \frac{\text{ord}_T(\mathcal{E}_k)}{k}.$$

If $\text{mult}_z N > 7/2$, then every sufficiently large permitted $\mathcal{E}_k$ contains an actual section of point order greater than $q_1 k$.

The last assertion produces a divisor of point-order-to-coefficient ratio greater than $6/5$ in the same complete family.

Proof. We compare the dimension supplied by the point-jet kernel with the dimension allowed by the entire fixed divisor.

Step 1. In this step, we prove the positive lower dimension bound for the complete intrinsic systems. Asymptotic Riemann–Roch and Serre vanishing for the ample rational class, after restricting to the permitted indices, give

$$h^0(Z, \mathcal{O}_Z(kqA)) = \frac{q^5 A^5}{120} k^5 + O(k^4).$$

The length of the point jets is $\text{length}(\mathcal{O}_{Z,z}/\mathfrak{m}_z^{\beta k}) = \binom{\beta k + 4}{5}$. The rank of evaluation is at most this length; surjectivity is not needed. Therefore

$$(8.3) \quad \dim \mathcal{E}_k \geq \frac{R_0}{120} k^5 + O(k^4), \quad R_0 = q^5 A^5 - \beta^5 \geq q^5 - \beta^5 > 480.$$

Indeed $q - \beta = 2/25$, and

$$q^5 - \beta^5 > 5(q - \beta)\beta^4 = \frac{2}{5}\beta^4 > 480, \quad \beta^4 = \frac{12117361}{10000} > 1200.$$

Step 2. In this step, we construct the limiting full fixed divisor and prove its coefficient infimum formulas. Let F_k be the full fixed divisor and put $r_T(k) = \text{ord}_T(\mathcal{E}_k)$ whenever $\mathcal{E}_k \neq 0$. Products of actual sections give

$$\mathcal{E}_k \mathcal{E}_l \subseteq \mathcal{E}_{k+l}, \quad r_T(k+l) \leq r_T(k) + r_T(l).$$

Fix one permitted k_0 with $\mathcal{E}_{k_0} \neq 0$, and a permitted K_0 after which all spaces are nonzero. For large permitted k , write $k = jk_0 + r$, with r in the finite permitted set $[K_0, K_0 + k_0)$. Then

$$F_k \leq jF_{k_0} + F_r.$$

The union of these finitely many supports contains every late fixed divisor. For any fixed eligible m_0 , division into the finitely many large residues modulo m_0 similarly gives

$$\limsup_{k \rightarrow \infty, d_0 | k} \frac{r_T(k)}{k} \leq \frac{r_T(m_0)}{m_0}.$$

The infimum of all the right sides bounds every term from below, proving the asserted limit and infimum. Primes outside the finite late support have limit zero. Thus N has finite support, and $F_k \geq kN$ in every eligible degree.

Step 3. In this step, we use the full fixed factors to force an actual section of order greater than $q_1 k$. Apply Proposition 7.7 on the fivefold, with the integral polarization $d_0 qA$, prescribed order $d_0 \beta$, and index k/d_0 . Its limiting fixed coefficients are $d_0 n_T$. Thus every actual nonzero section $a \in \mathcal{E}_k$, of exact point order ℓ , satisfies

$$(8.4) \quad \text{ord}_T(a) \geq \frac{\ell}{\beta} n_T \quad \text{for every } T \in \text{Supp } N.$$

Write $\eta = \text{mult}_z N$. Point multiplicities of effective Cartier divisors add in the regular local ring at z . Applying (8.4) to one nonzero section gives $\ell \geq (\ell/\beta)\eta$, hence $\eta \leq \beta$. Under the hypothesis of the lemma, put

$$\alpha = 1 - \frac{\eta}{\beta}, \quad 0 \leq \alpha < \frac{24}{59} < \frac{41}{100}.$$

For integral $\ell \geq \beta k$, set $\mathcal{E}_{k,\ell} = H^0(Z, \mathcal{O}_Z(kqA) \otimes \mathfrak{m}_z^\ell)$. The same padding proposition, now using $\eta \leq \beta$, bounds the actual initial image by

$$\dim(\mathcal{E}_{k,\ell}/\mathcal{E}_{k,\ell+1}) \leq \binom{\lfloor \alpha \ell \rfloor + 4}{4}.$$

This bound includes the full product of the local equations of all fixed primes, with their integral multiplicities. Coincident tangent factors are not removed. If $\mathcal{E}_k$ contained no section of order greater than $q_1 k$, its filtration would terminate at $\lfloor q_1 k \rfloor$. Summing the displayed dimensions would give

$$\dim \mathcal{E}_k \leq \frac{\alpha^4(q_1^5 - \beta^5)}{120} k^5 + O(k^4).$$

For $\alpha = 0$, the sum has only $O(k)$ terms and the same inequality holds. But

$$q_1 < \frac{36}{5}, \quad q_1^5 < 19400, \quad \beta^5 = \frac{714924299}{100000} > 7000, \quad \left(\frac{41}{100}\right)^4 < \frac{3}{100},$$

where $41^4 = 2825761 < 3000000$. Hence

$$\alpha^4(q_1^5 - \beta^5) < \frac{3}{100}(19400 - 7000) = 372 < 480.$$

This contradicts (8.3) for every sufficiently large permitted k . □

Theorem 8.5. *Let Z be a smooth complex projective fivefold, z a closed point and A an ample rational Cartier divisor such that $A^{\dim T} \cdot T \geq 1$ for every positive-dimensional integral subvariety T through z , including Z . There is a point boundary $P \sim_{\mathbb{Q}} vA$ at z with $0 < v < 299/50$.*

The construction uses only degrees of subvarieties through the specified point.

Proof. We first show that a divisor of point-order-to-coefficient ratio at least $6/5$ suffices. We then use the complete systems of Lemma 8.4 to produce one unless a threshold already gives the desired endpoint. Throughout this proof put

$$q = \frac{299}{50}, \quad \beta = \frac{59}{10}, \quad \gamma = \frac{q}{\beta} = \frac{299}{295}.$$

Step 1. In this step, we construct the required point boundary from a divisor with point-order-to-coefficient ratio at least $6/5$. Suppose $G_0 \sim_{\mathbb{Q}} cA$ is effective with $\text{mult}_z G_0/c \geq 6/5$. Threshold its full divisor against the zero boundary, and let ξ_0 be the resulting deficit. Since the initial deficit is five, Lemma 8.3 gives an added coefficient

$$u_0 \leq \frac{5}{6}(5 - \xi_0).$$

A point threshold has $u_0 \leq 25/6$. Otherwise, the last inequality of Lemma 8.2 gives

$$u_0 + C_A \leq \frac{25}{6} + \frac{5567}{3360} = \frac{19567}{3360} < \frac{47}{8}.$$

Choose total cutting error less than $1/100$. The resulting point boundary has positive coefficient less than

$$\frac{47}{8} + \frac{1}{100} = \frac{1177}{200} < \frac{299}{50}.$$

Step 2. In this step, we reduce every threshold without an affordable point continuation to a double or triple fourfold center. Suppose, for contradiction, that no such point boundary exists. Use the complete spaces $\mathcal{E}_k$ from Lemma 8.4. Let N be their limiting full fixed divisor from that lemma. Threshold their actual base ideals against zero and choose sufficiently general actual averages as in Lemma 8.3. If τ_k is the threshold, $u_k = qk\tau_k$ its coefficient, and ξ its deficit, then

$$u_k \leq \gamma(5 - \xi).$$

For any uniform scalar bound K_V on the entire tail from this threshold, the contrary assumption forces $u_k + K_V \geq q$. If it were strictly smaller, the subsequent errors could be chosen inside that margin.

For centers of embedding dimension at most four, including a smooth fourfold and threefolds of multiplicity at most two, Lemma 8.2 gives

$$u_k + C_A \leq 5\gamma + \frac{4}{5} < \frac{59}{10} < q,$$

because $\gamma > 1$. For a cubic threefold, the bound is

$$u_k + C_A \leq \frac{10}{3}\gamma + \frac{21817}{8400} = \frac{2961603}{495600} = q - \frac{139}{33040} < q - \frac{1}{500}.$$

Here $21817/8400 = (577/400)(5/3) + 811/4200$. For threefold multiplicities four, five, and six, the bounds are respectively

$$4\gamma + \frac{131}{80}, \quad \frac{22}{5}\gamma + \frac{21}{20}, \quad \frac{14}{3}\gamma + \frac{2}{3},$$

each less than $29/5$. For a surface of multiplicity j , the bound is 5γ if $q_j \leq \gamma$, and otherwise is

$$5\gamma + (q_j - \gamma)\frac{2}{j} < \frac{29}{5},$$

with q_j from Lemma 8.2. Curves give at most 5γ . A fourfold of multiplicity four gives $4\gamma + 41/24 < 29/5$. A point threshold has $u_k \leq 5\gamma < q$. The multiplicity lists are exhaustive by that lemma. Thus every sufficiently late threshold has a fourfold minimal center of multiplicity two or three.

Step 3. In this step, we force enough fixed multiplicity to obtain an actual higher-order section and contradict failure of the point boundary. Let V be one of these divisorial centers at level k ,

and write $n_k = \text{ord}_V(\mathcal{E}_k)/k$. The moving coefficients in the chosen average are less than one, so V is a fixed prime of $\mathcal{E}_k$. Its coefficient-one identity is

$$\tau_k k n_k = 1, \quad u_k = \frac{q}{n_k}, \quad \xi \leq 5 - \frac{\beta}{n_k}.$$

For multiplicity two we may weaken the constant 1871/3360 to 3/5 in the fourfold tail estimate. Failure of the endpoint then gives

$$q \leq u_k + \frac{6}{5}\xi + \frac{3}{5} \leq \frac{33}{5} - \frac{(6/5)\beta - q}{n_k}.$$

Since $(6/5)\beta - q = 11/10$ and $33/5 - q = 31/50$,

$$n_k \geq \frac{55}{31}, \quad 2n_k \geq \frac{110}{31} > \frac{7}{2}.$$

For multiplicity three the corresponding calculation is

$$q \leq \frac{20}{3} + \frac{3}{5} - \frac{(4/3)\beta - q}{n_k}, \quad n_k \geq \frac{283}{193}, \quad 3n_k \geq \frac{849}{193} > \frac{7}{2}.$$

The finite support in Lemma 8.4 contains a prime that occurs as such a center at arbitrarily large levels. Its multiplicity is fixed. Passing to its coefficient limit gives $\text{mult}_z N > 7/2$. The same lemma now produces an actual section in $\mathcal{E}_k$ of order greater than $(6/5)qk$. Its full divisor divided by k has class qA and point-order-to-coefficient ratio greater than $6/5$. The first part of the proof gives the desired point boundary, contradicting the assumption. $\square$

Theorem 8.6. *Let Z be a smooth complex projective fivefold, z a closed point and A an ample rational Cartier divisor with $A^{\dim T} \cdot T \geq 1$ for every positive-dimensional integral subvariety T through z , including Z . Let D be effective rational and klt near z , and set $b = \text{lt}_z((Z, D); \mathfrak{m}_z)$. Suppose $3 \leq b \leq 5$. For any real $s \geq 0$ with $s \leq (65/69)(6 - b) + e$, $e < 1/128$, there is an effective rational $H \sim_{\mathbb{Q}} vA$ such that $(Z, D + H)$ is log canonical near z with minimal center $\{z\}$ and $s + v < 7 - 1/256$. No proportionality between D and A is assumed.*

Here s is an independently specified cost. The construction adds only H and retains every component of D .

Proof. We use the same complete systems, now thresholded against D . If their first thresholds do not give the endpoint, their fixed divisors force a better actual section. We threshold that section against the same D and pay its entire cutting chain. Put

$$\lambda = \frac{65}{69}, \quad p_2 = \frac{1871}{3360}, \quad q = \frac{299}{50}, \quad \beta = \frac{59}{10}, \quad \gamma = \frac{299}{295}.$$

The point blow-up on Z gives

$$(8.5) \quad \text{mult}_z D \leq 5 - b.$$

Step 1. In this step, we retain D at each relative threshold and exclude all centers except double or triple fourfolds. Take the complete $\mathcal{E}_k$ of Lemma 8.4. Let N be their limiting full fixed divisor from that lemma. Let τ_k be the relative threshold of its actual base ideal against D , and choose a sufficiently general actual average as in Lemma 8.3. If ξ is the new deficit and $u_k = qk\tau_k$ is the added coefficient, that lemma gives

$$(8.6) \quad u_k \leq \gamma(b - \xi).$$

The threshold is positive and at most $b/(\beta k)$. Thus the moving coefficients can be chosen less than one.

Suppose that the conclusion of the theorem fails. If the new center is not a fourfold of multiplicity two or three, Lemma 8.2(iv) gives

$$\begin{aligned} s + u_k + C_A &\leq \lambda(6 - b) + e + \gamma b + \frac{91}{100} \\ &\leq \lambda + 5\gamma + \frac{91}{100} + e < \frac{174}{25} + \frac{1}{128} < 7 - \frac{1}{64}. \end{aligned}$$

Here $\gamma > \lambda$, $b \leq 5$, $\lambda < 19/20$, and $\gamma < 51/50$. The last strict comparison follows from $7 - 174/25 - 1/128 = 103/3200 > 1/64$. A point threshold satisfies the same bound with $C_A = \xi = 0$. The cutting errors could be chosen within the fixed margin to $7 - 1/256$. Hence every sufficiently late threshold has a fourfold minimal center of multiplicity two or three.

Step 2. In this step, we use the retained coefficient of D to force fixed point multiplicity greater than $7/2$. For such a center V , put

$$m = \text{mult}_z V \in \{2, 3\}, \quad c_V = \text{coeff}_V D, \quad n_k = \frac{\text{ord}_V(\mathcal{E}_k)}{k}.$$

Since D is klt near z , $0 \leq c_V < 1$. The moving coefficients are less than one, so V is a fixed prime and the exact coefficient-one equation is

$$(8.7) \quad \tau_k k n_k = 1 - c_V.$$

The effectivity of the entire D and (8.5) give

$$(8.8) \quad m c_V \leq 5 - b, \quad m(1 - c_V) \geq b + m - 5.$$

No relation between the divisor class of D and A enters either inequality.

Set

$$(a_2, a_3) = \left(\frac{6}{5}, \frac{4}{3}\right), \quad p_3 = \frac{3}{5}, \quad T_0 = 7 - \frac{1}{256}.$$

The fourfold bounds and (8.6)–(8.7) imply

$$s + u_k + C_A \leq 6\lambda + (a_m - \lambda)b + p_m + e - \frac{(a_m\beta - q)(1 - c_V)}{n_k}.$$

If this right side were less than T_0 , actual cuts with sufficiently small errors would prove the theorem. Define

$$\Delta_m = (a_m - \lambda)b + p_m + 6\lambda - 7 + e + \frac{1}{256}.$$

The contrary assumption thus forces $\Delta_m > 0$, since the subtracted term is positive. Rearranging and using (8.8), we obtain

$$(8.9) \quad m n_k \geq \frac{(a_m\beta - q)(b + m - 5)}{\Delta_m}.$$

Both right sides are strictly greater than $7/2$. We give the uniform comparisons explicitly. For $m = 2$,

$$a_2\beta - q = \frac{11}{10}, \quad \Delta_2|_{b=3} < -\frac{3307}{618240}.$$

The affine function

$$\frac{11}{10}(b - 3) - \frac{7}{2}\Delta_2$$

is positive at $b = 3$ and has slope

$$\frac{11}{10} - \frac{7}{2} \left(\frac{6}{5} - \frac{65}{69} \right) = \frac{68}{345} > 0.$$

For $m = 3$,

$$a_3\beta - q = \frac{283}{150}, \quad \Delta_3|_{b=3} < \frac{49}{115} + \frac{3}{256} < \frac{9}{20}.$$

The corresponding affine function is

$$\frac{283}{150}(b-2) - \frac{7}{2}\Delta_3.$$

At $b = 3$ it is greater than $283/150 - (7/2)(9/20) = 187/600 > 0$, and its slope is

$$\frac{283}{150} - \frac{7}{2} \left(\frac{4}{3} - \frac{65}{69} \right) = \frac{892}{1725} > 0.$$

These comparisons use only $e < 1/128$ and hold for every $3 \leq b \leq 5$. For the fixed b, e , the positive gap from $7/2$ in (8.9) is independent of the level and of the center: there are just the two values of m . An alternative with $\Delta_m \leq 0$ has already been excluded.

Step 3. In this step, we threshold the improved actual section against D and pay its entire continuation. The late fixed support of $\mathcal{E}_k$ is finite. A prime occurs as the actual center along an unbounded subsequence. Passing (8.9), with its fixed strict gap, to the coefficient limit gives $\text{mult}_z N > 7/2$. Lemma 8.4 then supplies an actual section of point order greater than $(6/5)qk$. Its full divisor, divided by k , has class qA . Threshold this divisor against D , and write w_0 for the added coefficient and ξ_* for the resulting deficit. Lemma 8.3 gives

$$w_0 \leq \frac{5}{6}(b - \xi_*).$$

Every possible new proper center, including both exceptional fourfold types, is covered by the second inequality of Lemma 8.2(iv). For a point center put $C_A = \xi_* = 0$. Thus

$$\begin{aligned} s + w_0 + C_A &\leq \lambda(6 - b) + e + \frac{5}{6}b + \frac{5567}{3360} \\ &\leq 3\lambda + \frac{5}{2} + \frac{5567}{3360} + e \\ &= 3\lambda + \frac{18}{5} + p_2 + e. \end{aligned}$$

The second inequality uses $b \geq 3$ and $5/6 < \lambda$. The exact remaining margin is

$$7 - \left(3\lambda + \frac{18}{5} + p_2 \right) = \frac{1319}{77280}, \quad \frac{1319}{77280} - \frac{1}{128} = \frac{2861}{309120} > \frac{1}{128}.$$

Reserve total cutting error less than $1/2048$. Since $1/128 - 1/2048 > 1/256$, the full added divisor H satisfies $s + v < 7 - 1/256$. All thresholds and all subsequent effective additions kept D unchanged. This contradicts the assumed failure and proves the theorem. $\square$

8.3. The ambient endpoint.

Theorem 8.7. *Let $\Gamma \sim_{\mathbb{Q}} sL$, $s \geq 0$, be log canonical near x , with a fivefold minimal center smooth at x . If*

$$s \leq \frac{65}{69} \text{mult}_x \Gamma + e, \quad e < \frac{1}{128},$$

then there is an effective rational divisor $Q \sim_{\mathbb{Q}} uL$ such that the entire $\Gamma + Q$ is a point boundary and $0 < s + u < 7 - 1/512$. No global smoothness of the center is required, and its residual boundary is retained.

The coefficient comparison in this theorem uses $65/69$. Its hypothesis must be checked for the actual boundary to which it is applied.

Proof. We use ambient cutting at small deficit and the relative intrinsic construction at larger deficit. Write

$$\lambda = \frac{65}{69}, \quad M = \text{mult}_x \Gamma, \quad b = \text{lct}_x((X, \Gamma); \mathfrak{m}_x).$$

Let W be the global prime closure of the minimal center. It is Cartier on the smooth X . As a divisorial lc center it has coefficient one, so

$$\Gamma = W + R, \quad R \geq 0, \quad W \not\subseteq \text{Supp } R.$$

Step 1. In this step, we identify the intrinsic deficit and the coefficient bounds for the full residual boundary. Every component of R restricts as an effective rational Cartier divisor on the integral W . Near x , where W is smooth, write $D_0 = R|_W$. Theorem 4.2 makes D_0 klt there: a non-klt center through x would be an lc center of Γ properly contained in W . The same adjunction applied to the point ideal gives

$$(8.10) \quad b = \text{lct}_x((W, D_0); \mathfrak{m}_{W,x}).$$

To check the ideal assertion from the divisor statement, resolve the point ideal together with R and W , and replace any fixed rational ideal exponent by a sufficiently large general average of local generators. Its restriction is the corresponding average for $\mathfrak{m}_x \mathcal{O}_W = \mathfrak{m}_{W,x}$. The resolved pairs have the discrepancies of the ideal pairs. Cartier adjunction therefore gives equivalence of log canonicity for every rational exponent. Taking suprema proves (8.10).

The point blow-ups on X and on the smooth germ of W give

$$(8.11) \quad \begin{aligned} 0 < b &\leq 5, & M + b &\leq 6, \\ \text{mult}_x D_0 + b &\leq 5, & s &\leq \lambda(6 - b) + e. \end{aligned}$$

These inequalities do not equate $\text{mult}_x D_0$ with $M - 1$.

Step 2. In this step, we construct an ambient additive point continuation when $b \leq 3$. If $b \leq 3$, the smooth-fivefold row of Proposition 5.5 bounds every entire ambient chain by $C \leq b + 3887/3360$. Hence

$$\begin{aligned} s + C &\leq \lambda(6 - b) + b + \frac{3887}{3360} + e \\ &\leq 3\lambda + \frac{18}{5} + \frac{1871}{3360} + e < 7 - \frac{1}{128}. \end{aligned}$$

The exact margin is the one computed in the preceding theorem. Proposition 3.7, with total allowance $1/2048$, constructs an effective additive point boundary of total coefficient less than $7 - 1/512$.

The multiplicity-eight and multiplicity-nine cases of the general six-dimensional bounds in Proposition 5.5 are covered by Lemma 5.1. In the row used here, all proper centers lie in the smooth germ of W , so their embedding dimension is at most five. The Hilbert argument in Lemma 8.2 then bounds every threefold descendant's multiplicity by six. In particular the two larger multiplicities cannot occur in this application.

Step 3. In this step, we construct and transfer an intrinsic point addition for $3 \leq b \leq 5$. Suppose $3 \leq b \leq 5$. Take a smooth projective resolution $f : \widetilde{W} \rightarrow W$ of the normalization, unchanged near x , and let y be the point above x . The effective rational divisor

$$D = f^*(R|_W)$$

is klt near y and has deficit b , by (8.10). Choose the ample perturbation A of Lemma 8.1, with

$$0 < \delta < \frac{1}{7 \cdot 2048}, \quad (1 + \delta)(1 - \theta) > 1.$$

Its degree bound holds for every integral subvariety through y . Theorem 8.6, applied with $Z = \widetilde{W}$ and $z = y$ to the same D and the cost s in (8.11), gives

$$\begin{aligned} H \sim_{\mathbb{Q}} vA, \quad & (\widetilde{W}, D + H) \text{ lc with point minimal center at } y, \\ s + v &< 7 - \frac{1}{256}. \end{aligned}$$

In particular $v < 7$. The correction by the effective divisor away from y costs $v\delta < 1/2048$. Apply Lemma 8.1 with $\varepsilon = 1/2048$. The conductor twist divided by a large power costs less

than this allowance, and gives an actual effective $Q \sim_{\mathbb{Q}} uL$ with point minimal center for $\Gamma + Q$. Its full coefficient satisfies

$$s + u < 7 - \frac{1}{256} + \frac{1}{2048} + \frac{1}{2048} = 7 - \frac{3}{1024} < 7 - \frac{1}{512}.$$

The effective divisor Γ is nonzero and has coefficient $s > 0$, since $\Gamma \cdot L^5 = sL^6 > 0$. Thus $0 < s + u$. Both cases retain the entire Γ , and the high-deficit case uses only a neighborhood of x on which W is smooth. $\square$

9. COMPLETE THREEFOLD ENDPOINTS

We retain the notation of Section 2. We prove that every threefold minimal center satisfying the stated coefficient bound admits an additive continuation to the point. The scalar estimates leave one case, with multiplicity four and degree one. We determine its full tangent cone and compare the number of intrinsic sections with the conditions imposed by their actual threshold centers.

Theorem 9.1 (Restricted-order deficit drop). *Let (Y, B) be klt, let $G \geq 0$ be a rational divisor such that $(Y, B + G)$ is lc at x , and let Z be its minimal lc center there. Suppose $D \geq 0$ is rational Cartier, does not contain Z , and $(Y, B + G + D)$ is lc at x with a properly smaller minimal center. Then its deficit is at most the old deficit minus $\text{ord}_x(D|_Z)$, with the deficit convention of the cited definition. Lemma 3.8 identifies that convention with the point-ideal convention used here.*

This is the local restriction inequality of [YZ20, Definition 2.5 and Lemma 2.10]. It measures the order on the old center; it requires no equality with the ambient point order.

Theorem 9.2 (Negativity). *Let $f : Y \rightarrow T$ be a projective birational morphism of normal varieties. If D is real Cartier and $-D$ is nef over T , then $D \geq 0$ if and only if $f_*D \geq 0$.*

We use [Fuj11, Lemma 4.16(1)] in this sign convention to compare a moduli trace with its nef trace on a higher model.

Theorem 9.3 (A split node in a flat family). *Let $f : Y \rightarrow S$ be locally of finite type over a locally Noetherian scheme, flat at y , and suppose that y is a split node of its fiber over s . Then there is $h \in \mathfrak{m}_{\widehat{\mathcal{O}}_{S,s}}$ with*

$$\widehat{\mathcal{O}}_{Y,y} \simeq \widehat{\mathcal{O}}_{S,s}[[u, v]]/(uv - h).$$

The successive lifting argument in [Sta26, Tag 0CBX, Lemma 53.20.11] also gives the complete local form used below. We check its flatness and split-node hypotheses at the attachment curves.

Definition 9.4. For integers $0 \leq a \leq b$, $a + b > 0$, the notation $S(a, b)$ denotes the image of $\text{Proj}_{\mathbb{P}^1}(\mathcal{O}(a) \oplus \mathcal{O}(b))$ under its tautological complete linear system, with the induced hyperplane polarization. The degree-four Veronese surface is $(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(2))$.

9.1. Reduction and the full tangent cone.

Proposition 9.5. *Let $\Gamma \sim_{\mathbb{Q}} sL$, $s \geq 0$, be log canonical near x with a threefold minimal center W . Suppose $s \leq (65/64)\text{mult}_x \Gamma + e$, $e < 1/2560$, and there is no effective additive point continuation of the entire Γ with total coefficient less than $7 - 1/4096$. Then*

$$\text{mult}_x W = 4, \quad \text{edim}_x W = 6, \quad L^3 \cdot W = 1, \quad \frac{6}{5} < b = \text{def}_x(\Gamma) \leq \frac{3}{2}.$$

Writing $\delta = 6 - \text{mult}_x \Gamma$, one has

$$0 \leq \delta - b < \frac{374b - 449}{650} \leq \frac{56}{325}.$$

The full tangent ring $G = \text{gr}_{\mathfrak{m}_{W,x}} \mathcal{O}_{W,x}$ is reduced and Cohen–Macaulay with Hilbert series $(1 + 3z)/(1 - z)^3$. Its polarized projectivization is one of the following:

- (i) An integral Veronese surface or $S(2, 2)$, $S(1, 3)$, or $S(0, 4)$.
- (ii) Two quadric cones joined scheme-theoretically along a reduced line.
- (iii) $S(0, 3)$ and a plane joined scheme-theoretically along a reduced ruling line through the cone vertex.

The ordinary point blow-up is normal and Cohen–Macaulay; it agrees near its exceptional divisor with the normalized blow-up.

The list concerns the entire tangent scheme. In particular, neither irreducibility nor a bound on the number of its components is assumed.

Proof. We first pay the numerical alternatives, then compare the ordinary and normal blow-ups. Effective adjunction proves reducedness, and intersection inequalities exclude the unwanted component arrangements. Put $M = \text{mult}_x \Gamma$, $\rho = 65/64$, and $\varepsilon_* = 1/4096$.

Step 1. In this step, we reduce failure of an additive continuation to the stated multiplicity-four threefold. Write $m = \text{mult}_x W$ and $g = L^3 \cdot W$. Theorems 4.5 and 4.6 give an Artinian reduction with

$$h_0 = 1, \quad h_1 = \text{edim}_x W - 3 \leq 3, \quad \sum_i h_i = m, \quad \bar{e}_1 \geq m - 1, \quad \bar{e}_1 \geq 2m - 2 - h_1 \geq 2m - 5.$$

Here and below the Hilbert coefficients refer to the point ideal of W . Proposition 4.10 and the ambient point blow-up yield

$$(9.1) \quad 2\bar{e}_1 \leq (3 - b)m, \quad 0 < b \leq 6 - M, \quad b \leq \frac{10}{m} - 1, \quad m \leq 9.$$

Also $\text{edim}_x W \leq m + 2$.

Every old lc place dominating W remains an lc place after an effective addition preserving log canonicity: its order on that addition must be zero. Thus W remains a normal lc center along every additive cutting chain, even after it ceases to be minimal. If such a chain visits a surface of multiplicity μ and deficit $\beta > 0$, Proposition 4.11 gives

$$(9.2) \quad \mu \leq (3 - \beta)m - 2\bar{e}_1.$$

By Corollary 4.13, $\mu \leq 5$, $\beta \leq 2/\mu$, and every later curve is smooth with radius at most one. A surface with radius at most $\alpha_1 \geq 1$ has entire remaining scalar cost at most $\alpha_1\beta$, since $\alpha_1(\beta - y) + y \leq \alpha_1\beta$ for its possible curve deficit y . Consequently, if the threefold radius is at most $\alpha_0 \geq 1$, the whole chain costs at most

$$(9.3) \quad \alpha_0 b + (\alpha_1 - \alpha_0)\beta$$

when it visits that surface, and at most $\alpha_0 b$ when it skips surfaces.

If $m \leq 2$, the embedding dimension is at most four, so Corollary 5.7 contradicts the hypothesis. For $m = 3$, Proposition 5.6(i) gives

$$s + C \leq 6\rho + \left(\frac{577}{400} - \rho\right) \frac{5}{3} + \frac{811}{4200} + e = 7 - \frac{19}{11200} + e.$$

The positive slope justifies using $b \leq 5/3$. Proposition 3.7, with total new allowance $1/2560$, gives a point coefficient less than $7 - 1/1280$, because $19/11200 > 4/2560$. For $m = 4$ and embedding dimension at most five, the first moment bound gives $b \leq 1$; again Corollary 5.7 applies.

For $m \geq 5$, (9.2) becomes $\mu \leq 10 - m - m\beta$. At $m = 5$, only a surface of multiplicity four can have radius greater than $7/4$, and then $\beta \leq 1/5$. At $m = 6, 7$, all radii are at most two. The remaining two rows are precisely the completed cases in Lemma 5.1: for $m = 8$, $b \leq 1/4$ and a positive-deficit surface successor is smooth; for $m = 9$, $b \leq 1/9$ and no such surface successor

exists. Thus

m	deficit bound used	whole cost bound
5	$b \leq 1$	$C \leq 7b/4 + 1/20$
6, 7, 8	$b \leq 2/3$	$C \leq 2b$
9	$b \leq 1/9$	$C \leq 9b/4$

The inequalities $(7/4)^3 > 5$ and $(9/4)^3 > 9$ verify the parent radius bounds. Nonincreasing deficits prove these bounds also for direct jumps to curves or the point. Combining the rows with (9.1) gives respectively

$$s + C \leq \frac{2201}{320} + e, \quad s + C \leq \frac{27}{4} + e, \quad s + C \leq \frac{195}{32} + \frac{79}{576} + e.$$

All three are less than 1101/160, using even $e < 1/320$. Adding a new allowance $1/2560$ gives the required contradiction. For the two large-multiplicity rows, the sharper deficit caps also give the following comparisons for the linear tail envelopes:

$$(9.4) \quad \begin{array}{c|c|c} m & C - \frac{2301}{2000}b & C - \frac{9}{10}b \\ \hline 8 & \leq \frac{1699}{8000} < \frac{3447}{4000} & \leq \frac{11}{40} < \frac{100663}{60000} \\ 9 & \leq \frac{733}{6000} < \frac{3447}{4000} & \leq \frac{3}{20} < \frac{100663}{60000} \end{array}$$

The first row follows by multiplying $2 - 2301/2000$ and $2 - 9/10$ by $1/4$; the second uses $9/4 - 2301/2000$ and $9/4 - 9/10$, multiplied by $1/9$. Thus both rows of Lemma 5.1, including their direct jumps, satisfy these envelope bounds.

It remains to consider $m = 4$, with embedding dimension six. Here $\bar{e}_1 \geq 3$, so $b \leq 3/2$ and $\mu \leq 6 - 4\beta$. The surface multiplicities 2, 3, 4, 5 have respective deficit caps $1, 2/3, 1/2, 1/4$. If $g \geq 2$, take $\alpha_0 = 19/15 > 2^{1/3}$ in (9.3) and surface radius bounds $17/12, 7/4, 2, 9/4$. Their maximal surpluses are

$$\frac{3}{20}, \quad \frac{29}{90}, \quad \frac{11}{30}, \quad \frac{59}{240}.$$

Hence

$$C \leq \frac{19}{15}b + \frac{11}{30}, \quad s + C \leq \frac{13127}{1920} + e < \frac{1101}{160}.$$

If instead $b \leq 6/5$, take $\alpha_0 = 8/5 > 4^{1/3}$. The only positive surface surpluses are at most $1/10, 1/5, 13/80$, so

$$C \leq \frac{8}{5}b + \frac{1}{5}, \quad s + C \leq \frac{1399}{200} + e = 7 - \frac{1}{200} + e.$$

The new allowance $1/2560$ again gives a point coefficient below $7 - 1/1280$, since $1/200 > 3/2560$. We have proved

$$(9.5) \quad m = 4, \quad \text{edim}_x W = 6, \quad g = 1, \quad \frac{6}{5} < b \leq \frac{3}{2}.$$

Step 2. In this step, we identify the ordinary and normalized point blow-ups and compute the full tangent Hilbert series. Set $R = \mathcal{O}_{W,x}$ and $\mathfrak{n} = \mathfrak{m}_{W,x}$. This is a normal excellent Cohen–Macaulay complex local domain. It is analytically unramified, has a canonical module, and has infinite residue field. Its multiplicity is $\text{edim } R - \dim R + 1$. By Theorem 4.14, a minimal reduction J satisfies $\mathfrak{n}^2 = J\mathfrak{n}$, and G is Cohen–Macaulay. Its quotient by three general linear parameters has Hilbert function $(1, 3)$. Therefore

$$\text{Hilb}_G(z) = \frac{1 + 3z}{(1 - z)^3}, \quad e_1(\mathfrak{n}) = 3.$$

Theorem 4.6 and Proposition 4.10 give $3 \leq \bar{e}_1(\mathfrak{n})$ and $2\bar{e}_1(\mathfrak{n}) \leq 12 - 4b < 8$. Integrality yields $\bar{e}_1(\mathfrak{n}) = 3$. The two lengths in the Rees criterion are

$$\text{length}(\mathfrak{n}/J) = 3, \quad \text{length}(\mathfrak{n}^2/(J \cap \mathfrak{n}^2)) = 0.$$

Thus $R[\mathbf{n}t]$ is Cohen–Macaulay, in particular S_2 . Theorem 4.16 implies that all powers of $\mathbf{n}$ are integrally closed. Since R is normal, this also makes the Rees algebra normal: its integral closure lies in $R[t]$, and its degree- j part is $\overline{\mathbf{n}^j t^j} = \mathbf{n}^j t^j$.

Let $f : Y_W \rightarrow \overline{W}$ be the normalized point blow-up of the normalization of W . Near the exceptional fiber it is the ordinary point blow-up. Its exceptional Cartier scheme and polarization are

$$(9.6) \quad E_W = \text{Proj } G, \quad \mathbf{n}\mathcal{O}_{Y_W} = \mathcal{O}_{Y_W}(-E_W), \quad H = \mathcal{O}_{Y_W}(-E_W)|_{E_W} = \mathcal{O}_{E_W}(1).$$

At this stage E_W is pure, arithmetically Cohen–Macaulay, of degree four in $\mathbb{P}^5$, and is not yet assumed reduced.

Step 3. In this step, we bound the deficit slack and derive the three strict inequalities needed for the component analysis. Put $a_0 = 4^{1/3}$ and $\sigma = \delta - b$. The point blow-up gives $\sigma \geq 0$. In any additive chain, a first surface successor of deficit β and multiplicity μ satisfies $\mu \leq 6 - 4\beta$; the whole cost is at most $a_0 b + \beta(\sqrt{\mu} - a_0)$. The surplus is nonpositive for $\mu \leq 2$. For $\mu = 3, 4, 5$ use respectively $\beta \leq 3/4, 1/2, 1/4$. The inequalities

$$3\sqrt{3} - a_0 < 4, \quad \sqrt{5} + a_0 < 4$$

show that the largest of these surpluses is at most $1 - a_0/2$, attained in the comparison for $\mu = 4$. For example, $\sqrt{3} < 7/4$, $3/2 < a_0 < 8/5$, and $\sqrt{5} < 9/4$ verify the two strict inequalities. Chains skipping surfaces have cost at most $a_0 b$. Hence

$$C \leq a_0 b + 1 - \frac{a_0}{2}.$$

By Proposition 3.7, failure of a point continuation below $7 - \varepsilon_*$ forces $s + 1 + a_0(b - 1/2) \geq 7 - \varepsilon_*$. Since $b - 1/2 > 0$ and $a_0 < 8/5$, this implies $s > 34/5 - (8/5)b - \varepsilon_*$. Combining it with $s \leq \rho(6 - \delta) + e$, and using $e + \varepsilon_* < 3/640$, gives

$$(9.7) \quad 0 \leq \sigma < \frac{374b - 449}{650} \leq \frac{56}{325} < \frac{1}{5}.$$

In particular,

$$(9.8) \quad 2b - \delta > 1, \quad 3b - \delta > 2, \quad 3b - 2\delta > 1.$$

For the last inequality, the sharper bound gives $b - 2\sigma > (449 - 49b)/325 \geq 751/650 > 1$.

Step 4. In this step, we prove reducedness of the full tangent scheme using the effective transformed pair. Let $\pi : X_1 = \text{Bl}_x X \rightarrow X$ have exceptional divisor F , and let Γ^{str} and W_1 be the strict transforms. Put $e_0 = \delta - 1$ and $\eta = 1 - \sigma$. Then $e_0 > 1/5$, $\eta > 4/5$, and

$$(9.9) \quad K_{X_1} + \Gamma^{\text{str}} = \pi^*(K_X + \Gamma) + e_0 F.$$

For rational $0 \leq \theta < \eta$, the effective pair $(X_1, \Gamma^{\text{str}} + \theta F)$ is lc near F . Indeed its discrepancy at a valuation ν is

$$A_{X, \Gamma}(\nu) - (e_0 + \theta)\nu(\mathbf{m}_x),$$

which is strictly positive when $\nu(\mathbf{m}_x) > 0$, because $e_0 + \theta < b$, and is unchanged otherwise.

The center W_1 is minimal at every point of its exceptional fiber. A proper center there cannot map to x by the strict discrepancy inequality. Any other such center would map to a proper old center inside W through x . The intersection statement in Theorem 3.1 also excludes another center meeting W_1 without containing it. The quotient of the ambient point Rees algebra onto the point Rees algebra of W embeds its ordinary blow-up in X_1 . The latter Rees algebra is a domain, so its projectivization is the closure of $W \setminus \{x\}$. Thus $W_1 = Y_W$ near E_W . Theorem 4.5 also gives rational singularities for Y_W there.

Write $E_W = \sum_A r_A A$ with integral $r_A > 0$. Apply Theorem 4.1 to $(X_1, \Gamma^{\text{str}})$ along Y_W . Its base discriminant is effective; write it as $\sum_A \gamma_A A + D_0$, where $D_0 \geq 0$ contains no component of E_W . Write its moduli trace as $\mathcal{M}_{Y_W} = h_* \mathcal{M}_V$, with $\mathcal{M}_V$ nef on a smooth higher projective

model. Addition of θF , by Proposition 4.3, and the generic threshold definition imply $\gamma_A + \theta r_A \leq 1$. Letting θ increase to η yields

$$0 \leq \gamma_A \leq 1 - r_A \eta.$$

Since $\eta > 4/5$, every r_A is one. The ring G is therefore generically reduced. A Cohen–Macaulay ring has no embedded associated primes, and the pure three-dimensional graded ring has no component supported only at the irrelevant ideal. A nonzero nilradical would have an associated minimal prime and survive there. Thus G and E_W are reduced. We have also proved

$$(9.10) \quad 0 \leq \gamma_A \leq \sigma, \quad K_{Y_W} + \sum_A \gamma_A A + D_0 + \mathcal{M}_{Y_W} \sim_{\mathbb{Q}} f^* \mathcal{L} + e_0 E_W,$$

where $\mathcal{L}$ is the pullback of $(K_X + \Gamma)|_W$ to $\overline{W}$. Its further pullback has degree zero on complete curves in E_W .

Step 5. In this step, we prove Gorensteinness at the generic points of exceptional curves. Choose rational $2/3 < \theta < \eta$. For any curve in E_W , take a general very ample ambient hypersurface through a general intersection point with that curve. On a common log resolution its pullback is transverse to every boundary stratum, so restriction of the crepant equality preserves the lc inequalities and coefficient-one strata. The resulting surface in Y_W is a minimal center for the restricted smooth ambient pair. A local equation of F lies in the point ideal. Log canonicity after adding θF therefore implies that the point deficit before that addition is at least θ .

For this normal Cohen–Macaulay surface of multiplicity m_S , Theorem 4.6 gives $\bar{e}_1 \geq m_S - 1$, and Proposition 4.10 gives $\theta m_S \leq 2$. Hence $m_S < 3$. A general two-parameter reduction has length m_S and embedding dimension $\text{edim } S - 2$, so $\text{edim } S \leq m_S + 1 \leq 3$. Its completed local ring is regular or a hypersurface. It is Gorenstein. Cohen–Macaulay type is unchanged on quotienting by the regular equation of the ambient hypersurface; hence Y_W is Gorenstein at the generic point of the original curve. Since E_W is Cartier, it is also Gorenstein there. The excluded subset of E_W is consequently finite.

Step 6. In this step, we determine the line attachments, their transverse node equations, and their conductor coefficients. The Hilbert series gives $\text{reg } G = 1$, so E_W is 2-regular in the ideal-sheaf convention. Theorem 6.2 supplies a linearly joined ordering of its minimal-degree components. If E_W is integral, the surface classification gives exactly the four types in (i). Otherwise, let the component degrees be $d_1, \dots, d_k$. Each spans $\mathbb{P}^{d_i+1}$. If l_i is the dimension of the intersection with the preceding span, then $l_i \leq 1$ and

$$5 = d_1 + 1 + \sum_{i=2}^k (d_i + 1 - l_i) = 5 + (k - 1) - \sum_{i=2}^k l_i.$$

Thus all $l_i = 1$: each attachment is a reduced line.

At the generic point of a line shared by k_0 components, their transverse directions are independent by linear joining. Over the function field K of the line the transverse ring of E_W is

$$K[[v_1, \dots, v_{k_0}]] / (v_i v_j : i \neq j).$$

Quotienting by the nonzerodivisor $\sum_i v_i$ gives a square-zero maximal ideal of socle dimension $k_0 - 1$. Gorensteinness forces $k_0 = 2$. Each new component therefore attaches along a line to exactly one predecessor, and the graph of line attachments is a tree; other intersections are at most points.

Fix an attachment line S , put $K = \mathbb{C}(S)$, and let u_0 be a local equation of the Cartier divisor E_W . The completed transverse ring of Y_W is torsion-free, hence flat, over the DVR $K[[u_0]]$; its quotient by u_0 is the split node $K[[v, w]] / (vw)$. Lift the two nodal generators. The kernel of the resulting power-series presentation is generated by a lift of vw , because u_0 is regular and the reduction of that kernel modulo u_0 is principal. Successive coordinate changes remove terms involving either lifted generator, leaving $vw - \psi(u_0)$. Completeness makes the successive changes

converge. This is the lifting argument of Theorem 9.3, applied to this complete local algebra. Excellence and normality of Y_W make the completion normal, so $\psi \neq 0$. Absorbing a unit gives

$$(9.11) \quad \widehat{\mathcal{O}}_{Y_W, \eta_S} \simeq K[[v, w, u_0]]/(vw - u_0^q), \quad q \geq 1.$$

The cover $v = a^q, w = c^q, u_0 = ac$, with action $(a, c) \mapsto (\zeta a, \zeta^{-1}c)$, shows that Weil divisors have Cartier multiples near the generic point of S : take norms of equations upstairs and descend Cartierness from the faithfully flat completion. Away from these lines and the isolated cone vertices, the Cartier quotient E_W is regular, so Y_W is regular. Moving curves below avoid the remaining finite exceptional set.

If components A, D meet along S , the node equation gives

$$(9.12) \quad D|_A = \frac{1}{q}S, \quad A|_D = \frac{1}{q}S, \quad (K_{Y_W} + E_W)|_A = K_A + \sum_{S \text{ incident to } A} S.$$

The last formula comes from the simple pole of a dualizing differential on each branch. Its conductor coefficient is one; the intersection coefficient $1/q$ has a different role.

Step 7. In this step, we use the moduli intersection sign to exclude all component arrangements outside the stated list. For a general complete curve F_A in one of the moving families used below,

$$(9.13) \quad (D_0 + \mathcal{M}_{Y_W}) \cdot F_A \geq 0.$$

Effectivity proves the assertion for D_0 . Near the curve the finitely many relevant Weil divisors are rational Cartier. On a higher model put $N = h^*\mathcal{M}_{Y_W} - \mathcal{M}_V$. Then $-N$ is h -nef and $h_*N = 0$. Theorem 9.2 gives $N \geq 0$. A general member of the moving family has strict transform not contained in N , and therefore

$$\mathcal{M}_{Y_W} \cdot F_A = \mathcal{M}_V \cdot F_A^{\text{str}} + N \cdot F_A^{\text{str}} \geq 0.$$

This proves the needed intersection sign without asserting nefness of the lower moduli trace.

Intersect (9.10) with F_A , and use (9.12). Writing $a_S = 1/q$ at each incident attachment gives

$$(\delta - \gamma_A)H \cdot F_A = -K_A \cdot F_A - \sum_{S=A \cap D} (1 + a_S(\gamma_D - \gamma_A))S \cdot F_A - (D_0 + \mathcal{M}_{Y_W}) \cdot F_A.$$

Since $0 \leq \gamma_A \leq \sigma$ and $0 < a_S \leq 1$,

$$(9.14) \quad bH \cdot F_A \leq -K_A \cdot F_A - (1 - \sigma) \sum_S S \cdot F_A.$$

A plane with at least two neighbors gives $b \leq 1 + 2\sigma$ on a general line, contradicting the last inequality in (9.8). A smooth quadric with any neighbor gives $b \leq 1 + \sigma$ on a line of the opposite ruling. A quadric cone with at least two neighbors gives the same inequality on a general hyperplane conic, whose hyperplane and anticanonical degrees are two and four. Both contradict $2b - \delta > 1$.

With at least three components the degree partition is $2 + 1 + 1$ or $1 + 1 + 1 + 1$. A vertex of valence at least two in its attachment tree would be one of the components just excluded. Thus there are two components. The partition $2 + 2$ consists of two quadric cones, as in (ii).

For the partition $3 + 1$, a smooth cubic scroll is $\mathbb{F}_1$, with

$$C_0^2 = -1, \quad C_0 \cdot \ell_0 = 1, \quad \ell_0^2 = 0, \quad H = C_0 + 2\ell_0, \quad K_{\mathbb{F}_1} = -2C_0 - 3\ell_0.$$

An attachment line is C_0 or a ruling ℓ_0 . In the first case a general ruling in (9.14) gives $b \leq 1 + \sigma$. In the second, a general member of $|C_0 + \ell_0|$ gives $2b \leq 2 + \sigma$, contradicting $3b - \delta > 2$. The remaining cubic surface is $S(0, 3)$, whose lines are rulings through its vertex. This proves (iii), and completes the classification and the proposition. $\square$

9.2. Intrinsic sections and their actual thresholds.

Theorem 9.6. *Let $\Gamma \sim_{\mathbb{Q}} sL$, $s \geq 0$, be log canonical near x with a threefold minimal center. If*

$$s \leq \frac{65}{64} \text{mult}_x \Gamma + e, \quad e < \frac{1}{2560},$$

then the entire Γ admits an effective additive point continuation of total coefficient less than $7 - 1/4096$. No hypothesis on its tangent cone, previous cutting history or chosen successor is imposed.

The addition retains the entire initial boundary. In particular, every cost below is charged to the same coefficient s .

Proof. Suppose that the asserted continuation does not exist. We bound the original discriminant on the exceptional components, then lift complete intrinsic spaces and take the thresholds of their actual members. A fixed surface at such a threshold imposes conditions on every leading jet in that space. The resulting bounds, summed over all jet levels, contradict the Hilbert polynomial.

Put $M = \text{mult}_x \Gamma$, $b = \text{def}_x(\Gamma)$, $\rho = 65/64$, and $\varepsilon_* = 1/4096$. By Proposition 9.5, the minimal center W has multiplicity four, embedding dimension six, and $L^3 \cdot W = 1$, with $6/5 < b \leq 3/2$. Let $f : Y_W \rightarrow \overline{W}$ be the point blow-up of its normalization, let E_W be its exceptional divisor, and set $H = \mathcal{O}_{Y_W}(-E_W)|_{E_W}$. The proposition identifies the ordinary and normalized blow-ups near E_W and proves that the full tangent ring is reduced and Cohen–Macaulay. Its proof, including (9.11) and (9.12), determines the transverse index and conductor at every attachment line.

Step 1. In this step, we bound the original discriminant and its changes under actual Cartier additions. Use Theorem 4.1 for the original pair (X, Γ) along W . In either reducible case write its discriminant trace on Y_W as

$$\beta_1 E_{W,1} + \beta_2 E_{W,2} + D_h,$$

where D_h is the effective strict transform of the base discriminant and contains no component of E_W . The β_i need not be nonnegative. Write $a_i = \deg_H E_{W,i}$ for the component degrees, so $(a_1, a_2) = (2, 2)$ or $(3, 1)$. In the latter case denote $E_{W,1} = E_{W,c} = S(0, 3)$ and $E_{W,2} = E_{W,p} = \mathbb{P}^2$.

Let F_i be a general hyperplane curve on $E_{W,i}$, and let q be the transverse index at the attachment line. The formulas proved in (9.12) give

$$(9.15) \quad E_W \cdot F_i = -a_i, \quad E_{W,j} \cdot F_i = \frac{1}{q}, \quad E_{W,i} \cdot F_i = -a_i - \frac{1}{q}, \quad K_{Y_W} \cdot F_i = -1 \quad (j \neq i).$$

For the last identity, a plane has anticanonical hyperplane degree three and a quadric cone has degree four on its hyperplane conic. On the resolution $\mathbb{F}_3$ of $S(0, 3)$, $H = C_0 + 3\ell_0$ and $K = -2C_0 - 5\ell_0$, so $K \cdot H = -5$. The general hyperplane curve misses the contracted negative section. In every case the conductor line contributes one, and $E_W \cdot F_i = -a_i$, giving the last equality.

The moduli intersection is nonnegative on each F_i : apply Theorem 9.2 to the original moduli data exactly as in (9.13). Intersecting the original adjunction identity, whose pullback term has degree zero, therefore gives, in the cubic-plane case,

$$(3q + 1)\beta_c - \beta_p \geq -q, \quad (q + 1)\beta_p - \beta_c \geq -q.$$

The coefficient matrix has determinant $q(3q + 4)$, and its inverse is

$$\frac{1}{q(3q + 4)} \begin{pmatrix} q + 1 & 1 \\ 1 & 3q + 1 \end{pmatrix}.$$

Its entries are nonnegative. On defining

$$(9.16) \quad B_c = \frac{4q + 6}{3q + 4}, \quad B_p = \frac{6q + 6}{3q + 4},$$

we obtain

$$\beta_i \geq 1 - B_i, \quad \frac{4}{3} < B_c \leq \frac{10}{7}, \quad 3B_c + B_p = 6.$$

For two quadric cones the inequalities are $(2q+1)\beta_i - \beta_j \geq -q$. The inverse matrix is again nonnegative and sends the two equal right sides to $-1/2$. Thus set $B_1 = B_2 = 3/2$ in that case.

If D is an actual effective rational Cartier divisor missing W , and $\Gamma + tD$ is lc with deficit b' , the original center W persists. The point-ideal comparison on an ambient lc place dominating W gives $\beta_i(t) \leq 1 - b'$: principalize the point ideal on that place and use log canonicity after its addition with coefficient b' . Its order along $E_{W,i}$ is one, by reducedness. The generic threshold definition in Theorem 4.1 then gives the stated inequality for its infimum over these lc places. Proposition 4.3 gives $\beta_i(t) = \beta_i + t \operatorname{ord}_{E_{W,i}}(D|_W)$. Consequently

$$(9.17) \quad b' \leq B_i - t \operatorname{ord}_{E_{W,i}}(D|_W).$$

The same comparison before addition gives $b \leq B_c$ in the cubic-plane case.

For a prime surface T through x , let $\mu_T = \operatorname{mult}_x T$ and let α_T be its coefficient in the original base discriminant. The horizontal contribution to Proposition 4.12 is $\sum_T \alpha_T \mu_T$. Since the other terms are nonnegative,

$$(9.18) \quad 0 \leq \alpha_T, \quad \alpha_T \mu_T \leq 6 - 4b < \frac{6}{5}.$$

If T is a smaller surface lc center for $\Gamma + tD$, the coefficient of its discriminant is one:

$$(9.19) \quad \alpha_T + t \operatorname{coeff}_T(D|_W) = 1.$$

Indeed, at the codimension-one generic point of T , the normal center is regular and higher birational models are isomorphisms. The lc condition bounds the coefficient above by one; the center detection statement of Theorem 4.1 bounds it below by one. Exact addition of the discriminant proves the formula.

Step 2. In this step, we lift every intrinsic jet level and reduce its actual threshold to a surface of multiplicity three, four, or five. Set $\gamma = 31/50$, $A = L|_W$, and

$$V_{p,\ell} = H^0(W, \mathcal{O}_W(pA) \otimes \mathfrak{m}_{W,x}^\ell), \quad J_{p,\ell} = V_{p,\ell}/V_{p,\ell+1}.$$

For every sufficiently large integer p , Serre vanishing for $\mathcal{I}_W(pL)$ makes

$$H^0(X, \mathcal{O}_X(pL)) \longrightarrow H^0(W, \mathcal{O}_W(pA))$$

surjective. Fix $p \geq 4$ in this range. This single restriction map supplies the inverse image of every $V_{p,\ell}$; its cutoff is independent of ℓ .

For $\ell \geq \lceil \gamma p \rceil$ with $J_{p,\ell} \neq 0$, choose a sufficiently general member of this actual inverse image, represented by a section h . Put

$$D = \frac{1}{p} \operatorname{div}(h), \quad w = \frac{\ell}{p}, \quad t = \operatorname{lct}_x((X, \Gamma); D).$$

Its nonzero restriction to W has point order exactly ℓ ; in particular D misses W . On a common log resolution, every old lc place through x has zero D -order, since its center contains W . All remaining positive-denominator discrepancy ratios are positive. At least one denominator is positive, since $h(x) = 0$. Thus t is positive, finite, rational, and attained. A new computing place cannot have center containing W . Intersect it with the persistent W , using Theorem 3.1; the new minimal center U is properly inside W .

Apply Theorem 9.1, with klt background $(X, 0)$, old addition Γ , and new addition tD . After taking Cartier multiples, powers of $h|_W$ have order at least the corresponding multiple of ℓ . Hence, with $b' = \operatorname{def}_x(\Gamma + tD)$,

$$(9.20) \quad 0 \leq b' \leq b - tw, \quad 0 < t \leq \frac{b}{w} \leq \frac{75}{31}.$$

This uses Lemma 3.8 to identify the two deficit conventions. No ambient order is substituted for the restricted order.

If U is a point, a curve, or a surface of multiplicity at most two, all its remaining radii are at most $\sqrt{2} < 50/31 = 1/\gamma$. If $b' > 0$, choose a total cutting allowance

$$0 < \varepsilon_{\text{tail}} < \left(\frac{50}{31} - \sqrt{2}\right) b'.$$

Proposition 3.7 then gives an actual point continuation whose total coefficient is strictly less than

$$s + t + \sqrt{2} b' + \varepsilon_{\text{tail}} < s + t + \frac{b'}{\gamma}.$$

If U is a point, then $b' = 0$ and no further cut is needed. In either case the total coefficient is bounded by

$$s + t + \frac{b'}{\gamma} \leq s + \frac{b}{\gamma} \leq \frac{27735}{3968} + e < 7 - \varepsilon_*.$$

Here $s \leq \rho(6 - b) + e$, $b \leq 3/2$, and $1/\gamma - \rho > 0$; the remaining gap is $41/3968$, larger than $e + \varepsilon_* < 3/640$. This contradicts the standing assumption. Therefore $U = T$ is a surface of multiplicity $\mu \in \{3, 4, 5\}$, where the upper bound is supplied by Corollary 4.13.

Step 3. In this step, we prove that the threshold surface is fixed in the complete intrinsic source and bound its order. Resolve the base ideal of $V_{p,\ell}$ on the normalization of W . Bertini's theorem makes the moving part of a general member reduced. If the selected T were moving, its coefficient in $D|_W$ would be $1/p$. Equations (9.18) and (9.20) would give

$$\alpha_T + \frac{t}{p} \leq \frac{6 - 4b}{3} + \frac{b}{4\gamma} = 2 - \frac{173}{186}b < \frac{137}{155} < 1,$$

contradicting (9.19). Thus T is fixed in $V_{p,\ell}$. Let

$$r_T = \min \text{ord}_T V_{p,\ell}, \quad \ell_i = \min \text{ord}_{E_{W,i}} V_{p,\ell}.$$

A sufficiently general h attains these finitely many minima simultaneously. From (9.18) and (9.19),

$$(9.21) \quad \alpha_T + \frac{tr_T}{p} = 1, \quad \frac{\mu r_T}{p} \geq \frac{\mu - 6 + 4b}{t}.$$

Each $\ell_i \geq \ell$, and $\min_i \ell_i = \ell$. For the last equality, a nonzero initial form in the reduced tangent ring is nonzero on some component of E_W ; the Cohen–Macaulay ring has no irrelevant torsion. All these minima belong to the same complete space.

Put

$$R_3 = \frac{7}{4}, \quad R_4 = 2, \quad R_5 = \frac{9}{4}, \quad c_0 = \frac{577}{640}, \quad L_0(b) = c_0 + \rho b,$$

and define $\Delta_\mu(b) = (R_\mu - \rho)b - c_0$. The number R_μ bounds every radius in the surface-and-curve tail from T . The absence of a point endpoint, applied to this actual threshold and Proposition 3.7, therefore forces

$$(9.22) \quad \begin{aligned} 7 - \varepsilon_* &\leq s + t + R_\mu b', \\ (R_\mu w - 1)t &\leq s + R_\mu b - 7 + \varepsilon_* < \Delta_\mu(b). \end{aligned}$$

The strict inequality uses $s \leq \rho(6 - b) + e$ and $e + \varepsilon_* < 3/640$. Since $R_\mu w - 1 > 0$, it follows that $\Delta_\mu(b) > 0$ and

$$(9.23) \quad \frac{1}{t} > \frac{R_\mu w - 1}{\Delta_\mu(b)}, \quad 7 - \varepsilon_* - s > L_0(b).$$

Step 4. In this step, we derive residual-degree bounds on every tangent component from the actual fixed cycles. Suppose first that E_W is integral. Each of its four types is normal and has at most an isolated singular vertex. At generic points of curves in E_W , the Cartier quotient E_W

is regular, so Y_W is regular. Let T' be the strict transform of T . The quotient of point Rees algebras embeds $\text{Bl}_x T$ as this strict transform: its Rees algebra is a domain, so its Proj is the closure of $T \setminus \{x\}$. Its exceptional Cartier scheme is the actual projective tangent scheme of T , of hyperplane degree μ . Thus $C_T = T'|_{E_W}$ is an effective integral Weil cycle of degree μ . It need not be reduced. The pullback of every source section contains $\ell E_W + r_T T'$. After subtracting ℓE_W and restricting, every degree- ℓ initial form vanishes along $r_T C_T$. Its nonnegative residual hyperplane degree d therefore satisfies

$$(9.24) \quad d \leq 4\ell - \mu r_T.$$

Now suppose $E_W = E_{W,1} \cup E_{W,2}$ has component degrees a_1, a_2 . The strict transform T' is rational Cartier near generic curve points of E_W , by the transverse node calculation and regularity away from the attachment and vertices. Its restrictions $C_i = T'|_{E_{W,i}}$ are effective rational Weil cycles. Put $c_i = H \cdot C_i$. The exceptional Cartier scheme of the actual $\text{Bl}_x T$ has degree μ ; symmetry of rational Cartier intersection along each generic curve therefore gives

$$(9.25) \quad c_i \geq 0, \quad c_1 + c_2 = \mu.$$

Fractions of denominator q along the attachment line are included in the c_i .

If an initial form is nonzero on $E_{W,i}$, then $\ell_i = \ell$. Subtracting ℓE_W from the full pullback and using (9.15) forces its zero divisor on $E_{W,i}$ to contain

$$(9.26) \quad r_T C_i + \frac{\ell_j - \ell}{q} S \quad (j \neq i).$$

One may take a Cartier multiple before restriction; each rational coefficient is then rounded up in the integral zero divisor. If $\ell_i > \ell$, every degree- ℓ initial form vanishes identically on $E_{W,i}$.

For the same general h , write its entire pullback divisor as

$$\ell_1 E_{W,1} + \ell_2 E_{W,2} + r_T T' + R_h,$$

with R_h effective horizontal. Intersection with F_j , whose degree against the pulled-back pA is zero, gives

$$(9.27) \quad r_T c_j \leq a_j \ell_j + \frac{\ell_j - \ell_i}{q}.$$

If both $\ell_i = \ell$, the remaining nonnegative degrees obey

$$(9.28) \quad d_1 + d_2 \leq 4\ell - \mu r_T.$$

If $\ell_i = \ell < \ell_j$, only $E_{W,i}$ carries leading jets. Using (9.25), (9.26), and (9.27) yields

$$(9.29) \quad d_i \leq a_i \ell + a_j \ell_j - \mu r_T.$$

The $1/q$ terms cancel at this step, after both component restrictions have been used.

Write $w_j = \ell_j/p$. Equations (9.17) and (9.22) imply

$$tw_j \leq B_j + \frac{t - (7 - \varepsilon_* - s)}{R_\mu}.$$

Combining this with (9.21) and (9.29) gives

$$(9.30) \quad \frac{d_i}{p} \leq a_i w + \frac{a_j}{R_\mu} - \frac{\mu - 6 + 4b - a_j B_j + a_j L_0(b)/R_\mu}{t}.$$

These are conditions on the whole image $J_{p,\ell}$. The fixed prime and all minimum orders were selected for its complete source, and attained by the same member.

Step 5. In this step, we bound the full leading image on each integral polarized quartic surface. For $\mu = 3, 4, 5$, the ratio $(\mu - 6 + 4b)/\Delta_\mu(b)$ is decreasing wherever $\Delta_\mu(b) > 0$. Its derivative has denominator $\Delta_\mu(b)^2$ and respective numerators

$$-\frac{898}{640}, \quad -\frac{1048}{640}, \quad -\frac{1518}{640}.$$

Use $b \leq 3/2$, together with (9.21) and (9.23). Since $\Delta_\mu(3/2) = 1/5, 23/40, 19/20$, respectively, we obtain

$$(9.31) \quad \begin{aligned} k_3(w) &= \frac{105}{4}w - 15, \\ \frac{\mu_{rT}}{p} &\geq k_\mu(w), \quad k_4(w) = \frac{320}{23}w - \frac{160}{23}, \\ k_5(w) &= \frac{225}{19}w - \frac{100}{19}. \end{aligned}$$

Only the actual positive-denominator interval is used; no sign is assumed for $\Delta_3(6/5)$.

On every integral polarized quartic in the proposition, a system with an imposed effective integral zero divisor and residual hyperplane degree at most $d \geq 0$ has dimension at most

$$\frac{d^2}{8} + O(d+1),$$

with a constant independent of the imposed divisor. We give the counts for each type.

On $S(0, 4) = \mathbb{P}(1, 1, 4)$, the hyperplane bundle has weighted degree four. Each prime curve is defined by a homogeneous irreducible polynomial; its weighted degree equals its hyperplane degree. Divide by the product of the imposed equations. At remaining weighted degree $d_0 \leq d$, the number of monomials is

$$\sum_{j=0}^{\lfloor d_0/4 \rfloor} (d_0 - 4j + 1) = \frac{d_0^2}{8} + O(d_0 + 1).$$

On the Veronese surface a residual class k times a line has hyperplane degree $2k$, and its dimension is $(k+1)(k+2)/2$, giving the same bound.

On $S(2, 2) = \mathbb{F}_0$, write $H = C_0 + 2\ell_0$, with $C_0^2 = \ell_0^2 = 0$ and $C_0 \cdot \ell_0 = 1$. An effective residual class $aC_0 + c\ell_0$ has $a, c \geq 0$, degree $2a + c \leq d$, and

$$h^0 = (a+1)(c+1), \quad ac \leq a(d-2a) \leq \frac{d^2}{8}.$$

The linear terms are bounded by a constant times $d+1$. On $S(1, 3) = \mathbb{F}_2$, write $H = C_0 + 3\ell_0$, with $C_0^2 = -2$. Remove the fixed part of the complete residual system. This preserves its dimension and decreases its hyperplane degree. The remaining class is nef: for each curve there is a member not containing it, and intersection with that member is nonnegative. Writing it as $aC_0 + c\ell_0$, we have $a \geq 0$, $c \geq 2a$, and $a + c \leq d$. Projection to $\mathbb{P}^1$ gives

$$h^0 = \sum_{i=0}^a (c - 2i + 1) = (a+1)(c+1-a).$$

Its quadratic term is $a(c-a) \leq a(d-2a) \leq d^2/8$, and its linear terms are $O(d+1)$. These projection formulas follow by writing the pushforward of $aC_0 + c\ell_0$ on $\mathbb{F}_e$ as $\bigoplus_{i=0}^a \mathcal{O}_{\mathbb{P}^1}(c - ie)$. They apply to every effective residual class, including those with fixed parts.

Define

$$(9.32) \quad \begin{aligned} \phi_3(w) &= \frac{15}{4} - \frac{89}{16}w, \\ \phi_4(w) &= \frac{40 - 57w}{23}, \\ \phi_5(w) &= \frac{100 - 149w}{76}. \end{aligned}$$

These are $(4w - k_\mu(w))/4$. Equations (9.24) and (9.31) imply

$$\dim J_{p,\ell} \leq 2p^2 \max\{\phi_3(w), \phi_4(w), \phi_5(w), 0\}^2 + O(p)$$

on every bounded interval of w . A negative residual-degree bound gives zero image exactly, without an error term.

Step 6. In this step, we bound the leading image on two quadric cones, including images carried by only one component. Polynomial division on $\mathbb{P}(1, 1, 2)$ gives component capacity

$$\frac{d_i^2}{4} + O(\ell + 1)$$

for nonnegative residual degree d_i . If both $\ell_i = \ell$, use (9.28), (9.31), and $d_1^2 + d_2^2 \leq (d_1 + d_2)^2$. The resulting bound is $p^2(4w - k_\mu(w))^2/4 + O(p)$.

If $\ell_i = \ell < \ell_j$, set

$$N_\mu(b) = \mu - 9 + 4b + \frac{2L_0(b)}{R_\mu}.$$

Equation (9.30), with $a_i = a_j = 2$ and $B_j = 3/2$, becomes

$$\frac{d_i}{p} \leq 2w + \frac{2}{R_\mu} - \frac{N_\mu(b)}{t}.$$

This numerator is positive. Indeed it is increasing in b , and $\mu \geq 3$, $R_\mu \leq 9/4$, and $L_0(6/5) = 1357/640$ give the lower bound $493/720 > 0$. The derivative numerator of N_μ/Δ_μ is

$$-6c_0 + (9 - \mu)(R_\mu - \rho),$$

whose values are $-642/640, -312/640, -302/640$. Using (9.23) and evaluating at $b = 3/2$ therefore gives

$$\frac{d_i}{p} \leq 4w - k_\mu(w).$$

To see the cancellation explicitly, put $L_0(3/2) = 97/40$. Then

$$\Delta_\mu(3/2) = \frac{3}{2}R_\mu - \frac{97}{40}, \quad N_\mu(3/2) = \mu - 3 + \frac{2(97/40)}{R_\mu},$$

so $\mu - N_\mu(3/2) = 2\Delta_\mu(3/2)/R_\mu$. Substitute this identity in $2w + 2/R_\mu - N_\mu(3/2)(R_\mu w - 1)/\Delta_\mu(3/2)$.

The leading image injects into the sum of the component images, since the tangent ring is reduced. We have proved, for both the integral quartics and the two-quadric case, the uniform bound

$$(9.33) \quad \dim J_{p,\ell} \leq 4p^2 \max\{\phi_3(w), \phi_4(w), \phi_5(w), 0\}^2 + O(p).$$

If all three affine functions are negative, the jet quotient is zero, including when it is supported on only one component.

Step 7. In this step, we obtain uniform leading-image bounds for a cubic cone joined to a plane. For the plane, round up the imposed rational zero divisor and divide by its equation. Residual degree $d_p \geq 0$ gives dimension at most $d_p^2/2 + O(\ell + 1)$. On $E_{W,c} = \mathbb{P}(1, 1, 3)$, each prime curve has an equation whose weighted degree is its hyperplane degree. After rounding up and division, residual weighted degree $d_0 \geq 0$ gives

$$\sum_{j=0}^{\lfloor d_0/3 \rfloor} (d_0 - 3j + 1) = \frac{d_0^2}{6} + O(d_0 + 1).$$

Thus the cubic component has capacity $d_c^2/6 + O(\ell + 1)$. Each negative residual degree forces its image to be zero. The constants are absolute; rounding only decreases the space. In particular, two nonzero component images have combined capacity at most

$$(9.34) \quad \frac{d_c^2}{6} + \frac{d_p^2}{2} + O(\ell + 1) \leq \frac{(d_c + d_p)^2}{2} + O(\ell + 1).$$

Put $U_0 = 10/7$. For a plane-only image, use $a_i = 1$, $a_j = 3$, and $B_j = B_c \leq U_0$ in (9.30). Set

$$N_{p,\mu}(b) = \mu - 6 + 4b - 3U_0 + \frac{3L_0(b)}{R_\mu}.$$

This is increasing in b . The bounds $\mu \geq 3$, $R_\mu \leq 9/4$, and $b \geq 6/5$ give $N_{p,\mu}(b) \geq 1147/3360 > 0$. Equation (9.23) gives

$$(9.35) \quad \frac{d_p}{p} \leq w + \frac{3}{R_\mu} - \frac{N_{p,\mu}(b)}{\Delta_\mu(b)}(R_\mu w - 1).$$

This right side also bounds the total residual degree divided by p when both components carry jets. Indeed (9.28), (9.21), and (9.23) bound that total by

$$4w - \frac{\mu - 6 + 4b}{\Delta_\mu(b)}(R_\mu w - 1).$$

Subtracting this expression from the right side of (9.35) gives

$$\frac{3(U_0 - b)(R_\mu w - 1)}{\Delta_\mu(b)} \geq 0,$$

because $b \leq B_c \leq U_0$.

The derivative numerator of $N_{p,\mu}/\Delta_\mu$ is

$$-7c_0 + (6 + 3U_0 - \mu)(R_\mu - \rho).$$

For $\mu = 3, 4, 5$ its values are $-4303/4480, -553/4480, 957/4480$. Thus its minimum occurs at $b = U_0$ for $\mu = 3, 4$, and at $b = 6/5$ for $\mu = 5$, always within the interval where the actual denominator is positive. Substitution in (9.35) gives

$$(9.36) \quad \begin{aligned} P_3(w) &= \frac{12160 - 18636w}{661}, \\ P_4(w) &= \frac{16640 - 24236w}{2261}, \\ P_5(w) &= \frac{13952 - 21004w}{2597}. \end{aligned}$$

For these substitutions,

$$\Delta_3(U_0) = \frac{661}{4480}, \quad \Delta_4(U_0) = \frac{2261}{4480}, \quad \Delta_5(6/5) = \frac{371}{640}, \quad N_{p,5}(6/5) = \frac{7867}{3360}.$$

These values, together with $\mu - 6 + 4U_0 = 19/7, 26/7$ for $\mu = 3, 4$, give the displayed affine expressions directly.

For $w \geq \gamma$, $P_5(w) \leq \max\{P_4(w), 0\}$. Its zero $3488/5251$ precedes the zero $4160/6059$ of P_4 , and P_4 has larger absolute slope:

$$3488 \cdot 6059 < 4160 \cdot 5251, \quad 21004 \cdot 2261 < 24236 \cdot 2597.$$

The difference $P_4 - P_5$ is decreasing and positive at the zero of P_5 , proving domination before that zero; afterwards $P_5 \leq 0$. It follows from (9.34) that plane-only and two-component images both have dimension at most

$$(9.37) \quad \frac{p^2}{2} \max\{P_3(w), P_4(w), 0\}^2 + O(p).$$

For a cubic-only image, retain $B = B_c$ and $B_p = 6 - 3B$. Equation (9.30) has $a_i = 3$, $a_j = 1$, and numerator

$$N_{c,\mu}(b, B) = \mu - 12 + 4b + 3B + \frac{L_0(b)}{R_\mu}.$$

It is increasing in b, B . At $b = 6/5$, $B = 4/3$, using $\mu \geq 3$ and $R_\mu \leq 9/4$ bounds it below by $1069/1440 > 0$. For fixed B , the derivative numerator of $N_{c,\mu}/\Delta_\mu$ is

$$-5c_0 + (12 - 3B - \mu)(R_\mu - \rho).$$

Its largest values, at $B = 4/3$, are $-535/640, -365/640, -515/640$. Hence this ratio decreases in b , and $b \leq B$ gives

$$(9.38) \quad \frac{d_c}{p} \leq 4w - \frac{\mu - 12 + 8B}{\Delta_\mu(B)}(R_\mu w - 1).$$

Here we used $N_{c,\mu}(B, B) = \mu - 12 + 8B - \Delta_\mu(B)/R_\mu$. All $\Delta_\mu(B)$ are positive for $4/3 \leq B \leq U_0$. The derivative numerator of $(\mu - 12 + 8B)/\Delta_\mu(B)$ is

$$-8c_0 + (12 - \mu)(R_\mu - \rho),$$

equal to $-386/640, 424/640, 914/640$. Evaluating at $B = U_0$ for $\mu = 3$ and at $B = 4/3$ for $\mu = 4, 5$ yields

$$(9.39) \quad \begin{aligned} C_3(w) &= \frac{10880 - 16396w}{661}, \\ C_4(w) &= \frac{5120 - 7084w}{789}, \\ C_5(w) &= \frac{7040 - 10124w}{1429}. \end{aligned}$$

The three numerator values before division in (9.38) are $17/7, 8/3, 11/3$; the denominator values are $661/4480, 263/640, 1429/1920$. They verify the coefficients in (9.39).

As before, $C_5(w) \leq \max\{C_4(w), 0\}$. The zero $1760/2531$ of C_5 precedes $1280/1771$, and the absolute slope of C_4 is larger, since

$$1760 \cdot 1771 < 1280 \cdot 2531, \quad 10124 \cdot 789 < 7084 \cdot 1429.$$

The same decreasing-difference argument proves the claim. Thus a cubic-only image has dimension at most

$$\frac{p^2}{6} \max\{C_3(w), C_4(w), 0\}^2 + O(p).$$

If its two displayed affine bounds are negative, its image is zero.

Set

$$Q_3 = \frac{7}{12}C_3, \quad Q_4 = \frac{7}{12}C_4.$$

Here $1/\sqrt{3} < 7/12$, and

$$Q_3(w) \leq \max\{P_3(w), P_4(w), 0\} \quad (w \geq \gamma).$$

On $[\gamma, 16/25]$, $P_3 - Q_3$ has negative slope, and at $16/25$ one has $P_3 > 7/20 > Q_3$. On $[16/25, \infty)$, $P_4 - Q_3$ has positive slope and is positive at $16/25$, where $P_4 > 49/100$. These endpoint and slope comparisons follow by subtracting the fractions in (9.36) and (9.39). Together with (9.37) this proves the uniform cubic-plane bound

$$(9.40) \quad \dim J_{p,\ell} \leq \frac{p^2}{2} \max\{P_3(w), P_4(w), Q_4(w), 0\}^2 + O(p).$$

The constant on a bounded interval is independent of T, r_T, ℓ_1, ℓ_2 , and the transverse index q . All those data may vary with p, ℓ .

Step 8. In this step, we sum all jet-level bounds and contradict the intrinsic Hilbert-polynomial lower bound. For the integral and two-quadric cases, all three ϕ_μ are negative when $w > 40/57$. On $[\gamma, 40/57]$,

$$\phi_4(w) - \phi_5(w) = \frac{740 - 905w}{1748} > 0.$$

The upper envelope of ϕ_3, ϕ_4 switches at $\lambda = 148/227$, and

$$\phi_3(\gamma) = \frac{241}{800}, \quad \phi_3(\lambda) = \phi_4(\lambda) = \frac{28}{227}.$$

The integral of the bound in (9.33), after removing the factor p^2 , is therefore

$$\begin{aligned}
 I_q &= 4 \int_{\gamma}^{\lambda} \phi_3(w)^2 dw + 4 \int_{\lambda}^{40/57} \phi_4(w)^2 dw \\
 (9.41) \quad &= \frac{64}{267} \left(\frac{241}{800} \right)^3 + \frac{4540}{15219} \left(\frac{28}{227} \right)^3 \\
 &< \frac{1}{4} \left(\frac{61}{200} \right)^3 + \frac{3}{10} \left(\frac{1}{8} \right)^3 = \frac{245731}{32000000} < \frac{1}{130}.
 \end{aligned}$$

For a decreasing affine function, its squared integral is the decrease of its cube divided by three times its absolute slope. This gives the equality. The strict bounds use

$$\frac{64}{267} < \frac{1}{4}, \quad \frac{4540}{15219} < \frac{3}{10}, \quad \frac{241}{800} < \frac{61}{200}, \quad \frac{28}{227} < \frac{1}{8},$$

and $245731 \cdot 130 = 31945030 < 32000000$.

For the cubic-plane case set

$$w_0 = \frac{31}{50}, \quad w_1 = \frac{79}{125}, \quad w_2 = \frac{327}{500}, \quad z_0 = \frac{1280}{1771}.$$

For $w > z_0$, the exact residual inequalities make every component image zero. On $[w_0, z_0]$, the maximum in (9.40) is bounded by

$$\begin{cases} L_1(w) = \frac{37}{40} - 28(w - w_0), & w_0 \leq w \leq w_1, \\ L_2(w) = \frac{119}{200} - \frac{21}{2}(w - w_1), & w_1 \leq w \leq w_2, \\ Q_4(w), & w_2 \leq w \leq z_0. \end{cases}$$

Here are the endpoint comparisons, which also make the integration checkable:

	P_3	P_4	Q_4
w_0	15142/16525	40342/56525	21231/39450
w_1	47756/82625	165356/282625	562548/1183500
w_2	< 0	394828/1130500	426181/1183500

Each entry in the first row is less than $37/40$; each in the second is less than $589/1000$. These are the endpoint values of L_1 , so affine interpolation proves its domination. At w_1 , $L_2 = 119/200 > 589/1000$. At w_2 ,

$$P_4 < \frac{7}{20}, \quad \frac{9}{25} < Q_4 < \frac{361}{1000} < \frac{91}{250} = L_2(w_2).$$

This proves domination by L_2 on the second interval. At w_2 , Q_4 exceeds P_3, P_4 , and its slope is less negative than either of theirs. It therefore dominates on the last interval and vanishes at z_0 . Finally

$$0 < z_0 - w_2 < \frac{69}{1000},$$

since $1280/1771 < 723/1000$, equivalently $1280000 < 1771 \cdot 723 = 1280433$.

For an affine function with endpoint values a, b on an interval of length l , the half-integral of its square is $l(a^2 + ab + b^2)/6$. Hence the half-integral I_c of the square of this three-piece majorant satisfies

$$\begin{aligned}
 (9.42) \quad I_c &< \frac{1747371}{500000000} + \frac{2578037}{1000000000} + \frac{2997383}{2000000000} \\
 &= \frac{15142941}{2000000000} < \frac{1}{130}.
 \end{aligned}$$

For the first term the endpoint numerators are 925, 589 over 1000, and $925^2 + 925 \cdot 589 + 589^2 = 1747371$; its factor $l/6$ is $1/500$. For the second the endpoint numerators are 595, 364, whose corresponding sum is 703101, and $l/6 = 11/3000$. The last term uses height less than $361/1000$ and length less than $69/1000$. The final comparison is $15142941 \cdot 130 = 1968582330 < 2000000000$.

For fixed p , the descending spaces $V_{p,\ell}$ eventually vanish. Their intersection is zero by Krull intersection at x , because a global section on integral W with zero germ is zero; finite dimensionality then gives eventual vanishing. Thus

$$\dim V_{p, [\gamma p]} = \sum_{\ell \geq [\gamma p]} \dim J_{p,\ell}.$$

In the integral and two-quadric cases the summands are exactly zero for $\ell > (40/57)p$; in the cubic-plane case they are exactly zero for $\ell > z_0 p$. Summing (9.33) or (9.40), using (9.41) or (9.42), yields

$$(9.43) \quad \dim V_{p, [\gamma p]} \leq I p^3 + O(p^2), \quad I < \frac{1}{130}.$$

There are $O(p)$ levels, each with a uniform $O(p)$ error. Rounding endpoints and summing each fixed affine-square profile also contribute $O(p^2)$. No last subspace or unbounded jet tail is omitted.

On the other hand $A^3 = 1$, and the point multiplicity of W is four. The ample Hilbert polynomial and the Hilbert–Samuel length give

$$h^0(W, pA) = \frac{p^3}{6} + O(p^2), \quad \text{length}(\mathcal{O}_{W,x}/\mathfrak{m}_{W,x}^\ell) = \frac{4\ell^3}{6} + O(\ell^2).$$

The kernel of evaluation at these jets therefore has dimension

$$(9.44) \quad \begin{aligned} \dim V_{p, [\gamma p]} &\geq \frac{1 - 4\gamma^3}{6} p^3 + O(p^2) \\ &= \frac{1459}{187500} p^3 + O(p^2). \end{aligned}$$

Surjectivity of this jet evaluation map is not needed for the kernel lower bound. Since $1459 \cdot 130 = 189670 > 187500$, its leading coefficient is strictly greater than $1/130$, contradicting (9.43) for all sufficiently large p .

The contradiction proves existence of an additive point continuation below $7 - \varepsilon_*$. At every use of a scalar tail, Proposition 3.7 chooses the positive cutting errors inside the strict margin for that actual boundary. The lifted divisor has its full price t , and every later addition retains both it and the original Γ . Thus the continuation has the coefficient and the retention asserted. $\square$

10. POINT BOUNDARIES FROM STRONGER COMPLETE SOURCES

We retain the ambient notation and the complete systems $D_r, V_r(k)$ of Proposition 7.5. We first show that a volume bound at $r = 371/5$ produces a point boundary. We also obtain point boundaries from a double fivefold at order 69, and from the dimension of one sufficiently late order-75 space. Every threshold below belongs to the complete space then under consideration.

When we realize a scalar tail by actual cuts, we use Proposition 3.7 with total coefficient error $\varepsilon < 1/65536$. Thus a scalar bound $s + C \leq 7 - \delta$ with $\delta > 1/4096 + 1/65536$ gives a point boundary with the required margin. The six error allowances are fixed before the chain; each local error is divided by one plus the radius of its actual center. Applications of an earlier point-continuation theorem use that theorem's stated margin directly.

Lemma 10.1. *Let $r > 0$ be rational and let k be a positive integer for which $V_r(k) \neq 0$. Put*

$$\tau = \text{lct}_x(X; \mathfrak{b}(V_r(k))),$$

and let Γ be a sufficiently general pure threshold average of this space. Write s for its coefficient, M for its point multiplicity, b for its deficit, and W for its minimal center through x . Then τ is positive rational, $\dim W \leq 5$, and

$$(10.1) \quad s = 65k\tau, \quad M \geq rk\tau, \quad s \leq \frac{65}{r}M, \quad M + b \leq 6.$$

If W is positive-dimensional, put $d = \dim W$ and $m = \text{mult}_x W$. Then

$$(10.2) \quad 0 < b \leq d - 2 + \frac{2}{m}, \quad \text{edim}_x W \leq d + m - 1.$$

If $\dim W = 5$, then W occurs in the full fixed divisor $F_r(k)$ with a positive integral coefficient ν , and

$$(10.3) \quad \tau\nu = 1, \quad s = \frac{65}{\nu/k}, \quad M \geq \frac{r}{\nu/k}, \quad m\nu \leq \text{mult}_x F_r(k).$$

The divisorial identity relates the coefficient of a center to the fixed part of its own complete system.

Proof. We realize the ideal threshold by an actual average and then compute its numerical data.

Step 1. In this step, we construct a pure average with the lc strata of the actual ideal threshold. Resolve the base ideal and the point ideal simultaneously. The pullback of the linear system is the sum of its fixed divisor and a base-point-free system. On this resolution, a sufficiently large general tuple of moving members has simple normal crossings with the fixed support. Choose the size of the tuple so that every moving coefficient in the threshold average is less than one. The discrepancy equalities and lc strata are then those of the ideal at its threshold. This proves that the average is lc but not klt at x , and that its minimal center is proper. The threshold is positive rational because a nonzero ideal on a smooth variety has its threshold computed on this finite resolution. It is finite since the ideal vanishes at x .

Step 2. In this step, we bound the coefficient, point multiplicity, deficit, and embedding dimension. Each member has class $65kL$ and point order at least $\lceil rk \rceil$. These facts give the first three assertions of (10.1). The exceptional divisor of the blow-up of x has log discrepancy six and point-ideal order one. Testing log canonicity of $(X, \Gamma; \mathfrak{m}_x^t)$ on it gives $M + t \leq 6$, and hence the last assertion.

For a positive-dimensional center, Theorem 4.5 ([dFH09, Corollary 7.12]) gives a normal Cohen–Macaulay local ring. In the notation of Theorem 4.6 ([Han26, Lemmas 4.2 and 4.4]),

$$\bar{e}_1 \geq \sum_{i \geq 1} ih_i \geq \sum_{i \geq 1} h_i = m - 1, \quad h_1 \leq m - 1.$$

Together with Proposition 4.10, these inequalities give (10.2).

Step 3. In this step, we identify a fivefold center as a full fixed prime and compute its coefficient-one relation. A fivefold center is a prime divisor, so its coefficient in an lc boundary is one. The moving members have coefficients less than one and are general. Such a center must therefore be a fixed prime of the system, with coefficient $\nu > 0$, and its coefficient in the average is $\tau\nu$. This proves $\tau\nu = 1$. The remaining assertions of (10.3) follow from (10.1) and the inclusion $\nu W \leq F_r(k)$. $\square$

Lemma 10.2. Fix rational numbers $0 < p < r < t$. Suppose that, at a positive integer k , every member of $V_r(k)$ contains an effective integral divisor G_k with $\text{mult}_x G_k \geq pk$. There is a constant $K \geq 0$, depending only on p, r, t , such that

$$(10.4) \quad \frac{720}{k^6} (\dim V_r(k) - \dim V_t(k)) \leq (t - p)^6 - (r - p)^6 + \frac{K}{k}.$$

The constant is independent of the divisor G_k .

The estimate bounds the loss between two nested complete spaces at the same degree.

Proof. Trivialize $\mathcal{O}_X(65kL)$ near x , and put

$$A_{\ell,k} = H^0(X, \mathcal{O}_X(65kL) \otimes \mathfrak{m}_x^\ell).$$

For $\lceil rk \rceil \leq \ell < \lceil tk \rceil$, every germ in $A_{\ell,k}$ is divisible by the full local equation of G_k . Let its point order be e_k . Since $\text{gr}_{\mathfrak{m}_x} \mathcal{O}_{X,x}$ is a polynomial ring in six variables, orders add under multiplication. Division by that equation injects the degree- ℓ initial image into the homogeneous polynomials of degree $\ell - e_k$. If this degree is negative, the image is zero. Consequently

$$\dim(A_{\ell,k}/A_{\ell+1,k}) \leq \binom{\lfloor \ell - pk \rfloor + 5}{5}.$$

No assertion of surjectivity onto the right-hand space is needed. Uniformly over these indices, the binomial is $(\ell - pk)^5/120 + O(k^4)$. Summing the increasing polynomial, and allowing for the two rounded endpoints, gives

$$\sum_{\ell=\lceil rk \rceil}^{\lceil tk \rceil-1} \binom{\lfloor \ell - pk \rfloor + 5}{5} \leq \frac{k^6}{720}((t-p)^6 - (r-p)^6) + O(k^5).$$

The sum on the left bounds the dimension difference exactly by the point filtration. Every error depends only on the three fixed rational numbers. This proves (10.4). $\square$

Lemma 10.3. *Assume $L^6 = 1$, and suppose there is no point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$. Let $0 < r < 92$ be rational and assume $\text{vol}_Y(D_r) > 0$. Let N_r be the limiting full fixed divisor of Proposition 7.5, and put $\eta_r = \text{mult}_x N_r$. Then*

$$(10.5) \quad 0 \leq \eta_r \leq r, \quad \text{vol}_Y(D_r) \leq \left(1 - \frac{\eta_r}{r}\right)^5 (92^6 - r^6).$$

If also $\text{vol}_Y(D_r) > 10000000$, then

$$(10.6) \quad \text{mult}_x F_r(k) < \frac{9}{10}rk \quad \text{for every sufficiently large } k.$$

Positive volume thus bounds the multiplicity of the entire fixed divisor, including fixed primes other than a chosen center.

Proof. We bound every point-order layer using all limiting fixed factors, and then sum to the last possible point order.

Step 1. In this step, we exclude every actual section of point order at least $92k$. An actual section of $65kL$ of point order at least $92k$ would give, after division of its divisor by k , an effective rational divisor of class $65L$ with order-to-coefficient ratio at least $92/65 > 140/99$. Proposition 5.8 would give a point boundary of coefficient less than $7 - 1/630$. Under the present assumption no such section exists, at any degree.

Step 2. In this step, we bound every initial layer using all limiting fixed factors with their full exponents. Write $N_r = \sum_T n_T T$ on its finite support. For a nonzero actual section $g \in V_r(k)$, put $d_g = \text{ord}_x g$. Proposition 7.7 gives

$$(10.7) \quad \text{ord}_T g \geq \frac{d_g n_T}{r} \quad \text{for every } T.$$

This also applies when rk is not integral: choose a positive integer q clearing the denominator of r , apply that proposition to g^q in degree qk , and divide the resulting orders by q . The limiting divisor on the admissible subsequence is the same N_r , by Proposition 7.5.

Summing the point orders of the factors in (10.7) gives $d_g \geq d_g \eta_r / r$; hence $\eta_r \leq r$. At any integer order ℓ , the entire initial image is divisible by the product of the initial equations of the primes T with exponents $\lceil \ell n_T / r \rceil$. This product has degree at least $\ell \eta_r / r$. All exponents

are retained even when different primes have coincident initial factors. With $\alpha = 1 - \eta_r/r$, its dimension is therefore at most

$$\binom{\lfloor \alpha \ell \rfloor + 5}{5}.$$

Step 3. In this step, we sum the complete filtration to bound the volume and the eventual fixed multiplicity. The space at order $92k$ is zero. Thus summing from $\lceil rk \rceil$ to $92k - 1$, as in the proof of Lemma 10.2, gives

$$\frac{720 \dim V_r(k)}{k^6} \leq \alpha^5(92^6 - r^6) + O(k^{-1}).$$

If $\alpha = 0$, each layer has dimension at most one, which also gives this inequality. The complete-space limit in Proposition 7.5 proves (10.5).

Finally, $\eta_r \geq 9r/10$ would imply

$$\text{vol}_Y(D_r) \leq \frac{92^6 - r^6}{100000} < \frac{606355001344}{100000} < 7000000.$$

This contradicts the additional volume hypothesis. Thus $\eta_r < 9r/10$, and convergence in point multiplicity of the full $F_r(k)/k$ proves (10.6). $\square$

Lemma 10.4. *Assume $L^6 = 1$. Let $r > 0$ be rational, let k be a positive integer with $V_r(k) \neq 0$, and let Γ be a sufficiently general pure threshold average of this space, with coefficient s , minimal center W , and deficit b . Then the following hold.*

- (i) *If $r \geq 371/5$ and at least one of $\dim W \leq 4$, smoothness of a fivefold W at x , or $b \leq 1$ holds, there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$.*
- (ii) *If W is a fivefold of multiplicity two, put $n = \text{ord}_W V_r(k)/k$. Every additive cutting chain from Γ to the point satisfies*

$$(10.8) \quad s + C \leq \frac{11353}{1440} - \frac{23r/20 - 65}{n}.$$

Only singular fivefold centers require a fixed-multiplicity estimate beyond the source's point order.

Proof. For (i), put $q_r = 65/r$. Lemma 10.1 gives

$$q_r \leq \frac{325}{371} < \frac{22}{25}, \quad s \leq q_r(6 - b).$$

Step 1. In this step, we obtain point boundaries for all center types in (i), including their complete lower-dimensional tails. A point center already gives the conclusion, since $s \leq 6q_r \leq 1950/371$. For a positive-dimensional center with $b \leq 1$, apply Corollary 5.7 with excess zero. The same corollary handles a smooth fourfold, whose embedding dimension is four. Theorem 9.6 handles every threefold, since $s \leq (65/64)M$. A smooth fivefold is handled by Theorem 8.7, since $s \leq (65/69)M$. These results retain the entire Γ .

Here are the remaining lower-dimensional cases. A curve or surface has $b \leq 2$. A double fourfold has $b \leq 3$ and embedding dimension at most five. A fourfold of multiplicity at least three has $b \leq 8/3$. These assertions follow from (10.2). Applying the corresponding rows of Proposition 5.5 gives

center	bound for $s + C$
$\dim W \leq 2$	$4q_r + 2 + \frac{2033}{1440} < \frac{347}{50}$
$\dim W = 4, \text{mult}_x W = 2$	$3q_r + 3 + \frac{3887}{3360} < \frac{34}{5}$
$\dim W = 4, \text{mult}_x W \geq 3$	$\frac{10}{3}q_r + \frac{8}{3} + \frac{2033}{1440} \leq 7 - \frac{2391}{1602720} < 7 - \frac{1}{1000}.$

For the first two comparisons one may replace q_r by $22/25$. For the last, substitute $q_r = 325/371$; its gap exceeds $1/1000$ because $2391000 > 1602720$. All three margins exceed $1/4096 + 1/65536$, so the reserved cutting errors give the stated point boundary.

Every tail estimate here includes the large threefold multiplicities of Lemma 5.1, also when the chain reaches them after a skipped dimension. Let z and m denote the deficit and multiplicity of such a threefold, and let C_{tail} be the scalar cost of the suffix starting there. That lemma gives $C_{\text{tail}} \leq 2z$ at multiplicity eight with $z \leq 1/4$, and $C_{\text{tail}} \leq 9z/4$ at multiplicity nine with $z \leq 1/9$. In particular, their contributions to the two affine envelopes used in Section 5 are bounded by

$$\begin{array}{c|cc} & m=8 & m=9 \\ \hline C_{\text{tail}} - \frac{2301}{2000}z & \frac{1699}{8000} < \frac{3447}{4000} & \frac{733}{6000} < \frac{3447}{4000} \\ C_{\text{tail}} - \frac{9}{10}z & \frac{11}{40} < \frac{100663}{60000} & \frac{3}{20} < \frac{100663}{60000} \end{array}.$$

Thus these uses of complete tails retain both rows.

Step 2. In this step, we bound the entire coefficient for a double fivefold in terms of its normalized fixed order. For (ii), Proposition 5.5 gives $C \leq 23b/20 + 1417/1440$. The identities (10.3) then give

$$s + C \leq \frac{65}{n} + \frac{23}{20} \left(6 - \frac{r}{n} \right) + \frac{1417}{1440} = \frac{11353}{1440} - \frac{23r/20 - 65}{n}.$$

This calculation uses no lower bound on r , so it also applies at $r = 69$. □

Theorem 10.5. Assume $L^6 = 1$. If

$$\text{vol}_Y \left(65\pi^*L - \frac{371}{5}E \right) > 336000000,$$

there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$.

This criterion requires only the volume of one fixed complete source.

Proof. We preserve positive volume along a finite sequence of point orders. At each order we bound the full fixed divisor and treat every threshold center. If no point boundary occurs, a double fivefold supplies a fixed factor controlling the loss to the next source. At the final order the required factor exceeds the entire available fixed multiplicity.

Step 1. In this step, we fix the finite sequence of point orders and the applicable complete-source cutoffs. Suppose that no point boundary with the asserted coefficient exists. Fix

$$R = \frac{371}{5}, \quad h = \frac{1}{1000}, \quad r_i = R + ih \quad (0 \leq i \leq 13800), \quad r_{13800} = 88.$$

These are finitely many fixed rational divisors D_{r_i} . Their complete-space limits, and the applicable full-fixed-part convergence cutoffs, all depend only on X, L, x . Whenever $\text{vol}_Y(D_{r_i}) > 10000000$, Lemma 10.3 gives

$$\text{mult}_x F_{r_i}(k) < \frac{9}{10} r_i k$$

at every sufficiently large degree. The maximum of the applicable cutoffs over this finite list may be fixed in advance. We will prove the volume hypothesis before using this bound at each new order.

Step 2. In this step, we construct a point boundary from every threshold center other than a double fivefold. Fix a mesh order r with this volume bound and a sufficiently large k , and form its own pure threshold average Γ . Write s, M, b for its coefficient, point multiplicity, and deficit. Lemma 10.4 handles point centers, every dimension at most four, and smooth fivefolds.

Suppose its center is a singular fivefold W of multiplicity m , and put $n = \text{ord}_W V_r(k)/k$. The full fixed bound and (10.3) give

$$(10.9) \quad mn < \frac{9}{10}r, \quad M \geq \frac{r}{n} > \frac{10}{9}m.$$

Since $M \leq 6$, one has $m \leq 5$.

For $m = 3$, we have $b < 8/3$. With $q_r = 65/r < 22/25$, the triple-fivefold row of Proposition 5.5 gives

$$s + C \leq q_r(6 - b) + \frac{5}{4}b + \frac{57}{80} < \frac{10}{3} \frac{22}{25} + \frac{8}{3} \frac{5}{4} + \frac{57}{80} = \frac{335}{48} = 7 - \frac{1}{48}.$$

For $m = 4$ or 5 , (10.9) gives $b < 14/9$. The initial radius is at most $5^{1/5} < 5/3$, and the universal affine row of Proposition 5.5 bounds the entire tail by $5b/3 + 113/480$. Consequently

$$s + C < \frac{40}{9} \frac{22}{25} + \frac{14}{9} \frac{5}{3} + \frac{113}{480} = \frac{29113}{4320} < \frac{27}{4}.$$

Both bounds give the required point after the reserved error. Thus, under its exclusion, every such threshold has a double fivefold center.

Step 3. In this step, we use the forced double fixed factor to bound the loss to the next complete source. Define, for $R \leq r \leq 88$,

$$\theta(r) = \frac{230r - 13000}{177}, \quad a(r) = r - 2\theta(r) = \frac{26000 - 283r}{177}.$$

Both functions are positive on this interval. If the normalized fixed coefficient $n = \nu/k$ of the double center satisfies $n \leq \theta(r)$, then (10.8) gives

$$s + C \leq \frac{11353}{1440} - \frac{177}{200} = 7 - \frac{7}{7200}.$$

The gap $7/7200$ exceeds $17/65536$, so this case also gives the required point. Its exclusion forces

$$(10.10) \quad \frac{\nu}{k} > \theta(r).$$

For $r = r_i$ with $i < 13800$, every intermediate complete space at this same degree between orders $\lceil rk \rceil$ and $\lceil (r + h)k \rceil$ contains the entire factor νW , of exact point order 2ν . Apply Lemma 10.2 with discount $p = 2\theta(r) < r$. It gives

$$\frac{720}{k^6} (\dim V_r(k) - \dim V_{r+h}(k)) \leq \mathcal{L}(r) + O(k^{-1}), \quad \mathcal{L}(r) = (a(r) + h)^6 - a(r)^6.$$

If no point exists, this inequality holds at every sufficiently large k : each such degree has its own nonzero complete source and its own threshold satisfying (10.10). The center may depend on k , but the bound and its error do not. Taking the two complete-space limits of Proposition 7.5, including a possibly zero volume on the right, yields

$$(10.11) \quad \text{vol}_Y(D_{r+h}) \geq \text{vol}_Y(D_r) - \mathcal{L}(r).$$

Step 4. In this step, we bound the total loss and prove positivity of every intermediate volume. Put $\kappa = 283/177 < 2$. Then $a(R + t) = a(R) - \kappa t$, $a(R) = 25007/885$, and $a(R) + h < 29$. For one interval,

$$\mathcal{L}(r) = \int_0^h 6(a(r) + t)^5 dt.$$

All arguments of these fifth powers and of $(a(r) - \kappa t)^5$ lie between zero and 29. The mean-value inequality therefore gives

$$\begin{aligned} 0 \leq \mathcal{L}(r) - \int_0^h 6(a(r) - \kappa t)^5 dt &\leq \int_0^h 30 \cdot 29^4 (1 + \kappa)t dt \\ &\leq 45 \cdot 29^4 h^2. \end{aligned}$$

Summing these errors over the 13800 intervals gives

$$45 \cdot 29^4(88 - R)h = \frac{439221501}{1000} < 500000.$$

The sum of the integrals is

$$\int_0^{88-R} 6a(R+t)^5 dt = \frac{a(R)^6 - a(88)^6}{\kappa} < \frac{177}{283} \left(\frac{25007}{885} \right)^6 < 324000000.$$

For the last comparison,

$$\begin{aligned} \frac{177}{283} &< \frac{63}{100}, & \frac{25007}{885} &< \frac{283}{10}, \\ \left(\frac{283}{10} \right)^3 &= \frac{22665187}{1000} < 22666, & 22666^2 &= 513747556 < 514000000. \end{aligned}$$

Thus the preceding integral is less than $(63/100)514000000 = 323820000$. We have proved

$$(10.12) \quad \sum_{i=0}^{13799} \mathcal{L}(r_i) < 324500000.$$

Starting from the stated volume at R , induct using (10.11). Every partial sum of the nonnegative losses is less than the full sum in (10.12). Therefore, before each new fixed-part estimate is invoked,

$$\text{vol}_Y(D_{r_i}) > 336000000 - 324500000 = 11500000 > 10000000.$$

In particular, each new complete space is nonzero in all sufficiently large degrees. This proves all hypotheses needed for the induction.

Step 5. In this step, we contradict the full fixed-multiplicity bound at order 88. At $r = 88$, the absence of the point still forces a double center, whose fixed coefficient satisfies

$$\frac{\nu}{k} > \theta(88) = \frac{7240}{177}.$$

The bound on the entire fixed divisor instead gives

$$2\nu \leq \text{mult}_x F_{88}(k) < \frac{9}{10}88k, \quad \frac{\nu}{k} < \frac{198}{5}.$$

These bounds concern the same actual factor and are incompatible:

$$\frac{7240}{177} - \frac{198}{5} = \frac{1154}{885} > 0.$$

Hence an actual point boundary occurs in one of the alternatives already proved. $\square$

Proposition 10.6. *Assume $L^6 = 1$ and $\text{vol}_Y(65\pi^*L - 69E) \geq 3250000000$. There is an integer J , depending only on X, L, x , such that if a pure threshold average of the complete $V_{69}(j)$ at one $j \geq J$ has a fivefold minimal center of multiplicity two, then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$.*

Here one sufficiently late occurrence of a double center supplies the higher complete volume.

Proof. Put $R = 371/5$. Fix the convergence cutoffs of Proposition 7.5 for D_{69} and D_R , and the shell-error cutoff of Lemma 10.2 for

$$p = \frac{649}{20}, \quad r = 69, \quad t = R.$$

Choose J so that the sum of the two absolute normalized dimension-to-volume errors and this normalized shell error is less than 1000000 at every $j \geq J$. All three quantities concern fixed complete systems or fixed numerical discounts. Thus J is chosen before the occurrence in the statement.

Step 1. In this step, we obtain the point boundary directly when the normalized double fixed coefficient is at most $649/40$. Let W be that occurrence's double fivefold and let $\nu = \text{ord}_W V_{69}(j)$, $n = \nu/j$. Lemma 10.1 gives

$$\tau\nu = 1, \quad s = \frac{65}{n}, \quad M \geq \frac{69}{n}, \quad M + b \leq 6.$$

Its complete double tail therefore satisfies

$$s + C \leq \frac{11353}{1440} - \frac{287/20}{n}.$$

If $n \leq 649/40$, this is at most

$$7 - \frac{383}{934560}.$$

Indeed

$$\frac{574}{649} - \frac{1273}{1440} = \frac{383}{934560}.$$

This gap exceeds $17/65536$, so the reserved cutting error gives the point boundary directly.

Step 2. In this step, we use the larger full fixed factor to obtain the volume required by the point-boundary criterion. Otherwise the full factor νW has exact point order $2\nu > (649/20)j$. Every intermediate complete space between orders $69j$ and $\lceil Rj \rceil$ contains it. Lemma 10.2 gives

$$\frac{720}{j^6}(\dim V_{69}(j) - \dim V_R(j)) \leq \left(\frac{835}{20}\right)^6 - \left(\frac{731}{20}\right)^6 + O(j^{-1}).$$

The leading loss is less than 2912320000. For an integer check, $835^3 = 582182875 < 582200000$ and $731^3 = 390617891 > 390600000$, so their squared difference is less than

$$(582200000 - 390600000)(582200000 + 390600000) = 186388480000000000.$$

Division by $20^6 = 64000000$ gives the asserted upper bound. The preselected combined error now yields

$$\text{vol}_Y(D_R) > 3250000000 - 2912320000 - 1000000 = 336680000 > 336000000.$$

The convergence theorem applies even if positivity of this volume was not known before the comparison. Theorem 10.5 therefore gives the required point boundary. $\square$

Theorem 10.7. Assume $L^6 = 1$. There is an integer J , depending only on X, L, x , such that

$$\frac{720 \dim V_{75}(j)}{j^6} > 600000000 \quad \text{for one } j \geq J$$

implies the existence of a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$.

The four thresholds in this construction are all taken at the same integer j .

Proof. We first choose cutoffs for the four fixed complete systems. At orders 75, 78, 80, every threshold either gives a point or has a double fixed factor that preserves enough sections at the next order. The resulting full fixed-part bound at order 84 pays every remaining center.

Step 1. In this step, we choose a common cutoff for the four complete dimensions, fixed-divisor bounds, and shell errors. Suppose that the required point boundary does not exist. For $r \in \{75, 78, 80, 84\}$, write

$$\mu_r(k) = \frac{720 \dim V_r(k)}{k^6}.$$

By Proposition 7.5, choose common convergence cutoffs for errors less than 5000000 at 75, and less than 1000000 at 78, 80, 84. We will use the following constants:

r	mass at j	volume	eventual mass	a_r
75	600000000	595000000	590000000	55
78	258000000	257000000	256000000	61
80	140000000	139000000	138000000	64
84	39000000	38000000	37000000	357/5

Every mass and volume entry denotes a strict lower bound. The last column will give $\text{mult}_x F_r(k) < a_r k$.

To select these latter cutoffs independently of j , consider each of the four fixed volumes. If it is at most its listed volume threshold, the listed mass at j cannot occur beyond the already chosen convergence cutoff. Otherwise the fixed complete space has positive volume, and its eventual mass exceeds the eventual-mass entry in that row. Lemma 10.3 applies to its full limiting divisor N_r . In each such row it forces the corresponding bound below, with $\eta_r = \text{mult}_x N_r$:

$$(10.13) \quad \eta_{75} < 55, \quad \eta_{78} < 61, \quad \eta_{80} < 64, \quad \eta_{84} < \frac{357}{5}.$$

Here are the comparisons establishing each strict bound.

If $\eta_{75} \geq 55$, the factor in (10.5) is at most

$$\left(\frac{4}{15}\right)^5 < \frac{27}{20000}, \quad 1024 \cdot 20000 = 20480000 < 20503125 = 27 \cdot 759375.$$

Since $92^6 - 75^6 = 428376485719 < 430000000000$, the volume would be less than $580500000 < 590000000$. If $\eta_{78} \geq 61$, use

$$\frac{17}{78} < \frac{11}{50}, \quad \left(\frac{11}{50}\right)^5 < \frac{13}{25000}, \quad 161051 \cdot 25000 < 13 \cdot 312500000.$$

Together with $92^6 - 78^6 = 381155400640 < 382000000000$, this gives a volume less than $198640000 < 256000000$. If $\eta_{80} \geq 64$, the upper bound is

$$\frac{92^6 - 80^6}{3125} < \frac{345000000000}{3125} = 110400000 < 138000000.$$

Finally, if $\eta_{84} \geq 357/5$, then

$$1 - \frac{\eta_{84}}{84} \leq \frac{3}{20}, \quad 92^6 - 84^6 = 255056969728 < 256000000000,$$

so the upper bound is

$$\left(\frac{3}{20}\right)^5 256000000000 = 19440000 < 37000000.$$

Each contradicts the relevant lower volume bound. Convergence in point multiplicity of the entire fixed divisor therefore gives

$$(10.14) \quad \text{mult}_x F_{75}(k) < 55k, \quad \text{mult}_x F_{78}(k) < 61k, \quad \text{mult}_x F_{80}(k) < 64k, \quad \text{mult}_x F_{84}(k) < \frac{357}{5}k$$

in every applicable row and at every sufficiently large k .

Choose J to include these conditional fixed-part cutoffs and the four dimension-to-volume cutoffs. Also include the three shell-error cutoffs for the fixed triples

$$(p, r, t) = (48, 75, 78), \quad (54, 78, 80), \quad (61, 80, 84).$$

If their constants in Lemma 10.2 are K_{75}, K_{78}, K_{80} , respectively, require

$$(10.15) \quad \frac{K_{75}}{j} < 420489, \quad \frac{K_{78}}{j} < 187200, \quad \frac{K_{80}}{j} < 9992 \quad (j \geq J).$$

This choice uses only the four fixed complete series and three numerical shell estimates. In particular, it precedes the one occurrence in the hypothesis and every choice of a threshold center. If the required point exists already, any J suffices.

Step 2. In this step, we reduce the first three thresholds to double fivefolds with the required full fixed coefficients. At a nonzero $V_r(j)$, take its own sufficiently general pure threshold average Γ_r , at threshold $\tau_r = \text{lct}_x(X; \mathfrak{b}(V_r(j)))$. Write s_r, M_r, b_r for its coefficient, point multiplicity, and deficit. Then

$$s_r \leq q_r M_r, \quad M_r + b_r \leq 6, \quad q_r = \frac{65}{r} \leq \frac{13}{15}.$$

Lemma 10.4 handles points, every threefold, smooth fourfolds, smooth fivefolds, and all $b_r \leq 1$ cases. In particular its use of complete tails includes the two large-multiplicity threefold rows displayed in its proof. For the other centers of dimension at most four, its three scalar comparisons specialize to

center	bound for $s_r + C$
$\dim W \leq 2$	$4\frac{13}{15} + 2 + \frac{2033}{1440} < \frac{69}{10}$
$\dim W = 4, \text{ mult}_x W = 2$	$3\frac{13}{15} + 3 + \frac{3887}{3360} < \frac{34}{5}$
$\dim W = 4, \text{ mult}_x W \geq 3$	$\frac{10}{3}\frac{13}{15} + \frac{8}{3} + \frac{2033}{1440} = 7 - \frac{47}{1440}$.

Each admits the reserved error.

For a singular fivefold center W_r , put

$$m_r = \text{mult}_x W_r, \quad \nu_r = \text{ord}_{W_r} V_r(j), \quad n_r = \frac{\nu_r}{j}.$$

Whenever the row of (10.14) is applicable, its own pure identity gives

$$(10.16) \quad \tau_r \nu_r = 1, \quad s_r = \frac{65}{n_r}, \quad M_r \geq \frac{r}{n_r}, \quad m_r n_r \leq \frac{\text{mult}_x F_r(j)}{j} < a_r.$$

The first inequality involving the prime contribution may be an equality; strictness comes from the full fixed-part bound.

At $r = 75, 78, 80$, respectively, $m_r \geq 4$ gives

$$M_{75} > \frac{300}{55}, \quad M_{78} > \frac{312}{61}, \quad M_{80} > \frac{320}{64}.$$

Thus $b_r < 1$ in each case, so these centers give a point. If $m_r = 3$, the corresponding bounds are

$$b_{75} < \frac{21}{11}, \quad b_{78} < \frac{132}{61}, \quad b_{80} < \frac{9}{4}.$$

They are all at most $9/4$. Proposition 5.5 therefore gives

$$s_r + C \leq \frac{26}{5} + \left(\frac{5}{4} - \frac{13}{15}\right) \frac{9}{4} + \frac{57}{80} = \frac{271}{40} < \frac{34}{5}.$$

Hence, at these three orders, only a double fivefold can remain.

For that double, (10.8) gives the following sufficient bounds:

r	condition on n_r	bound for $s_r + C$
75	$n_{75} \leq 24$	$7 - \frac{1}{720}$
78	$n_{78} \leq 27$	$7 - \frac{133}{4320}$
80	$n_{80} \leq 61/2$	$7 - \frac{107}{87840}$.

Every displayed gap is greater than $1/1000$, and

$$\frac{1}{1000} > \frac{1}{1024} + \frac{1}{65536} = \frac{65}{65536}.$$

Thus absence of the required point forces, whenever its source row has been established,

$$(10.17) \quad \nu_{75} > 24j, \quad \nu_{78} > 27j, \quad \nu_{80} > \frac{61}{2}j.$$

Step 3. In this step, we obtain the complete source and fixed-divisor bound at order 78. Now take the single integer $j \geq J$ in the hypothesis. Its mass $\mu_{75}(j) > 600000000$, together with the preselected error at 75, gives

$$\text{vol}_Y(D_{75}) > 595000000, \quad \mu_{75}(k) > 590000000 \quad (k \geq J).$$

Consequently the 75-row of (10.14) holds at this same j . The preceding center alternatives force a double factor with $\nu_{75} > 24j$.

All intermediate complete spaces from point order $75j$ to $78j$ contain this full factor, whose exact point order $2\nu_{75}$ exceeds $48j$. Applying Lemma 10.2 gives

$$\begin{aligned} \mu_{78}(j) &> 600000000 - (30^6 - 27^6) - \frac{K_{75}}{j} \\ &= 258420489 - \frac{K_{75}}{j} > 258000000. \end{aligned}$$

Here $30^6 = 729000000$ and $27^6 = 387420489$; the last inequality uses (10.15). The preselected 78-error gives

$$\text{vol}_Y(D_{78}) > 257000000, \quad \mu_{78}(k) > 256000000 \quad (k \geq J).$$

Thus $\text{mult}_x F_{78}(j) < 61j$. The newly formed threshold belongs to the nonzero complete $V_{78}(j)$. Its only remaining center is a double with $\nu_{78} > 27j$.

Step 4. In this step, we obtain the complete source and fixed-divisor bound at order 80. Every intermediate space between $78j$ and $80j$ contains this second full factor, of point order greater than $54j$. Hence

$$\begin{aligned} \mu_{80}(j) &> 258000000 - (26^6 - 24^6) - \frac{K_{78}}{j} \\ &= 140187200 - \frac{K_{78}}{j} > 140000000, \end{aligned}$$

where $26^6 = 308915776$ and $24^6 = 191102976$. It follows that

$$\text{vol}_Y(D_{80}) > 139000000, \quad \mu_{80}(k) > 138000000 \quad (k \geq J).$$

Therefore $\text{mult}_x F_{80}(j) < 64j$. The own threshold of this nonzero complete $V_{80}(j)$ must have a double factor with $\nu_{80} > (61/2)j$.

Step 5. In this step, we obtain the complete source and fixed-divisor bound at order 84. This third full factor has point order greater than $61j$ and divides all intervening spaces up to order $84j$. Consequently

$$\begin{aligned} \mu_{84}(j) &> 140000000 - (23^6 - 19^6) - \frac{K_{80}}{j} \\ &= 39009992 - \frac{K_{80}}{j} > 39000000. \end{aligned}$$

Here $23^6 = 148035889$ and $19^6 = 47045881$. Thus

$$\text{vol}_Y(D_{84}) > 38000000, \quad \mu_{84}(k) > 37000000 \quad (k \geq J), \quad \text{mult}_x F_{84}(j) < \frac{357}{5}j.$$

All three shell constants were fixed before j , using discounts $48j, 54j, 61j$. They are independent of the three selected primes. Every volume comparison has been made for a complete space, and every new threshold has been taken only after its nonzero space was established.

Step 6. In this step, we construct a point boundary from every possible threshold center at order 84. Take the own pure threshold average of $V_{84}(j)$. The lower-center alternatives already apply. For a singular fivefold use (10.16) with $a_{84} = 357/5$. If $m_{84} \geq 5$, then

$$M_{84} > \frac{100}{17}, \quad b_{84} < \frac{2}{17} < 1,$$

and the small-deficit endpoint gives the point. For $m_{84} = 3$, we have $M_{84} > 60/17$ and $b_{84} < 42/17$. The entire triple tail satisfies

$$s_{84} + C < \frac{65}{14} + \left(\frac{5}{4} - \frac{65}{84}\right) \frac{42}{17} + \frac{57}{80} = \frac{65}{14} + \frac{20}{17} + \frac{57}{80} < \frac{33}{5}.$$

For $m_{84} = 4$, we have $M_{84} > 80/17$ and $b_{84} < 22/17$. The initial radius is at most $4^{1/5} < 4/3$, since $1024 > 972$. After the first cut the center has dimension at most four. If b' denotes its deficit, the remaining scalar tail is at most $b' + 2033/1440$, by Proposition 5.5. Therefore

$$C \leq \frac{4}{3}(b_{84} - b') + b' + \frac{2033}{1440} \leq \frac{4}{3}b_{84} + \frac{2033}{1440}.$$

A direct jump to the point satisfies the same bound with $b' = 0$. It follows that

$$s_{84} + C < \frac{65}{14} + \frac{47}{84} \frac{22}{17} + \frac{2033}{1440} < \frac{34}{5}.$$

For the last comparison, the three terms are respectively less than $4643/1000$, $725/1000$, and $1412/1000$, whose sum is $6780/1000 < 34/5$.

Finally, a double fivefold has $n_{84} < 357/10$, by the full fixed-part bound. Equation (10.8) gives

$$s_{84} + C < \frac{11353}{1440} - \frac{316}{357} = 7 - \frac{193}{171360}.$$

Since $193/171360 > 1/1000 > 1/1024 + 1/65536$, this case also gives the required point after the reserved error. Thus every possible final center has been covered, contradicting the exclusion of the point boundary. $\square$

11. POLARIZATIONS OF VOLUME AT LEAST TWO

We retain the notation of Section 2. We prove that $L^6 \geq 2$ suffices for generation at every point. The complete sources used here have point order $6k$. Their degrees are $5k$ when $L^6 \geq 3$, and $(549/100)k$ when $L^6 = 2$.

Theorem 11.1 (Thresholds of sums of ideals). *For coherent ideals I, J on a smooth complex variety, $\text{lct}_x(I + J) \leq \text{lct}_x(I) + \text{lct}_x(J)$. In a regular local ring of dimension d , the finite primary approximation consequently satisfies $\text{lct}(I + \mathfrak{m}^N) \leq \text{lct}(I) + d/N$. The generic coefficient-field passage in its use is part of the application, and the chosen approximation retains its strict margin.*

This is [DK01, Theorem 2.9]. We will use it on the regular local surface at the generic point of an actual fourfold center.

11.1. Tail estimates at the two source degrees.

Lemma 11.2. *Let B be an effective rational boundary on X , log canonical near x , with positive-dimensional minimal center W and deficit b . Let C be the scalar cost of any additive cutting chain from B to the point. Then the following hold.*

- (i) *If $\dim W \leq 3$, or if W is a fourfold smooth at x , then $C \leq b + 87/80$.*
- (ii) *For a fourfold of multiplicity $n = 2, \dots, 6$, the bounds are*

n	upper bound for b	a_n	p_n	$C \leq a_n b + p_n.$
2	3	6/5	19/32	
3	8/3	79/60	817/1440	
4	2	17/12	181/480	
5	8/5	3/2	3/10	
6	4/3	8/5	13/50	

A fourfold of multiplicity at least seven has $b \leq 6/7$.

- (iii) *If $\dim W = 4$, $\text{mult}_x W = 3$, and $L^4 \cdot W \geq 2$, then $C \leq (6/5)b + 347/480$.*

(iv) A fivefold has multiplicity at most five and $b \leq 6 - \text{mult}_x W$. For multiplicities four and five, respectively,

$$C \leq \frac{4}{3}b + \frac{791}{960}, \quad C \leq \frac{7}{5}b + \frac{113}{160}.$$

These estimates bound the entire continuation. In particular, they apply when the first cut skips a dimension.

Proof. We first record the multiplicity bounds, then bound the possible surface and threefold contributions to a chain.

Step 1. In this step, we bound center multiplicities, deficits, and codimension-one successors. By Theorem 4.5, from [dFH09, Corollary 7.12], the local rings of the minimal centers are normal and Cohen–Macaulay. They are excellent and analytically unramified. Write h_i for the Hilbert function in Theorem 4.6, namely [Han26, Lemmas 4.2 and 4.4], and put $\mu = \sum_i i h_i$. For a center of dimension d and multiplicity m , that theorem and Proposition 4.10 give

$$h_1 \leq 6 - d, \quad m - 1 \leq \mu \leq e_1 \leq \bar{e}_1, \quad 2\bar{e}_1 \leq (d - b)m.$$

For threefolds, $\mu \geq 2m - 2 - h_1 \geq 2m - 5$. For fourfolds, $\mu \geq 2m - 4$; also $h_2 \leq 3$, so $\mu \geq 3m - 3 - 2h_1 - h_2 \geq 3m - 10$. Together with integrality of $\bar{e}_1$, these inequalities give the following deficit caps:

m	1	2	3	4	5	6	7	8	9
dim $W = 3$	3	2	$5/3$	$3/2$	1	$2/3$	$3/7$	$1/4$	$1/9$
dim $W = 4$	4	3	$8/3$	2	$8/5$	$4/3$	$6/7$	$1/2$	$2/9$

No larger multiplicity has positive deficit. For example, the fourfold inequality with $3m - 10$ gives $b \leq 20/m - 2$. For a fivefold, $h_1 \leq 1$. Its Artinian reduction is either a field or has Hilbert function a string of m ones. Thus $\bar{e}_1 \geq m(m - 1)/2$, giving the fivefold assertions about multiplicity and deficit. A multiplicity-one Cohen–Macaulay local ring here is regular: its Artinian minimal reduction has length one, so the maximal ideal is generated by a system of parameters.

An old lc place remains an lc place under an effective addition for which the pair stays lc. Indeed, a positive order on that addition would give it negative discrepancy. Consequently, at a later boundary of deficit $y > 0$, Proposition 4.11 applies to an earlier center of dimension d and multiplicity m . A codimension-one successor of multiplicity t satisfies

$$t \leq (d - y)m - 2\bar{e}_1.$$

The deficit in this formula belongs to the later boundary. All radii are at most the corresponding multiplicity roots, because their L -degrees are positive integers. Embedding dimension cannot increase on passage to a contained center.

Step 2. In this step, we compute the complete surface and threefold tails. Corollary 4.13 bounds a surface of multiplicity j by deficit $2/j$. Its whole tail costs at most $q_j y$, where

$$(q_1, q_2, q_3, q_4, q_5) = (1, 17/12, 7/4, 2, 9/4).$$

These numbers dominate the square roots of $1, \dots, 5$, and curve radii are at most one. For $1 \leq a \leq 9/4$,

$$\max_{1 \leq j \leq 5} \frac{2}{j} (q_j - a)_+ \leq \frac{9}{10} - \frac{2}{5}a.$$

For $j = 2, 3, 4$, subtracting the untruncated candidate from the right side gives $3a/5 - 31/60$, $4(a - 1)/15$, and $(a - 1)/10$, respectively. The fifth candidate is equal to the right side, and the first vanishes. Both sides remain nonnegative after truncation. Thus a chain whose stages of dimension at least three have radius at most a costs at most $ab + 9/10 - 2a/5$.

For a threefold of multiplicity m , we obtain

$$(11.4) \quad C \leq c_m b + d_m, \quad \begin{array}{c|cccccccccc} m & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline c_m & 1 & 19/15 & 13/9 & 127/80 & 7/4 & 2 & 2 & 2 & 9/4 \\ d_m & 1/2 & 29/120 & 11/54 & 33/160 & 1/20 & 0 & 0 & 0 & 0 \end{array}$$

The coefficients c_m dominate the cube roots. For $m = 2$, the embedding dimension is at most four. A surface successor has $j \leq 3$, and (11.2) gives $j \leq 4 - 2y$. The possible positive surpluses over $(19/15)y$ are at most $3/20$ and $29/120$. For $m = 3$, the embedding dimension is at most five; here $j \leq 4$ and $j \leq 5 - 3y$. The positive surpluses are $11/54$ and $5/27$. For $m = 4$, one has $j \leq 6 - 4y$; the surpluses for $j = 3, 4, 5$ are $13/120, 33/160, 53/320$. For $m \geq 5$, the bound becomes $j \leq 10 - m - my$. At $m = 5$, only $j \leq 4$ occurs, and $j = 4$ forces $y \leq 1/5$, with surplus $1/20$. At $m = 6, 7$, all surface radii are at most two. The first row follows from (11.3) with $a = 1$.

The last two rows use Lemma 5.1: at $m = 8$, only smooth positive-deficit surface successors are possible, and at $m = 9$, none is possible. Their deficit caps are $1/4$ and $1/9$, respectively. In particular the two rows contribute at most $1/4$ and $5/36$ to $C - b$. Their contributions at the slopes $2301/2000$ and $9/10$ also satisfy the comparisons

$$\begin{array}{c|cc} m & C - (2301/2000)b & C - (9/10)b \\ \hline 8 & 1699/8000 < 3447/4000 & 11/40 < 100663/60000 \\ 9 & 733/6000 < 3447/4000 & 3/20 < 100663/60000 \end{array}$$

where the entries are upper bounds. Direct jumps to curves or points are included, since their radii are at most one. The rowwise upper bounds for $C - b$ in (11.4) are

$$\frac{1}{2}, \frac{31}{40}, \frac{17}{18}, \frac{87}{80}, \frac{4}{5}, \frac{2}{3}, \frac{3}{7}, \frac{1}{4}, \frac{5}{36}.$$

This proves (i) in dimension at most three. A smooth fourfold has radius at most one, so the same bound applies after its first cut.

Step 3. In this step, we bound fourfold tails using their successor restrictions and the degree-sensitive radius. For the five rows of (ii), put $D_n = 6, 8, 8, 8, 8$, respectively. These are the bounds for $4n - 2\bar{e}_1$ just obtained. If a first threefold successor has multiplicity m and deficit y , then

$$y \leq \min\{B_m, (D_n - m)/n\},$$

where B_m is the threefold row of (11.1). The numbers a_n dominate the fourth roots; in the closest comparison, $79^4 = 38950081 > 3 \cdot 60^4 = 38880000$. When $c_m > a_n$, the surplus is bounded by $(c_m - a_n) \min\{B_m, (D_n - m)/n\} + d_m$. When $c_m \leq a_n$, use (11.3). This also covers direct surface successors. The candidates, with the surface bound first, are

n	surface bound and positive-slope candidates	m
2	21/50; 3/8, 77/135, 19/32, 13/40	2, 3, 4, 5
3	28/75; 5/12, 817/1440, 29/60, 41/90, 41/180	3, 4, 5, 6, 7
4	1/3; 103/432, 181/480, 3/10, 7/24, 7/48	3, 4, 5, 6, 7
5	3/10; 221/800, 1/5, 1/5, 1/10	4, 5, 6, 7
6	13/50; 1/8, 2/15, 1/15	5, 6, 7

Their maxima are p_n . Multiplicities eight and nine cannot occur at positive deficit in these successor rows, because $m < D_n \leq 8$. This proves (ii).

If $n = 3$ and the degree is at least two, the initial radius is at most $(3/2)^{1/4} < 6/5$. Keep the flag $m \leq 8 - 3y$ and use reference radius $6/5$. The candidates are

$$\frac{21}{50}; \frac{3}{8}, \frac{11}{18}, \frac{347}{480}, \frac{3}{5}, \frac{8}{15}, \frac{4}{15} \quad (m = 2, \dots, 7).$$

Their maximum is $347/480$, proving (iii).

Step 4. In this step, we bound the complete tails of fivefolds of multiplicities four and five. For a proper center of a fivefold, Proposition 5.5 gives both

$$C_{\text{tail}} \leq \frac{5}{3}y + \frac{113}{480}, \quad C_{\text{tail}} \leq y + \frac{113}{80},$$

where the second follows from $2033/1440 < 113/80$. Subtracting $(4/3)y$ gives the minimum of $y/3 + 113/480$ and $-y/3 + 113/80$, bounded by $791/960$. Subtracting $(7/5)y$ gives the minimum of $(4/15)y + 113/480$ and $-(2/5)y + 113/80$, bounded by $113/160$. In both cases the two lines meet at $y = 113/64$. The initial radii are at most $4/3$ and $7/5$, respectively. Adding the initial deficit drop proves (iv). $\square$

11.2. Complete sources and full fixed divisors.

Lemma 11.3. *Let $q > 0$ be rational, let d_0 be a positive integer with $d_0q \in \mathbb{Z}$, and suppose $R_q = L^6q^6 - 6^6 > 0$. For positive k divisible by d_0 , set*

$$\mathcal{V}_k = V_k(q, 6), \quad \mathbf{a}_k = \mathbf{b}(\mathcal{V}_k).$$

Call k eligible when $\mathcal{V}_k \neq 0$, and let F_k be its full fixed divisor. Then:

(i) *All sufficiently large permitted k are eligible, and*

$$\dim \mathcal{V}_k \geq \frac{R_q}{720}k^6 + O(k^5).$$

At each eligible index, $0 < \tau_k = \text{lct}_x(\mathbf{a}_k) \leq 1/k$. A sufficiently general pure threshold average has the same lc centers near x as the ideal threshold pair. If its coefficient, point multiplicity and deficit are s_k, M_k, b_k , then

$$s_k = qk\tau_k, \quad M_k \geq 6k\tau_k, \quad b_k + M_k \leq 6.$$

(ii) *The supports of the F_k for all sufficiently large permitted indices lie in a finite set of primes. For every prime P ,*

$$n_P = \lim_{k \rightarrow \infty, d_0|k} \frac{\text{ord}_P(\mathbf{a}_k)}{k} = \inf_{k \text{ eligible}} \frac{\text{ord}_P(\mathbf{a}_k)}{k}.$$

The divisor $N = \sum_P n_P P$ has finite support and $0 \leq \eta := \text{mult}_x N \leq 6$.

(iii) *Put $\alpha = 1 - \eta/6$. For every eligible k and every integer $\ell \geq 6k$, set*

$$\mathcal{U}_{k,\ell} = H^0(X, \mathcal{O}_X(qkL) \otimes \mathbf{m}_x^\ell).$$

Then

$$\dim(\mathcal{U}_{k,\ell}/\mathcal{U}_{k,\ell+1}) \leq \binom{\lfloor \alpha \ell \rfloor + 5}{5}.$$

For fixed $Q > 6$, if $\mathcal{V}_k$ contains no nonzero section of order greater than Qk , then

$$\dim \mathcal{V}_k \leq \frac{\alpha^5(Q^6 - 6^6)}{720}k^6 + O(k^5).$$

The constant in this last error depends on the fixed source and Q , and is independent of k .

The fixed divisor in this lemma includes every component with its full multiplicity. The leading-image bound therefore retains repeated tangent factors.

Proof. We establish the actual source dimension, the fixed-divisor limits, and then sum the point-order filtration.

Step 1. In this step, we prove the source dimension bound and realize its threshold by an actual pure average. Asymptotic Riemann–Roch and Serre vanishing give

$$h^0(X, qkL) = \frac{L^6q^6}{720}k^6 + O(k^5), \quad \text{length}(\mathcal{O}_{X,x}/\mathbf{m}_x^{6k}) = \binom{6k+5}{6} = \frac{6^6}{720}k^6 + O(k^5).$$

The dimension of the kernel of the jet evaluation map is at least the difference. This proves the asserted lower bound and eventual eligibility. The point blow-up has log discrepancy six and sees $\text{ord}_x \mathbf{a}_k \geq 6k$, giving $\tau_k \leq 1/k$. Positivity and rationality follow by resolving the nonzero ideal.

On a log resolution, the actual system is its full fixed divisor plus a basepoint-free moving system. Choose a sufficiently large general tuple whose moving divisors, together with the fixed and exceptional support, have simple normal crossings. In the threshold average every moving coefficient is less than one. The coefficient-one strata are therefore precisely those of the ideal threshold pair. This proves the assertion about lc centers. The source degree gives s_k , and its prescribed point order gives $M_k \geq 6k\tau_k$. Applying the point-blow-up valuation to the ideal pair defining the deficit gives $b_k + M_k \leq 6$.

Step 2. In this step, we prove finite support and convergence of the full fixed coefficients. Actual multiplication gives

$$(11.5) \quad \mathcal{V}_k \mathcal{V}_l \subseteq \mathcal{V}_{k+l}, \quad \mathbf{a}_k \mathbf{a}_l \subseteq \mathbf{a}_{k+l}.$$

Thus $f_P(k) := \text{ord}_P(\mathbf{a}_k)$ is subadditive. Fix an eligible k_0 and a permitted H so large that every permitted index at least H is eligible. Every sufficiently large permitted k can be written $jk_0 + r$, with $H \leq r \leq H + k_0 - d_0$. Consequently

$$F_k \leq jF_{k_0} + F_r.$$

The union of the finitely many supports on the right contains every late fixed support. For any eligible m , the same bounded-residue argument gives

$$\limsup_{k \rightarrow \infty} \frac{f_P(k)}{k} \leq \frac{f_P(m)}{m}.$$

Taking the infimum over m proves the limit formula. Along multiples of k_0 , one has $F_{jk_0} \leq jF_{k_0}$, so $n_P = 0$ outside $\text{Supp } F_{k_0}$.

Step 3. In this step, we bound and sum all actual point-order layers using the entire limiting fixed divisor. Apply Proposition 7.7 with $A = d_0 qL$, point-order slope $6d_0$, and index k/d_0 . Its limiting fixed divisor in that indexing is $d_0 N$. Thus every actual section $\sigma \in \mathcal{V}_k$ of order $\nu \geq 6k$ satisfies

$$(11.6) \quad \text{ord}_P(\sigma) \geq \frac{\nu}{6} n_P \quad (P \in \text{Supp } N).$$

Additivity of local divisor orders on the smooth X gives $\nu \geq (\nu/6)\eta$, proving $\eta \leq 6$. At order ℓ , the integral Cartier divisor retained by every section is

$$D_\ell = \sum_{P \in \text{Supp } N} \left\lceil \frac{n_P \ell}{6} \right\rceil P.$$

Its order at x is at least $\eta\ell/6$. The degree- ℓ initial image is divisible by the full initial form of its local equation. Proposition 7.7 therefore gives the displayed binomial bound, including all components of D_ℓ . If its order exceeds ℓ , the image is zero.

If there is no section of order greater than Qk , the filtration terminates after $\lfloor Qk \rfloor$. For $\alpha > 0$, sum

$$\binom{\lfloor \alpha \ell \rfloor + 5}{5} = \frac{\alpha^5 \ell^5}{120} + O((\ell + 1)^4)$$

from $6k$ to $\lfloor Qk \rfloor$. The leading sum is the integral of $\alpha^5 t^5/120$, up to $O(k^5)$, proving (iii). If $\alpha = 0$, every summand is at most one and the sum is $O(k)$, which also satisfies the assertion. $\square$

Theorem 11.4. *If $L^6 \geq 3$, then at every closed point x there is a point boundary $B \sim_{\mathbb{Q}} uL$ with $0 < u < 7 - 1/1000$.*

The degree-five source suffices because failure of all affordable tails forces a fixed divisor whose leading-image capacity is too small.

Proof. We first exclude every threshold-center type except a double fivefold. We then use its full fixed multiplicity to force an actual section of higher point order.

Step 1. In this step, we pay every threshold-center type except a double fivefold. Use Lemma 11.3 with $q = 5$ and $d_0 = 1$, and retain its notation $\mathcal{V}_k, F_k, N, \tau_k$. The source mass satisfies

$$R_5 = 5^6 L^6 - 6^6 \geq 3 \cdot 15625 - 46656 = 219.$$

For a pure threshold average with positive-dimensional minimal center, write b for its deficit. Its coefficient satisfies $s \leq 5 - (5/6)b$. Lemma 11.2 and Proposition 5.5 give the following upper bounds for $s + C$:

minimal center	upper bound for $s + C$
$\dim W \leq 3$	527/80
$\dim W = 4, m(W) = 1$	1621/240
$\dim W = 4, m(W) = 2$	1071/160
$\dim W = 4, m(W) = 3$	1097/160
$\dim W = 4, m(W) = 4$	1047/160
$\dim W = 4, m(W) = 5$	191/30
$\dim W = 4, m(W) = 6$	2827/450
$\dim W = 4, m(W) \geq 7$	$5 + 5/7 + 113/480$
$\dim W = 5, m(W) = 1$	$7 - 11/1120$
$\dim W = 5, m(W) = 3$	557/80
$\dim W = 5, m(W) = 4$	$6 + 791/960$
$\dim W = 5, m(W) = 5$	3011/480

For example, the smooth fivefold entry is $5 + 5/6 + 3887/3360 = 7 - 11/1120$; the fourfold entries are $5 + (a_n - 5/6)b + p_n$ evaluated at their deficit caps. Every entry is at most $7 - 11/1120$. Since $11/1120 > 2/1000$, Proposition 3.7 with total error less than $1/1000$ gives a point boundary of coefficient less than $7 - 1/1000$ in every listed case. A point minimal center already has coefficient at most five.

Step 2. In this step, we force a limiting fixed point multiplicity greater than $30/7$. Suppose no point boundary of the asserted coefficient exists, and put $\delta = 1/1000$. Every eligible level must therefore have a fivefold minimal center P_k of multiplicity two. It is a prime divisor, and its coefficient in the threshold average is one. It belongs to the actual fixed part, since the moving coefficients are less than one. Put

$$r_k = \text{ord}_{P_k}(F_k), \quad n_k = r_k/k.$$

The coefficient-one relation is $\tau_k r_k = 1$, so $s_k = 5/n_k$ and $b_k \leq 6 - 6/n_k$. The double-fivefold row of Proposition 5.5 gives

$$7 - \delta \leq s_k + \frac{23}{20}b_k + \frac{1417}{1440} \leq \frac{69}{10} + \frac{1417}{1440} - \frac{19}{10n_k}.$$

Otherwise Proposition 3.7, with errors inside the resulting strict margin, would give a forbidden point boundary. It follows that

$$n_k \geq n_{\text{crit}} := \frac{19/10}{1273/1440 + 1/1000} > \frac{15}{7}.$$

Indeed $133/150 - 1273/1440 = 19/7200 > 1/1000$, and $(19/10)/(133/150) = 15/7$.

Fix an eligible k_0 . For the levels jk_0 , every selected P_{jk_0} lies in the finite support of F_{k_0} . One such prime occurs infinitely often. The coefficient limit in Lemma 11.3 gives

$$\eta = \text{mult}_x N \geq 2n_{\text{crit}} > \frac{30}{7}.$$

Thus $0 \leq \alpha = 1 - \eta/6 < 2/7$.

Step 3. In this step, we obtain an actual higher-order section and the required point boundary. Set $Q = 700/99$. Since $490000 < 50 \cdot 9801 = 490050$, we have $Q^2 < 50$. Hence

$$\alpha^5(Q^6 - 6^6) < \frac{32}{16807}(125000 - 46656) = \frac{2507008}{16807} < 219 \leq R_5,$$

where $219 \cdot 16807 = 3680733 > 2507008$. The lower and upper dimension bounds in Lemma 11.3 are therefore incompatible for large k unless the same $\mathcal{V}_k$ contains a nonzero section σ with $\text{ord}_x \sigma > Qk$. Its entire divisor

$$D = \frac{1}{k} \text{div}(\sigma) \sim_{\mathbb{Q}} 5L$$

satisfies $\text{mult}_x D/5 > 140/99$. Proposition 5.8 gives a point boundary of coefficient less than $7 - 1/630 < 7 - 1/1000$, a contradiction. $\square$

11.3. The surviving centers in volume two.

Lemma 11.5. *Assume $L^6 = 2$ and that $K_X + 7L$ is not generated at x . Put $q = 549/100$, and use $\mathcal{V}_k = V_k(q, 6)$, $100 \mid k$, with the base ideals, thresholds and limiting full fixed divisor N of Lemma 11.3. Then $\text{mult}_x N < 3$. For every sufficiently large permitted k , every sufficiently general pure threshold average, of deficit b , has a fourfold minimal center W with*

$$\text{mult}_x W = 3, \quad L^4 \cdot W = 1, \quad \text{edim } \mathcal{O}_{W,x} = 6, \quad \frac{6787}{2892} \leq b \leq \frac{8}{3}.$$

For $R_0 = \mathcal{O}_{W,x}$ and its maximal ideal $\mathfrak{n}_0$, one has

$$e_1(\mathfrak{n}_0) = \bar{e}_1(\mathfrak{n}_0) = 2, \quad \text{Hilb}(\text{gr}_{\mathfrak{n}_0} R_0; z) = \frac{1 + 2z}{(1 - z)^4}.$$

The associated graded ring is reduced and Cohen–Macaulay.

Both the center and its tangent ring in this statement come from the degree-549/100 source at an actual occurrence.

Proof. We first bound the full fixed mass, then exclude all other centers and prove the assertions about the surviving local ring.

Step 1. In this step, we bound the limiting full fixed point multiplicity by three. Since $q^2 = 301401/10000 > 30$,

$$(11.7) \quad R_q = 2q^6 - 6^6 > 2 \cdot 30^3 - 46656 = 7344.$$

Lemma 11.3 applies. Put $Q = 427/55 = q(140/99) > 6$. We have $Q^2 < 61$, since $427^2 = 182329 < 61 \cdot 55^2 = 184525$. If $\eta = \text{mult}_x N \geq 3$, then $0 \leq \alpha = 1 - \eta/6 \leq 1/2$, and

$$\alpha^5(Q^6 - 6^6) \leq \frac{Q^6 - 6^6}{32} < \frac{61^3 - 46656}{32} = \frac{180325}{32} < 7344 < R_q.$$

Here $180325 < 32 \cdot 7344 = 235008$. The dimension comparison in Lemma 11.3 forces an actual $\sigma \in \mathcal{V}_k$ with $\text{ord}_x \sigma > Qk$ for large permitted k . Its entire divisor divided by k has coefficient q and multiplicity-to-coefficient ratio greater than $140/99$. Propositions 5.8 and 3.4 contradict nongeneration. Therefore

$$(11.8) \quad \eta < 3.$$

Step 2. In this step, we exclude all uniformly affordable threshold-center types. Set $q_0 = 11/2$. A threshold average of deficit b has

$$s \leq q - \frac{q}{6}b \leq q_0 - \frac{q_0}{6}b,$$

because $b \leq 6$. Under nongeneration, every uniform whole-tail bound K must satisfy $s + K \geq 7$, by Proposition 3.7. The estimates of Lemma 11.2 give the following forbidden upper bounds:

(11.9)	minimal center	upper bound for $s + C$
	$\dim W \leq 3$	$547/80$
	$\dim W = 4, m(W) = 1$	$1661/240$
	$\dim W = 4, m(W) = 2$	$1111/160$
	$\dim W = 4, m(W) = 4$	$3301/480$
	$\dim W = 4, m(W) = 5$	$101/15$
	$\dim W = 4, m(W) = 6$	$1501/225$
	$\dim W = 4, m(W) \geq 7$	$11/2 + 9/14 + 113/480$
	$\dim W = 4, m(W) = 3, g(W) \geq 2$	$7 - 31/1440$
	$\dim W = 5, m(W) = 5$	$3211/480$

Every entry is strictly less than seven. The fourfold rows are obtained by substituting the deficit cap in $q_0 + (a_n - q_0/6)b + p_n$. The degree-sensitive row is

$$\frac{11}{2} + \frac{178}{60 \cdot 3} + \frac{347}{480} = 7 - \frac{31}{1440}.$$

For the multiplicity-five fivefold, the bound is $11/2 + 29/60 + 113/160 = 3211/480$. A point center is already paid because $s \leq q < 7$. Thus only fivefolds of multiplicity one through four and a fourfold of multiplicity three and degree one remain.

Step 3. In this step, we exclude fivefold centers at every sufficiently large permitted level. Suppose fivefold minimal centers occur at infinitely many permitted levels. Each is an actual fixed prime P , with multiplicity $m \in \{1, 2, 3, 4\}$. If $r = \text{ord}_P(F_k)$ and $n = r/k$, its coefficient-one relation gives

$$\tau_k r = 1, \quad s = q/n \leq q_0/n, \quad b \leq 6 - 6/n.$$

Use the following whole-tail coefficients, from Proposition 5.5 and Lemma 11.2:

m	a	d	forced lower bound for mn	$C \leq ab + d.$
1	1	3887/3360	1680/527	
2	23/20	1417/1440	4032/1273	
3	5/4	57/80	480/97	
4	4/3	791/960	9600/1751	

Indeed,

$$7 \leq s + ab + d \leq 6a + d - \frac{6a - q_0}{n}, \quad mn \geq \frac{m(6a - q_0)}{6a + d - 7}.$$

The denominators are positive, and substitution gives the last column. All four entries exceed three:

$$1680 > 3 \cdot 527, \quad 4032 > 3 \cdot 1273, \quad 480 > 3 \cdot 97, \quad 9600 > 3 \cdot 1751.$$

By the finite-support assertion of Lemma 11.3, a single prime P occurs at infinitely many of these levels. Passing to its existing coefficient limit gives $\eta \geq mn_P > 3$, contrary to (11.8). Thus every sufficiently large permitted level has only the stated fourfold minimal center. This argument applies to any choice of sufficiently general averages at the offending levels, so the cutoff excludes all such averages.

Step 4. In this step, we prove the Hilbert coefficient and reduced tangent-ring assertions for each surviving cubic fourfold. For the remaining fourfold, the sharp source coefficient bound and the multiplicity-three row of Lemma 11.2 give

$$7 \leq q + \left(\frac{79}{60} - \frac{q}{6} \right) b + \frac{817}{1440}.$$

Here

$$\frac{79}{60} - \frac{q}{6} = \frac{241}{600}, \quad 7 - q - \frac{817}{1440} = \frac{6787}{7200}.$$

Consequently $b \geq 6787/2892 = 7/3 + 13/964$; the upper bound $b \leq 8/3$ was recorded in (11.1). Since $b > 2$, Proposition 4.10 implies

$$2 \leq \bar{e}_1(\mathfrak{n}_0) \leq \frac{(4-b)3}{2} < 3.$$

Integrality gives $\bar{e}_1(\mathfrak{n}_0) = 2$, and Theorem 4.6 then gives $e_1(\mathfrak{n}_0) = \sum_i ih_i = 2$. If $\text{edim } R_0 \leq 5$, its multiplicity-three Artinian reduction would have Hilbert function $(1, 1, 1)$, of first moment three. It follows that $\text{edim } R_0 = 6$.

The normal-blow-up defect satisfies

$$0 \leq (4-b)3 - 2\bar{e}_1(\mathfrak{n}_0) = 8 - 3b \leq \frac{925}{964} < 1.$$

Proposition 4.12 makes every exceptional coefficient one. We now pass from this conclusion to the full associated graded ring.

The ring R_0 is normal and Cohen–Macaulay by Theorem 4.5. It has minimal multiplicity $3 = 6 - 4 + 1$. Theorem 4.14, using [Ver08, Theorem 2.7], gives a minimal reduction J_0 with $\mathfrak{n}_0^2 = J_0\mathfrak{n}_0$ and a Cohen–Macaulay associated graded ring. Its Artinian linear reduction has Hilbert function $(1, 2)$; hence its Hilbert series is $(1 + 2z)/(1 - z)^4$. Moreover,

$$\text{length}(\mathfrak{n}_0/J_0) = 2, \quad \mathfrak{n}_0^j \subseteq J_0 \quad (j \geq 2).$$

The sum in the Rees criterion of Theorem 4.14 is therefore exactly two, equal to $e_1(\mathfrak{n}_0)$. By [Ver08, Corollary 6.6], the ordinary Rees algebra is Cohen–Macaulay, in particular S_2 .

The ring R_0 is excellent and normal, hence analytically unramified. As a local ring of a complex variety it has infinite residue field and, being Cohen–Macaulay, has a canonical module. Together with $e_1 = \bar{e}_1$, these are all the hypotheses of Theorem 4.16, namely [PUV05, Corollary 2.5]. Every power of $\mathfrak{n}_0$ is therefore integrally closed. The normalization of the Rees algebra of an ideal in a normal ring has degree- j piece $\overline{\mathfrak{n}_0^j}$; thus this Rees algebra is normal. The ordinary and normalized point blow-ups agree.

The exceptional scheme is now $\text{Proj}(\text{gr}_{\mathfrak{n}_0} R_0)$, and it is reduced. To spell out the scheme assertion, its ideal on the normal blow-up is the invertible ideal of a divisor whose prime coefficients are all one. The divisor is generically reduced, and the quotient by its local nonzerodivisor satisfies S_1 , since the ambient ring satisfies S_2 . It has no embedded associated primes, so it is reduced. Finally, the graded ring is Cohen–Macaulay, and all its minimal primes have positive dimension. They are visible on Proj , where reducedness makes the ring generically reduced. Any nonzero nilradical would have an associated prime among these minimal primes, a contradiction. The full associated graded ring is reduced. $\square$

11.4. Finite recurring centers and generic thresholds.

Lemma 11.6. *Assume $L^6 = 2$ and that $K_X + 7L$ is not generated at x . Put $q = 549/100$, and use $\mathcal{V}_k = V_k(q, 6)$, $100 \mid k$, with base ideal $\mathfrak{a}_k$ and threshold τ_k at eligible indices. There are a real number T , a permitted eligible index K , and finitely many integral fourfolds $W_1, \dots, W_r$ through x , with $r \geq 1$, such that the following hold.*

(i) *Along all eligible permitted indices,*

$$\lim_{k \rightarrow \infty} k\tau_k = \sup_{k \text{ eligible}} k\tau_k = T, \quad 0 < T \leq \frac{10565}{17352} < \frac{11}{18}.$$

(ii) *Every sufficiently general pure threshold average at every sufficiently large level mK has minimal center among the W_i . Each W_i occurs at infinitely many such levels and satisfies the local conclusions of Lemma 11.5.*

(iii) *If η_i is the generic point of W_i , then every eligible $\mathfrak{a}_k$ is nonunit there, and*

$$\lim_{k \rightarrow \infty} k \text{ lct}_{\eta_i}(\mathfrak{a}_k) = \sup_{k \text{ eligible}} k \text{ lct}_{\eta_i}(\mathfrak{a}_k) = T.$$

For every eligible k and every nonzero actual subspace $U \subseteq \mathcal{V}_k$, let $\nu(U) = \min_{0 \neq \sigma \in U} \text{ord}_x \sigma$, and let $J_{U,i}$ be its generated ideal in $\mathcal{O}_{X,\eta_i}$. Then

$$\text{lct}_{\eta_i}(J_{U,i}) \leq \frac{6T}{\nu(U)}.$$

The same finite list controls all sufficiently general averages on one cofinal ray. The last inequality holds at every eligible source level, including levels outside that ray.

Proof. We form the normalized threshold limit, capture the centers in a single multiplier-ideal cosupport, and identify each generic threshold by recurrence. Actual multiplication and ample padding then give the subspace assertion. For each prime P , write $f_P(k) = \text{ord}_P(\mathfrak{a}_k)$ and $n_P = \lim f_P(k)/k$, and put $N = \sum_P n_P P$, as in Lemma 11.3.

Step 1. In this step, we construct the normalized threshold limit and prove its strict upper bound. Set $\beta_k = 1/\tau_k$. The valuative formula is

$$\beta_k = \sup_v \frac{v(\mathfrak{a}_k)}{A_X(v)},$$

where v ranges over divisorial valuations with centers containing x , and $A_X(v)$ is the log discrepancy. The quantities are positive and finite. The point blow-up gives $\beta_k \geq k$. By (11.5),

$$\beta_{k+l} \leq \beta_k + \beta_l.$$

The bounded-residue argument in Lemma 11.3 applies to these numbers as well: for each eligible m , write large permitted $k = jm + \rho_k$ with ρ_k in a fixed finite interval of eligible residues. Thus

$$\beta_\infty := \lim \frac{\beta_k}{k} = \inf_{k \text{ eligible}} \frac{\beta_k}{k}, \quad 1 \leq \beta_\infty < \infty.$$

Put $T = 1/\beta_\infty$. Reciprocation gives the limit and supremum in (i). At every sufficiently large permitted level, Lemma 11.5 and the point-blow-up inequality give

$$6k\tau_k \leq 6 - b_k \leq 6 - \frac{6787}{2892}, \quad k\tau_k \leq \frac{10565}{17352}.$$

Take the limit. The strict comparison with $11/18$ follows from $10565 < 10604 = 11 \cdot 964$.

Step 2. In this step, we obtain a fixed multiplier ideal on a cofinal ray and prove that its divisorial components are smooth at x . Let $S = \mathcal{O}_{X,x}$ and define

$$\mathcal{I}_k = \mathcal{J}(X, \mathfrak{a}_k^{T/k})_x.$$

For clarity, the properties we need follow directly from the log-resolution formula

$$\mathcal{J}(\mathfrak{a}^c) = f_* \mathcal{O}_{\tilde{X}}(K_{\tilde{X}/X} - \lfloor cG \rfloor), \quad \mathfrak{a} \mathcal{O}_{\tilde{X}} = \mathcal{O}_{\tilde{X}}(-G).$$

Localization commutes with this formula. Replacing the ideal by a larger ideal increases its multiplier ideal, and increasing c decreases it. The formula also gives $\mathcal{J}((\mathfrak{a}^m)^c) = \mathcal{J}(\mathfrak{a}^{mc})$. Every $\mathcal{I}_k$ is nonzero, and it is proper because $T/k \geq \tau_k$. Multiplication of actual sections gives

$$\mathcal{I}_k \subseteq \mathcal{I}_{mk} \quad (m \geq 1).$$

Any finite set of eligible indices has an eligible common multiple. The family is therefore directed under divisibility.

The ideal $\mathcal{I}_\infty = \sum_k \mathcal{I}_k$ in the noetherian local ring S is generated by finitely many elements, each belonging to a finite sum of the $\mathcal{I}_k$. Choose an eligible common multiple K of the resulting finite set of indices. Directedness yields

$$(11.10) \quad \mathcal{I}_\infty = \mathcal{I}_K = \mathcal{I}_{mK} \quad (m \geq 1).$$

This ideal is nonzero and proper: all summands lie in the maximal ideal of S .

Let a divisorial component through x of $V(\mathcal{I}_\infty)$ be the germ of a prime divisor P . Localizing (11.10) at its generic point gives a proper ideal in a regular discrete valuation ring, whose exponent is

$$\left\lfloor \frac{Tf_P(mK)}{mK} \right\rfloor.$$

It follows that $Tf_P(mK)/(mK) \geq 1$ for every m , and the coefficient limit gives $Tn_P \geq 1$. If $\text{mult}_x P \geq 2$, then

$$\text{mult}_x N \geq n_P \text{mult}_x P \geq 2/T > 3,$$

because $T < 2/3$. This contradicts Lemma 11.5. Therefore every such divisorial component is smooth at x .

Step 3. In this step, we capture every sufficiently late threshold center in a finite recurring list. At a sufficiently large ray level mK , let W be the minimal center of any sufficiently general threshold average. By Lemma 11.3, it is also the minimal center of the actual ideal threshold pair. Since $\tau_{mK} \leq T/(mK)$,

$$\mathcal{I}_\infty \subseteq \mathcal{J}(X, \mathfrak{a}_{mK}^{\tau_{mK}})_x.$$

Thus the germ of W lies in $V(\mathcal{I}_\infty)$. By Lemma 11.5, it is irreducible, four-dimensional, and has embedding dimension six. An irreducible component of the cosupport containing it has dimension four or five, since $\mathcal{I}_\infty \neq 0$. A five-dimensional component would be smooth at x , so its local ring, and every quotient of that ring, would have embedding dimension at most five. It cannot contain W . Hence W itself is a four-dimensional irreducible component germ of $V(\mathcal{I}_\infty)$.

There are only finitely many such germs. Their closures in X are integral projective fourfolds; two integral subvarieties agreeing as germs have the same closure. Retain those occurring at infinitely many ray levels, and enlarge the starting level past all occurrences of the others. At least one remains. This gives (ii). Equality (11.10) concerns only these multiplier-ideal stalks; the argument requires no equality of base ideals at different levels.

Step 4. In this step, we identify the generic threshold limit at every recurring center. Fix i , and put $R_i = \mathcal{O}_{X, \eta_i}$, a regular local ring of dimension two. Define

$$c_{k,i} = \frac{1}{\text{lct}_{\eta_i}(\mathfrak{a}_k)},$$

with value zero if $\mathfrak{a}_k R_i = R_i$. The reciprocal valuative formula and multiplication make these nonnegative finite numbers subadditive. The bounded-residue argument therefore gives

$$c_i := \lim_k \frac{c_{k,i}}{k} = \inf_{k \text{ eligible}} \frac{c_{k,i}}{k}.$$

At any actual occurrence of W_i as an lc center, a zero-discrepancy divisor centered on W_i remains after localization. Log canonicity near x also gives log canonicity at η_i . Together these facts give $\text{lct}_{\eta_i}(\mathfrak{a}_k) = \tau_k$ at that occurrence. Along its infinite subsequence of occurrences,

$$\frac{c_{k,i}}{k} = \frac{1}{k\tau_k} \longrightarrow \frac{1}{T}.$$

The existing limit forces $c_i = 1/T > 0$. Consequently every $c_{k,i}$ is positive; every eligible ideal is nonunit at η_i . Reciprocation proves the first assertion of (iii). No divisor computing one threshold has been chosen for another level.

Step 5. In this step, we amplify actual subspaces to bound their generic thresholds at every eligible level. Fix $0 \neq U \subseteq \mathcal{V}_k$ and put $\nu = \nu(U)$. If $\nu = 6k$, then $J_{U,i} \subseteq \mathfrak{a}_k R_i$, so

$$\text{lct}(J_{U,i}) \leq \text{lct}(\mathfrak{a}_k R_i) \leq T/k = 6T/\nu.$$

If $\nu > 6k$, choose a sufficiently large positive integer A divisible by 600, and set $k' = A\nu/6$. Then $100 \mid k'$, and $q(k' - Ak)$ is a positive integer tending to infinity with A . Ampleness makes $\mathcal{O}_X(q(k' - Ak)L)$ globally generated for sufficiently large A . Choose a section z nonzero at x

and generically nonzero on all the finitely many W_i . This is possible by avoiding finitely many proper linear subspaces in its section space.

The span of all products of A members of U , multiplied by z , is an actual subspace of $\mathcal{V}_{k'}$. Its point order is at least $A\nu = 6k'$, and its ideal at η_i is $J_{U,i}^A$, since z is a unit there. Therefore

$$J_{U,i}^A \subseteq \mathfrak{a}_{k'} R_i, \quad \frac{1}{A} \text{lct}(J_{U,i}) \leq \text{lct}(\mathfrak{a}_{k'} R_i).$$

Multiplying by k' yields

$$\frac{\nu}{6} \text{lct}(J_{U,i}) \leq k' \text{lct}(\mathfrak{a}_{k'} R_i) \leq T.$$

This proves the assertion for every actual subspace, with the same T for all of them. $\square$

Theorem 11.7. *For every smooth complex projective sixfold X and every ample Cartier divisor L with $L^6 \geq 2$, the line bundle $\mathcal{O}_X(K_X + 7L)$ is globally generated.*

The remaining volume-two case follows by comparing the complete source dimension with the uniform cubic capacity at one recurring center.

Proof. For $L^6 \geq 3$, Theorem 11.4 and Proposition 3.4 prove generation at every point. It remains to consider $L^6 = 2$. Suppose $K_X + 7L$ is not generated at x . We apply cubic capacity to every actual extra-jet level and sum its uniform error before constructing the final point boundary.

Step 1. In this step, we choose a recurring cubic center satisfying every tangent hypothesis of the capacity estimate. Set $q = 549/100$ and $\mathcal{V}_k = V_k(q, 6)$, with $100 \mid k$. Fix one center W and the number T from Lemma 11.6. Put

$$S = \mathcal{O}_{X,x}, \quad \mathfrak{m} = \mathfrak{m}_{X,x}, \quad P = \mathcal{I}_{W,x}, \quad R = S_P, \quad \mathfrak{n} = PR.$$

The ideal P is a height-two prime. By Lemma 11.5, the local ring S/P has embedding dimension six, so $P \subseteq \mathfrak{m}^2$. Its full associated graded ring is reduced and Cohen–Macaulay with Hilbert series $(1 + 2z)/(1 - z)^4$. These are precisely the center hypotheses of Theorem 6.17.

Step 2. In this step, we approximate each actual generic ideal by a primary ideal while preserving the strict threshold margin. For an integer $\ell \geq 6k$, let

$$\mathcal{U}_{k,\ell} = H^0(X, \mathcal{O}_X(qkL) \otimes \mathfrak{m}_x^\ell).$$

If this space is nonzero, let $A_{k,\ell} \subset R$ be the ideal generated by its actual germs. The subspace $\mathcal{U}_{k,\ell}$ has minimum point order at least ℓ . Lemma 11.6 gives

$$\text{lct}_R(A_{k,\ell}) \leq \frac{6T}{\ell}, \quad \frac{11}{3} - 6T \geq \frac{13}{964} > 0.$$

Choose a finite integer $N_{k,\ell} > 1928\ell/13$. Theorem 11.1 gives

$$(11.11) \quad \text{lct}_R(A_{k,\ell} + \mathfrak{n}^{N_{k,\ell}}) \leq \text{lct}_R(A_{k,\ell}) + \frac{2}{N_{k,\ell}} < \frac{11}{3\ell}.$$

Here the local ring has residue field $\mathbb{C}(W)$, rather than $\mathbb{C}$. The generic application in Theorem 11.1 uses the same resolution formula for these ideals: completion and extension of the characteristic-zero coefficient field preserve the discrepancy and ideal-order integers. Equivalently, spread a common log resolution over a smooth open subset of W and shrink it until its strata are smooth over that subset. Restriction to a general transverse smooth surface computes the same generic thresholds and gives (11.11). The maximal-ideal power has threshold $2/N_{k,\ell}$ on this regular surface.

The contraction

$$I_{k,\ell} = (A_{k,\ell} + \mathfrak{n}^{N_{k,\ell}}) \cap S$$

is P -primary. It contains every actual section germ in $\mathcal{U}_{k,\ell}$, including all its divisorial factors. Thus the leading-image map injects $\mathcal{U}_{k,\ell}/\mathcal{U}_{k,\ell+1}$ into $(\text{in}_{\mathfrak{m}} I_{k,\ell})_\ell$. Apply Theorem 6.17 with $n = \ell$

and $\Lambda = 11/18$. Its constant is independent of the ideal, the primary exponent and the order level, so

$$(11.12) \quad \begin{aligned} \dim(\mathcal{U}_{k,\ell}/\mathcal{U}_{k,\ell+1}) &\leq \frac{2(5/11)^5}{120} \ell^5 + C_2(\ell+1)^4 \\ &\leq \frac{\ell^5}{25 \cdot 120} + C_2(\ell+1)^4. \end{aligned}$$

The last inequality uses $2(5/11)^5 = 6250/161051 < 1/25$, since $156250 < 161051$. If $\mathcal{U}_{k,\ell} = 0$, the same bound holds with zero on the left. The primary approximation is used only as an upper bound on these actual leading images.

Step 3. In this step, we compare the source mass with the summed capacities to obtain a point boundary and contradict nongeneration. Let $Q = 427/55 = q(140/99)$. Suppose $\mathcal{V}_k$ has no section of order greater than Qk . Sum (11.12) from $6k$ to $\lfloor Qk \rfloor$. The filtration then terminates, and the same constant C_2 at every level gives

$$\dim \mathcal{V}_k \leq \frac{Q^6 - 6^6}{25 \cdot 720} k^6 + O(k^5).$$

There are $O(k)$ error terms, each $O(k^4)$. But $Q^2 < 61$, so

$$\frac{Q^6 - 6^6}{25} < \frac{61^3 - 46656}{25} = \frac{180325}{25} = 7213 < 7344 < R_q.$$

This contradicts the lower source dimension in Lemma 11.3 and (11.7) for every sufficiently large permitted k .

Therefore an actual $\sigma \in \mathcal{V}_k$ has $\text{ord}_x \sigma > Qk$. Its entire effective rational divisor

$$D = \frac{1}{k} \text{div}(\sigma) \sim_{\mathbb{Q}} qL, \quad \frac{\text{mult}_x D}{q} > \frac{140}{99},$$

meets Proposition 5.8. The resulting point boundary has coefficient less than $7 - 1/630$, and Proposition 3.4 gives generation at x . This contradicts the chosen base point. There is consequently no base point when $L^6 = 2$, completing the proof. $\square$

12. ORIGINAL FOURFOLD CENTERS OF HIGHER MULTIPLICITY

We retain the notation of Section 2. We first bound the complete tails of fourfold centers. We then study the ordinary exceptional scheme of a center for which these bounds do not produce a point. The source argument uses the complete spaces $V_r(k)$ of Proposition 7.5.

We use two geometric inputs to construct connected sections of the ordinary exceptional scheme.

Theorem 12.1 (Stein factorization). *Let $f : T \rightarrow S$ be a proper morphism with S locally Noetherian. There is a factorization $T \xrightarrow{g} S' \rightarrow S$ such that $S' = \text{Spec}_S(f_* \mathcal{O}_T)$, the second morphism is finite, g is proper with geometrically connected fibers, and $g_* \mathcal{O}_T = \mathcal{O}_{S'}$.*

This is [Sta26, Theorem 37.53.4, Tag 03H0]. For the ordinary blow-up below, normality of the base identifies the finite factor with the base itself.

Theorem 12.2 (Bertini irreducibility). *Let T be an integral projective complex variety of dimension at least two, with a fixed closed immersion into projective space. A general hyperplane section of T is geometrically irreducible.*

This is the very ample specialization of [Sta26, Lemma 37.32.3, Tag 0G4F]. We apply it to each irreducible component of the exceptional scheme; connectedness of their sections will be proved separately.

Lemma 12.3. *Let Γ be an effective rational boundary on X , log canonical near x , with a fourfold minimal center W . Put $b = \text{def}_x(\Gamma)$ and $m = \text{mult}_x W$. If $m \geq 7$, then $b \leq 1$. If $m \in \{3, 4, 5, 6\}$, every additive cutting chain from Γ to the point satisfies*

m	upper bound for b	a_m	p_m	$C \leq a_m b + p_m.$
3	8/3	13161/10000	34087/60000	
4	2	17/12	181/480	
5	8/5	3/2	3/10	
6	4/3	8/5	13/50	

For every $m \geq 4$, the condition $\text{edim}_x W \leq 5$ also implies $b \leq 1$.

These estimates depend on the current boundary and its actual descendants. They impose no condition on the source of that boundary.

Proof. We bound the Hilbert moments, enumerate the threefold tails, and then impose the inequality supplied by the fourfold parent.

Step 1. In this step, we derive the fourfold deficit bounds from the Artinian Hilbert moments. The local rings of the centers are normal and Cohen–Macaulay by Theorem 4.5. They are excellent and analytically unramified, with residue field $\mathbb{C}$. Thus Theorem 4.6 applies. For a fourfold, its Artinian reduction has $h_1 \leq 2$ and $h_2 \leq 3$, so

$$\bar{e}_1 \geq m - 1, \quad \bar{e}_1 \geq 2m - 4, \quad \bar{e}_1 \geq 3m - 10.$$

Together with Proposition 4.10, the second inequality gives $b \leq 8/m$, and the third gives $b \leq 20/m - 2$. For $m = 7$ the latter is $6/7$; for $m \geq 8$ the former is at most one. If the embedding dimension is at most five, a nontrivial Artinian reduction has embedding dimension one and Hilbert function consisting of m ones. Its first moment is $m(m-1)/2$, whence $b \leq 5 - m \leq 1$.

Step 2. In this step, we bound every complete threefold tail, including multiplicities eight and nine. For a threefold of multiplicity u and deficit $y > 0$, the same argument gives $\bar{e}_1 \geq u - 1$ and $\bar{e}_1 \geq 2u - 5$. Write B_u for the resulting deficit cap. We will use the following complete-tail bounds:

$$(12.1) \quad \begin{array}{c|c|c|c} u & B_u & c_u & d_u \\ \hline 1 & 3 & 1 & 1/2 \\ 2 & 2 & 19/15 & 29/120 \\ 3 & 5/3 & 13/9 & 11/54 \\ 4 & 3/2 & 127/80 & 33/160 \\ 5 & 1 & 7/4 & 1/20 \\ 6 & 2/3 & 2 & 0 \\ 7 & 3/7 & 2 & 0 \\ 8 & 1/4 & 2 & 0 \\ 9 & 1/9 & 9/4 & 0 \end{array} \quad C_3 \leq c_u y + d_u.$$

Here C_3 denotes the scalar cost from this threefold onward. The moment bounds exclude $u \geq 10$.

An earlier lc place has value zero on every subsequent effective addition that preserves log canonicity. Its center therefore remains an lc center. If a surface of multiplicity t and deficit $z > 0$ occurs after the threefold, then Proposition 4.11 gives

$$t \leq (3 - z)u - 2\bar{e}_1(\mathbf{m}_{W_3, x}),$$

where W_3 is that threefold. By Corollary 4.13, $t \leq 5$, $z \leq 2/t$, and the remaining curve centers are smooth. More precisely, the surface inequalities give $t - 1 \leq \sum_i i h_i \leq \bar{e}_1 < t$, so $\sum_i i h_i = t - 1$. Every Artinian term after degree one vanishes. Hence $t = 1 + h_1 = \text{edim}_x W_2 - 1$ for that surface W_2 ; this will also bound its multiplicity inside a parent of smaller embedding dimension. Set

$$(q_1, q_2, q_3, q_4, q_5) = (1, 17/12, 7/4, 2, 9/4).$$

These numbers dominate the square roots of the corresponding multiplicities. The surface tail is at most $q_t z$. For $1 \leq A \leq 9/4$,

$$(12.2) \quad \max_{1 \leq t \leq 5} \frac{2}{t} (q_t - A)_+ \leq \frac{9}{10} - \frac{2}{5}A.$$

For $t = 2, 3, 4$, subtracting the untruncated expression on the left from the right gives respectively

$$\frac{3}{5}A - \frac{31}{60}, \quad \frac{4}{15}(A - 1), \quad \frac{1}{10}(A - 1).$$

They are nonnegative. The fifth expression is equality, and the first contributes zero. The right side is nonnegative on the specified interval, so truncation preserves the comparisons.

For $u = 2$, the embedding dimension is at most four, so $t \leq 3$; the flag bound is $t \leq 4 - 2z$. The positive surpluses over $(19/15)z$ are at most $3/20$ and $29/120$. For $u = 3$, the embedding dimension is at most five, so $t \leq 4$; the flag bound $t \leq 5 - 3z$ gives surpluses $11/54$ and $5/27$ over $(13/9)z$. For $u = 4$, $t \leq 6 - 4z$, and the surpluses for $t = 3, 4, 5$ are $13/120, 33/160, 53/320$. For $u = 5$, $t \leq 5 - 5z$; only $t = 4$ gives a positive surplus over $(7/4)z$, and it is at most $1/20$. For $u = 6, 7$, respectively $t \leq 4 - 6z$ and $t \leq 3 - 7z$, so every radius in the tail is at most two. The smooth threefold uses (12.2) with $A = 1$. The displayed c_u dominate the cube roots: the relevant comparisons are

$$19^3 > 2 \cdot 15^3, \quad 13^3 > 3 \cdot 9^3, \quad 127^3 > 4 \cdot 80^3, \quad (7/4)^3 > 5, \quad 2^3 = 8.$$

This proves the first seven rows of (12.1), including direct jumps to curves and points.

The last two rows are precisely Lemma 5.1. At $u = 8$, the inequalities are $y \leq 1/4$, $\bar{e}_1 \geq 11$, and $t \leq 2 - 8z < 2$. At $u = 9$, they are $y \leq 1/9$, $\bar{e}_1 \geq 13$, and $t \leq 1 - 9z < 1$. Thus the first case has only smooth surface successors and the second has none; $9^{1/3} < 9/4$ bounds the remaining radii. For completeness, their contributions to the two linear envelopes are

u	$C_3 - (2301/2000)y$	$C_3 - (9/10)y$
8	$\leq 1699/8000$	$\leq 11/40$
9	$\leq 733/6000$	$\leq 3/20$

and

$$\begin{aligned} \frac{1699}{8000} &< \frac{6894}{8000} = \frac{3447}{4000}, & \frac{733}{6000} &= \frac{1466}{12000} < \frac{10341}{12000} = \frac{3447}{4000}, \\ \frac{11}{40} &= \frac{16500}{60000} < \frac{100663}{60000}, & \frac{3}{20} &= \frac{9000}{60000} < \frac{100663}{60000}. \end{aligned}$$

These rows are included whenever the earlier complete-tail estimates are used below.

Step 3. In this step, we impose the fourfold-parent restriction and maximize the permitted successor surpluses. For each of $m = 3, 4, 5, 6$, the moment bound is $\bar{e}_1 \geq 2m - 4$. A threefold successor of multiplicity u and deficit y consequently satisfies

$$(12.3) \quad u \leq 8 - my, \quad 0 < y \leq \min\{B_u, (8 - u)/m\}.$$

In particular $u \leq 7$; the last two rows above cannot occur under this particular parent.

For $m = 4, 5, 6$, use $A = a_m$. Its fourth power exceeds m . If the threefold radius is at most A , every segment before a surface costs at most A times its deficit drop; (12.2) controls what remains. Otherwise the surplus is bounded by

$$(c_u - A) \min\{B_u, (8 - u)/m\} + d_u.$$

The direct-surface term followed by these candidates is

m	surplus bounds
4	$1/3; 103/432, 181/480, 3/10, 7/24, 7/48$
5	$3/10; 221/800, 1/5, 1/5, 1/10$
6	$13/50; 1/8, 2/15, 1/15.$

Their maxima are p_4, p_5, p_6 .

For $m = 3$, put $c = 13161/10000$. We have $c^2 > 17321/10000$ and $17321^2 = 300017041 > 3 \cdot 10^8$, so $c > 3^{1/4}$. The direct-surface term is less than $1/2$. The other candidates are

$$\left(\frac{13}{9} - c\right)\frac{5}{3} + \frac{11}{54}, \quad \left(\frac{127}{80} - c\right)\frac{4}{3} + \frac{33}{160}, \quad \frac{9}{5} - c, \quad \frac{2}{3}(2 - c), \quad \frac{1}{3}(2 - c).$$

The second is $34087/60000 > 1/2$; all the others are less than $1/2$, using $c > 13/10$. This proves the cubic row. Nonincreasing deficits justify every radius replacement. A direct curve or point has no positive surplus, so all dimension jumps have been included. $\square$

Lemma 12.4. *Let Γ be an effective rational boundary on X , log canonical near x , with a fourfold minimal center W . Put*

$$b = \text{def}_x(\Gamma), \quad \sigma = 6 - \text{mult}_x \Gamma - b.$$

Assume $b > 1$ and $0 \leq \sigma < 1/2$. Then the ordinary point blow-up of W is normal and has rational singularities near its exceptional fiber. Its full exceptional Cartier scheme is reduced and Cohen–Macaulay.

This concerns the ordinary exceptional scheme before any assumption about the full associated graded ring.

Proof. Write $\pi_1 : X_1 = \text{Bl}_x X \rightarrow X$, let E_1 be its exceptional divisor, and let Γ^{str} be the effective strict transform. Set $e_0 = b + \sigma - 1 > 0$. Then

$$K_{X_1} + \Gamma^{\text{str}} = \pi_1^*(K_X + \Gamma) + e_0 E_1.$$

Step 1. In this step, we prove log canonicity of the effective transformed pair and minimality of the strict transform above x . For $0 \leq \theta < 1 - \sigma$, the discrepancy of a valuation v for $\Gamma^{\text{str}} + \theta E_1$ is

$$A_{X,\Gamma}(v) - (e_0 + \theta)v(\mathfrak{m}_x).$$

When $v(\mathfrak{m}_x) > 0$, this is positive because $A_{X,\Gamma}(v) \geq b v(\mathfrak{m}_x)$ and $e_0 + \theta < b$. When the point-ideal value is zero, it is the original nonnegative discrepancy. A common log resolution therefore gives an lc pair on a neighborhood of the exceptional fiber.

The strict transform W_1 is an lc center, as follows from an unchanged lc place over the generic point of W . It is minimal at every point above x . Indeed, by Theorem 3.1, the minimal center there is contained in W_1 . A smaller center cannot map to x , by the strict positivity just proved. If its image has positive dimension, its lc place has point-ideal value zero and descends to an original lc center properly contained in W and containing x . Properness makes this image closed, contradicting minimality of W .

Step 2. In this step, we identify the ordinary blow-up with the transformed minimal center and prove Cohen–Macaulayness of its exceptional scheme. For $R = \mathcal{O}_{W,x}$ and $\mathfrak{n} = \mathfrak{m}_{W,x}$, the surjection of point Rees algebras embeds $\text{Bl}_{\mathfrak{n}} \text{Spec } R$ in the ambient blow-up. The Rees algebra of the domain R is a domain, so this is the closure of $W \setminus \{x\}$, namely W_1 . By Theorem 4.5, it is normal and has rational singularities near the exceptional fiber. In particular it is Cohen–Macaulay. The invertible ideal $\mathfrak{n}\mathcal{O}_{W_1}$ defines a Cartier scheme E_W , with

$$E_W = \text{Proj}(\text{gr}_{\mathfrak{n}} R), \quad E_1|_{W_1} = E_W.$$

It is Cohen–Macaulay as a Cartier divisor in a Cohen–Macaulay scheme.

Step 3. In this step, we use the effective discriminant to prove reducedness of every exceptional component. Apply Theorem 4.1 to the new effective pair $(X_1, \Gamma^{\text{str}})$ and its normal center W_1 . Write $E_W = \sum_A r_A A$. Each r_A is a positive integer. For every lc place dominating W_1 , adding θE_1 preserves the induced sub-lc pair. At the generic point of A , it adds θr_A times the pullback of A . The threshold definition and effectivity of the base discriminant $\mathcal{B}_{W_1}^{\text{str}}$ give

$$0 \leq \text{coeff}_A \mathcal{B}_{W_1}^{\text{str}} \leq 1 - \theta r_A.$$

Letting θ increase to $1 - \sigma > 1/2$ forces $r_A = 1$. Thus E_W is generically reduced. Its Cohen–Macaulay property excludes embedded associated primes; a nonzero nilradical would survive at an associated minimal prime. Hence the full scheme E_W is reduced. $\square$

Lemma 12.5. *Let $\Gamma \sim_{\mathbb{Q}} sL$, $s \geq 0$, be log canonical near x , with a fourfold minimal center W of multiplicity $m \geq 4$. Suppose*

$$s \leq \frac{65}{64} \text{mult}_x \Gamma + e, \quad e < \frac{1}{2560},$$

and suppose its entire boundary has no additive point continuation of coefficient less than $7 - 1/16384$. Put $b = \text{def}_x(\Gamma)$, $\sigma = 6 - \text{mult}_x \Gamma - b$, and $\bar{e} = \bar{e}_1(\mathfrak{m}_{W,x})$. Then $\text{edim}_x W = 6$, and exactly one of these rows applies:

m	b	σ	$\bar{e}$
4	$13/10 < b \leq 2$	$0 \leq \sigma < 27/100$	4 or 5
5	$5/4 < b \leq 8/5$	$0 \leq \sigma < 1/6$	6
6	$11/10 < b \leq 4/3$	$0 \leq \sigma < 2/15$	8.

In the subcase $m = 4, \bar{e} = 5$, one has $7/5 < b \leq 3/2$ and $\sigma < 1/32$. The ordinary point blow-up of W is normal and rational near its exceptional fiber, and its full exceptional Cartier scheme is reduced and Cohen–Macaulay.

Only the value $\bar{e} = 5$ remains exceptional in the multiplicity-four row.

Proof. Corollary 5.7 gives $b > 1$. Lemma 12.3 then gives $m \in \{4, 5, 6\}$ and $\text{edim}_x W = 6$. The point blow-up gives $\sigma \geq 0$. Set $\eta = e + 1/16384 < 1/2048$.

Step 1. In this step, we derive the three numerical rows from the failure of an affordable additive continuation. If $s + a_m b + p_m < 7 - 1/16384$, Proposition 3.7, with total error smaller than this strict margin, would give the prohibited continuation. Consequently

$$\frac{65}{64}(6 - b - \sigma) + e + a_m b + p_m \geq 7 - \frac{1}{16384}.$$

Solving for σ yields

$$(12.4) \quad \begin{array}{c|c} m & \text{upper bound for } \sigma \\ \hline 4 & (385b - 508)/975 + (64/65)\eta \\ 5 & (155b - 194)/325 + (64/65)\eta \\ 6 & (935b - 1034)/1625 + (64/65)\eta. \end{array}$$

At $b = 13/10, 5/4, 11/10$, respectively, the right sides are strictly less than

$$-\frac{1}{130} + \frac{1}{2080}, \quad -\frac{1}{1300} + \frac{1}{2080}, \quad -\frac{11}{3250} + \frac{1}{2080},$$

all negative. Substituting the upper bounds $b \leq 8/m$ in (12.4) gives

$$\frac{262}{975} + \frac{1}{2080} < \frac{27}{100}, \quad \frac{54}{325} + \frac{1}{2080} < \frac{1}{6}, \quad \frac{638}{4875} + \frac{1}{2080} < \frac{2}{15}.$$

Integrality, $\bar{e} \geq 2m - 4$, and $2\bar{e} \leq (4 - b)m$ now give the three displayed possibilities for $\bar{e}$.

Step 2. In this step, we sharpen the exceptional Hilbert-coefficient-five subcase and deduce the asserted blow-up properties. If $m = 4$ and $\bar{e} = 5$, the deficit bound improves to $b \leq 3/2$, and the parent inequality becomes $u \leq 6 - 4y$. Using the same threefold table with slope $17/12$, the surplus candidates are

$$\frac{1}{3}; \quad \frac{97}{432}, \frac{7}{24}, \frac{2}{15}.$$

Their maximum is $1/3$. The preceding budget argument gives

$$\sigma \leq \frac{77b - 110}{195} + \frac{64}{65}\eta.$$

At $b \leq 7/5$ this is negative, while at $b \leq 3/2$ it is less than $11/390 + 1/2080 < 1/32$. All rows satisfy the hypotheses of Lemma 12.4, which proves the assertions about the ordinary blow-up and its exceptional scheme. $\square$

Lemma 12.6. *Let $(R, \mathfrak{n})$ be a normal Cohen–Macaulay local complex domain, essentially of finite type, with residue field $\mathbb{C}$, dimension four, and embedding dimension six. Suppose its multiplicity m is four, five, or six, its first normal Hilbert coefficient is $2m - 4$, and $\text{Proj}(\text{gr}_{\mathfrak{n}} R)$ is reduced. Then $\text{gr}_{\mathfrak{n}} R$ is reduced and Cohen–Macaulay, with*

$$\text{Hilb}(\text{gr}_{\mathfrak{n}} R; z) = \frac{1 + 2z + (m - 3)z^2}{(1 - z)^4}.$$

If $R = S/P$, where S is a regular local ring of dimension six with maximal ideal $\mathfrak{m}$, then $P \subseteq \mathfrak{m}^2$. For $m = 6$, one has $P \subseteq \mathfrak{m}^3$.

The conclusion concerns the full tangent ring, including its vertex.

Proof. Choose a general parameter reduction J of $\mathfrak{n}$. Theorem 4.6 and the assumed equality force the Artinian quotient to have Hilbert function $(1, 2, m - 3)$. All terms after degree two vanish, and

$$e_1(\mathfrak{n}) = \bar{e}_1(\mathfrak{n}) = 2m - 4 = \sum_{i=1}^3 \text{length}(\mathfrak{n}^i / (J \cap \mathfrak{n}^i)).$$

Step 1. In this step, we prove Cohen–Macaulayness of the full tangent ring and compute its Hilbert series. Theorem 4.14 makes the Rees algebra $\mathcal{R} = R[\mathfrak{n}t]$ Cohen–Macaulay of dimension five. We spell out the passage to its associated graded ring. At the homogeneous maximal ideal, the exact sequence

$$0 \longrightarrow \mathcal{R}_+ \longrightarrow \mathcal{R} \longrightarrow R \longrightarrow 0$$

gives depth five for $\mathcal{R}_+$. Multiplication by t identifies $\mathfrak{n}\mathcal{R}$, up to a grading shift, with $\mathcal{R}_+$. Thus the sequence

$$0 \longrightarrow \mathfrak{n}\mathcal{R} \longrightarrow \mathcal{R} \longrightarrow \text{gr}_{\mathfrak{n}} R \longrightarrow 0$$

gives depth at least four for the four-dimensional last term. It is therefore Cohen–Macaulay. The initial forms of the four parameters are a homogeneous system of parameters and hence a regular sequence. Theorem 4.15 identifies the quotient by them with the associated graded Artinian reduction, giving the asserted Hilbert series.

Step 2. In this step, we prove reducedness of the full tangent ring and exclude the stated low-order ambient equations. Every homogeneous minimal prime of this Cohen–Macaulay ring has dimension four. Its generic localization is reduced because its projectivization is reduced. There are no embedded associated primes, so the nilradical is zero. Finally embedding dimension six means that P has no linear initial equation. In multiplicity six, the degree-two Hilbert function is

$$\binom{5}{3} + 2\binom{4}{3} + 3\binom{3}{3} = 10 + 8 + 3 = 21,$$

the full dimension of the quadratic forms in six variables. There is no quadratic initial equation either. $\square$

Lemma 12.7. *Let $(R, \mathfrak{n})$ be a normal Cohen–Macaulay local complex domain, essentially of finite type, with residue field $\mathbb{C}$, dimension four, multiplicity four, and embedding dimension at most six. Suppose its ordinary blow-up $f : T \rightarrow \text{Spec } R$ is normal, and its full exceptional Cartier scheme E_R is reduced and Cohen–Macaulay. Then*

$$\bar{e}_1(\mathfrak{n}) \in \{4, 6\}.$$

In particular, the normal coefficient cannot be five. No Cohen–Macaulay or reduced hypothesis on the full tangent ring is needed here.

Proof. We construct connected reduced curve sections of the full exceptional scheme, identify their spans, and compute their sectional genus.

Step 1. In this step, we prove connectedness of the exceptional scheme and of its graph of divisorial component intersections. The blow-up is integral and proper birational over $\text{Spec } R$. The coherent R -algebra $H^0(T, \mathcal{O}_T)$ is finite over R and embeds in its fraction field. Normality of R therefore gives $f_*\mathcal{O}_T = \mathcal{O}_{\text{Spec } R}$. Theorem 12.1 makes the exceptional fiber connected. That fiber is the full Cartier scheme E_R , since $\mathfrak{n}\mathcal{O}_T = \mathcal{O}_T(-E_R)$. As it is reduced and projective over $\mathbb{C}$,

$$H^0(E_R, \mathcal{O}_{E_R}) = \mathbb{C}.$$

Indeed a regular function is constant on each integral projective component, and these constants agree wherever components meet.

The scheme E_R is pure of dimension three. Consider its finite graph of irreducible components, joining two components when their intersection has dimension two. This graph is connected. Otherwise partition its vertices into two nonempty groups with no edge between them, and let Z be the intersection of the corresponding unions. Then Z has codimension at least two in E_R . On $E_R \setminus Z$, the function equal to zero on one union and one on the other is a nonconstant idempotent. The Cohen–Macaulay scheme E_R satisfies S_2 , so this function extends uniquely across Z . To see the extension directly, the local cohomology sequence for restriction has kernel and cokernel given by $H_Z^0(\mathcal{O}_{E_R})$ and $H_Z^1(\mathcal{O}_{E_R})$; both vanish because the local depths along Z are at least two. The extension contradicts $H^0(E_R, \mathcal{O}_{E_R}) = \mathbb{C}$.

Step 2. In this step, we construct actual connected reduced Cohen–Macaulay curve sections of degree four. Put $H = \mathcal{O}_{E_R}(1)$. The tautological embedding realizes $E_R = \text{Proj}(\text{gr}_{\mathfrak{n}} R)$ as a closed subscheme of $\mathbb{P}^{\dim R - 1}$, of degree four. Choose two successive sufficiently general hyperplanes for this embedding, and write S_R and C_R for their actual scheme-theoretic sections. Each equation avoids the associated primes of the preceding Cohen–Macaulay scheme and is a nonzerodivisor. Thus S_R and C_R are Cohen–Macaulay of pure dimensions two and one.

Theorem 12.2 applies to every integral component at the first cut, and then to every integral component at the second cut. These are fixed very ample embeddings over $\mathbb{C}$, and the component dimensions before the respective cuts are three and two. The resulting sections of each component are irreducible. Choose the hyperplanes also transverse on the dense smooth loci and so that the intersections of distinct components have the expected dimensions. Such choices form a nonempty open set: for a smooth locus of dimension d in $\mathbb{P}^q$, the incidence of a point and a hyperplane containing its tangent space has dimension $d + (q - d - 1) = q - 1$, smaller than the dual projective space. A general hyperplane also contains no component of any of the finitely many singular or intersection loci. The generic points of the sections then lie in the smooth locus of a single component, where the cut is reduced. Hence the full sections are generically reduced. Their Cohen–Macaulay property excludes embedded primes, so the full schemes S_R and C_R are reduced.

Every edge in the original component graph contains a projective surface of positive H -degree. Its intersection with the first hyperplane is a nonempty curve, and with both hyperplanes is a nonempty finite scheme. Thus the component sections remain joined along every original edge. Each component section is irreducible, so S_R and C_R are connected. We have obtained an actual connected reduced Cohen–Macaulay curve C_R of degree four, without replacing either section by its reduction or normalization.

Step 3. In this step, we determine the span of the curve section and express the normal Hilbert coefficient through its arithmetic genus. Write $\mathbb{P}^r$ for the span of E_R . Its dimension and degree imply $4 \leq r \leq 5$: a reduced pure three-dimensional closed subscheme spanning $\mathbb{P}^3$ would be all of $\mathbb{P}^3$, of degree one.

A connected reduced projective complex scheme T_0 has $H^0(T_0, \mathcal{O}_{T_0}) = \mathbb{C}$. If it spans $\mathbb{P}^q$, then $H^1(\mathbb{P}^q, \mathcal{I}_{T_0}) = 0$ and $H^0(\mathbb{P}^q, \mathcal{I}_{T_0}(1)) = 0$. For a regular hyperplane section T_1 , the restriction

sequence

$$0 \longrightarrow \mathcal{I}_{T_0} \longrightarrow \mathcal{I}_{T_0}(1) \longrightarrow \mathcal{I}_{T_1/\mathbb{P}^{q-1}}(1) \longrightarrow 0$$

shows that T_1 spans that hyperplane. Applying this twice shows that C_R spans $\mathbb{P}^{r-2}$, either a plane or a three-dimensional projective space.

The normalized Rees algebra is finite over the ordinary Rees algebra, and its projectivization is T , by normality of the ordinary blow-up. Its tautological sheaf is $\mathcal{O}_T(-E_R)$. The eventual graded section identification and relative Serre vanishing give, for all sufficiently large k ,

$$H^0(T, \mathcal{O}_T(-kE_R)) = \overline{\mathbf{n}}^k, \quad H^i(T, \mathcal{O}_T(-kE_R)) = 0 \quad (i > 0).$$

Taking sections of the Cartier restriction sequence therefore identifies the Hilbert polynomial of E_R with the first difference of the normal Hilbert polynomial. Two more differences, for the actual regular hyperplane sections, give

$$P_{C_R}(k) = 4(k+1) - \bar{e}_1(\mathbf{n}).$$

Since this also equals $4k+1 - p_a(C_R)$,

$$(12.5) \quad \bar{e}_1(\mathbf{n}) = 3 + p_a(C_R).$$

Step 4. In this step, we compute the possible sectional genera and exclude normal Hilbert coefficient five. If C_R spans a plane, it is a reduced quartic plane curve. The hypersurface sequence gives $P_{C_R}(k) = 4k - 2$, so $p_a(C_R) = 3$ and $\bar{e}_1 = 6$.

If it spans $\mathbb{P}^3$, a general hyperplane meets it in four distinct points Z_R : it avoids the finite singular and component-intersection locus and is transverse at its smooth intersection points. The same restriction argument shows that these points span the hyperplane. Four distinct noncollinear plane points have Hilbert function $1, 3, 4, 4, \dots$. Indeed, for each specified point a product of two linear forms vanishes on the other three and not on that point. Such a product exists because some pair of the other three has a joining line avoiding the specified point; otherwise all four would be collinear. A linear form avoiding the four points propagates degree-two surjectivity to higher degrees.

Let A_C be the homogeneous coordinate ring of C_R . The hyperplane equation h is a nonzerodivisor, and A_C/hA_C maps onto the homogeneous coordinate ring of Z_R . Keeping any saturation torsion in this quotient gives

$$\dim(A_C)_N = \sum_{i=0}^N \dim(A_C/hA_C)_i \geq 1 + 3 + 4(N-1) = 4N \quad (N \geq 1).$$

Its eventual Hilbert polynomial is $4N + 1 - p_a(C_R)$, so $p_a(C_R) \leq 1$. Finally Theorem 4.6, with $h_1 \leq 2$ and total length four, gives

$$\bar{e}_1 \geq \sum_i i h_i \geq 2(4-1) - h_1 \geq 4.$$

Equation (12.5) now forces $\bar{e}_1 = 4$. □

Lemma 12.8. *Assume $L^6 = 1$ and*

$$\text{vol}_Y(65\pi^*L - 69E) > 3250000000.$$

Then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$.

The assertion uses an arbitrary sufficiently late pure threshold of this complete source. Its center need not coincide with any earlier center.

Proof. We first control the full fixed point multiplicity, then treat every possible center of a pure threshold. A remaining cubic fourfold supplies the volume needed for Theorem 10.5.

Step 1. In this step, we bound the entire fixed point multiplicity of the order-69 source. By Proposition 7.5, the spaces $V_{69}(k)$ have normalized dimensions greater than 3250000000 for every

sufficiently large k . Their full fixed divisors $F_{69}(k)$ converge after division by k to a divisor N_{69} on a fixed finite support. Put $\eta = \text{mult}_x N_{69}$.

An actual section of point order at least $92k$ yields, after division of its entire divisor by k , an effective divisor of class $65L$ and point-to-class ratio at least $92/65 > 140/99$. Proposition 5.8 then gives the asserted conclusion, with a larger margin. We may therefore assume there is no such section in any sufficiently large degree.

Proposition 7.7 applies to the complete family with $A = 65L$ and $r = 69$. Applied to any actual nonzero section, its prime-order inequalities first give $\eta \leq 69$. Its bound for every point-order layer and telescoping through order $92k$ give

$$\frac{720 \dim V_{69}(k)}{k^6} \leq \left(1 - \frac{\eta}{69}\right)^5 (92^6 - 69^6) + O(k^{-1}).$$

If $\eta \geq 48$, the leading expression is less than 13800000000. Indeed

$$(7/23)^5 < 1/380, \quad 380 \cdot 7^5 = 6386660 < 6436343 = 23^5,$$

$$92^6 - 69^6 = 498436838263 < 5000000000000, \quad \frac{5000000000000}{380} < 13800000000.$$

This contradicts the complete-source mass. Thus $\eta < 48$, and coefficientwise convergence gives

$$(12.6) \quad \text{mult}_x F_{69}(k) < 48k$$

for every sufficiently large k . All fixed companions and their repeated tangent factors are included in this inequality.

Step 2. In this step, we treat every order-69 threshold center except a cubic fourfold. Choose k beyond these cutoffs and the cutoff in Proposition 10.6. Let τ be the threshold of the full base ideal of $V_{69}(k)$. On a common resolution, take a sufficiently large sufficiently general tuple of actual members and form its pure threshold average $\Gamma \sim_{\mathbb{Q}} sL$. This construction has the same ideal-threshold lc places: the individual moving coefficients are less than one and the moving divisors meet the resolved boundary transversely. Put $M = \text{mult}_x \Gamma$ and, for a positive-dimensional minimal center, $b = \text{def}_x(\Gamma)$. Then

$$(12.7) \quad s = 65k\tau \leq \frac{65}{69}M, \quad M + b \leq 6.$$

Write $q = 65/69$. The center has dimension at most five. A point already has $s \leq 6q < 7 - 1/4096$. A center with $b \leq 1$, or a smooth fourfold, is treated by Corollary 5.7; its excess at slope $65/64$ is nonpositive. Every threefold is treated by Theorem 9.6 with the same zero excess. The threefold assertion includes the multiplicities eight and nine of Lemma 5.1.

A curve or surface has $b \leq 2$ and complete tail at most $b + 1/2$, by (12.2) with $A = 1$. Its full coefficient is therefore at most $4q + 5/2 < 63/10$, before approximation errors. A double fourfold has embedding dimension at most five and $b \leq 3$ by Theorem 4.6 and Proposition 4.10. Proposition 5.5 gives

$$s + C \leq 3q + 3 + \frac{3887}{3360} = 7 - \frac{1319}{77280} < \frac{699}{100}.$$

For fourfold multiplicities four, five, and six, Lemma 12.3 gives respectively

$$\begin{aligned} s + C &\leq 4q + \frac{17}{6} + \frac{181}{480} = 7 - \frac{79}{3680}, \\ s + C &\leq \frac{22}{5}q + \frac{27}{10} = \frac{286}{69} + \frac{27}{10} < \frac{69}{10}, \\ s + C &\leq \frac{14}{3}q + \frac{359}{150} = \frac{910}{207} + \frac{359}{150} < \frac{69}{10}. \end{aligned}$$

Higher fourfold multiplicities have $b \leq 1$.

A smooth fivefold is treated by Theorem 8.7, since its excess at slope $65/69$ is zero. A double fivefold is treated by Proposition 10.6, using the volume hypothesis for this very complete family.

More specifically, when $b \leq 2$, Proposition 5.9 gives an additive continuation of the entire boundary.

For a singular fivefold Z of multiplicity $m \geq 3$, its coefficient in Γ is one. It is an actual fixed prime: sufficiently small general moving coefficients cannot produce a coefficient-one prime. If $\nu = \text{ord}_Z V_{69}(k)$, then

$$\tau\nu = 1, \quad M \geq \frac{69k}{\nu}, \quad m\nu \leq \text{mult}_x F_{69}(k) < 48k.$$

For $m \geq 4$, this gives $M > 69m/48 \geq 23/4 > 5$ and hence $b < 1$. For $m = 3$, it gives $b < 6 - 69/16 = 27/16 < 9/5$. The triple-fivefold row of Proposition 5.5 gives

$$s + C < 6q + \left(\frac{5}{4} - q\right) \frac{9}{5} + \frac{57}{80} = 7 - \frac{149}{1840}.$$

This proves every fivefold case.

In all scalar cases just listed, fix a total cutting allowance $h = 1/65536$. Split it into six positive allowances of sum less than h , and after each radius r_i is known choose the local error below its allowance divided by $1 + r_i$. Proposition 3.7 gives actual point coefficients less than $s + C + h < 7 - 1/4096$. The cited endpoint theorems already include their own errors. It remains to consider a cubic fourfold for which there is no such additive continuation.

Step 3. In this step, we prove the reduced Cohen–Macaulay tangent-ring properties of the remaining cubic fourfold. Put $c = 13161/10000$, $p = 34087/60000$, and $\sigma = 6 - M - b$. Lemma 12.3 gives $b \leq 8/3$ and $C \leq cb + p$. If $b \leq 33/16$, then

$$s + C \leq q \frac{63}{16} + c \frac{33}{16} + p < q \frac{63}{16} + c \frac{33}{16} + p + \frac{1}{128} = 7 - \frac{829}{2208000}.$$

The displayed gap exceeds $1/4096 + 1/65536$, because $829 \cdot 65536 = 54329344 > 37536000 = 17 \cdot 2208000$. It would give the prohibited continuation. Thus $b > 33/16$. If $\sigma \geq 1/4$, then

$$s + C \leq q \frac{37}{12} + \frac{244663}{60000} < \frac{407}{140} + \frac{244663}{60000} + \frac{1}{128} < 7 - \frac{1}{256},$$

using $q < 33/35$. This again leaves room for the same errors, so $\sigma < 1/4$.

The Hilbert inequalities now give $e_1 = \bar{e}_1 = 2$. Embedding dimension at most five would give the Artinian Hilbert function $(1, 1, 1)$, whose first moment is three. Hence the embedding dimension is six and the Artinian Hilbert function is $(1, 2)$. The minimal-multiplicity clause of Theorem 4.14 gives a Cohen–Macaulay full tangent ring with Hilbert series $(1 + 2z)/(1 - z)^4$. Lemma 12.4 gives its reduced projectivization. Cohen–Macaulayness excludes embedded primes, and each minimal prime appears in that projectivization; thus the full tangent ring is reduced.

Step 4. In this step, we use the cubic center to obtain the order-75 volume and the required point boundary. Write $S = \mathcal{O}_{X,x}$, let $\mathfrak{m}$ be its maximal ideal, let P_W be the prime of the cubic center, and put $A = S_{P_W}$. This is a regular local surface. The prime satisfies $P_W \subseteq \mathfrak{m}^2$, and its full tangent quotient is reduced and Cohen–Macaulay. If I_{69} is the full base ideal of $V_{69}(k)$, its ordinary threshold at A equals τ . Indeed an lc place dominating the minimal center computes the ideal threshold, and log canonicity near x supplies the reverse inequality after localization. Consequently

$$\text{lct}_A(I_{69}A) = \tau \leq \frac{6 - b}{69k} < \frac{4}{69k}.$$

For $69k \leq \ell < 75k$, let $T_\ell = H^0(X, 65kL \otimes \mathfrak{m}_x^\ell)$. Its ideal of germs J_ℓ in A is contained in $I_{69}A$. For a nonzero such space, we choose a closed-point-centered divisorial valuation w with

$$\frac{A_A(w)}{w(J_\ell)} < \frac{4}{69k}.$$

If a curve divisor D computes the threshold, take a log resolution dominating the blow-up of the closed point, on which its strict transform meets an exceptional component F in a simple

normal crossing. The primitive monomial valuation of weights $(N, 1)$ on local equations of D, F is centered at the closed point of $\text{Spec } A$ and has

$$\frac{A_A(w_N)}{w_N(J_\ell)} = \frac{NA_A(D) + A_A(F)}{N \text{ord}_D(J_\ell) + \text{ord}_F(J_\ell)} \rightarrow \frac{A_A(D)}{\text{ord}_D(J_\ell)}.$$

The equality follows by pulling back the canonical divisor and the resolved principal ideal in these two coordinates. The strict threshold bound permits a finite choice of N . If the computing divisor is already centered at the closed point, take it for w .

Let $\mathfrak{p} = P_W A$, and choose d_ℓ so that $d_\ell w(\mathfrak{p}) \geq w(J_\ell)$. Set

$$Q_\ell = (J_\ell + \mathfrak{p}^{d_\ell}) \cap S.$$

This is P_W -primary and contains every actual germ of T_ℓ . The chosen valuation has $w(Q_\ell A) = w(J_\ell)$, so the generic threshold of Q_ℓ remains strictly less than $4/(69k)$.

Proposition 6.18 applies since $75k \cdot 4/(69k) = 100/23 < 11/2$. Taking initial forms gives

$$\dim(T_\ell/T_{\ell+1}) \leq \frac{(\ell - 69k/2)^5}{120} + C_4(\ell + 1)^4.$$

Zero spaces satisfy the same bound. Summing the actual shells has a uniform error $O(k^5)$, and its normalized leading loss is

$$(81/2)^6 - (69/2)^6 = \frac{21813921675}{8} < 2727000000.$$

Since the initial normalized mass exceeds 3250000000, choosing the shell-error cutoff in advance gives

$$\frac{720 \dim V_{75}(k)}{k^6} > 500000000.$$

Also choose k beyond the fixed convergence cutoff making the difference between this normalized dimension and $\text{vol}_Y(65\pi^*L - 75E)$ less than 5000000. Proposition 7.5 then gives

$$\text{vol}_Y(65\pi^*L - 75E) > 495000000.$$

The complete order- $(371/5)$ spaces contain the order-75 spaces. Their volume is at least this number, hence exceeds 336000000. Theorem 10.5 supplies the point boundary. This final step covers the remaining cubic case, while the previous cases cover every other possible center. All cutoffs concern fixed complete families or absolute capacity errors and depend only on X, L, x . $\square$

Proposition 12.9. *Assume $L^6 = 1$. There is an integer J , depending only on X, L, x , with the following property. Suppose a pure threshold average from U_j , at one $j \geq J$, has a fourfold minimal center W of embedding dimension six and deficit greater than $11/10$. If its full tangent ring $\text{gr}_{\mathfrak{m}_{W,x}} \mathcal{O}_{W,x}$ is reduced and Cohen–Macaulay, then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$.*

One sufficiently late original occurrence determines the stronger complete source; no recurrence of its center is required.

Proof. Let Γ_0 be the specified pure average, b_0 its deficit, and τ_0 the threshold of the full base ideal $\mathfrak{q}_j$. Write $S = \mathcal{O}_{X,x}$, $\mathfrak{m} = \mathfrak{m}_x S$, $P_0 = \mathcal{I}_{W,x}$, and $A_0 = S_{P_0}$.

Step 1. In this step, we bound the generic thresholds of full primary ideals containing every actual layer. Embedding dimension six gives $P_0 \subseteq \mathfrak{m}^2$. The common-resolution argument for a pure threshold identifies $\text{lct}_{A_0}(\mathfrak{q}_j A_0) = \tau_0$. The point blow-up and the actual order of every source member give

$$64j\tau_0 \leq 6 - b_0 < \frac{49}{10}, \quad \text{lct}_{A_0}(\mathfrak{q}_j A_0) < \frac{49}{640j} < \frac{7}{90j}.$$

For each $64j \leq \ell < 69j$, set $U_{j,\ell} = H^0(X, 65jL \otimes \mathfrak{m}_x^\ell)$. Its generic ideal is contained in $\mathfrak{q}_j A_0$. As in the preceding proof, adjoining a sufficiently high generic maximal-ideal power and contracting gives a P_0 -primary ideal containing all actual germs and of generic threshold less than $7/(90j)$. The strict bound is preserved by choosing a closed-point-centered valuation first, including when a curve valuation computes the original threshold.

Step 2. In this step, we sum the two-unit capacity bounds to obtain a uniform lower dimension bound at order 69. Since $69 \cdot 7/90 = 161/30 < 11/2$, Proposition 6.18 gives uniformly

$$\dim(U_{j,\ell}/U_{j,\ell+1}) \leq \frac{(\ell - 180j/7)^5}{120} + C_4(\ell + 1)^4.$$

All full base ideals and fixed components remain in these actual spaces. Asymptotic Riemann–Roch and Serre vanishing, together with the exact length $\binom{64j+5}{6}$ of the regular point jet target, give

$$\frac{720 \dim U_j}{j^6} \geq 65^6 - 64^6 - O_{X,L,x}(j^{-1}) = 6699413889 - O_{X,L,x}(j^{-1}).$$

The normalized loss through the $5j$ shells is at most

$$(303/7)^6 - (268/7)^6 + O_{X,L,x}(j^{-1}) = \frac{403330656423905}{117649} + O_{X,L,x}(j^{-1}).$$

The leading fraction is less than 3430000000, since $117649 \cdot 3430000000 = 403536070000000$. Thus, for a uniformly chosen sufficiently large j ,

$$\frac{720 \dim V_{69}(j)}{j^6} > 3260000000.$$

Step 3. In this step, we pass from the single source occurrence to a fixed volume bound and a point boundary. Choose J also beyond the convergence cutoff of Proposition 7.5 for the fixed divisor $65\pi^*L - 69E$, with tolerance 5000000. The single occurrence then implies

$$\text{vol}_Y(65\pi^*L - 69E) > 3255000000.$$

Lemma 12.8 gives the asserted point boundary. Uniformity of the primary-ideal capacity error and fixedness of the complete-volume sequence make J independent of the original center. $\square$

Theorem 12.10. *Assume $L^6 = 1$. There is an integer J , depending only on X, L, x , such that if at one $j \geq J$ a pure threshold average from U_j has a fourfold minimal center of multiplicity at least four, then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/16384$.*

The point boundary in this conclusion need not be an additive continuation of the original threshold average.

Proof. Choose J at least the cutoff in Proposition 12.9. Let Γ_0 be the specified pure threshold average at a level $j \geq J$, let W be its fourfold minimal center, and let τ_0 be its threshold.

Step 1. In this step, we reduce failure of the additive alternative to the three numerical rows and their ordinary exceptional schemes. Every actual source member has point order at least $64j$. Thus, writing $\Gamma_0 \sim_{\mathbb{Q}} s_0L$ and $M_0 = \text{mult}_x \Gamma_0$, we have

$$s_0 = 65j\tau_0 \leq \frac{65}{64}M_0.$$

If the entire Γ_0 has an additive point continuation of coefficient less than $7 - 1/16384$, this is the required point boundary. Suppose there is no such continuation.

Lemma 12.5, with $e = 0$, then gives $m = \text{mult}_x W \in \{4, 5, 6\}$, $\text{edim}_x W = 6$, and $b_0 = \text{def}_x(\Gamma_0) > 11/10$. In particular, every larger original multiplicity and every smaller embedding dimension have already been treated by the additive alternative. Put $R = \mathcal{O}_{W,x}$ and $\mathfrak{n} = \mathfrak{m}_{W,x}$. This is a normal Cohen–Macaulay local complex domain with residue field $\mathbb{C}$. Its ordinary blow-up is normal near its exceptional fiber, and normal elsewhere because it is an isomorphism

away from the closed point of $\text{Spec } R$. Its full exceptional Cartier scheme is reduced and Cohen–Macaulay.

Step 2. In this step, we determine the normal Hilbert coefficient, prove the tangent hypotheses, and obtain the point boundary. If $m = 4$, Lemma 12.7 gives $\bar{e}_1(\mathfrak{n}) \in \{4, 6\}$. The numerical reduction restricts it to $\{4, 5\}$, so it equals four. For $m = 5, 6$, the same numerical reduction gives respectively six and eight. Thus in all three cases

$$\bar{e}_1(\mathfrak{n}) = 2m - 4.$$

Lemma 12.6 proves that the full tangent ring is reduced and Cohen–Macaulay, with Hilbert series $(1 + 2z + (m - 3)z^2)/(1 - z)^4$. All hypotheses of Proposition 12.9 now hold for the original occurrence at j . It supplies a point boundary with

$$0 < v < 7 - \frac{1}{4096} < 7 - \frac{1}{16384}.$$

The additive alternative treats the discarded local rows, the quartic-section lemma treats the exceptional multiplicity-four coefficient, and the complete-source proposition treats the three remaining tangent rings. This exhausts the original hypothesis. $\square$

13. FULL FIXED DIVISORS AND DOUBLE FIVEFOLDS

We retain the complete-source notation of Section 7. We first bound the entire fixed divisor of the original source at a hypothetical base point. We then construct a point boundary from an original double fivefold. This second argument uses the complete sources at orders 69 and 75, with their own thresholds and all their fixed components.

Proposition 13.1. *Suppose $K_X + 7L$ is not generated at x , and let F_j be the full fixed divisor of U_j . Then $L^6 = 1$, and for all sufficiently large j ,*

$$F_j \cdot L^5 < 22j, \quad \text{mult}_x F_j < \frac{75}{2}j.$$

There is a single cutoff depending only on X, L, x such that every pure threshold average at a later level with a fivefold minimal center W satisfies

$$\text{mult}_x W \in \{1, 2\}, \quad L^5 \cdot W = 1.$$

The two bounds apply to the sum of all fixed components, including components away from x in the intersection-degree bound.

Proof. We bound the degree of the limiting fixed divisor by volume concavity, bound its point multiplicity by the dimensions of the complete jet layers, and then use the coefficient of an actual divisorial center.

Step 1. In this step, we bound the intersection degree of the entire original fixed divisor by volume concavity. The nongeneration hypothesis and Theorem 11.7 give $L^6 = 1$, since L^6 is a positive integer. Riemann–Roch and the length of the point jets give

$$(13.1) \quad \dim U_j \geq \frac{R_0}{720}j^6 + O(j^5), \quad R_0 = 65^6 - 64^6 = 6699413889.$$

Indeed, the dimension of the kernel of the restriction map to $\mathcal{O}_{X,x}/\mathfrak{m}_x^{64j}$ is at least the dimension of its domain minus the length of its target. No surjectivity of that map is required.

Proposition 7.5 gives an effective real divisor N_{64} , with fixed finite support, such that $F_j \geq jN_{64}$ and $F_j/j \rightarrow N_{64}$ coefficientwise. Choose effective rational divisors $N' \leq N_{64}$ on this support approaching N_{64} . For integers j clearing their denominators, division by jN' embeds the actual space U_j into

$$H^0(X, \mathcal{O}_X(j(65L - N'))).$$

It follows from (13.1) and volume continuity that

$$\text{vol}_X(65L - N_{64}) \geq R_0.$$

By Theorem 7.2, as stated in [BFJ09, Theorem A and Theorems 3.1–3.2], the function $f(t) = \text{vol}_X(65L - tN_{64})^{1/6}$ lies below its tangent at zero. The segment in question is big, and differentiation at the ample class $65L$ gives $f'(0) = -N_{64} \cdot L^5$. Consequently

$$R_0^{1/6} \leq f(1) \leq 65 - N_{64} \cdot L^5.$$

Since $R_0 > 43^6 = 6321363049$, we have $N_{64} \cdot L^5 < 22$. Coefficientwise convergence proves the asserted degree bound for every sufficiently large j .

Step 2. In this step, we bound the original fixed point multiplicity by counting every complete jet layer. Nongeneration also excludes every nonzero section in any U_j of order at least $92j$. Its entire divisor would have order to coefficient ratio at least

$$\frac{92}{65} > \frac{140}{99}, \quad 92 \cdot 99 = 9108 > 9100 = 140 \cdot 65,$$

and Propositions 5.8 and 3.4 would give generation at x .

Write $N_{64} = \sum_T n_T T$ and $\eta = \text{mult}_x N_{64}$. The order assertion in Proposition 7.7 applies to every actual section of point order $\ell \geq 64j$. It gives $\text{ord}_T(h) \geq \ell n_T / 64$ for each fixed prime T . Taking point orders in the product of their local equations shows $\ell \geq \ell \eta / 64$; hence $0 \leq \eta \leq 64$. Set $\alpha = 1 - \eta / 64$. The same proposition bounds the entire degree- ℓ initial image by

$$\binom{\lfloor \alpha \ell \rfloor + 5}{5}.$$

This includes every repeated or coincident tangent factor. Summing the filtration from $64j$ to $92j - 1$ gives

$$(13.2) \quad \dim U_j \leq \frac{\alpha^5(92^6 - 64^6)}{720} j^6 + O(j^5).$$

For $\alpha > 0$, use $\binom{\lfloor \alpha \ell \rfloor + 5}{5} = \alpha^5 \ell^5 / 120 + O(\ell^4)$. For $\alpha = 0$, the sum is $O(j)$, which satisfies the same bound. All constants here concern the fixed divisor N_{64} .

If $\eta \geq 75/2$, then

$$\alpha \leq \frac{53}{128} < \frac{83}{200}, \quad 81 \cdot 83^5 = 319062292083 < 320000000000 = 200^5.$$

Thus $\alpha^5 < 1/81$, whereas

$$92^6 - 64^6 = 537635524608 < 540000000000, \quad \frac{540000000000}{81} = \frac{200000000000}{3} < R_0.$$

This contradicts (13.1) and (13.2). Therefore $\eta < 75/2$, and convergence gives $\text{mult}_x F_j < (75/2)j$ for all sufficiently large j .

Step 3. In this step, we restrict the multiplicity and degree of every sufficiently late divisorial threshold center. Choose j beyond both fixed-divisor cutoffs, and let Γ be a pure threshold average from U_j with fivefold minimal center W . Let τ be its ideal threshold, write $\Gamma \sim_{\mathbb{Q}} sL$, and put $M = \text{mult}_x \Gamma$, $b = \text{def}_x(\Gamma)$, and

$$\nu = \text{ord}_W \mathfrak{q}_j, \quad p = \text{ord}_x \mathfrak{q}_j, \quad m = \text{mult}_x W, \quad g = L^5 \cdot W.$$

At the generic point of W the boundary coefficient is one. The sufficiently general members attain the orders of the full base ideal, so

$$\tau\nu = 1, \quad s = \frac{65j}{\nu}, \quad M = \frac{p}{\nu}, \quad 64j \leq p < 92j, \quad p \geq m\nu.$$

In particular $\nu > 0$: W is an actual fixed prime. Proposition 7.6 gives $s \leq \frac{65}{64}M$ and $b \leq 6 - M$. Under the nongeneration assumption, Corollary 5.7 and Proposition 3.4 force $b > 1$. Thus $M < 5$ and $\nu/j > 64/5$. The full fixed divisor of this same source satisfies

$$m\nu \leq \text{mult}_x F_j < \frac{75}{2}j, \quad g\nu \leq F_j \cdot L^5 < 22j.$$

If $m \geq 3$, the first inequality contradicts $m\nu > 192j/5 > 75j/2$. If $g \geq 2$, the second contradicts $g\nu > 128j/5 > 22j$. Both are positive integers, so $m \in \{1, 2\}$ and $g = 1$. Every cutoff used before choosing the average depends only on X, L, x . $\square$

We next record the chain estimates needed for the new 69-jet source. They apply to entire additive chains and do not prescribe the next center.

Lemma 13.2. *Let Γ be an effective rational boundary on X , log canonical near x , with a fourfold minimal center W . Put $m = \text{mult}_x W$ and $b = \text{def}_x(\Gamma)$. Every additive cutting chain from Γ to a point has the following bounds:*

(i) *If $m = 3$, then $b \leq 8/3$ and*

$$C \leq \frac{79}{60}b + \frac{817}{1440}.$$

(ii) *If $m \geq 4$ and $b > 1$, then $m \in \{4, 5, 6\}$, $b \leq 8/m$, and*

m	upper bound for C
4	$\frac{17}{12}b + \frac{181}{480}$
5	$\frac{3}{2}b + \frac{3}{10}$
6	$\frac{8}{5}b + \frac{13}{50}$

The additive constants account for all later increases in radius, including direct jumps to surfaces.

Proof. We first bound the threefold and surface tails, and then impose the fourfold-parent inequality. Minimal centers are normal and Cohen–Macaulay by Theorem 4.5 [dFH09, Corollary 7.12]. Their excellent normal local rings are analytically unramified and have infinite residue field. Theorem 4.6 [Han26, Lemmas 4.2 and 4.4] therefore applies.

Step 1. In this step, we bound all complete surface and threefold tails, including both large-multiplicity rows. For a surface of multiplicity u , write

$$q_1 = 1, \quad q_2 = \frac{17}{12}, \quad q_3 = \frac{7}{4}, \quad q_4 = 2, \quad q_5 = \frac{9}{4}.$$

These numbers dominate $\sqrt{u}$; the nonintegral comparisons are $289 > 288$, $49 > 48$, and $81 > 80$. Corollary 4.13 bounds its deficit w by $2/u$, and its later curve is smooth. Its Artinian function has no nonzero term of degree at least two: such a term would give $\bar{e}_1 \geq u$, whereas the positive deficit gives $2\bar{e}_1 \leq (2 - w)u < 2u$. Thus $u = 1 + h_1 = \text{edim} - 1$, which also bounds u when its embedding dimension is smaller than six. Its entire tail is at most $q_u w$. For $1 \leq A \leq 9/4$,

$$(13.3) \quad \max_{1 \leq u \leq 5} \frac{2}{u} (q_u - A)_+ \leq \frac{9}{10} - \frac{2A}{5}.$$

For $u = 2, 3, 4$, subtracting the untruncated expression on the left from the right gives, respectively, $3A/5 - 31/60$, $4(A - 1)/15$, and $(A - 1)/10$. They are nonnegative. The right side is nonnegative throughout the stated interval; the cases $u = 1, 5$ give zero and equality. In particular every surface tail is at most $w + 1/2$.

For a threefold of multiplicity t and deficit z , the Hilbert moments give

$$\bar{e}_1 \geq t - 1, \quad \bar{e}_1 \geq 2t - 5.$$

The second inequality follows by assigning weight at least two to all terms after h_1 , where $h_1 \leq 3$. Proposition 4.10 supplies the deficit caps in the following table. The entire threefold tail is at

most $c_t z + d_t$:

(13.4)

t	upper bound for z	c_t	d_t
1	3	1	1/2
2	2	19/15	29/120
3	5/3	13/9	11/54
4	3/2	127/80	33/160
5	1	7/4	1/20
6	2/3	2	0
7	3/7	2	0
8	1/4	2	0
9	1/9	9/4	0

Here $c_t \geq t^{1/3}$. For the four nonintegral comparisons, one can cube to obtain $19^3 > 2 \cdot 15^3$, $13^3 > 3 \cdot 9^3$, $127^3 > 4 \cdot 80^3$, and $7^3 > 5 \cdot 4^3$.

To verify the additive constants, an old lc center persists under each effective addition preserving log canonicity: its zero-place valuation must have order zero on that addition. For a surface successor of multiplicity u and deficit $w > 0$, Proposition 4.11 thus gives

$$u \leq (3 - w)t - 2\bar{e}_1(\mathfrak{m}_{W_3, x}),$$

where W_3 is this threefold. Also $w \leq 2/u$. If $q_u > c_t$, its possible surplus is at most $(q_u - c_t)w$; otherwise it gives no surplus. For $t = 2$, the threefold has embedding dimension at most four, so $u \leq 3$, and $u \leq 4 - 2w$. The two positive surpluses are $3/20$ and $29/120$. For $t = 3$, its embedding dimension is at most five; hence $u \leq 4$ and $u \leq 5 - 3w$, giving $11/54$ and $5/27$. For $t = 4$, the inequality $u \leq 6 - 4w$ gives $13/120, 33/160, 53/320$. For $t = 5$, the inequality $u \leq 5 - 5w$ leaves only the positive surplus $1/20$. For $t = 6, 7$, the surface inequalities are respectively $u \leq 4 - 6w < 4$ and $u \leq 3 - 7w < 3$, so every subsequent radius is at most two. The smooth row follows from (13.3) with $A = 1$. These calculations prove the first seven rows, including jumps straight to a curve or a point.

The last two rows, and the exclusion of $t > 9$, are Lemma 5.1. In particular their contributions to the two universal envelopes used in Proposition 5.5 satisfy

$$\begin{array}{c|c|c} t & C - \frac{2301}{2000}z & C - \frac{9}{10}z \\ \hline 8 & \leq \frac{1699}{8000} < \frac{3447}{4000} & \leq \frac{11}{40} < \frac{100663}{60000} \\ 9 & \leq \frac{733}{6000} < \frac{3447}{4000} & \leq \frac{3}{20} < \frac{100663}{60000} \end{array}$$

For example the first row uses $(2 - 2301/2000)/4 = 1699/8000$; the other entries follow by substituting $(c_t, z) = (2, 1/4)$ or $(9/4, 1/9)$.

Step 2. In this step, we impose the fourfold-parent restrictions to obtain the stated complete-tail bounds. For a cubic fourfold, $\bar{e}_1 \geq 2$ and $b \leq 8/3$. At its threefold successor, of multiplicity t and deficit $z > 0$, the flag inequality gives $t \leq 8 - 3z < 8$. Thus only the first seven rows of (13.4) occur here. Set $A = 79/60$. We have $79^4 = 38950081 > 38880000 = 3 \cdot 60^4$. For $t = 3, \dots, 7$, substitute the smaller of the displayed deficit cap and $(8 - t)/3$ into $(c_t - A)z + d_t$. The results are

$$\frac{5}{12}, \frac{817}{1440}, \frac{29}{60}, \frac{41}{90}, \frac{41}{180}.$$

When $c_t \leq A$, all cost before the surface is bounded by the slope A , and (13.3) bounds the remaining surplus by $28/75$. It also bounds a direct surface jump. The largest surplus is $817/1440$, proving (i).

Now suppose $m \geq 4$ and $b > 1$. Embedding dimension at most five would give a one-variable Artinian reduction with Hilbert function a string of m ones, so $\bar{e}_1 \geq m(m - 1)/2$ and $b \leq 5 - m \leq 1$. Thus $h_1 = 2$. The moments give $\bar{e}_1 \geq 2m - 4$ and $b \leq 8/m$. If $m \geq 7$, then $h_2 \leq 3$ also gives $\bar{e}_1 \geq 3m - 10$ and $b \leq 20/m - 2 \leq 6/7$, a contradiction. Only $m = 4, 5, 6$ remain. Their threefold successors satisfy $t \leq 8 - mz < 8$.

Use slopes $17/12, 3/2, 8/5$, respectively; their fourth powers exceed 4, 5, 6. For each slope, the direct-surface surplus and all positive threefold surpluses are as follows:

m	direct surface	threefold surpluses
4	1/3	103/432, 181/480, 3/10, 7/24, 7/48
5	3/10	221/800, 1/5, 1/5, 1/10
6	13/50	1/8, 2/15, 1/15

Each entry is $(c_t - A) \min\{\text{cap}_t, (8 - t)/m\} + d_t$ when $c_t > A$. The cases $c_t \leq A$ use (13.3) as before. The row maxima are the constants in (ii). Since deficits decrease along the same chain, these surplus bounds telescope with its first radius and bound its entire scalar cost. $\square$

The remaining cubic case requires the full tangent scheme. The next lemma establishes its reducedness before any capacity estimate is applied.

Lemma 13.3. *Let Γ be an effective rational boundary on X , log canonical near x , with a fourfold minimal center W of multiplicity three. Put*

$$M = \text{mult}_x \Gamma, \quad b = \text{def}_x(\Gamma), \quad \sigma = 6 - M - b.$$

Let $S = \mathcal{O}_{X,x}$, $P = I_{W,x}$, $R = S/P$, and $\mathfrak{n} = \mathfrak{m}_x R$. If $b > 33/16$ and $0 \leq \sigma < 1/4$, then $P \subseteq \mathfrak{m}_x^2$, and $\text{gr}_{\mathfrak{n}} R$ is reduced and Cohen–Macaulay with Hilbert series $(1 + 2z)/(1 - z)^4$.

The assertion concerns the entire associated graded ring, including its possible embedded components at the vertex.

Proof. We first identify the associated graded ring by minimal multiplicity. We then use the effective transformed pair to control every generic multiplicity of its projectivization.

Step 1. In this step, we determine the full tangent Hilbert series and its Cohen–Macaulay property. Theorem 4.6 and Proposition 4.10 give

$$2 \leq e_1(\mathfrak{n}) \leq \bar{e}_1(\mathfrak{n}) \leq \frac{3(4 - b)}{2} < \frac{93}{32} < 3.$$

Both Hilbert coefficients are therefore two. If $\text{edim } R \leq 5$, its length-three Artinian reduction would have Hilbert function $(1, 1, 1)$, with first moment three. Hence $\text{edim } R = 6$, $P \subseteq \mathfrak{m}_x^2$, and its Artinian function is $(1, 2)$. The normality theorem makes R a Cohen–Macaulay domain. Choose a general minimal reduction $J = (y_1, \dots, y_4)$ whose generators form an R -regular sequence and have independent initial linear forms. Minimal multiplicity and Theorem 4.14 [Ver08, Theorem 2.7] give $\mathfrak{n}^2 = J\mathfrak{n}$ and the Cohen–Macaulay property of $\text{gr}_{\mathfrak{n}} R$. The independent parameters and the quotient function $(1, 2)$ give the asserted Hilbert series.

Step 2. In this step, we prove reducedness of the exceptional divisor and then of the full associated graded ring. On $Y = \text{Bl}_x X$, let Γ^{str} be the effective strict transform and put $e = 5 - M = b + \sigma - 1 > 0$. Then

$$K_Y + \Gamma^{\text{str}} = \pi^*(K_X + \Gamma) + eE.$$

For $0 \leq \theta < 1 - \sigma$ and any divisorial valuation v with $v(\mathfrak{m}_x) > 0$, the transformed discrepancy is

$$A_{(Y, \Gamma^{\text{str}} + \theta E)}(v) = A_{(X, \Gamma)}(v) - (e + \theta)v(\mathfrak{m}_x) > 0,$$

because $e + \theta < b$. If $v(\mathfrak{m}_x) = 0$, it is the old nonnegative discrepancy. Thus this effective pair is lc near the exceptional fiber and has no lc center mapping to x .

Let $\widetilde{W}$ be the strict transform of W . It remains an lc center. At a point of $\widetilde{W}$ over x , Theorem 3.1 [GL24, Lemma 2.7] puts the minimal center inside $\widetilde{W}$. A smaller such center cannot map to x by the strict discrepancy inequality. If its image has positive dimension, its defining valuation has point-ideal order zero and gives an old lc center properly inside W through x . This contradicts minimality of W . Thus $\widetilde{W}$ is minimal at every point of that fiber. Theorem 4.5 makes it normal and Cohen–Macaulay there.

The Rees quotient embeds the ordinary point blow-up of W into Y . Its Rees algebra is a domain, and the image is the closure of $W \setminus \{x\}$; hence the image is $\widetilde{W}$. Its effective Cartier exceptional scheme is

$$E_W = E|_{\widetilde{W}} = \text{Proj}(\text{gr}_{\mathfrak{n}} R).$$

Write $E_W = \sum_A r_A A$ at its generic points, with positive integers r_A . Apply Theorem 4.1 [FH23, Theorem 1.2 and Definition 1.3] to the center $\widetilde{W}$ of (Y, Γ^{str}) . Its discriminant coefficient β_A is nonnegative. For each ambient lc place T dominating $\widetilde{W}$, adding θE adds θr_A times the pullback of A at its generic point. The sub-lc threshold defining the discriminant on that place is therefore at least θr_A . Taking the infimum over those actual places gives

$$0 \leq \beta_A \leq 1 - \theta r_A.$$

Let θ tend to $1 - \sigma > 3/4$. This implies $r_A < 4/3$, so every $r_A = 1$.

The Cartier scheme E_W is Cohen–Macaulay and generically reduced, hence reduced. The graded ring is Cohen–Macaulay of pure dimension four, and its homogeneous minimal primes are detected on this projectivization. In particular its local rings at those minimal primes are reduced. It has no embedded prime, including at the vertex. It is therefore reduced, as required. $\square$

Theorem 13.4. *Assume $L^6 = 1$. There is an integer J , depending only on X, L, x , such that if at one $j \geq J$ a pure threshold average from U_j has a fivefold minimal center of multiplicity two, then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/4096$.*

The point boundary need not be an additive continuation of the original average. Each new source below is complete, and its threshold is taken afresh. We do not assume nongeneration in proving this theorem.

Proof. We first obtain a lower bound for the volume at order 69 from the full original fixed coefficient. We bound the fixed part of this new source, account for every possible center at its own threshold, and use the remaining cubic case to obtain sufficient volume at order 75.

Put $h = 1/65536$. Each time a scalar bound is used below, Proposition 3.7 realizes it with total coefficient error less than h , allocating the allowances after the actual radii are known. An endpoint already supplied by an earlier theorem carries its own error budget. All universal threefold estimates used here include Lemma 5.1; their last two rows and the four envelope comparisons are displayed in the proof of Lemma 13.2.

Suppose, for contradiction, that there is no point boundary with the asserted coefficient. This excludes every actual section in any $V_r(k)$ of point order at least $92k$, by Proposition 5.8, since $1/630 > 1/4096$.

Step 1. In this step, we use the original double fixed factor to obtain a lower volume bound at order 69. Let $\Gamma_{64} \sim_{\mathbb{Q}} s_{64}L$ be the average in the hypothesis, and put $M_{64} = \text{mult}_x \Gamma_{64}$ and $b_{64} = \text{def}_x(\Gamma_{64})$. Purity and the entire point order give

$$s_{64} \leq \frac{65}{64} M_{64}, \quad M_{64} + b_{64} \leq 6.$$

The first radius is at most $2^{1/5} < 5/3$. After one cut the center has dimension at most four, so Proposition 5.5 bounds the complete scalar cost by $5b_{64}/3 + 113/480$. If $b_{64} \leq 41/40$,

$$s_{64} + C \leq 6 \frac{65}{64} + \left(\frac{5}{3} - \frac{65}{64} \right) \frac{41}{40} + \frac{113}{480} = 7 - \frac{9}{2560}.$$

This leaves more than $h + 1/4096$, contradicting our assumption. Hence $b_{64} > 41/40$ and $M_{64} < 199/40$.

Let Z be the original double fivefold and let $\nu = \text{ord}_Z \mathfrak{q}_j$. At its generic point, $\tau\nu = 1$, where τ is this original ideal threshold. In particular the full fixed factor νZ has point order 2ν , and

$$M_{64} \geq \frac{64j}{\nu}, \quad \frac{\nu}{j} > \frac{2560}{199}, \quad \frac{2\nu}{j} > \frac{5120}{199} > \frac{257}{10}.$$

Every section in the nested source layers between $64j$ and $69j$ contains this same entire factor. The associated graded ring of $\mathcal{O}_{X,x}$ is a polynomial ring, so multiplication by the initial equation of the factor is injective. The degree- ℓ image consequently has dimension at most

$$\binom{\lfloor \ell - (257/10)j \rfloor + 5}{5} \quad (64j \leq \ell < 69j).$$

Summation yields the uniform estimate

$$(13.5) \quad \frac{720}{j^6}(\dim U_j - \dim V_{69}(j)) \leq \left(\frac{433}{10}\right)^6 - \left(\frac{383}{10}\right)^6 + O(1/j) < 3435000000 + O(1/j).$$

For the last comparison, $433^3 = 81182737 < 81183000$ and $383^3 = 56181887 > 56181000$. The difference of the squares, divided by 10^6 , is less than $25002 \cdot 137364 = 3434374728$. The error is independent of Z and ν , because the displayed fixed lower shift alone was used.

Choose J so that the normalized errors in (13.1), (13.5), and the fixed D_{69} volume limit total less than 4000000 for every $j \geq J$. This is possible by Proposition 7.5, before the average or its center is chosen. The single occurrence in the hypothesis now gives

$$(13.6) \quad \text{vol}_Y(D_{69}) > 6699413889 - 3435000000 - 4000000 = 3260413889 > 3260000000.$$

Step 2. In this step, we bound the full fixed point order of the new source by $45k$. The positive volume in (13.6) makes every sufficiently large $V_{69}(k)$ nonzero. Let N_{69} be its limiting full fixed divisor, given by Proposition 7.5, and put $\eta_{69} = \text{mult}_x N_{69}$. Proposition 7.7, applied to this new family with $r = 69$, gives $\eta_{69} \leq 69$ and bounds each initial image by $\binom{\lfloor (1-\eta_{69}/69)\ell \rfloor + 5}{5}$. There are no sections of order $92k$, by our contradiction assumption. Summing from $69k$ to $92k - 1$ and passing to the complete volume limit gives

$$\text{vol}_Y(D_{69}) \leq \left(1 - \frac{\eta_{69}}{69}\right)^5 (92^6 - 69^6).$$

If $\eta_{69} \geq 45$, the right side is at most $(8/23)^5(92^6 - 69^6) < 2544900000$: here $(8/23)^5 < 51/10000$ and $92^6 - 69^6 < 499000000000$. This contradicts (13.6). Convergence of the full fixed divisors therefore gives

$$(13.7) \quad \text{mult}_x F_{69}(k) < 45k \quad \text{for every sufficiently large } k.$$

Step 3. In this step, we treat every center of the new threshold except a cubic fourfold. Choose k beyond this cutoff and the fixed cutoffs needed below. Take a pure threshold average Γ_{69} from the complete space $V_{69}(k)$, and write s, M, b for its entire coefficient, point multiplicity and deficit. Put $q = 65/69$. The same ideal-threshold construction gives

$$(13.8) \quad s = 65k\tau_{69} \leq qM, \quad M + b \leq 6.$$

For completeness, on a common log resolution a general large averaging tuple has the ideal's fixed orders; its moving coefficients are less than one. The resulting pair is lc, and the center computing the ideal threshold remains an lc center. The blow-up at x gives the second inequality.

A point center is already paid by $s \leq 6q$. For a positive-dimensional center with $b \leq 1$, Corollary 5.7 applies with zero excess. The same corollary pays a smooth fourfold. A threefold of any multiplicity is paid by Theorem 9.6, again with zero excess relative to $65/64$. Its exhaustive coverage includes multiplicities eight and nine by Lemma 5.1. For curves and surfaces the surface estimate above gives

$$s + C \leq 4q + \frac{5}{2} < \frac{63}{10}.$$

A double fourfold has $h_1 = 1$, embedding dimension five, and $b \leq 3$. The corresponding row of Proposition 5.5 gives

$$s + C \leq 3q + 3 + \frac{3887}{3360} < \frac{6984}{1000}.$$

If a fourfold has multiplicity at least four and $b > 1$, Lemma 13.2 leaves only the following bounds:

m	upper bound for $s + C$	strict upper bound
4	$\frac{260}{69} + \frac{17}{6} + \frac{181}{480}$	6981/1000
5	$\frac{286}{69} + \frac{12}{5} + \frac{3}{10}$	6845/1000
6	$\frac{910}{207} + \frac{32}{15} + \frac{13}{50}$	6791/1000

These follow by maximizing $6q + (A - q)b + d$ at the respective caps $2, 8/5, 4/3$. Termwise bounds in thousandths are (3769, 2834, 378), (4145, 2400, 300), and (4397, 2134, 260). Every displayed margin exceeds $h + 1/4096$.

A smooth fivefold is paid by Theorem 8.7, applied to the full boundary in (13.8) with zero excess. For a singular fivefold T , put $m_T = \text{mult}_x T$ and $\nu_T = \text{ord}_T \mathfrak{b}(V_{69}(k))$. Purity gives $\tau_{69}\nu_T = 1$. Equation (13.7) gives

$$m_T \nu_T < 45k, \quad M \geq \frac{69k}{\nu_T} > \frac{69}{45} m_T.$$

Thus $m_T \geq 4$ contradicts $M \leq 6$. For $m_T = 3$, we have $M > 23/5$, $b < 7/5$, and the triple-fivefold row of Proposition 5.5 gives

$$s + C < \frac{13}{3} + \frac{7}{4} + \frac{57}{80} = \frac{1631}{240} < \frac{68}{10}.$$

For $m_T = 2$, Proposition 10.6 applies by (13.6). Since T is Cartier in the smooth sixfold, its initial local equation is a nonzero quadric in six variables. The proposition covers every rank of that equation: rank one, rank two, and ranks three through six, including the nondegenerate ordinary quadratic case. Neither the original double fivefold nor this new one was assumed to have reduced tangent cone. No fixed coefficient has been removed in applying the proposition.

The only center not yet paid is a cubic fourfold at this own 69-jet threshold. No relation with the original center is required.

Step 4. In this step, we prove the full reduced tangent properties and generic threshold bound of the remaining cubic center. Let W be this cubic fourfold and set $\sigma = 6 - M - b \geq 0$. Lemma 13.2 gives $b \leq 8/3$ and $C \leq 79b/60 + 817/1440$. If $b \leq 33/16$, then

$$s + C \leq q \frac{63}{16} + \frac{79}{60} \frac{33}{16} + \frac{817}{1440} < \frac{371}{100} + \frac{2716}{1000} + \frac{568}{1000} = \frac{6994}{1000}.$$

If $\sigma \geq 1/4$, maximizing at $b = 8/3$ instead gives

$$s + C \leq q \frac{37}{12} + \frac{158}{45} + \frac{817}{1440} < \frac{291}{100} + \frac{3512}{1000} + \frac{568}{1000} = \frac{699}{100}.$$

Both margins exceed $h + 1/4096$. Thus failure forces $b > 33/16$ and $\sigma < 1/4$. Lemma 13.3 now proves every tangent hypothesis needed below.

Let $S = \mathcal{O}_{X,x}$, $\mathfrak{p} = I_{W,x}$, and let λ_{69} be the threshold of the 69-source ideal in the regular two-dimensional local ring $S_{\mathfrak{p}}$. It equals τ_{69} . Indeed a zero place dominating W for the general pure average has the same fixed orders as the ideal and computes this ratio; lc near x gives the opposite inequality at the generic point. Hence

$$(13.9) \quad 69k\lambda_{69} \leq M < \frac{63}{16} < 4.$$

Step 5. In this step, we obtain the order-75 volume from the cubic capacity and construct the point boundary. For $69k \leq \ell < 75k$, put

$$\mathcal{U}_{\ell} = H^0(X, \mathcal{O}_X(65kL) \otimes \mathfrak{m}_x^{\ell}).$$

A zero space contributes no layer. Otherwise let $\mathfrak{a}_{\ell}$ be the ideal generated by all its actual germs in $S_{\mathfrak{p}}$. It is contained in the 69-source ideal, so its threshold is at most $\lambda_{69} < 4/(69k)$. Choose

an integer N_ℓ sufficiently large that

$$\lambda_\ell = \text{lct}_{S_{\mathfrak{p}}}(\mathfrak{a}_\ell + (\mathfrak{p}S_{\mathfrak{p}})^{N_\ell}) < \frac{4}{69k}.$$

This follows from Theorem 11.1 [DK01, Theorem 2.9], since the added error is at most $2/N_\ell$. The strict margin in (13.9) is preserved. Here $S_{\mathfrak{p}}$ is a regular local ring of dimension two, with maximal ideal $\mathfrak{p}S_{\mathfrak{p}}$, so the stated regular-local form applies. Let I_ℓ be the contraction of this primary ideal to S . It is $\mathfrak{p}$ -primary and contains every actual germ from $\mathcal{U}_\ell$, including all fixed factors.

We have $\mathfrak{p} \subseteq \mathfrak{m}_x^2$, and the full tangent ring is reduced and Cohen–Macaulay. Moreover

$$\ell\lambda_\ell < \frac{75 \cdot 4}{69} = \frac{100}{23} < \frac{11}{2}.$$

The full two-unit estimate of Proposition 6.18 therefore gives an absolute constant C_4 with

$$\dim_{\mathbb{C}}(\text{in}_{\mathfrak{m}_x} I_\ell)_\ell \leq \frac{(\ell - 2/\lambda_\ell)_+^5}{120} + C_4(\ell + 1)^4 \leq \frac{(\ell - 69k/2)^5}{120} + C_4(\ell + 1)^4.$$

Local trivialization embeds the actual quotient $\mathcal{U}_\ell/\mathcal{U}_{\ell+1}$ in this initial image. The estimate is uniform in W , in the primary approximation, and in ℓ . Summing all layers gives

$$\begin{aligned} \frac{720}{k^6} (\dim V_{69}(k) - \dim V_{75}(k)) &\leq \frac{81^6 - 69^6}{64} + O(1/k) \\ (13.10) \qquad \qquad \qquad &= \frac{174511373400}{64} + O(1/k) \\ &< 2727000000 + O(1/k). \end{aligned}$$

This is a comparison at the same k ; it requires no persistence of the cubic center at a later threshold.

The two volume limits and the error in (13.10) have cutoffs depending only on the fixed families and the absolute capacity constant. Choose the auxiliary k beyond them so that their combined normalized error is below 1000000, as well as beyond the earlier 69-source cutoffs. Then

$$\text{vol}_Y(D_{75}) > 3260000000 - 2727000000 - 1000000 = 532000000 > 336000000.$$

For every k , the complete source at order $75k$ is contained in that at order $\lceil (371/5)k \rceil$. Proposition 7.5 therefore gives $\text{vol}_Y(D_{371/5}) \geq \text{vol}_Y(D_{75})$. Theorem 10.5 supplies the required point boundary. Its conclusion imposes no condition on subsequent own thresholds or on their center types, so it accounts for every later source choice. This contradicts our assumption and completes the cubic case. Together with the preceding list of 69-source centers, it proves the theorem. $\square$

14. ORIGINAL SMOOTH FIVEFOLD CENTERS

We retain the notation of Section 2. We treat a smooth fivefold center of the original complete source U_j . The small-deficit argument takes place on a resolution of the center and retains its entire adjoint boundary. For larger deficit, the fixed order of the same center controls the complete point-order filtration on X .

14.1. Intrinsic thresholds and their complete tails.

Lemma 14.1. *Let Z be a smooth complex projective fivefold, let ζ be a closed point of Z , and let A be an ample rational Cartier divisor such that $A^{\dim T} \cdot T \geq 1$ for every positive-dimensional integral subvariety T through ζ . Let Δ be an effective rational boundary, log canonical but not klt at ζ , with positive-dimensional minimal center V . Put*

$$z = \text{lct}_\zeta((Z, \Delta); \mathfrak{m}_\zeta), \quad m = \text{mult}_\zeta V.$$

For an additive cutting chain on Z ending at ζ , let C denote its entire scalar cost, with radii computed using A . Then the following hold:

(i) Unless V is a fourfold of multiplicity two or three,

$$C - \frac{65}{64}z \leq \frac{10131}{11200}.$$

(ii) If $\dim V = 4$ and $m = 2$, then $z \leq 3$ and

$$C \leq \frac{6}{5}z + \frac{1871}{3360}.$$

If also $A^4 \cdot V \geq 2$, then $C \leq z + 2879/3360$.

(iii) If $\dim V = 4$ and $m = 3$, then $z \leq 2$ and

$$C \leq \frac{79}{60}z + \frac{41}{120}.$$

If also $A^4 \cdot V \geq 2$, then $C \leq (10/9)z + 8/15$.

These bounds include chains that omit an intermediate dimension.

The two exceptional fourfold rows require their own slopes. All other proper centers fit a single bound relative to $65/64$.

Proof. We first bound the local Hilbert data, then compute the costs of threefold and fourfold tails. Put $d = \dim V$.

Step 1. In this step, we derive the local Hilbert bounds and the multiplicity bound for codimension-one successors. The same estimates will apply to each successor with its own current data. By Theorems 4.5 and 4.6 ([dFH09, Corollary 7.12] and [Han26, Lemmas 4.2 and 4.4]), its local ring is normal and Cohen–Macaulay, and a general Artinian minimal reduction satisfies

$$h_0 = 1, \quad h_1 \leq 5 - d, \quad m = \sum_i h_i, \quad T_1 := \sum_i i h_i \leq \bar{e}_1(\mathfrak{m}_{V,\zeta}).$$

Propositions 4.10 and 4.11 give

$$(14.1) \quad 2\bar{e}_1(\mathfrak{m}_{V,\zeta}) \leq (d - z)m, \quad m' \leq (d - w)m - 2\bar{e}_1(\mathfrak{m}_{V,\zeta}).$$

The second inequality concerns a codimension-one successor with current deficit w . The earlier center remains an lc center under each effective lc addition: a place with discrepancy zero cannot acquire positive order from that addition. Its normal germ therefore remains available for this inequality.

Step 2. In this step, we bound the complete curve and surface tails on the smooth fivefold. A normal curve is smooth, and its radius is at most one. For a surface, $T_1 \geq m - 1$, whereas $2T_1 \leq (2 - z)m < 2m$. Integrality gives $T_1 = m - 1$, so $h_i = 0$ for $i \geq 2$. Thus $m = 1 + h_1 \leq 4$ and $z \leq 2/m$. All subsequent radii are at most one. Consequently its whole tail is bounded by $q_m z$, where

$$(14.2) \quad (q_1, q_2, q_3, q_4) = \left(1, \frac{17}{12}, \frac{97}{56}, 2\right).$$

These are upper bounds for $\sqrt{m}$; for example, $17^2 > 2 \cdot 12^2$ and $97^2 > 3 \cdot 56^2$.

Step 3. In this step, we compute every permitted threefold tail and its deficit cap. Now $h_1 \leq 2$, and the Artinian moment bounds give

$$T_1 \geq m - 1, \quad T_1 \geq 2m - 4, \quad T_1 \geq 3m - 10.$$

For the last two inequalities, use respectively $2m - T_1 \leq 2h_0 + h_1 \leq 4$ and $3m - T_1 \leq 3h_0 + 2h_1 + h_2 \leq 10$. Equation (14.1) implies $m \leq 6$ and yields the following bounds:

$$(14.3) \quad \begin{array}{c|cc} m & \text{upper bound for } z & \text{upper bound for } C \\ \hline 1 & 3 & z + 5/12 \quad \text{and} \quad 17z/12 \\ 2 & 2 & 19z/15 + 29/120 \quad \text{and} \quad 7z/4 \\ 3 & 5/3 & 577z/400 + 811/4200 \\ 4 & 1 & 8z/5 + 3/80 \\ 5 & 3/5 & 7z/4 \\ 6 & 1/3 & 2z \end{array}$$

Here is the calculation of the entire tails. If a parent radius is at most a , and a successor at deficit w has tail bound $T(w)$, the combined cost is at most

$$(14.4) \quad a(z - w) + T(w) = az + (T(w) - aw).$$

One therefore bounds the surplus in parentheses for every possible successor. Direct passage to a curve or the point contributes no positive surplus when $a \geq 1$.

A smooth threefold has embedding dimension three; its surface successors have multiplicity at most two. Their largest surplus over slope one is $(17/12 - 1) \cdot 1 = 5/12$, and all its radii are at most $17/12$. For a double threefold, (14.1) gives surface multiplicity $h \leq 4 - 2w$. The only surface with radius exceeding the chosen parent bound $19/15$ by more than the double-surface contribution has $h = 3$ and $w \leq 1/2$. Using $7/4$ for its radius gives

$$\frac{1}{2} \left(\frac{7}{4} - \frac{19}{15} \right) = \frac{29}{120}.$$

All radii in this chain are at most $7/4$, giving its second bound.

For a triple threefold, $h \leq 5 - 3w$. The surface cases $h = 3, 4$ have respectively $w \leq 2/3, 1/3$. Their surpluses over $577/400$ are bounded by

$$\frac{2}{3} \left(\frac{97}{56} - \frac{577}{400} \right) = \frac{811}{4200}, \quad \frac{1}{3} \left(2 - \frac{577}{400} \right) < \frac{811}{4200}.$$

The double-surface radius is smaller than $577/400$. For $m = 4, 5, 6$, the inequality $h \leq 8 - m - mw$ restricts positive-deficit surfaces to $h \leq 3, 2, 1$, respectively. At $m = 4, h = 3$, one has $w \leq 1/4$, and the surplus over $8/5$ is at most $(7/4 - 8/5)/4 = 3/80$. In the last two rows the chosen parent radius dominates every descendant radius. The parent comparisons follow from

$$19^3 > 2 \cdot 15^3, \quad 577^3 > 3 \cdot 400^3, \quad 8^3 > 4 \cdot 5^3, \quad 7^3 > 5 \cdot 4^3, \quad 2^3 > 6.$$

This proves every row of (14.3).

Step 4. In this step, we bound fourfold tails using the restrictions on their actual successors. For a fourfold, $h_1 \leq 1$, so its Artinian Hilbert function has one in each degree from zero to $m - 1$. Hence $\bar{e}_1(\mathbf{m}_{V,\zeta}) \geq m(m-1)/2$, $m \leq 4$, and $z \leq 5 - m$. A smooth fourfold has radius at most one and descendants of embedding dimension at most four. In that embedding dimension a threefold has multiplicity at most three. The double row has surplus over slope one at most $31/40$. A triple threefold has normal first Hilbert coefficient at least three, deficit at most one, and only smooth or double surface successors. Its entire cost is at most $13/9$ times its deficit. The smooth threefold and surface rows give smaller surpluses. Thus a smooth fourfold has $C \leq z + 4/5$.

For a double fourfold, a threefold successor of multiplicity l and deficit w satisfies $l \leq 6 - 2w$. Subtracting $6w/5$ from (14.3), with this restriction and the deficit cap of that row, gives

$$\begin{array}{c|ccccc|c} l & 1 & 2 & 3 & 4 & 5 & \text{direct surface} \\ \hline \text{surplus} & 5/12 & 3/8 & 1871/3360 & 7/16 & 11/40 & 2/5 \end{array}$$

For example, the triple-threefold entry is

$$\left(\frac{577}{400} - \frac{6}{5} \right) \frac{3}{2} + \frac{811}{4200} = \frac{1871}{3360}.$$

No positive-deficit row $l \geq 6$ occurs. The largest entry is $1871/3360$, and $(6/5)^4 > 2$, proving (ii).

For a triple fourfold, the successor condition is $l \leq 6 - 3w$. For $l = 1, 2$, take the minimum of the two corresponding bounds in (14.3). Their surpluses over $79w/60$ peak at $w = 1, 1/2$, respectively. The full list is

l	1	2	3	4	5	direct surface
surplus	$1/10$	$13/60$	$893/2800$	$163/720$	$13/90$	$41/120$.

For $l = 3, 4, 5$, the bounds on w are $1, 2/3, 1/3$. The last entry comes from $h = 4, w = 1/2$ in (14.2); the other surface entries are smaller. Since $(79/60)^4 > 3$, the maximum $41/120$ proves (iii).

For a multiplicity-four fourfold, use initial radius $17/12$. The condition $l \leq 4 - 4w$ permits only $l = 1, 2, 3$. The first has no positive surplus, the second contributes at most $(7/4 - 17/12)/2 = 1/6$, and the third contributes at most $(2 - 17/12)/4 = 7/48$. A direct surface contributes at most $(2 - 17/12)/2 = 7/24$. Thus $C \leq 17z/12 + 7/24$, with $z \leq 1$.

Step 5. In this step, we bound the surplus relative to $65/64$ for every nonexceptional center type. For smooth and multiplicity-four fourfolds the surpluses are at most $4/5$ and $133/192$. For the six threefold rows they are at most

$$\frac{5}{12}, \quad \frac{119}{160}, \quad \frac{10131}{11200}, \quad \frac{199}{320}, \quad \frac{141}{320}, \quad \frac{21}{64}.$$

The third value is exactly

$$\left(\frac{577}{400} - \frac{65}{64}\right) \frac{5}{3} + \frac{811}{4200} = \frac{10131}{11200}.$$

The surface rows contribute at most $63/128$; the triple-surface entry is $107/224$. Curves and points contribute no positive surplus. This proves (i).

Step 6. In this step, we sharpen the double- and triple-fourfold tails when their degrees are at least two. For a double fourfold of A -degree at least two, the initial radius is at most one. The successor restriction $l \leq 6 - 2w$ and the threefold table give surpluses over slope one at most

$$\frac{5}{12}, \quad \frac{31}{40}, \quad \frac{2879}{3360}, \quad \frac{51}{80}, \quad \frac{3}{8}.$$

The direct-surface maximum is $1/2$. The largest entry proves the remaining assertion in (ii). For a triple fourfold of degree at least two, its radius is less than $10/9$, since $(10/9)^4 > 3/2$. Under $l \leq 6 - 3w$, the corresponding threefold surplus bounds are

$$\frac{5}{12}, \quad \frac{97}{216}, \quad \frac{1193}{3600} + \frac{811}{4200}, \quad \frac{44}{135} + \frac{3}{80}, \quad \frac{23}{108}.$$

Each is less than $8/15$, and direct surfaces contribute at most $4/9$. This proves the last assertion of (iii). At each use of (14.4), all lower dimensions were included, so the bounds also cover direct jumps. $\square$

Lemma 14.2. *Let $\Gamma \sim_{\mathbb{Q}} sL$, $s \geq 0$, be an effective rational boundary log canonical near x , with fivefold minimal center W smooth at x . Put $b = \text{def}_x(\Gamma)$, and let $e \in \mathbb{Q}$ satisfy $s \leq \frac{65}{64} \text{mult}_x \Gamma + e$. Write $\Gamma = W + R$, let $f : \widetilde{W} \rightarrow W$ be a smooth projective resolution of the normalization, isomorphic near x , and let y be the point above x . Put $D = f^*(R|_W)$, and choose an ample rational divisor A as in Lemma 8.1. For all sufficiently large integers k clearing the denominator of A , the spaces*

$$\mathcal{E}_k = H^0(\widetilde{W}, \mathcal{O}_{\widetilde{W}}(65kA) \otimes \mathfrak{m}_y^{64k})$$

are nonzero. Let τ_k be the threshold of their actual base ideal against D . For a sufficiently large sufficiently general tuple of actual divisors $G_{k,i}$ from $\mathcal{E}_k$, put

$$\Delta_k = D + \frac{\tau_k}{N_k} \sum_{i=1}^{N_k} G_{k,i},$$

and set

$$v_k = 65k\tau_k, \quad z_k = \text{lct}_y((\widetilde{W}, \Delta_k); \mathfrak{m}_y).$$

Then D is effective and klt near y , its deficit is b , $\tau_k > 0$, and

$$(14.5) \quad v_k \leq \frac{65}{64}(b - z_k), \quad s + v_k \leq 6\frac{65}{64} - \frac{65}{64}z_k + e.$$

The charge s records all of the ambient boundary, while the threshold on $\widetilde{W}$ is taken against all of D .

Proof. The divisorial center W has coefficient one in Γ . Since X is smooth, its global prime closure is Cartier, so R is effective and does not contain W .

Step 1. In this step, we identify the deficit of the full intrinsic boundary D with the ambient deficit. Cartier adjunction, Theorem 4.2 ([FH23, Corollary 1.4]), identifies the full different near x with $R|_W$. Minimality makes it klt: a non-klt center of the different would give a smaller ambient lc center. Applying the same inversion of adjunction to the point ideal with each nonnegative exponent identifies the two deficits. Pullback to $\widetilde{W}$ preserves the germ and effectivity. The point blow-ups on X and on W give

$$(14.6) \quad 0 < b \leq 5, \quad \text{mult}_x \Gamma + b \leq 6, \quad \text{mult}_y D \leq 5 - b.$$

There is no identification of $\text{mult}_y D$ with $\text{mult}_x \Gamma - 1$.

Step 2. In this step, we construct the complete intrinsic source and bound its actual relative threshold cost. The choice of A gives $A^5 \geq 1$. Riemann–Roch and the length of the point jets give the kernel estimate

$$\dim \mathcal{E}_k \geq \frac{65^5 A^5 - 64^5}{120} k^5 + O(k^4), \quad 65^5 - 64^5 = 86548801 > 0.$$

This requires no surjectivity onto those jets. The base ideal is contained in $\mathfrak{m}_y^{64k}$. On a common log resolution of D , this ideal, and $\mathfrak{m}_y$, its relative threshold is the minimum of the finitely many ratios of the discrepancy of D to the positive base-ideal order, over resolved primes whose images contain y . Klt of D makes the threshold positive, and the point-blowup valuation makes it finite. Pullback separates every actual member into the full fixed divisor of the resolved ideal and a member of a base-point-free system. Choose these moving members generally, transverse to the resolved boundary, and take N_k so large that $\tau_k/N_k < 1$. The resulting simple-normal-crossing pair has the same threshold and lc centers as the ideal pair, and its point-ideal threshold is also unchanged. For every valuation of positive point-ideal order, the added order is at least $64k\tau_k$ times that point-ideal order. Taking the infimum of discrepancy divided by point-ideal order gives $z_k \leq b - 64k\tau_k$. This proves the first inequality in (14.5); the second follows from $s \leq \frac{65}{64}(6 - b) + e$. $\square$

Lemma 14.3. *Let $\Gamma \sim_{\mathbb{Q}} sL$, $s \geq 0$, be effective rational and log canonical near x , with a fivefold minimal center smooth at x . Put $b = \text{def}_x(\Gamma)$, and let $e \in \mathbb{Q}$ satisfy $s \leq \frac{65}{64} \text{mult}_x \Gamma + e$. Assume either*

$$b \leq \frac{15}{8}, \quad e < \frac{1}{65536}, \quad \text{or} \quad b \leq \frac{187}{100}, \quad e < \frac{1}{2560}.$$

Then there is an effective rational $Q \sim_{\mathbb{Q}} uL$ such that the entire $\Gamma + Q$ is a point boundary and $0 < s + u < 7 - 1/16384$.

Both alternatives retain the original boundary. The second allows more initial excess at the cost of a smaller deficit range.

Proof. Use the intrinsic source of Lemma 14.2. Write v, z for its added coefficient and deficit, and C for the full scalar cost after that threshold.

Step 1. In this step, we bound the total scalar cost in both stated deficit ranges. If it already has point minimal center, take $z = C = 0$. Otherwise apply Lemma 14.1. In the nonexceptional cases,

$$s + v + C \leq 6\frac{65}{64} + \frac{10131}{11200} + e = 7 - \frac{19}{11200} + e.$$

For a double fourfold, $z \leq b \leq 15/8$ gives

$$s + v + C \leq 6\frac{65}{64} + \left(\frac{6}{5} - \frac{65}{64}\right) \frac{15}{8} + \frac{1871}{3360} + e = 7 - \frac{199}{53760} + e.$$

For a triple fourfold in the first alternative, one obtains

$$s + v + C \leq 6\frac{65}{64} + \left(\frac{79}{60} - \frac{65}{64}\right) \frac{15}{8} + \frac{41}{120} + e = 7 - \frac{1}{7680} + e.$$

In the second alternative, $15/8 - 187/100 = 1/200$ and $79/60 - 65/64 = 289/960$. The corresponding bound is

$$s + v + C \leq 7 - \frac{1}{7680} - \frac{289}{192000} + e < 7 - \frac{239}{192000}.$$

All three bounds in that alternative have margin greater than $1/1000$.

Step 2. We now realize the scalar costs and transfer the added divisor. Apply Theorem 3.5 ([GL24, Proposition 3.2 and Remark 3.3]) on the smooth projective fivefold $\widetilde{W}$. After each radius r_i is known, choose its approximation error below $a_i/(1+r_i)$. The theorem gives actual added cost at most $(z_i - z_{i+1})r_i + r_i\epsilon_i$. At most four cuts follow the intrinsic threshold, so the sum of their errors can be made smaller than $\sum_i a_i$, as in Proposition 3.7.

Choose the ample perturbation before constructing these sources. Its parameter δ , the cutting errors, and the final conductor-and-lifting allowance ϵ_{lift} can satisfy

$$7\delta + \sum_i a_i + \epsilon_{\text{lift}} < \frac{1}{32768}.$$

The constructed intrinsic addition has coefficient less than seven. Lemma 8.1 therefore adds less than $7\delta + \epsilon_{\text{lift}}$ to its cost and lifts it to an actual ambient divisor, retaining every component of R . For the first alternative the smallest final margin is greater than

$$\frac{1}{7680} - \frac{1}{65536} - \frac{1}{32768} = \frac{83}{983040} > \frac{1}{16384}.$$

For the second it is greater than $1/1000 - 1/32768 > 1/16384$. This proves both conclusions. $\square$

14.2. The residual boundary and the first normal image.

Proposition 14.4. *Suppose $K_X + 7L$ is not generated at x . For all sufficiently large j , let a pure threshold average Γ from U_j have a fivefold minimal center W smooth at x . Let $\nu = \text{ord}_W U_j$, write $\Gamma = W + R \sim_{\mathbb{Q}} sL$, and put $b = \text{def}_x(\Gamma)$. Let σ_W be the canonical section of $\mathcal{O}_X(W)$. Set*

$$A_0 = L|_W, \quad N_0 = \mathcal{O}_W(W),$$

and define

$$\mathcal{I}_j = \text{Im} \left[U_j / \sigma_W^\nu \longrightarrow H^0(W, \mathcal{O}_W(65jA_0) \otimes N_0^{-\nu}) \right].$$

Let $f : \widetilde{W} \rightarrow W$ be a smooth projective resolution of the normalization, isomorphic near x , let y be the point above x , and set $D = f^*(R|_W)$. Then $L^6 = L^5 \cdot W = 1$ and $15/8 < b < 34/11$. There are ample rational divisors A arbitrarily close to f^*A_0 , satisfying Lemma 8.1, with the following property. For every sufficiently large permitted k , threshold the full base ideal of

$$\mathcal{E}_k = H^0(\widetilde{W}, \mathcal{O}_{\widetilde{W}}(65kA) \otimes \mathfrak{m}_y^{64k})$$

against this same D , and write τ_k for that threshold. The following hold for every sufficiently general actual threshold average:

- (i) Its minimal center V is a fourfold. With $\bar{V} = f(V)$ and $m = \text{mult}_y V$, one has $m \in \{2, 3\}$ and $L^4 \cdot \bar{V} = 1$.
- (ii) For this prime V , put

$$c = \text{coeff}_V D, \quad n = \frac{\text{ord}_V \mathcal{E}_k}{k}, \quad \epsilon = \frac{1}{65536},$$

and define, for $t \in \mathbb{R}$,

$$a_2 = \frac{6}{5}, \quad a_3 = \frac{79}{60}, \quad \chi_2(t) = \frac{59t}{320} - \frac{587}{1680}, \quad \chi_3(t) = \frac{289t}{960} - \frac{271}{480}.$$

Then

$$\begin{aligned} \frac{1}{2} < c < 1, \quad 0 < n < \frac{1053}{40}, \quad \tau_k k n = 1 - c, \\ \chi_m(b) + \epsilon > 0, \quad n \geq \frac{(64a_m - 65)(1 - c)}{\chi_m(b) + \epsilon}. \end{aligned}$$

- (iii) The prime $\bar{V}$ is an actual fixed prime of $\mathcal{I}_j$, with fixed order exactly $\nu c > \nu/2$.

The fixed prime is shared by the complete intrinsic source and the actual first normal restriction image. This assertion keeps the line bundle of that image and the full coefficient of the residual boundary.

Proof. We first eliminate the intrinsic centers whose entire tails fit the budget. A bound for the full intrinsic fixed divisor then forces a large residual coefficient at each remaining prime.

Step 1. In this step, we bound the original coefficient and deficit before choosing the intrinsic perturbation. Proposition 13.1 gives $L^6 = L^5 \cdot W = 1$ and $\nu < 22j$. Let τ be the threshold of U_j . Since W is the divisorial center of the pure average, its coefficient gives $\tau\nu = 1$. Thus

$$\Gamma = \frac{1}{\nu N} \sum_{i=1}^N H_i = W + R, \quad s = \frac{65j}{\nu}, \quad \text{mult}_x \Gamma = \frac{\text{ord}_x U_j}{\nu} \geq \frac{64j}{\nu}.$$

In particular $s \leq \frac{65}{64} \text{mult}_x \Gamma$, and (14.6) gives

$$b \leq 6 - \frac{64j}{\nu} < 6 - \frac{64}{22} = \frac{34}{11}.$$

Lemma 14.3 and Proposition 3.4 exclude $b \leq 15/8$.

Choose A sufficiently close to f^*A_0 . All cutting and transfer errors below will have total less than ϵ . This is possible uniformly in the intrinsic level: if a scalar total is below seven, its added coefficient is less than seven, so the ample-perturbation cost is at most 7δ . The conductor error is chosen after the constructed section.

Step 2. In this step, we reduce the intrinsic thresholds to degree-one fourfolds of multiplicity two or three. For any sufficiently late $\mathcal{E}_k$, Lemma 14.2 gives $s + v_k \leq 6\frac{65}{64} - \frac{65}{64}z_k$. A point, or any center covered by part (i) of Lemma 14.1, would give an ambient point boundary below seven and contradict Proposition 3.4. Thus V is a fourfold of multiplicity two or three.

The morphism f is an isomorphism near y , so V maps birationally to $\bar{V}$. Its integer degree $g = L^4 \cdot \bar{V}$ is positive, and the construction of A gives $A^4 \cdot V \geq g$. If $g \geq 2$, the degree refinements in the same lemma give, respectively,

$$s + v_k + C \leq 6\frac{65}{64} + \frac{2879}{3360} < 7 - \frac{1}{100},$$

or, using $z_k \leq 2$,

$$s + v_k + C \leq 6\frac{65}{64} + 2\left(\frac{10}{9} - \frac{65}{64}\right) + \frac{8}{15} < 7 - \frac{1}{10}.$$

These also contradict nongeneration after the prescribed errors. Hence $g = 1$.

Step 3. In this step, we bound the degree of the entire intrinsic fixed divisor and the normalized order of each remaining prime. Let $\mathcal{F}_k$ be the full fixed divisor of $\mathcal{E}_k$. Products of sections lie in $\mathcal{E}_{k+l}$. Choose one nonzero section at an allowed index k_0 , and one at a large representative of each residue class modulo k_0 on the permitted index lattice. Products of these sections show that all sufficiently late $\mathcal{F}_k$ have support in one finite set $\mathcal{S}$. For each $T \in \mathcal{S}$, subadditivity and the same residue classes give

$$n_T = \lim_k \frac{\text{ord}_T \mathcal{E}_k}{k} = \inf_{\mathcal{E}_k \neq 0} \frac{\text{ord}_T \mathcal{E}_k}{k}.$$

Indeed, for any fixed nonzero allowed index q , write late indices as $tq + d$. Subadditivity bounds the limsup by the normalized order at q , and taking its infimum gives the limit. This also includes earlier nonzero indices. Set $N_A = \sum_{T \in \mathcal{S}} n_T T$. Then $\mathcal{F}_k \geq kN_A$ and $\mathcal{F}_k/k \rightarrow N_A$ coefficientwise.

Every effective rational approximation $N' \leq N_A$ on this finite support divides the actual spaces in sufficiently divisible degrees. Their kernel estimate therefore gives $\text{vol}(65A - N') \geq 65^5 A^5 - 64^5$. Continuity of volume and Theorem 7.2 ([BFJ09, Theorems 2.16, 3.1, and 3.2]) give

$$\text{vol}(65A - N_A) \geq 65^5 A^5 - 64^5.$$

Put $V_A = A^5$. The concave function $t \mapsto \text{vol}(65A - tN_A)^{1/5}$ lies below its tangent at zero, whose slope is $-N_A \cdot A^4/V_A^{4/5}$, obtained by differentiating the ample intersection polynomial. Hence

$$(14.7) \quad N_A \cdot A^4 \leq 65V_A - V_A^{4/5}(65^5 V_A - 64^5)^{1/5}.$$

At $V_A = 1$, this is strictly less than $1053/40$, since $65 - 1053/40 = 1547/40$ and $(1547/40)^5 < 86548801$. To check the last comparison, expand $(77/2 + 7/40)^5$. Its six terms are respectively

$$84588000, \quad 1923000, \quad 17477, \quad 80, \quad 1, \quad 1;$$

their sum is $86528559 < 86548801$. Since $(f^* A_0)^5 = 1$, continuity allows A to be chosen so that the right side of (14.7) remains strictly below $1053/40$. Coefficientwise convergence on $\mathcal{S}$ then gives

$$\mathcal{F}_k \cdot A^4 < \frac{1053}{40} k$$

at every sufficiently late permitted level. All components have positive A -degree, and $A^4 \cdot V \geq 1$. Thus $n < 1053/40$, using the full fixed divisor.

Step 4. In this step, we use the exact retained coefficient to prove $c > 1/2$ and the stated lower bound for n . Klt of D gives $0 \leq c < 1$. At the prime V , the new boundary has coefficient one, and its actual general members have the exact base order. Therefore

$$(14.8) \quad \tau_k k n = 1 - c, \quad n > 0, \quad z_k \leq b - \frac{64(1 - c)}{n}.$$

Write $P_2 = 1871/3360$ and $P_3 = 41/120$. The two complete-tail bounds give

$$\begin{aligned} s + v_k + C &\leq s + a_m b + P_m - \frac{(64a_m - 65)(1 - c)}{n} \\ &\leq 7 + \chi_m(b) - \frac{(64a_m - 65)(1 - c)}{n}. \end{aligned}$$

If the last expression were below $7 - \epsilon$, the actual cuts and transfer would contradict nongeneration. Since its subtracted term is positive, this forces $\chi_m(b) + \epsilon > 0$ and the asserted lower bound for n .

At the upper endpoint for b , one has

$$\chi_2(34/11) = \frac{8149}{36960}, \quad \chi_3(34/11) = \frac{161}{440}.$$

For $m = 2$, $\chi_2(b) + \epsilon < 221/1000$, and

$$\frac{1053}{40} \frac{221}{1000} = \frac{232713}{40000} < \frac{59}{10} = \frac{64a_2 - 65}{2}.$$

For $m = 3$, the exact margins are

$$\frac{289}{30} - \frac{1053}{40} \frac{161}{440} = \frac{41}{52800}, \quad \frac{1053}{40} \epsilon = \frac{1053}{2621440} < \frac{41}{52800}.$$

Thus in both cases $n(\chi_m(b) + \epsilon) < (64a_m - 65)/2$. Combining this with the lower bound for n gives $1 - c < 1/2$.

Step 5. In this step, we identify the remaining prime as an actual fixed prime of the first normal image with order exactly νc . Division by σ_W^ν is defined for every section of U_j , and the restricted image $\mathcal{I}_j$ is nonzero by the definition of ν . Near the normal germ at x , remove its full divisorial fixed part. General members of the remaining linear system have reduced moving divisors at their generic points. One can check this on a pencil with no common moving prime: away from the denominator divisor, its fibers are levels of a nonconstant rational function. In characteristic zero the differential is nonzero, and a multiple fiber component must be one of its finitely many divisorial zeros. General levels avoid these components. There is nothing to check if the moving system has dimension zero. Having selected one general member, avoidance of each of its finitely many moving primes is an open condition on the next member. Thus a sufficiently general tuple has no moving prime shared by two members, and each moving prime in $R|_W$ has coefficient at most $1/(\nu N)$. The original sufficiently large averaging length has $N \geq 3$, so this is less than $49/100$ without changing the original Γ .

The prime V passes through y , so it is not exceptional for f . Its coefficient $c > 1/2$ cannot arise from those moving divisors. Hence $\bar{V}$ is fixed in $\mathcal{I}_j$. If its fixed order there is l , every general restricted member has order exactly l , so $c = l/\nu$. This proves $l = \nu c > \nu/2$ with the original image and its line bundle. It makes no assertion about higher normal images or an ambient fixed divisor supported on $\bar{V}$. $\square$

14.3. Fixed-prime amplification and the original endpoint.

Lemma 14.5. *Assume $L^6 = 1$. Let W be a prime divisor on X smooth at x . Suppose the limiting order of the original complete source along this prime satisfies*

$$n_W = \lim_{k \rightarrow \infty} \frac{\text{ord}_W U_k}{k} > 15.$$

Then

$$\text{vol}_Y(65\pi^*L - 75E) > 1418890625.$$

The estimate uses the initial linear equation of this fixed prime on every actual point-order layer below $75k$.

Proof. We obtain a supporting line for the fixed-prime order, bound each complete initial image, and integrate the resulting bound for all possible supporting slopes.

Step 1. In this step, we obtain a supporting line for the order along the fixed smooth prime. For $w = \text{ord}_W$, one has $A_X(w) = 1$. Proposition 7.9, with $h_0 = 15$, gives a finite slope $a > 15/64$ such that every actual section $h \in H^0(X, 65kL)$ of point order ℓ satisfies

$$(14.9) \quad \text{ord}_W h \geq k \max\{0, 15 + a(\ell/k - 64)\}.$$

The hypothesis concerns this single prime in the complete source. The supporting inequality holds on both sides of $64k$, and there is no upper restriction on a .

Step 2. In this step, we bound every complete initial image by its forced power of the prime's linear tangent equation. For integers $k \geq 1$ and $\ell \geq 0$, put

$$\mathcal{W}_{k,\ell} = H^0(X, \mathcal{O}_X(65kL) \otimes \mathfrak{m}_x^\ell).$$

The quotient $\mathcal{W}_{k,\ell}/\mathcal{W}_{k,\ell+1}$ injects into the degree- ℓ part of the polynomial ring $\mathrm{gr}_{\mathfrak{m}_x}\mathcal{O}_{X,x}$ in six variables. Because W is smooth at x , its local equation has a nonzero linear initial form. Every initial form in the actual image is divisible by this linear form to at least the power

$$r_{k,\ell} = \lceil k \max\{0, 15 + a(\ell/k - 64)\} \rceil.$$

The image is zero if $r_{k,\ell} > \ell$. Otherwise its dimension is at most $\binom{\ell - r_{k,\ell} + 5}{5}$. Define

$$g_a(t) = \max\{0, t - \max\{0, 15 + a(t - 64)\}\}.$$

Uniformly for $0 \leq \ell < 75k$, the binomial estimate yields

$$\dim(\mathcal{W}_{k,\ell}/\mathcal{W}_{k,\ell+1}) \leq \frac{k^5}{120} g_a(\ell/k)^5 + O(k^4).$$

The error is uniform because rounding changes the argument of a degree-five binomial polynomial by a bounded amount, and its nonnegative argument is at most $75k$. Summing all these layers and using the Riemann sums of the continuous piecewise linear function g_a , we obtain

$$(14.10) \quad \limsup_{k \rightarrow \infty} \frac{720}{k^6} (h^0(X, 65kL) - \dim \mathcal{W}_{k,75k}) \leq I(a) := 6 \int_0^{75} g_a(t)^5 dt.$$

Step 3. In this step, we bound the integrated loss for supporting slopes between $15/64$ and one. For $t \leq 64$, the supporting line is at least $t - 49$, so $g_a(t) \leq \min\{t, 49\}$. For $t \geq 64$, it is at least $15t/64$, so $g_a(t) \leq 49t/64$. Therefore

$$\begin{aligned} I(a) &\leq 49^6 + 6(64 - 49)49^5 + \left(\frac{49}{64}\right)^5 (75^6 - 64^6) \\ &= 139 \cdot 49^5 + \left(\frac{49}{64}\right)^5 (75^6 - 64^6). \end{aligned}$$

Use

$$49^5 = 282475249 < 283000000, \quad (49/64)^5 < (77/100)^5 < 271/1000, \quad 75^6 - 64^6 < 110000000000.$$

This gives $I(a) < 39337000000 + 29810000000 = 69147000000 < 74000000000$.

Step 4. In this step, we bound the integrated loss for supporting slopes between one and two. Below 64 , the supporting line is at least $2t - 113$. Hence $g_a(t) \leq t$ up to $113/2$, and $g_a(t) \leq 113 - t$ from $113/2$ to 64 . Above 64 , the line is at least $t - 49$, so $g_a(t) \leq 49$. Consequently

$$I(a) \leq 2(113/2)^6 - 49^6 + 66 \cdot 49^5 = 2(113/2)^6 + 17 \cdot 49^5.$$

Since $113/2 < 57$, $57^6 = 34296447249 < 34300000000$, and $49^5 < 283000000$, it follows that $I(a) < 68600000000 + 4811000000 = 73411000000 < 74000000000$.

Step 5. In this step, we bound the integrated loss for every supporting slope at least two. Set $l = 64 - 15/a > 0$. Then $g_a(t) = t$ for $0 \leq t \leq l$, and $g_a(t) = \max\{0, al - (a - 1)t\}$ for $t \geq l$. Extend the nonnegative integral to infinity to get

$$I(a) \leq l^6 + \frac{l^6}{a - 1} = 64^6 \frac{(1 - (15/64)u)^6}{1 - u}, \quad u = 1/a \in (0, 1/2].$$

The function $F(u) = (1 - (15/64)u)^6$ is convex on this interval, with $F(0) = 1$ and

$$F(1/2) = (113/128)^6 < (8/9)^6 < 1/2.$$

The final comparison is $524288 < 531441$. Its chord therefore gives $F(u) < 1 - u$ for $0 < u \leq 1/2$. Thus $I(a) < 64^6 = 68719476736 < 74000000000$.

Step 6. In this step, we subtract the loss from the complete Hilbert growth to prove the strict order-75 volume bound. The three intervals exhaust all supporting slopes. Since $L^6 = 1$, Riemann–Roch gives $720h^0(X, 65kL)/k^6 \rightarrow 65^6$. Equation (14.10) implies

$$\liminf_{k \rightarrow \infty} \frac{720 \dim \mathcal{W}_{k,75k}}{k^6} \geq 65^6 - I(a)$$

$$> 75418890625 - 74000000000 = 1418890625.$$

These are exactly the complete sections of $k(65\pi^*L - 75E)$ on Y . Their volume has the asserted lower bound; no choice of a subsequence is needed. $\square$

Theorem 14.6. *Assume $L^6 = 1$. There is an integer J , depending only on X, L, x , such that if at one $j \geq J$ a pure threshold average Γ from U_j has a fivefold minimal center smooth at x , then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/16384$. If $\text{def}_x(\Gamma) > 187/100$, then*

$$\text{vol}_Y(65\pi^*L - 75E) > 1418890625.$$

A single occurrence at a sufficiently late level suffices. The point boundary in the large-deficit case comes from a stronger complete source.

Proof. We choose the cutoff before the center is selected, pay the small-deficit case intrinsically, and apply the fixed-prime volume bound in the remaining case.

Step 1. In this step, we choose a cutoff controlling every limiting fixed coefficient before selecting the center. The actual point-jet kernel estimate gives

$$\dim U_k \geq \frac{65^6 - 64^6}{720} k^6 + O(k^5), \quad 65^6 - 64^6 = 6699413889 > 0.$$

Thus $65\pi^*L - 64E$ is big, and Proposition 7.5 applies to this complete family. There is a finite set $\mathcal{S}_0$ containing every sufficiently late fixed prime. For each $T \in \mathcal{S}_0$, put $r_T(k) = \text{ord}_T U_k$ and $n_T = \lim_k r_T(k)/k$. Choose J , depending only on X, L, x , so that the spaces are nonzero, their fixed support lies in $\mathcal{S}_0$, and, simultaneously for all $T \in \mathcal{S}_0$,

$$(14.11) \quad \left| \frac{r_T(j)}{j} - n_T \right| < \frac{1}{8} \quad (j \geq J).$$

This uses the general complete-source convergence and no nongeneration assumption.

Step 2. In this step, we record the full pure-threshold data and construct the small-deficit continuation. Let W be the center in the statement, let $\tau = \text{lct}_x(X; \mathfrak{q}_j)$, and put $\nu = \text{ord}_W U_j$, $p = \text{ord}_x U_j$, $s = 65j\tau$, $M = \text{mult}_x \Gamma$, and $b = \text{def}_x(\Gamma)$. The divisorial coefficient of the actual average is one, so $\tau\nu = 1$. In particular $\nu > 0$, and W belongs to $\mathcal{S}_0$. The pure source formulas in Proposition 7.6 give

$$s = \frac{65j}{\nu}, \quad M = \frac{p}{\nu}, \quad p \geq 64j, \quad s \leq \frac{65}{64}M, \quad M + b \leq 6.$$

If $b \leq 187/100$, apply Lemma 14.3 with $e = 0$. This gives the claimed point boundary as an additive continuation.

Step 3. In this step, we obtain the large-deficit volume bound and construct the point boundary from the stronger complete source. Suppose $b > 187/100$. Then $M < 413/100$, and

$$\frac{\nu}{j} \geq \frac{64}{M} > \frac{6400}{413} > \frac{61}{4}.$$

The last comparison is $25600 > 25193$. Equation (14.11) now gives

$$n_W > \frac{61}{4} - \frac{1}{8} = \frac{121}{8} > 15.$$

Lemma 14.5 proves the volume assertion in the statement. Since $371/5 < 75$, inclusion of the complete point-order spaces implies

$$\text{vol}_Y \left(65\pi^*L - \frac{371}{5}E \right) \geq \text{vol}_Y(65\pi^*L - 75E) > 1418890625 > 336000000.$$

Theorem 10.5 therefore supplies a point boundary with $v < 7 - 1/4096 < 7 - 1/16384$.

The complete ambient threefold tails used in that theorem include the multiplicity-eight and multiplicity-nine cases of Lemma 5.1. Their bounds are $C \leq 2z$ for $z \leq 1/4$ and $C \leq 9z/4$

for $z \leq 1/9$, including direct jumps. In the two ambient envelopes, with $\gamma = 2301/2000$, their contributions satisfy

$$\begin{array}{c|cc} & m=8 & m=9 \\ \hline C - \gamma z & \leq 1699/8000 & \leq 733/6000 \\ C - (9/10)z & \leq 11/40 & \leq 3/20 \end{array}$$

The first row is strictly below $3447/4000$, and the second strictly below $100663/60000$. The intrinsic enumeration in Lemma 14.1 has ambient dimension five and proves $m \leq 6$ directly. $\square$

15. ORIGINAL CUBIC FOURFOLD CENTERS

We retain the notation of Section 2. We first bound the complete cutting chains below a cubic fourfold. We then derive the geometry of its full tangent scheme from the failure of these bounds to give a point boundary. In embedding dimension five we use the complete-source profile of Section 7. In embedding dimension six we construct the complete sources of orders 69 and 75.

We use the rational constants

$$\kappa = \frac{13161}{10000}, \quad p_0 = \frac{34087}{60000}, \quad p_5 = \frac{6839}{20000}.$$

Thus $\kappa > 3^{1/4}$ and $p_5 = (2 - \kappa)/2$. All costs below are the scalar costs of entire additive chains. Proposition 3.7 converts a strict scalar margin into an actual point boundary, with all cutting errors included.

15.1. Scalar bounds below a cubic fourfold.

Lemma 15.1. *Let an effective rational boundary on X be log canonical near x , with a threefold minimal center of multiplicity m and deficit $y > 0$. For $l = 1, \dots, 9$, define $\beta_l, \gamma_l, \delta_l$ by*

l	β_l	γ_l	δ_l
1	3	1	1/2
2	2	19/15	29/120
3	5/3	13/9	11/54
4	3/2	127/80	33/160
5	1	7/4	1/20
6	2/3	2	0
7	3/7	2	0
8	1/4	2	0
9	1/9	9/4	0

Then $m \leq 9$, $y \leq \beta_m$, and every additive cutting chain from this boundary to the point satisfies $C \leq \gamma_m y + \delta_m$.

The last two rows include chains that jump directly to a curve or to the point. They will also distinguish this unrestricted list from the shorter lists forced by a persistent fourfold parent.

Proof. We bound the possible surfaces after the threefold and then include their complete curve tails.

Step 1. In this step, we bound the possible threefold multiplicities and their deficits. Theorem 4.5 [dFH09, Corollary 7.12] makes these local rings normal and Cohen–Macaulay. They are excellent, hence analytically unramified, and have infinite residue field $\mathbb{C}$. Theorem 4.6 [Han26, Lemmas 4.2 and 4.4] and Proposition 4.10 therefore apply. If h_i is the Hilbert function of its Artinian reduction, then

$$h_0 = 1, \quad h_1 \leq 3, \quad \sum_i h_i = m, \quad \bar{e}_1 \geq \sum_i i h_i \geq \max\{m - 1, 2m - 5\}.$$

Here the reduction uses a system of parameters forming a regular sequence in the Cohen–Macaulay local ring. Consequently

$$y \leq \min \left\{ 1 + \frac{2}{m}, \frac{10}{m} - 1 \right\}.$$

For $m = 1$, the ring is regular and $y \leq 3$. These inequalities give all the displayed deficit bounds and exclude $m \geq 10$.

Step 2. In this step, we bound the complete surface surplus using the persistent threefold and inherited embedding dimension. An earlier minimal center remains an lc center after every effective addition preserving log canonicity: an old place of discrepancy zero must have order zero on the addition. Thus Proposition 4.11 applies to the same threefold at every later surface. If that surface has multiplicity t and deficit $z > 0$, it gives

$$(15.1) \quad t \leq (3 - z)m - 2\bar{e}_1.$$

By Corollary 4.13, its full surface-and-curve tail is bounded by $q_t z$, where

$$(q_1, q_2, q_3, q_4, q_5) = \left(1, \frac{17}{12}, \frac{7}{4}, 2, \frac{9}{4} \right), \quad z \leq \frac{2}{t}.$$

The square of each q_t is at least t . Similarly $\gamma_m^3 \geq m$. Thus the surplus over $\gamma_m y$ is bounded by the largest allowed value of $(q_t - \gamma_m)_+ z$; a direct curve or point adds no surplus. For the embedding-dimension restrictions used below, let a_i be the Artinian Hilbert function of a surface of multiplicity t and positive deficit z . Then

$$t - 1 \leq \sum_i i a_i \leq \frac{(2 - z)t}{2} < t.$$

Its integral moment is $t - 1$, so every a_i for $i \geq 2$ vanishes. Hence that surface has embedding dimension $t + 1$. Embedding dimensions cannot increase under passage to a contained center, since its local ring is a quotient.

Step 3. In this step, we compute the complete-tail constants for multiplicities one through seven. For $m = 1$, the threefold is smooth. Its surface descendants have embedding dimension at most three and multiplicity at most two. Their surplus over z is at most $5/12 < 1/2$. For $m = 2$, the embedding dimension is at most four, so $t \leq 3$, and (15.1) gives $t \leq 4 - 2z$. The positive surpluses for $t = 2, 3$ are at most $3/20, 29/120$. For $m = 3$, the embedding dimension is at most five, so $t \leq 4$ and $t \leq 5 - 3z$. The positive surpluses for $t = 3, 4$ are at most $11/54, 5/27$.

For $m = 4$, the bound is $t \leq 6 - 4z$. At $t = 3, 4, 5$, the corresponding caps on z are $2/3, 1/2, 1/4$, giving surpluses $13/120, 33/160, 53/320$. For $m = 5$, the bound is $t \leq 5 - 5z$. Only $t = 4$ has radius above $7/4$, and then $z \leq 1/5$, giving surplus $1/20$. For $m = 6, 7$, the bounds are $t \leq 4 - 6z$ and $t \leq 3 - 7z$. Every allowed surface radius is at most two.

Step 4. In this step, we include both large-multiplicity threefold tails and their direct jumps. For $m = 8, 9$, Lemma 5.1 gives the stated complete tails. More explicitly, at $m = 8$ one has $\bar{e}_1 \geq 11$ and $t \leq 2 - 8z < 2$, so every surface successor is smooth and all radii are at most two. At $m = 9$, one has $\bar{e}_1 \geq 13$ and $t \leq 1 - 9z < 1$, so no positive-deficit surface occurs; $9^{1/3} < 9/4$ bounds every remaining radius. Nonincreasing deficits give the displayed full costs, including direct jumps. This proves every row. $\square$

Lemma 15.2. *Let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be an effective rational boundary log canonical near x , with a fourfold minimal center W of multiplicity three. Put $b = \text{def}_x(B)$ and $g = L^4 \cdot W$. Then the following hold for every additive cutting chain to the point:*

- (i) *One has $b \leq 8/3$ and $C \leq \kappa b + p_0$.*
- (ii) *If $\text{edim}_x W = 5$, then $b \leq 2$ and $C \leq \kappa b + p_5$.*
- (iii) *If $\text{edim}_x W = 5$ and $g \geq 2$, then $C \leq (10/9)b + 29/54$.*

The improved intercept in (ii) uses the embedding dimension of every descendant, as well as the stronger adjacent-center bound from the original fourfold.

Proof. We compare each possible first lower center with the fourfold radius.

Step 1. In this step, we bound the possible surface surplus relative to the fourfold radius. For any $1 \leq a \leq 9/4$, the surface bounds in the preceding proof give

$$(15.2) \quad \max_{1 \leq t \leq 5} \frac{2}{t} (q_t - a)_+ \leq \frac{9}{10} - \frac{2a}{5}.$$

For $t = 2, 3, 4$, the difference between the right side and the untruncated expression on the left is, respectively,

$$\frac{3a}{5} - \frac{31}{60}, \quad \frac{4(a-1)}{15}, \quad \frac{a-1}{10}.$$

These numbers are nonnegative; $t = 5$ is equality and $t = 1$ contributes zero. The right side is nonnegative throughout the stated interval.

Step 2. In this step, we prove the general cubic-fourfold bound using its persistent-parent restriction. The Artinian moment is at least two, so $\bar{e}_1(\mathfrak{m}_{W,x}) \geq 2$ and $b \leq 8/3$. At a later threefold of multiplicity m and deficit $y > 0$, persistence of W and Proposition 4.11 give

$$(15.3) \quad m \leq 8 - 3y, \quad y \leq \min\{\beta_m, (8 - m)/3\}.$$

In particular $m \leq 7$. Thus the multiplicities eight and nine in Lemma 15.1 cannot occur below this parent. If the threefold radius is at most κ , use (15.2) at $a = \kappa$. For the other rows, Lemma 15.1 and (15.3) bound the surplus over κb by one of

$$\left(\frac{13}{9} - \kappa\right) \frac{5}{3} + \frac{11}{54}, \quad \left(\frac{127}{80} - \kappa\right) \frac{4}{3} + \frac{33}{160}, \quad \frac{9}{5} - \kappa, \quad \frac{2}{3}(2 - \kappa), \quad \frac{1}{3}(2 - \kappa).$$

The second expression is p_0 . Each other expression, and the direct-surface bound $9/10 - 2\kappa/5$, is less than $1/2 < p_0$. This proves (i), including chains with no threefold or no surface.

Step 3. In this step, we sharpen the cubic-fourfold intercept when every descendant has embedding dimension at most five. The Artinian reduction of W has Hilbert function $(1, 1, 1)$. Hence $\bar{e}_1(\mathfrak{m}_{W,x}) \geq 3$, $b \leq 2$, and the persistent-parent inequality improves to

$$(15.4) \quad m \leq 6 - 3y.$$

Every descendant has embedding dimension at most five. A surface therefore has multiplicity at most four, by the surface moment calculation. Its surplus over κz is at most $(2 - \kappa)/2 = p_5$.

For a smooth threefold, all subsequent radii are at most $17/12$. For threefold multiplicities two, three and five we use Lemma 15.1. A positive-deficit threefold of multiplicity four and embedding dimension at most four would have Artinian moment six, contradicting $2\bar{e}_1 \leq (3 - y)4$. It consequently has embedding dimension five, and Proposition 5.6(ii) gives $C_3 \leq (127/80)y + 7/192$. Using (15.4), the five threefold surpluses over κy are bounded by

$$\left(\frac{17}{12} - \kappa\right) \frac{5}{3}, \quad \frac{29}{120}, \quad \frac{89}{54} - \kappa, \quad \left(\frac{127}{80} - \kappa\right) \frac{2}{3} + \frac{7}{192}, \quad \frac{7/4 - \kappa}{3} + \frac{1}{20}.$$

Each is less than p_5 . For the third, the comparison is $\kappa > 35/27$; the other four follow by substituting $\kappa = 13161/10000$. This proves (ii).

Step 4. In this step, we obtain the degree-sensitive cubic-fourfold bound in embedding dimension five. If $g \geq 2$, the parent radius is less than $a = 10/9$, since $(10/9)^4 > 3/2$. The direct-surface surplus is at most $4/9$. The smooth, double and triple threefold surpluses are, respectively,

$$\left(\frac{17}{12} - \frac{10}{9}\right) \frac{5}{3} = \frac{55}{108}, \quad \left(\frac{19}{15} - \frac{10}{9}\right) \frac{4}{3} + \frac{29}{120} = \frac{97}{216}, \quad \frac{13}{9} - \frac{10}{9} + \frac{11}{54} = \frac{29}{54}.$$

For multiplicities four and five the bounds are

$$\left(\frac{127}{80} - \frac{10}{9}\right) \frac{2}{3} + \frac{7}{192}, \quad \left(\frac{7}{4} - \frac{10}{9}\right) \frac{1}{3} + \frac{1}{20},$$

both less than $29/54$. This proves (iii). $\square$

Lemma 15.3. *Let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be an effective rational boundary log canonical near x , with a fourfold minimal center W of multiplicity three, deficit b , and point multiplicity M . Suppose $s \leq (65/64)M + e$ for a rational $e < 1/320$. Then the following hold:*

- (i) *If $b \leq 6/5$, the entire B has an additive point continuation of total coefficient less than $7 - 1/128$.*
- (ii) *If $\text{edim}_x W = 5$ and either $b \leq 37/20$ or $L^4 \cdot W \geq 2$, the entire B has an additive point continuation of total coefficient less than $7 - 1/512$.*

These conclusions retain the whole initial boundary. No restriction on the dimension of its first successor is needed.

Proof. The point blow-up gives $M + b \leq 6$, so $s \leq (65/64)(6 - b) + e$.

Step 1. In this step, we construct the additive point continuation for $b \leq 6/5$. If $b \leq 1$, Corollary 5.7 proves (i). Suppose $1 < b \leq 6/5$. For a direct surface, use (15.2); for a threefold use its row in Lemma 15.1 with (15.3). The bounds for $s + C - e$, with the first successor indicated, are

$$\begin{array}{l|l} \dim Z \leq 2 & 6\frac{65}{64} + \left(\kappa - \frac{65}{64}\right) \frac{6}{5} + \frac{9}{10} - \frac{2\kappa}{5} < \frac{69}{10} \\ \dim Z = 3, m = 1, 2 & 6\frac{65}{64} + \left(\kappa - \frac{65}{64}\right) \frac{6}{5} + \frac{1}{2} < \frac{174}{25} \\ \dim Z = 3, m = 3 & 6\frac{65}{64} + \left(\frac{13}{9} - \frac{65}{64}\right) \frac{6}{5} + \frac{11}{54} < \frac{69}{10} \\ \dim Z = 3, m = 4 & 6\frac{65}{64} + \left(\frac{127}{80} - \frac{65}{64}\right) \frac{6}{5} + \frac{33}{160} = 7 - \frac{11}{800} \\ \dim Z = 3, m = 5 & 6\frac{65}{64} + \left(\kappa - \frac{65}{64}\right) \frac{6}{5} + \frac{7}{4} - \kappa + \frac{1}{20} < \frac{139}{20} \\ \dim Z = 3, m = 6 & 6\frac{65}{64} + \left(\kappa - \frac{65}{64}\right) \frac{6}{5} + \frac{2}{3}(2 - \kappa) < \frac{139}{20}. \end{array}$$

For $m = 7$, the last surplus is replaced by $(2 - \kappa)/3$, which is smaller. The largest bound is $7 - 11/800$. Choose the total cutting error less than $1/10000$. Proposition 3.7 then gives an actual final coefficient less than

$$7 - \frac{11}{800} + \frac{1}{320} + \frac{1}{10000} < 7 - \frac{1}{128}.$$

Step 2. In this step, we construct the continuation under either additional hypothesis in embedding dimension five. For (ii), Lemma 15.2(ii) gives, when $b \leq 37/20$,

$$s + C < 6\frac{65}{64} + \left(\kappa - \frac{65}{64}\right) \frac{37}{20} + p_5 + \frac{1}{320} = \frac{699470375}{100000000}.$$

If instead $L^4 \cdot W \geq 2$, its part (iii), with $b \leq 2$, gives

$$s + C < 4\frac{65}{64} + \frac{20}{9} + \frac{29}{54} + \frac{1}{320} < \frac{683}{100}.$$

Either bound, with a total cutting error less than $1/100000$, is below $7 - 1/512$. $\square$

15.2. The full tangent scheme.

Lemma 15.4. *Let B be an effective rational boundary on X , log canonical near x , with a fourfold minimal center W of multiplicity three. Set*

$$b = \text{def}_x(B), \quad M = \text{mult}_x B, \quad \sigma = 6 - M - b.$$

Suppose $b > 1$ and $0 \leq \sigma < 1/2$. Then the ordinary point blow-up $W^+ = \text{Bl}_x W$ is normal and has rational singularities near its exceptional Cartier scheme. The full ring $G = \text{gr}_{\mathfrak{m}_{W,x}} \mathcal{O}_{W,x}$ is reduced and Cohen–Macaulay. Moreover:

- (i) If $\text{edim}_x W = 5$, then $G \simeq \mathbb{C}[z_1, \dots, z_5]/(f_3)$, where f_3 is a nonzero homogeneous cubic.
- (ii) If $\text{edim}_x W = 6$, then the ideal of W in $\mathcal{O}_{X,x}$ is contained in $\mathfrak{m}_x^2$ and G has Hilbert series $(1 + 2z)/(1 - z)^4$.

Reducedness here concerns the entire tangent scheme, with every component retained. Neither integrality nor smoothness of its components is asserted.

Proof. We first identify the ordinary blow-up as a minimal center of an effective pair. We then use its discriminant to control every exceptional multiplicity.

Step 1. In this step, we identify the ordinary blow-up as a minimal center of the effective transformed pair. Let B^{str} be the strict transform on $\pi : Y \rightarrow X$, and put $e_X = b + \sigma - 1 = 5 - M > 0$. The point blow-up formula is

$$K_Y + B^{\text{str}} = \pi^*(K_X + B) + e_X E.$$

For every rational $0 \leq \theta < 1 - \sigma$, a divisorial valuation w with $w(\mathfrak{m}_x) > 0$ has discrepancy

$$A_{X,B}(w) - (e_X + \theta)w(\mathfrak{m}_x) > 0,$$

since $A_{X,B}(w) \geq bw(\mathfrak{m}_x)$ and $e_X + \theta < b$. If $w(\mathfrak{m}_x) = 0$, its discrepancy is unchanged. Thus $(Y, B^{\text{str}} + \theta E)$ is log canonical near the exceptional fiber.

The strict transform of W is still an lc center, since π is an isomorphism at its generic point. At any point of this transform over x , a smaller lc center cannot map to x , by the strict discrepancy inequality. A smaller center with positive-dimensional image would give an old lc center properly inside W and through x . This contradicts its original minimality. The intersection statement of Theorem 3.1 [GL24, Lemma 2.7] shows that the strict transform is minimal at every such point.

The quotient map on point Rees algebras embeds the ordinary $\text{Bl}_x W$ in Y . Its Rees algebra is a domain, so its image is precisely the strict transform. Applying Theorem 4.5 to the effective pair (Y, B^{str}) proves that this ordinary blow-up W^+ is normal and has rational singularities near its exceptional scheme

$$E_W = E|_{W^+} = \text{Proj } G.$$

In particular W^+ is Cohen–Macaulay and E_W is Cartier.

Step 2. In this step, we use the effective discriminant to prove reducedness of every exceptional component. Apply Theorem 4.1 [FH23, Theorem 1.2 and Definition 1.3] to the new effective pair (Y, B^{str}) and its normal center W^+ . Write $E_W = \sum_T r_T T$, with positive integer Cartier multiplicities. The base discriminant trace is effective. Since adding θE preserves log canonicity, its generic threshold definition, or Proposition 4.3, gives

$$0 \leq \text{coeff}_T \mathcal{B}_{W^+} \leq 1 - \theta r_T \quad (0 \leq \theta < 1 - \sigma).$$

Choose a rational $\theta > 1/2$ in this interval. Then $r_T = 1$ for every component. The Cartier scheme E_W is Cohen–Macaulay and generically reduced, hence reduced.

Step 3. In this step, we determine the full tangent ring in both embedding dimensions and prove its reducedness. The original local ring of W is normal and Cohen–Macaulay. Its length-three Artinian reduction bounds its embedding dimension by six. Embedding dimension four would give a regular ring of multiplicity one. Thus the embedding dimension is five or six. If its embedding dimension is five, its completion is a hypersurface $\mathbb{C}[[z_1, \dots, z_5]]/(f)$, with $\text{ord } f = 3$. Its full tangent ring is consequently $\mathbb{C}[z_1, \dots, z_5]/(f_3)$, which is Cohen–Macaulay. If its embedding dimension is six, it has minimal multiplicity. Theorem 4.14 [Ver08, Theorem 2.7] gives Cohen–Macaulayness of the full associated graded ring. Its Artinian reduction has Hilbert function $(1, 2)$, so its Hilbert series is $(1 + 2z)/(1 - z)^4$. Embedding dimension six also says that its ambient ideal has no linear initial equation.

In both cases G is Cohen–Macaulay of dimension four, with no embedded associated prime. All its homogeneous minimal primes are detected by the reduced scheme $\text{Proj } G = E_W$. Thus G is generically reduced at every minimal prime. A nonzero nilradical would have an associated prime among these primes, which is impossible. This proves reducedness of the full ring, including at its vertex. $\square$

15.3. The centers of the order-69 source.

Lemma 15.5. *Let $B \sim_{\mathbb{Q}} sL$, $s \geq 0$, be an effective rational boundary log canonical near x , with a fourfold minimal center of multiplicity $m \geq 4$. Suppose $s \leq (65/69) \text{mult}_x B$. Then the entire B has an additive point continuation of total coefficient less than $7 - 1/16384$.*

This estimate applies directly to the boundary selected from the order-69 source. Its proof uses that boundary’s Hilbert data and its own coefficient.

Proof. Put $b = \text{def}_x(B)$ and $q = 65/69$.

Step 1. In this step, we reduce to fourfold multiplicities four, five, and six in embedding dimension six. If $b \leq 1$, use Corollary 5.7, since $q < 65/64$. Suppose $b > 1$. For a fourfold in X , the Artinian Hilbert function has $h_1 \leq 2$. Theorem 4.6 gives

$$\bar{e}_1 \geq 2m - 4, \quad b \leq \frac{8}{m}.$$

Thus $m \leq 7$. At $m = 7$, the bound $h_2 \leq 3$ improves the moment to $3m - 10 = 11$, giving $b \leq 6/7$. Hence $m = 4, 5, 6$. Embedding dimension at most five would give an Artinian string of m ones, and then $b \leq 5 - m \leq 1$. Thus the current center has embedding dimension six.

Step 2. In this step, we bound the complete tails using the restrictions imposed by the persistent fourfold. At any subsequent threefold of multiplicity l and deficit $y > 0$, Proposition 4.11 gives

$$l \leq (4 - y)m - 2\bar{e}_1 \leq 8 - my, \quad y \leq \min\{\beta_l, (8 - l)/m\}.$$

In particular $l \leq 7$; the larger rows of Lemma 15.1 are excluded by this same-parent inequality. Use the parent radius bounds

$$a_4 = \frac{17}{12}, \quad a_5 = \frac{3}{2}, \quad a_6 = \frac{8}{5}.$$

Their fourth powers dominate the corresponding multiplicities. For threefold radius at most a_m , use (15.2). For every other threefold, use its row in Lemma 15.1. The resulting surplus candidates, with the direct-surface bound first, are

m	surplus candidates	p'_m
4	1/3, 103/432, 181/480, 3/10, 7/24, 7/48	181/480
5	3/10, 221/800, 1/5, 1/5, 1/10	3/10
6	13/50, 1/8, 2/15, 1/15	13/50.

Each noninitial entry is $(\gamma_l - a_m) \min\{\beta_l, (8 - l)/m\} + \delta_l$ for a threefold row with radius greater than a_m . The last column is the maximum, so $C \leq a_m b + p'_m$.

Step 3. In this step, we pay the actual cuts inside the coefficient margin of the order-69 boundary. Since $a_m > q$, the point inequality and $b \leq 8/m$ give the following bounds for $s + C$:

$$\begin{array}{l|l} m = 4 & 4q + 2(17/12) + 181/480 = 7 - 79/3680 \\ m = 5 & (22/5)q + (8/5)(3/2) + 3/10 = 286/69 + 27/10 < 69/10 \\ m = 6 & (14/3)q + (4/3)(8/5) + 13/50 = 910/207 + 359/150 < 69/10. \end{array}$$

All three leave the asserted margin after a total cutting error less than $1/65536$, by Proposition 3.7. $\square$

Lemma 15.6. *Let k, a be positive integers and suppose $\mathcal{V} = H^0(X, 65kL \otimes \mathfrak{m}_x^a) \neq 0$. Let $S = \mathcal{O}_{X,x}$, and let $\mathfrak{p}$ be a height-two prime of S contained in $\mathfrak{m}_x^2$, with $\text{gr}_{\mathfrak{m}_x}(S/\mathfrak{p})$ reduced and Cohen–Macaulay. Let $\mathfrak{a}$ be the local base ideal of $\mathcal{V}$. Suppose*

$$0 < \lambda = \text{lct}_{S_{\mathfrak{p}}}(\mathfrak{a}S_{\mathfrak{p}}) < \Lambda.$$

Let C_4 be the constant in Proposition 6.18. For integers $n \geq a$, put $\mathcal{V}_n = H^0(X, 65kL \otimes \mathfrak{m}_x^n)$. If $n\Lambda \leq 11/2$, then

$$\dim(\mathcal{V}_n/\mathcal{V}_{n+1}) \leq \frac{(n - 2/\Lambda)_+^5}{120} + C_4(n+1)^4,$$

The estimate concerns the image of the actual complete space in the point-order layer. It includes every fixed component.

Proof. In the regular local surface $S_{\mathfrak{p}}$, choose an integer N large enough that $\lambda + 2/N < \Lambda$. By Theorem 11.1 [DK01, Theorem 2.9], the primary ideal

$$\mathfrak{a}S_{\mathfrak{p}} + (\mathfrak{p}S_{\mathfrak{p}})^N$$

has threshold $\lambda_N < \Lambda$. Its contraction I_N to S is $\mathfrak{p}$ -primary and contains all actual section germs from $\mathcal{V}_n$. Hence their degree- n initial image is contained in $(\text{in}_{\mathfrak{m}_x} I_N)_n$. Proposition 6.18 applies because $n\lambda_N < n\Lambda \leq 11/2$. Its bound, and $2/\lambda_N > 2/\Lambda$, give the assertion. Its error constant is independent of N , of the prime, and of the section space. $\square$

15.4. The original cubic source.

Theorem 15.7. *Assume $L^6 = 1$. There is an integer J , depending only on X, L, x , such that if at one $j \geq J$ a pure threshold average Γ from U_j has a fourfold minimal center W of multiplicity three, then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/16384$. If $\text{edim}_x W = 5$ and the entire Γ has no additive point continuation of total coefficient less than $7 - 1/16384$, then*

$$\text{vol}_Y(65\pi^*L - 75E) > 1418890625.$$

The point boundary may be constructed from a stronger complete source. The additional volume assertion requires failure of an additive continuation only in embedding dimension five.

Proof. We separate the two possible embedding dimensions. The first case gives the stated volume estimate by a uniform valuation comparison. In the second case we first obtain the complete order-69 source, examine all of its threshold centers, and obtain the complete order-75 source from each remaining case.

Put $\tau = \text{lct}_x(X; \mathfrak{q}_j)$, and write

$$\Gamma \sim_{\mathbb{Q}} sL, \quad M = \text{mult}_x \Gamma, \quad b = \text{def}_x(\Gamma), \quad \sigma = 6 - M - b.$$

Proposition 7.6 gives

$$(15.5) \quad s = 65j\tau, \quad M \geq 64j\tau, \quad s \leq \frac{65}{64}M, \quad b > 0, \quad \sigma \geq 0.$$

The local ring of W is Cohen–Macaulay. Its Artinian reduction has length three, so its embedding dimension is at most six. Embedding dimension four would make it regular and of multiplicity one. Thus it is five or six.

Step 1. In this step, we determine the tangent geometry forced by failure of an additive continuation in embedding dimension five. If the entire Γ has an additive point continuation below $7 - 1/16384$, use that continuation. Otherwise Lemma 15.3(ii) gives

$$g = L^4 \cdot W = 1, \quad \frac{37}{20} < b \leq 2.$$

If $\sigma \geq 1/25$, Lemma 15.2(ii) would give

$$s + C < 6\frac{65}{64} + 2\left(\kappa - \frac{65}{64}\right) + p_5 + \frac{1}{320} - \frac{65}{64} \frac{1}{25} = \frac{139983}{20000}.$$

Even after a total cutting error less than $1/100000$, this is below $7 - 1/16384$. Hence $\sigma < 1/25$. The moment of $(1, 1, 1)$ is three, and

$$3 \leq e_1(\mathfrak{m}_{W,x}) \leq \bar{e}_1(\mathfrak{m}_{W,x}) \leq \frac{3(4-b)}{2} < \frac{129}{40} < 4.$$

Both Hilbert coefficients are therefore three. Lemma 15.4 now shows that the full tangent is a reduced Cohen–Macaulay cubic hypersurface in five variables. Its linear normal equation in the ambient six-dimensional tangent space remains part of its ideal.

Step 2. In this step, we obtain the strict asymptotic valuation bound and the resulting volume and point boundary in embedding dimension five. Choose a divisorial lc place w of the original ideal pair with center W . Such a place is obtained on a log resolution of $\mathfrak{q}_j$: for a sufficiently large general tuple, the moving coefficients are less than one, and its lc strata are those of the threshold ideal pair. In particular

$$\tau w(\mathfrak{q}_j) = A_X(w).$$

This valuation is centered at the maximal ideal of the regular local surface $\mathcal{O}_{X,W}$. From (15.5) and $b > 37/20$,

$$(15.6) \quad \frac{w(\mathfrak{q}_j)}{jA_X(w)} = \frac{1}{j\tau} \geq \frac{64}{6-b} > \frac{1280}{83} > \frac{1260}{83}.$$

Let C_3 be the uniform constant in Proposition 7.8. Choose J larger than its cutoff and than $10C_3$. This choice is made from X, L, x , before the occurrence and its valuation. Combining it with (15.6) gives

$$\frac{\alpha(w)}{A_X(w)} \geq \frac{w(\mathfrak{q}_j)}{jA_X(w)} - \frac{C_3}{j} > \frac{1260}{83} - \frac{1}{10} > 15.$$

Proposition 7.10 applies to this height-two center. It uses every actual point-order layer from zero to $75k$, and the full profile for every possible supporting slope. It gives

$$\text{vol}_Y(65\pi^*L - 75E) > 1418890625.$$

This proves the additional assertion. Inclusion of complete spaces, since $371/5 < 75$, gives volume greater than 336000000 at order $371/5$. Theorem 10.5 supplies a point boundary with coefficient less than $7 - 1/4096$, as required.

Step 3. In this step, we use the original center in embedding dimension six to obtain the complete order-69 mass. Assume now that $\text{edim}_x W = 6$, and suppose for a contradiction that there is no point boundary with coefficient less than $7 - 1/16384$. By Lemma 15.3(i), one has $b > 6/5$. If $\sigma \geq 1/2$, then Lemma 15.2(i) gives

$$s + C < \frac{65}{64} \frac{11}{2} + \left(\kappa - \frac{65}{64} \right) \frac{8}{3} + p_0 + \frac{1}{320} < \frac{697}{100}.$$

An adaptive continuation contradicts the point exclusion. Thus $\sigma < 1/2$, and Lemma 15.4 gives a reduced Cohen–Macaulay full tangent with Hilbert series $(1 + 2z)/(1 - z)^4$. Its ambient ideal $\mathfrak{p}$ is contained in $\mathfrak{m}_x^2$.

The generic threshold of the full original ideal at $\mathfrak{p}$ is τ . Indeed localizing an lc ideal pair gives one inequality, and an lc place with center W gives the reverse inequality. Equation (15.5) gives the strict bound

$$\tau \leq \frac{6-b}{64j} < \frac{3}{40j} < \frac{1}{13j}.$$

For integers $\ell \geq 0$, put $\mathcal{U}_\ell = H^0(X, 65jL \otimes \mathfrak{m}_x^\ell)$. Lemma 15.6, with $\Lambda = 1/(13j)$, applies for $64j \leq \ell < 69j$, since $69/13 < 11/2$. It gives

$$\dim(\mathcal{U}_\ell/\mathcal{U}_{\ell+1}) \leq \frac{(\ell - 26j)^5}{120} + C_4(\ell + 1)^4.$$

Riemann–Roch and the length of the smooth point-jet quotient give

$$\frac{720 \dim U_j}{j^6} \geq 65^6 - 64^6 - O(j^{-1}) = 6699413889 - O(j^{-1}).$$

Summing all the layers from $64j$ to $69j - 1$, and bounding the sum of the increasing fifth powers by its integral, gives a constant K_0 , independent of the occurrence, such that

$$(15.7) \quad \frac{720 \dim V_{69}(j)}{j^6} \geq 6699413889 - (43^6 - 38^6) - \frac{K_0}{j} = 3388987224 - \frac{K_0}{j} > 3380000000.$$

Here $43^6 = 6321363049$ and $38^6 = 3010936384$. The error is uniform because the local capacity constant is independent of every primary approximation used in its proof.

Step 4. In this step, we bound the full order-69 fixed divisor with cutoffs chosen before the original occurrence. Proposition 7.5 applies to the fixed family $V_{69}(k)$. Choose its dimension-to-volume error below 5000000 for every $k \geq J$. Then (15.7) implies

$$\text{vol}_Y(65\pi^*L - 69E) > 3375000000, \quad \frac{720 \dim V_{69}(k)}{k^6} > 3370000000 \quad \text{for all sufficiently large } k.$$

Let N_{69} be its limiting full fixed divisor and put $\eta = \text{mult}_x N_{69}$. Our point exclusion implies $V_{92}(k) = 0$ for every positive integer k : an actual member would satisfy $92/65 > 140/99$, and Proposition 5.8 would give a point boundary below $7 - 1/630$.

For a nonzero actual section of point order $\ell \geq 69k$, Proposition 7.7 gives divisorial order at least $\ell N_{69}/69$. Taking point multiplicities shows $0 \leq \eta \leq 69$. Its full initial-image bound, summed through order $92k - 1$, yields

$$\text{vol}_Y(65\pi^*L - 69E) \leq (1 - \eta/69)^5(92^6 - 69^6).$$

If $\eta \geq 44$, the right side is less than

$$\frac{499000000000}{157} < 3370000000,$$

because $25/69 < 4/11$, $(4/11)^5 < 1/157$, and $92^6 - 69^6 < 499000000000$. This is a contradiction. Therefore $\eta < 44$. Coefficientwise convergence of the full fixed divisors gives

$$(15.8) \quad \text{mult}_x F_{69}(k) < 44k \quad \text{for all sufficiently large } k.$$

These cutoffs can all be chosen before the original occurrence. The order-69 volume is a number attached to X, L, x . If it is at most 3375000000, convergence already excludes (15.7) at every sufficiently large level. If it is larger, its fixed-divisor convergence and (15.8) have cutoffs attached to this same fixed family. Increase J to include them. Thus (15.8) holds at the single original j ; recurrence of its center is unnecessary.

Step 5. In this step, we treat all order-69 threshold centers except the remaining cubic fourfold and double fivefold. Take a pure threshold average $\widehat{\Gamma}$ of the actual nonzero space $V_{69}(j)$. Denote its threshold, coefficient, point multiplicity and deficit by $\widehat{\tau}, \widehat{s}, \widehat{M}, \widehat{b}$, and put $q = 65/69$, $\widehat{\sigma} = 6 - \widehat{M} - \widehat{b}$ when its center is positive-dimensional. On a log resolution of its full base ideal, choose enough general members that every moving coefficient is less than one. The resulting pair is lc and has the threshold ideal's lc strata. In particular its entire cost satisfies

$$(15.9) \quad \widehat{s} = 65j\widehat{\tau}, \quad \widehat{M} \geq 69j\widehat{\tau}, \quad \widehat{s} \leq q\widehat{M}, \quad \widehat{M} + \widehat{b} \leq 6.$$

If its minimal center is a point, its coefficient is at most $6q < 7 - 1/16384$, a contradiction. For the remaining centers reserve a total adaptive cutting error less than $1/65536$.

Curves, smooth surfaces and smooth fourfolds have embedding dimension at most four and are covered by Corollary 5.7. A singular surface has deficit at most one, so the same corollary covers it and, more generally, every center with $\widehat{b} \leq 1$. Every threefold is covered by Theorem 9.6 with zero $65/64$ excess. This is the unrestricted threefold endpoint: its exhaustive tails include

the multiplicity-eight and multiplicity-nine cases of Lemma 5.1, also displayed in Lemma 15.1. Every smooth fivefold is covered by Theorem 8.7, with zero 65/69 excess.

A double fourfold has embedding dimension at most five and $\widehat{b} \leq 3$. Proposition 5.5 and (15.9) give

$$\widehat{s} + C \leq 3q + 3 + \frac{3887}{3360} < 7 - \frac{1}{100}.$$

Every fourfold of multiplicity at least four is covered by Lemma 15.5.

For a singular fivefold center Z , let $\nu = \text{ord}_Z V_{69}(j)$, $n = \nu/j$, and $m_Z = \text{mult}_x Z$. Its coefficient in the threshold boundary is one, so

$$\widehat{\tau}\nu = 1, \quad \widehat{s} = \frac{65}{n}, \quad \widehat{M} \geq \frac{69}{n}, \quad m_Z n < 44.$$

The last inequality follows from the full fixed-divisor bound (15.8). If $m_Z \geq 4$, then $\widehat{M} > 276/44 > 6$, which is impossible. If $m_Z = 3$, then $\widehat{b} < 57/44 < 25/16$. The triple-fivefold bound of Proposition 5.5 gives

$$\widehat{s} + C \leq 6q + \left(\frac{5}{4} - q\right) \frac{25}{16} + \frac{57}{80} < \frac{69}{10}.$$

Finally suppose $m_Z = 2$. If $\nu \leq 69j/4$, then $\widehat{M} \geq 4$ and $\widehat{b} \leq 2$. Proposition 5.9 applies with zero excess and contradicts the point exclusion. Thus the only remaining fivefold case satisfies

$$(15.10) \quad \nu > \frac{69j}{4}.$$

The remaining centers are precisely a cubic fourfold or this double fivefold. Every coefficient used so far belongs to $\widehat{\Gamma}$ itself.

Step 6. In this step, we bound the order-75 layers using the tangent ring of the new cubic fourfold. Suppose the minimal center $\widehat{W}$ of $\widehat{\Gamma}$ is a cubic fourfold. Lemma 15.2(i) and (15.9) force

$$\widehat{b} > \frac{33}{16}, \quad \widehat{\sigma} < \frac{1}{4}.$$

For the first inequality, if $\widehat{b} \leq 33/16$, the enlarged scalar bound satisfies

$$q \frac{63}{16} + \kappa \frac{33}{16} + p_0 + \frac{1}{128} = 7 - \frac{829}{2208000}.$$

Its margin exceeds $1/16384$, even after the reserved cutting error. For the second, if $\widehat{\sigma} \geq 1/4$, use $\widehat{b} \leq 8/3$ to obtain

$$\widehat{s} + C \leq q \frac{37}{12} + \kappa \frac{8}{3} + p_0 < 7 - \frac{1}{256}.$$

This also contradicts the point exclusion.

The moment inequality now gives

$$2 \leq e_1(\mathbf{m}_{\widehat{W},x}) \leq \bar{e}_1(\mathbf{m}_{\widehat{W},x}) < \frac{93}{32} < 3,$$

so both coefficients are two. Embedding dimension five would give moment three. Thus the own cubic has embedding dimension six. Lemma 15.4 applies to this actual boundary and gives its reduced Cohen–Macaulay full tangent and an ambient ideal in $\mathbf{m}_x^2$. Its generic threshold equals $\widehat{\tau}$, by the same log-resolution argument as for the original source, and

$$\widehat{\tau} \leq \frac{6 - \widehat{b}}{69j} < \frac{63}{16 \cdot 69j} < \frac{4}{69j}.$$

Lemma 15.6, with $\Lambda = 4/(69j)$, bounds every actual layer $69j \leq \ell < 75j$ by

$$\frac{(\ell - 69j/2)^5}{120} + C_4(\ell + 1)^4,$$

since $75 \cdot 4/69 < 11/2$.

Step 7. In this step, we bound the order-75 layers using the full fixed factor of the new double fivefold. In the other remaining case, every actual member of $V_{69}(j)$ contains the whole fixed factor νZ . Its point order is $2\nu > 69j/2$, by (15.10). In the polynomial initial ring of the smooth ambient point, division by this fixed initial form bounds every layer $69j \leq \ell < 75j$ by

$$\binom{\lfloor \ell - 69j/2 \rfloor + 5}{5}.$$

This retains the factor with its full multiplicity. Its fifth-degree leading term is the same as in the cubic case, with a uniform $O(j^4)$ error per layer.

Step 8. In this step, we sum the layer bounds at the original level and apply the finite-source point-boundary criterion. In both remaining cases, summing the actual layers from $69j$ through $75j - 1$ and using (15.7) gives a fixed constant K_1 such that

$$\frac{720 \dim V_{75}(j)}{j^6} > 3380000000 - \frac{81^6 - 69^6}{64} - \frac{K_1}{j} > 650000000$$

for $j \geq J$, after increasing the uniform cutoff. Indeed

$$81^6 = 282429536481, \quad 69^6 = 107918163081, \quad \frac{81^6 - 69^6}{64} < 2727000000.$$

The remaining difference is greater than 653000000, so a uniform error smaller than 3000000 suffices. This is the same complete $V_{75}(j)$ at the original level, with normalized dimension greater than 600000000. Include the cutoff of Theorem 10.7 in J . That theorem gives a point boundary of coefficient less than $7 - 1/4096$, contradicting the point exclusion.

The embedding-dimension-five argument proves the additional volume assertion and its point endpoint. The embedding-dimension-six argument covers every center of the own order-69 source and proves its point endpoint. All constants defining J depend only on the fixed X, L, x , so these two cases prove the theorem with the required single cutoff. $\square$

16. ORIGINAL DOUBLE FOURFOLD CENTERS

We retain the notation of Section 2. We first bound the entire scalar cost below a double fourfold. This gives an additive point continuation when the deficit is at most $15/8$. A larger deficit gives a valuation estimate for the original complete source, to which the volume criterion of Section 7 applies.

Lemma 16.1. *Let B be an effective rational divisor on X , log canonical near x , with a four-dimensional minimal center W of multiplicity two. Put $b = \text{def}_x(B)$. Then $0 < b \leq 3$, and every additive cutting chain from B to the point satisfies*

$$C \leq \frac{6}{5}b + \frac{1871}{3360}.$$

The bound includes chains that pass directly from the fourfold to a surface, a curve, or the point.

Proof. We bound the possible successors using the Hilbert coefficients of their persisting parents, estimate each threefold tail, and compare the resulting costs with the slope $6/5$.

Step 1. In this step, we bound all successors using persistent centers, Hilbert moments, and inherited embedding dimension. Every positive-dimensional minimal center in the chain is normal and Cohen–Macaulay near x , by Theorem 4.5 [dFH09, Corollary 7.12]. Its local ring is excellent and normal, hence analytically unramified, and its residue field is $\mathbb{C}$. Theorem 4.6 [Han26, Lemmas 4.2 and 4.4] therefore applies to these rings. In particular, the Artinian reduction of $\mathcal{O}_{W,x}$ has length two and Hilbert function $(1, 1)$. Thus

$$\text{edim}_x W = 5, \quad \bar{e}_1(\mathfrak{m}_{W,x}) \geq 1.$$

Proposition 4.10 gives

$$0 < b, \quad 2 \leq 2\bar{e}_1(\mathbf{m}_{W,x}) \leq 2(4 - b),$$

so $b \leq 3$. Every later center is contained in W . The surjections of local rings show that its embedding dimension is at most five. Deficits are nonincreasing under effective addition.

An earlier lc center remains an lc center after each effective lc addition. Indeed, if F is a zero-discrepancy place for an earlier boundary B_i , then, for a later boundary B_j ,

$$0 \leq A_{X,B_j}(F) = A_{X,B_i}(F) - \text{ord}_F(B_j - B_i) = -\text{ord}_F(B_j - B_i) \leq 0.$$

Thus F remains a zero place with the same center. Applying Proposition 4.11 to the persisting normal fourfold W , a threefold successor T of multiplicity m and current deficit $y > 0$ satisfies

$$(16.1) \quad m \leq 2(4 - y) - 2\bar{e}_1(\mathbf{m}_{W,x}) \leq 6 - 2y < 6.$$

Consequently $1 \leq m \leq 5$. In particular, the multiplicity-eight and multiplicity-nine cases of Lemma 5.1 cannot occur in this local chain.

Let S be a surface in the chain, of multiplicity t and deficit $z > 0$. Write h_i for the Hilbert function of its Artinian reduction. The Hilbert moment and deficit inequalities give

$$\sum_i ih_i \leq \bar{e}_1(\mathbf{m}_{S,x}) \leq \frac{(2 - z)t}{2} < t, \quad \sum_i ih_i - t = -1 + \sum_{i \geq 2} (i - 1)h_i.$$

If any h_i with $i \geq 2$ were nonzero, the last sum would be at least one, contradicting the strict inequality. The Hilbert function is therefore $(1, h_1)$. It follows that

$$(16.2) \quad t = 1 + h_1 = \text{edim}_x S - 1 \leq 4, \quad z \leq \frac{2}{t}.$$

Every curve in the chain is smooth at x , by Corollary 4.13.

For every positive-dimensional integral center Z , the degree $g(Z)$ is a positive integer. Hence $r(Z) \leq \text{mult}_x(Z)^{1/\dim Z}$, and every curve radius is at most one. If C_S denotes the entire cost starting at the surface S , and ζ is the deficit of its curve successor (zero when that dimension is skipped), then

$$(16.3) \quad C_S \leq r(S)(z - \zeta) + \zeta \leq \max\{1, r(S)\}z \leq \sqrt{t}z.$$

This also covers a direct cut from the surface to the point.

Step 2. In this step, we compute the complete tails of the five possible threefold successors. For a threefold T in the chain, let C_T be the cost of the part of the same chain beginning at T . With m and y as above, we claim the following bounds:

$$(16.4) \quad \begin{array}{c|cc} m & \text{upper bound for } C_T & \text{deficit bound used} \\ \hline 1 & y + \frac{5}{12} \min\{y, 1\} & y \leq \frac{5}{2} \\ 2 & \frac{63}{50}y + \frac{661}{2800} & y \leq 2 \\ 3 & \frac{400}{127}y + \frac{4200}{7} & y \leq \frac{3}{2} \\ 4 & \frac{80}{7}y + \frac{192}{7} & y \leq 1 \\ 5 & \frac{7}{4}y & y \leq \frac{1}{2} \end{array}$$

For $m = 1$, the Hilbert function of the Artinian reduction is (1) , so T is smooth at x . Its descendants have embedding dimension at most three. Thus every surface descendant has multiplicity at most two by the argument giving (16.2). A double surface has $z \leq 1$ and radius less than $17/12$. If it occurs, then $z \leq y$, and telescoping gives

$$C_T \leq (y - z) + \frac{17}{12}z \leq y + \frac{5}{12} \min\{y, 1\}.$$

When it does not occur, all the radii are at most one and $C_T \leq y$.

For $m = 2$, the Artinian reduction has function $(1, 1)$. Thus $\text{edim}_x T = 4$ and $\bar{e}_1(\mathfrak{m}_{T,x}) \geq 1$. Its surface descendants have multiplicity at most three. At such a surface, the flag inequality for the persisting parent T gives

$$t \leq 2(3 - z) - 2 = 4 - 2z.$$

The deficit caps for a double and a triple surface are therefore 1 and $1/2$, respectively. Using the initial radius bound $2^{1/3} < 63/50$, and the surface bounds $\sqrt{2} < 17/12$, $\sqrt{3} < 97/56$, we obtain

$$\begin{aligned} C_T - \frac{63}{50}y &\leq \max \left\{ 0, \frac{17}{12} - \frac{63}{50}, \frac{1}{2} \left(\frac{97}{56} - \frac{63}{50} \right) \right\} \\ &= \max \left\{ 0, \frac{47}{300}, \frac{661}{2800} \right\} = \frac{661}{2800}. \end{aligned}$$

Here $y \leq 2$ follows from $2 \leq 2\bar{e}_1(\mathfrak{m}_{T,x}) \leq 2(3 - y)$. The rational radius comparisons follow from $63^3 = 250047 > 2 \cdot 50^3$, $17^2 = 289 > 2 \cdot 12^2$, and $97^2 = 9409 > 3 \cdot 56^2$.

For $m = 3$, Proposition 5.6(i) gives the stated bound for C_T . The stronger incoming cap $y \leq 3/2$ follows from (16.1).

For $m = 4$, embedding dimension at most four would force the Artinian Hilbert function $(1, 1, 1, 1)$. Indeed, a standard graded Artinian algebra with one generator in degree one has Hilbert function a string of ones until it vanishes; zero generators would give length one. The moment would then be six, implying

$$12 \leq 2\bar{e}_1(\mathfrak{m}_{T,x}) \leq 4(3 - y),$$

contrary to $y > 0$. Hence $\text{edim}_x T = 5$, and Proposition 5.6(ii) supplies both the fourth row and $y \leq 1$. We use its intrinsic tail inequality, whose hypotheses impose no condition on the boundary's source.

For $m = 5$, embedding dimension at most four would similarly force $(1, 1, 1, 1, 1)$, whose moment ten contradicts $2\bar{e}_1(\mathfrak{m}_{T,x}) \leq 5(3 - y) < 15$. Thus the embedding dimension is five. Its Artinian reduction has degree-one value two, with two further units of length in degrees at least two. Its moment is at least six. At any surface descendant, Proposition 4.11 now gives

$$t \leq 5(3 - z) - 2\bar{e}_1(\mathfrak{m}_{T,x}) \leq 3 - 5z < 3.$$

All subsequent radii are at most $\sqrt{2}$, whereas the initial radius is at most $5^{1/3} < 7/4$, since $5 \cdot 4^3 < 7^3$. Telescoping all deficit drops gives $C_T \leq (7/4)y$. Equation (16.1) gives the cap $y \leq 1/2$. The same equation supplies the first row's cap. Every row remains valid if the surface or curve dimension is skipped: the corresponding contribution is then absent.

Step 3. In this step, we compare every successor tail with the fourfold radius and include all skipped dimensions. Put

$$c = \frac{6}{5}, \quad d_0 = \frac{1871}{3360}.$$

The initial radius satisfies $r(W) \leq 2^{1/4} < c$, because $2 \cdot 5^4 < 6^4$. Subtracting cy from the five bounds in (16.4) gives, respectively,

m	upper bound for $C_T - cy$
1	$\frac{13}{60}$
2	$\frac{3}{25} + \frac{661}{2800} = \frac{997}{2800}$
3	$\frac{3}{2} \left(\frac{577}{400} - \frac{6}{5} \right) + \frac{811}{4200} = \frac{1871}{3360}$
4	$\frac{80}{127} - \frac{5}{6} + \frac{192}{7} = \frac{407}{960}$
5	$\frac{1}{2} \left(\frac{7}{4} - \frac{6}{5} \right) = \frac{11}{40}$

For the first entry, the function $-y/5 + (5/12)\min\{y, 1\}$ increases up to $y = 1$ and decreases thereafter, with maximum $13/60$. All entries other than the third are less than $1/2$, while $d_0 > 1/2$. If the first successor is a threefold, its full tail therefore gives

$$C = r(W)(b - y) + C_T \leq c(b - y) + cy + d_0 = cb + d_0.$$

It remains to include first successors of lower dimension. If the first successor is a surface, use (16.3), the caps (16.2), and the radius bounds $1, 17/12, 7/4, 2$ for $t = 1, 2, 3, 4$. They give

$$C_S - cz \leq \begin{cases} 0, & t = 1, \\ 13/60, & t = 2, \\ 11/30, & t = 3, \\ 2/5, & t = 4. \end{cases}$$

Thus $C \leq c(b - z) + C_S \leq cb + 2/5 < cb + d_0$. For a direct curve successor of deficit ζ , $C \leq c(b - \zeta) + \zeta \leq cb$. For a direct point successor, $C = r(W)b \leq cb$. These cases together prove the bound for every chain. $\square$

Theorem 16.2. *Assume $L^6 = 1$. There is an integer J , depending only on X, L, x , such that if at one $j \geq J$ a pure threshold average Γ from U_j has a fourfold minimal center of multiplicity two, then there is a point boundary $P \sim_{\mathbb{Q}} vL$ with $0 < v < 7 - 1/16384$. If $b = \text{def}_x(\Gamma) \leq 15/8$, the point can be obtained by an additive continuation of the entire Γ . If $b > 15/8$, then*

$$\text{vol}_Y(65\pi^*L - 75E) > 1418890625.$$

No recurrence or tangent-cone hypothesis is imposed on the fourfold center.

The small-deficit conclusion retains the entire threshold average. In the other branch the complete source supplies a new point boundary.

Proof. We first fix a cutoff independent of the chosen center. The scalar bound then gives the small-deficit continuation. For larger deficit we extract a computing place of the actual average, pass to its asymptotic valuation order, and apply the two earlier volume results.

Step 1. In this step, we fix a uniform cutoff and record the full coefficient and deficit of the specified source. Let C_3 and J_0 be the constant and cutoff of Proposition 7.8. Choose an integer J large enough that $J \geq J_0$, that $U_j \neq 0$ for every $j \geq J$, and that

$$(16.5) \quad \frac{C_3}{j} < \frac{1}{10} \quad (j \geq J).$$

Nonemptiness follows from Proposition 7.6. These choices involve only X, L, x .

Fix one such j and an average as in the statement. Denote its fourfold minimal center by W , and write

$$\tau = \text{lct}_x(X; \mathfrak{q}_j), \quad \Gamma = \frac{\tau}{N} \sum_{\nu=1}^N H_\nu, \quad H_\nu \in U_j.$$

As usual a nonzero section here denotes its actual effective divisor. With s, M the entire coefficient and point multiplicity of Γ , Proposition 7.6 gives

$$(16.6) \quad s = 65j\tau > 0, \quad M \geq 64j\tau, \quad s \leq \frac{65}{64}M, \quad M + b \leq 6.$$

Lemma 16.1 gives $0 < b \leq 3$.

Step 2. In this step, we construct the additive point continuation for $b \leq 15/8$ with every approximation error included. For every additive chain starting with this same Γ , Lemma 16.1

and (16.6) give

$$\begin{aligned} s + C &\leq \frac{65}{64}(6 - b) + \frac{6}{5}b + \frac{1871}{3360} \\ &= \frac{195}{32} + \frac{59}{320}b + \frac{1871}{3360} \\ &\leq \frac{195}{32} + \frac{59}{320} \frac{15}{8} + \frac{1871}{3360} = 7 - \frac{199}{53760}. \end{aligned}$$

Fix the total new allowance $h = 1/65536$ before cutting, and apply Proposition 3.7. At each stage its approximation error is chosen using the current radius, with the sum of the weighted errors less than h . The resulting effective point boundary retains every coefficient of Γ , and its actual coefficient satisfies

$$0 < v < 7 - \frac{199}{53760} + \frac{1}{65536} < 7 - \frac{1}{16384}.$$

This proves the asserted additive conclusion with the approximation errors included.

Step 3. In this step, we obtain a divisorial place centered exactly on W that computes the actual pure threshold. Suppose $b > 15/8$. We claim that a divisorial valuation w with center exactly W satisfies

$$(16.7) \quad \tau w(\mathbf{q}_j) = A_X(w).$$

To see this for the actual average, take a log resolution $\mu : \tilde{X} \rightarrow X$ of its complete base ideal. Write

$$\mathbf{q}_j \mathcal{O}_{\tilde{X}} = \mathcal{O}_{\tilde{X}}(-F_j), \quad \mu^* H_\nu = F_j + G_\nu.$$

Here F_j is the full fixed divisor on the resolution, and the G_ν are sufficiently general members of the resulting basepoint-free moving system. We may take their union with the fixed and exceptional supports to be simple normal crossing, and they contain no stratum of the fixed and exceptional supports. Their coefficients τ/N are less than one.

If $K_{\tilde{X}} + \Delta_{\tilde{X}} = \mu^*(K_X + \Gamma)$, then

$$\Delta_{\tilde{X}} = \tau F_j - K_{\tilde{X}/X} + \frac{\tau}{N} \sum_{\nu=1}^N G_\nu.$$

Every coefficient is at most one near $\mu^{-1}(x)$. Since the moving coefficients are less than one, the lc centers are images of strata formed by coefficient-one components of $\tau F_j - K_{\tilde{X}/X}$. There is such a stratum whose image is W . Its divisorial component, or the exceptional divisor of the blow-up of the stratum at its generic point, gives a zero-discrepancy place w centered exactly on W . The moving divisors do not contain this stratum, so their orders at w vanish. Therefore

$$A_X(w) = w(\Gamma) = \tau w(F_j) = \tau w(\mathbf{q}_j),$$

as claimed. This argument retains the full ideal and every fixed component of the actual source.

Step 4. In this step, we pass from the specified threshold level to a strict asymptotic valuation bound. The point-order inequality in (16.6) gives

$$64j\tau \leq M \leq 6 - b < \frac{33}{8}, \quad \frac{1}{j\tau} > \frac{512}{33}.$$

Apply Proposition 7.8 to the place just chosen. Its error bound holds for all divisorial valuations with center containing x , so the choice of w after the choice of the average does not change C_3 or J . From (16.7) and (16.5), we obtain

$$\begin{aligned} (16.8) \quad \frac{\alpha(w)}{A_X(w)} &\geq \frac{w(\mathbf{q}_j)}{jA_X(w)} - \frac{C_3}{j} \\ &= \frac{1}{j\tau} - \frac{C_3}{j} > \frac{512}{33} - \frac{1}{10} = 15 + \frac{137}{330} > 15. \end{aligned}$$

This is a strict bound obtained from the one specified level.

Step 5. In this step, we obtain the complete order-75 volume and construct a point boundary from the stronger source. The center of w has codimension two on the smooth sixfold. Proposition 7.10 applies to (16.8), since $L^6 = 1$, and gives

$$\text{vol}_Y(65\pi^*L - 75E) > 1418890625.$$

This is the asserted volume conclusion. The proposition controls all point-order layers of the complete source, including those below $64k$.

For each positive integer k divisible by five, the actual complete spaces satisfy

$$H^0(Y, k(65\pi^*L - 75E)) \subseteq H^0\left(Y, k\left(65\pi^*L - \frac{371}{5}E\right)\right).$$

Taking their normalized complete-series limits as in Proposition 7.5 gives

$$\text{vol}_Y\left(65\pi^*L - \frac{371}{5}E\right) \geq \text{vol}_Y(65\pi^*L - 75E) > 1418890625 > 336000000.$$

Theorem 10.5 therefore supplies a point boundary $P \sim_{\mathbb{Q}} vL$ with

$$0 < v < 7 - \frac{1}{4096} < 7 - \frac{1}{16384}.$$

That criterion covers all centers of its own subsequent sources; its general threefold estimates include the multiplicity-eight and multiplicity-nine cases in Lemma 5.1. The boundary obtained in this branch is separately constructed and need not contain Γ . The additive construction covers $b \leq 15/8$, and the volume criterion covers $b > 15/8$, proving all the conclusions. $\square$

17. FREENESS IN DIMENSION SIX

We retain the notation of Section 2. We first show that a failure of generation forces every sufficiently late pure threshold average from U_j to have a fourfold minimal center of multiplicity two. Theorem 16.2 then provides a point boundary of coefficient less than seven from one such average.

Lemma 17.1. *Let B be an effective rational divisor on a smooth complex projective sixfold X , log canonical near a closed point x , and let W be its positive-dimensional minimal log canonical center through x . If $\text{mult}_x W = 1$, then W is smooth at x and $\text{edim}_x W = \dim W$.*

This identifies the multiplicity-one cases with the smooth cases in the earlier endpoint statements.

Proof. Set $R = \mathcal{O}_{W,x}$ and $d = \dim R$. Theorem 4.5 [dFH09, Corollary 7.12] makes R normal and Cohen–Macaulay. It is an excellent local ring over $\mathbb{C}$, so normality also makes it analytically unramified. Its residue field is $\mathbb{C}$. Thus Theorem 4.6 [Han26, Lemmas 4.2 and 4.4] applies to its maximal ideal. For the Hilbert function (h_i) of a general Artinian minimal reduction, it gives

$$h_0 = 1, \quad h_i \geq 0, \quad \sum_i h_i = \text{mult}_x W = 1, \quad h_1 = \text{edim } R - d.$$

Consequently $h_1 = 0$ and $\text{edim } R = d$. This is the regularity criterion for the local ring R . Over $\mathbb{C}$, regularity at the closed point is equivalent to smoothness there. $\square$

Proposition 17.2. *Suppose $K_X + 7L$ is not generated at x . Then $L^6 = 1$, and there is an integer J , depending only on X, L, x , such that $U_j \neq 0$ for every $j \geq J$ and every pure threshold average Γ_j from U_j has a fourfold minimal center W_j of multiplicity two. Writing $\Gamma_j \sim_{\mathbb{Q}} s_j L$, $M_j = \text{mult}_x \Gamma_j$, and $b_j = \text{def}_x(\Gamma_j)$, one has*

$$1 < b_j \leq 6 - M_j, \quad 0 < s_j \leq \frac{65}{64}M_j \leq \frac{195}{32} < 7.$$

The cutoff is chosen before the level and its sufficiently general tuple; the centers need not recur.

The conclusion concerns the original complete source at each level. In particular, it supplies the precise center required by Theorem 16.2.

Proof. We choose a common cutoff for the original-source results, exclude the possible centers using their complete point endpoints, and retain the numerical bounds for the remaining case.

Step 1. In this step, we choose a common original-source cutoff and one pure average satisfying all required generality conditions. The integer L^6 is positive because L is ample Cartier. If $L^6 \geq 2$, Theorem 11.7 gives generation at x , contrary to the hypothesis. Hence $L^6 = 1$.

Keep X, L, x fixed. Choose an integer J_0 as in Proposition 7.6, so that $U_j \neq 0$ for every $j \geq J_0$. Under the assumed failure of generation, Proposition 13.1 gives an integer J_F for both its full fixed-divisor bounds and its restriction on fivefold centers. Let

$$J_{\geq 4}, \quad J_{\text{cub}}, \quad J_{\text{dbl},5}, \quad J_{\text{sm},5}$$

be the cutoffs in Theorems 12.10, 15.7, 13.4, and 14.6, respectively. Each depends only on the fixed X, L, x . We choose

$$(17.1) \quad J = \max\{J_0, J_F, J_{\geq 4}, J_{\text{cub}}, J_{\text{dbl},5}, J_{\text{sm},5}\}.$$

The small-deficit and threefold endpoint statements apply to any boundary satisfying their displayed hypotheses and require no additional cutoff in the original level.

Now fix any $j \geq J$. Use the full base ideal $\mathfrak{q}_j$ of the complete space U_j , and put

$$\tau_j = \text{let}_x(X; \mathfrak{q}_j), \quad \Gamma_j = \frac{\tau_j}{N} \sum_{i=1}^N H_i, \quad H_i \in U_j,$$

where the H_i are actual divisors and the tuple is sufficiently large and sufficiently general. All the generality requirements in the finitely many source statements can be imposed together. At this fixed level, take a common principalization of $\mathfrak{q}_j$, choose N above their lower bounds, and choose the moving members generally on it. Their required transversality and order conditions give nonempty open subsets of the same irreducible parameter space $\mathbb{P}(U_j)^N$. Their finite intersection is nonempty. The full fixed part of every H_i is included in Γ_j . Thus this is one pure average in the sense of Definition 2.5, with the same ideal threshold for every source theorem used below.

Proposition 7.6 gives that τ_j is positive rational and that Γ_j is log canonical near x , but not klt at x . Its minimal center W_j has dimension at most five. The same proposition gives

$$(17.2) \quad \begin{aligned} s_j &= 65j\tau_j > 0, & M_j &\geq 64j\tau_j, \\ s_j &\leq \frac{65}{64}M_j \leq \frac{195}{32} < 7, \\ 0 < b_j &\leq 6 - M_j \quad \text{if } \dim W_j > 0. \end{aligned}$$

In particular, every endpoint that assumes $s_j \leq (65/64)M_j + e$ applies to the entire Γ_j with $e = 0$, whenever its center hypothesis is satisfied.

Step 2. In this step, we exclude point centers and small deficits by point evaluation. Whenever an earlier result produces a point boundary $P \sim_{\mathbb{Q}} vL$ on X with $v < 7$, Proposition 3.4, with integer 7, makes the evaluation map at x surjective. Thus every such endpoint contradicts our hypothesis. If $W_j = \{x\}$, Γ_j itself is such a boundary by (17.2). If $\dim W_j > 0$ and $b_j \leq 1$, Corollary 5.7 applies with $e = 0 < 1/320$ and produces an additive point continuation of coefficient less than $7 - 1/100$. Consequently every remaining center satisfies

$$(17.3) \quad b_j > 1, \quad M_j \leq 6 - b_j < 5.$$

For completeness, the following table records every dimension and multiplicity of the original center. In the table, $m = \text{mult}_x W_j$, and an upper bound for a point coefficient is strict. The threefold multiplicities are at most nine by Lemma 5.1; the surface multiplicities are at most five by Corollary 4.13.

Original minimal center	Earlier result and consequence
Point	Proposition 3.4, since $s_j < 7$.
Any positive dimension, $b_j \leq 1$	Corollary 5.7: point coefficient $< 7 - 1/100$.
Curve	Corollary 4.13: $b_j \leq 1$, so the preceding row applies.
Surface, $m = 1$	Lemma 17.1 and Corollary 5.7: point coefficient $< 7 - 1/32$.
Surface, $2 \leq m \leq 5$	Corollary 4.13: $b_j \leq 2/m \leq 1$.
Threefold, $1 \leq m \leq 7$	Theorem 9.6: point coefficient $< 7 - 1/4096$.
Threefold, $m = 8$	Lemma 5.1: $b_j \leq 1/4$, followed by Corollary 5.7.
Threefold, $m = 9$	Lemma 5.1: $b_j \leq 1/9$, followed by Corollary 5.7.
Fourfold, $m = 1$	Lemma 17.1 and Corollary 5.7: point coefficient $< 7 - 1/32$.
Fourfold, $m = 2$	The remaining case, subject to (17.3).
Fourfold, $m = 3$	Theorem 15.7: point coefficient $< 7 - 1/16384$.
Fourfold, $m \geq 4$	Theorem 12.10: point coefficient $< 7 - 1/16384$.
Fifold, $m = 1$	Lemma 17.1 and Theorem 14.6: point coefficient $< 7 - 1/16384$.
Fifold, $m = 2$	Theorem 13.4: point coefficient $< 7 - 1/4096$.
Fifold, $m \geq 3$	Proposition 13.1 excludes this case at $j \geq J_F$.

We check the hypotheses in the positive-dimensional rows.

Step 3. In this step, we exclude every center of dimension at most three, including both large threefold multiplicities. For a curve, Corollary 4.13 gives $b_j \leq 1$, contradicting (17.3). If W_j is a surface of multiplicity at least two, the same corollary gives $b_j \leq 2/m \leq 1$. If it has multiplicity one, Lemma 17.1 gives $\text{edim}_x W_j = 2$. The embedding-dimension clause of Corollary 5.7 therefore applies to Γ_j , with $e = 0$, and gives an additive point continuation of coefficient less than $7 - 1/32$.

For a threefold, Theorem 9.6 applies directly to the effective rational boundary Γ_j : the pair is log canonical near x , its minimal center has dimension three, and

$$s_j \leq \frac{65}{64} \text{mult}_x \Gamma_j + 0, \quad 0 < \frac{1}{2560}.$$

It provides an additive point continuation of the entire boundary with coefficient less than $7 - 1/4096$. This treats every threefold, including every surface or curve that may occur during its continuation.

The enumeration here also explicitly includes the two largest threefold multiplicities. Lemma 5.1 gives

m	deficit bound	entire scalar tail
8	$b_j \leq 1/4$	$C \leq 2b_j$
9	$b_j \leq 1/9$	$C \leq (9/4)b_j$.

Both bounds include direct jumps to a curve or to the point. Thus these two original-center rows already satisfy the small-deficit endpoint, and their possible occurrence in the scalar envelopes used above is covered by Lemma 5.1 in Section 5. In particular, no threefold multiplicity is omitted from the exhaustion.

Step 4. In this step, we exclude every fourfold multiplicity other than two. If $\dim W_j = 4$ and $m = 1$, Lemma 17.1 gives $\text{edim}_x W_j = 4$. The embedding-dimension clause of Corollary 5.7, again with $e = 0$, provides a point continuation of coefficient less than $7 - 1/32$.

If $m = 3$, apply Theorem 15.7. We have $L^6 = 1$, $j \geq J_{\text{cub}}$, and Γ_j is the pure threshold average of the complete U_j chosen above. Its original minimal center is precisely this fourfold of multiplicity three. The theorem therefore gives a point boundary of coefficient less than $7 - 1/16384$. If $m \geq 4$, the same source satisfies Theorem 12.10, since $j \geq J_{\geq 4}$. This gives a point boundary with the same strict bound. Each point boundary contradicts failure of generation. The positive integers 1, 2, 3 and those at least four exhaust all fourfold multiplicities, leaving only $m = 2$.

Step 5. In this step, we exclude every fivefold center using the original-source fixed-divisor bounds and point endpoints. Suppose $\dim W_j = 5$. The standing failure of generation and $j \geq J_F$ allow us to apply Proposition 13.1 to this original average. It gives

$$\text{mult}_x W_j \in \{1, 2\}, \quad L^5 \cdot W_j = 1.$$

In particular, it excludes every other fivefold multiplicity and degree at all levels under consideration.

For multiplicity one, Lemma 17.1 makes W_j smooth at x . All hypotheses of Theorem 14.6 now hold: the polarization has volume one, the level satisfies $j \geq J_{\text{sm},5}$, and the original pure average has a smooth fivefold minimal center. It yields a point boundary of coefficient less than $7 - 1/16384$, for either deficit range in that theorem.

For multiplicity two, use Theorem 13.4 at $j \geq J_{\text{dbl},5}$. The center is a fivefold of multiplicity two for the same pure threshold of the same complete U_j . The theorem yields a point boundary of coefficient less than $7 - 1/4096$. Both fivefold cases therefore contradict the assumed failure of generation.

Step 6. In this step, we identify the remaining double fourfold and retain its coefficient and deficit bounds uniformly over all later levels. The point and small-deficit arguments exclude dimension zero and every positive-dimensional center with $b_j \leq 1$. The low-dimensional endpoints exclude dimensions one, two and three; the original-source endpoints exclude dimension five and every fourfold multiplicity except two. Hence

$$\dim W_j = 4, \quad \text{mult}_x W_j = 2.$$

Combining (17.2) with (17.3) gives all the claimed inequalities.

Each endpoint used above returns an actual point boundary with its entire coefficient bounded strictly below seven. Its approximation errors have already been included in that bound, so Proposition 3.4 applies directly. The integer in (17.1) was fixed before j , and $j \geq J$ and its qualifying tuple were arbitrary. The conclusion therefore holds for every such level and pure threshold average, independently of how its minimal center varies. $\square$

Theorem 17.3. *For every smooth complex projective sixfold X and every ample Cartier divisor L , the line bundle $\mathcal{O}_X(K_X + 7L)$ is globally generated.*

We apply the reduction and the double-fourfold endpoint to a single threshold average, and then pass from generation at closed points to generation of the line bundle.

Proof. We use one common source to obtain a point boundary, apply point evaluation, and then eliminate the cokernel of sheaf evaluation.

Step 1. In this step, we apply the reduction and the double-fourfold endpoint to one pure threshold average. Fix X and L as in the statement, and write $\mathcal{A} = \mathcal{O}_X(K_X + 7L)$. Let x be an arbitrary closed point. Suppose that

$$\text{ev}_x : H^0(X, \mathcal{A}) \longrightarrow \mathcal{A} \otimes k(x)$$

is not surjective.

Proposition 17.2 gives $L^6 = 1$ and an integer J_{red} , depending only on X, L, x , such that its conclusion holds for every level at least J_{red} . For these same fixed data, Theorem 16.2 supplies an integer $J_{\text{dbl},4}$. First put

$$J_* = \max\{J_{\text{red}}, J_{\text{dbl},4}\},$$

and then choose one integer $j \geq J_*$. This choice puts j beyond every cutoff entering the reduction as well as the cutoff for the last fourfold case.

The complete space U_j is nonzero. With its full base ideal $\mathfrak{q}_j$, set

$$\tau_j = \text{lct}_x(X; \mathfrak{q}_j), \quad \Gamma_j = \frac{\tau_j}{N} \sum_{i=1}^N H_i, \quad H_i \in U_j.$$

Choose one N sufficiently large for both results and one tuple of actual divisors sufficiently general for both. On a common principalization of $\mathfrak{q}_j$, their generality conditions have a nonempty intersection, as in the proof of Proposition 17.2. Both results therefore apply to this same pure average, formed with the ordinary threshold of the full ideal $\mathfrak{q}_j$.

By Proposition 17.2, the effective rational boundary Γ_j is log canonical near x and its minimal center W_j satisfies

$$\dim W_j = 4, \quad \text{mult}_x W_j = 2.$$

We have now met every hypothesis of Theorem 16.2: $L^6 = 1$, the chosen level is at least $J_{\text{dbl},4}$, and the pure threshold average from the complete U_j has a fourfold minimal center of multiplicity two. That theorem gives an effective rational divisor P on X with

$$P \sim_{\mathbb{Q}} vL, \quad 0 < v < 7 - \frac{1}{16384},$$

such that (X, P) is log canonical near x and has minimal center $\{x\}$.

Step 2. In this step, we use the constructed point boundary to prove surjectivity of evaluation at x . Apply Proposition 3.4 to $(Z, A, z, \Delta, c) = (X, L, x, P, v)$ and the positive integer 7. Its strict inequality is $v < 7$, supplied by the displayed bound. It follows that ev_x is surjective, a contradiction. The point boundary and the line bundle in this application are both on the original X .

Step 3. In this step, we deduce global generation by eliminating the coherent cokernel of sheaf evaluation. Thus ev_x is surjective for every closed point x . Let $\mathcal{C}$ be the coherent cokernel of the sheaf evaluation map

$$H^0(X, \mathcal{A}) \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow \mathcal{A}.$$

Right exactness of tensoring with $k(x)$ shows that $\mathcal{C} \otimes k(x) = 0$ at every closed point. Nakayama's lemma gives $\mathcal{C}_x = 0$ there. If $\mathcal{C}$ were nonzero, its nonempty closed support on the complex projective variety X would contain a closed point, which is impossible. Hence $\mathcal{C} = 0$, and $\mathcal{A}$ is globally generated. $\square$

APPENDIX A. DETAILS OF THE USE OF GENERATIVE AI

The results of this paper were obtained using Danus [Liu⁺26], an automated mathematical reasoning system built on Rethlas [Ju⁺26]. For transparency we record the experiment in detail: how the system was run (§A.1), the problem exactly as it was posed (§A.2), and the guidance the authors gave during the run (§A.3).

A.1. How the system was run. The six-dimensional research run used `gpt-6-astra`. The main agent was configured at reasoning effort `ultra`. There were seven research workers: four at `max` and three at `xhigh`; their launch records confirm this allocation. The retained research-verifier configuration specifies `gpt-6-astra` at `max`. These are recorded settings and launch observations, not an independent audit of every historical backend request.

The project instance was created on 7 September 2026 at 09:58:13 UTC. Its main startup was recorded at 10:32:57 UTC, and all seven workers were confirmed running by 11:03 UTC. After an interruption, the same research thread resumed on 8 September at 21:34 UTC. The terminal checkpoint on 9 September at 09:00 UTC recorded acceptance of the unconditional sixfold statement and closure of its dependency chain. The completion receipt followed at 09:02 UTC, with all seven workers confirmed stopped. The interval from project creation to the terminal checkpoint is about 47 hours and 2 minutes. It is calendar time, including the interruption; it excludes earlier inherited research and is not a measurement of uninterrupted work or CPU time. The checkpoint records acceptance to the minute. No controlled comparison run with an outcome is recorded for this experiment.

The research system stored candidate arguments as a dependency graph and admitted them after model-based textual review. A subsequent independent cold review on 9 September used six

fresh branch reviewers and a root reviewer to examine the target’s support, its earlier foundations, the principal endpoint families, and the final assembly. It found an incomplete enumeration in the earlier threefold-tail argument: multiplicities eight and nine had been omitted from a table used as exhaustive. The supplied completion follows from the existing Hilbert and successor inequalities. Multiplicity eight has $z \leq 1/4$ and complete tail $C \leq 2z$; multiplicity nine has $z \leq 1/9$ and complete tail $C \leq 9z/4$. The proof, including direct jumps and the restrictions on surface successors, is given in Lemma 5.1. The resulting four additional envelope comparisons preserve every original budget constant. The historical source records were left unchanged, and the completed cases were included in the manuscript and in its review obligations.

After this completion, the cold review reported no other unresolved load-bearing objection within its stated scope. It also corrected author names, distinguished a regular sequence of parameters from regularity of the ring, and required complete extraction of proofs whose internal source headings had caused truncation. Historical comparisons with revoked or differently scoped results were excluded from the proof premises. This review was not a formal proof-assistant verification or a certification of every historical sentence.

Manuscript production used the native Danus writing workflow, with a main-owned outline and fifteen parallel section-writing jobs, each with its own verifier. The writing jobs used `gpt-6-astra` at `max`; the whole-paper verifier used the same model and effort. All fifteen sections passed. The first whole-paper review identified two conflicts between ambient and intrinsic model symbols. After coordinated notation corrections, the second whole-paper review passed before the final readability and front-matter pass. Its checks explicitly included the completed multiplicity-eight and multiplicity-nine cases and their later uses. These manuscript checks are distinct from research acceptance and from the independent cold review. The displayed finite arithmetic, integral estimates, and error bounds in the body are given as checkable textual arguments; no unreported numerical experiment serves as a proof premise.

A.2. The problem as posed. The following is the complete original problem file, including its scope and authorization text; a direct user quotation and a host name are replaced by bracketed notices. Its final sentence records the research-stage status; the later writing authorization is described in §A.3.

```
# Current objective: six-dimensional Fujita freeness

The sole active mathematical objective is to prove the following statement:

> For every smooth complex projective sixfold X and every ample Cartier divisor
↪ L on X, the line bundle  $\mathcal{O}_X(K_X+7L)$  is globally generated.

Equivalently, for every such X and L and every closed point x in X, the
↪ evaluation map

\[\mathcal{H}^0(X, \mathcal{O}_X(K_X+7L)) \xrightarrow{\quad} \mathcal{O}_X(K_X+7L) \otimes k(x)\]

is surjective. No global-generation assumption on L, special construction class
↪ for X, or additional hypothesis on a hypothetical lc center is part of the
↪ target.

## Completion standard

Success requires an unconditional proof covering every X, L and x above. The
↪ proof must discharge every added hypothesis and connect its intermediate
↪ statements to this evaluation map.
```

The known six-dimensional K_X+8L result is a literature baseline, not
 ↪ completion. Conditional qCI exclusions, a special center classification, a
 ↪ transport theorem for an already chosen section, or a finite table of
 ↪ first-cut survivors are also not completion. In particular, both the
 ↪ initially unaffordable branches and the actual successors of an affordable
 ↪ first cut remain part of the sixfold problem.

The project may use any sound proof strategy. The existing center-cutting,
 ↪ full-chain budget, qCI and rooted-obstruction materials are available
 ↪ tools; none is imposed as the exclusive route.

Current scope and authorization

Operator's stated problem, preserved verbatim:

> [118 bytes of discussion have been omitted because they contain private
 ↪ information.]

The user has explicitly authorized starting or resuming research in the
 ↪ existing `fujita` project on [host name omitted] for this sixfold
 ↪ objective. This authorization replaces the earlier initial-pause
 ↪ instruction for this project. It does not authorize launching other
 ↪ projects.

All-dimensional Fujita freeness and Fujita very ampleness (FVA) remain
 ↪ recoverable historical research. They are not active objectives and must
 ↪ not receive separate research assignments during this run. An archived tool
 ↪ may be used when it directly supports the sixfold theorem, without
 ↪ reopening its broader campaign.

This narrowing is a user-directed change of scope, not a mathematical reduction
 ↪ from higher dimensions. No $n \geq 7$ -to-six reduction has been established by
 ↪ these materials. Likewise, known smooth low-dimensional Fujita theorems
 ↪ cannot automatically be applied to singular lc/qlc successors.

Read [current guidance](expert_guidance/GUIDANCE.md) and the [starting
 ↪ ledger](materials/STARTING_LEDGER.md). Original facts, source bytes and
 ↪ revocations remain unchanged. This objective update creates no facts or
 ↪ verification events and does not initiate manuscript drafting.

A.3. Guidance given during the run. The dated instruction record contains the following three verbatim instructions. The account after each quotation records its effect.

7 September 2026, launch.

[118 bytes of discussion have been omitted because they contain private
 ↪ information.]

This restricted the active objective to six-dimensional freeness in the existing project. Earlier all-dimensional and very-ampleness work remained historical material. No reduction from those broader questions to the sixfold theorem was assumed.

7 September 2026, 10:32 UTC..

You are the main agent for Danus project fujita (this directory). Read
 ↪ AGENTS.md and begin.

This started coordination under the existing project contract and fixed objective, with persistent records and verifier-gated workers. The instruction ledger also records, at 11:12 UTC, that the Han preprint named in the standing launch guidance had been fetched and read in its public

first version. That entry records fulfillment of an earlier source direction, not a fourth verbatim instruction. The normal Hilbert bounds and Artinian moment statements from that source enter Section 4.

8 September 2026, 21:34 UTC..

The operator has explicitly instructed: [24 bytes of discussion have been
 ↪ omitted because they contain private information.]. Resume this SAME
 ↪ existing Fujita main thread and its original Goal, whose paused status has
 ↪ been natively restored to active. The sole fixed target remains
 ↪ unconditional sixfold Fujita freeness: K_X+7L is globally generated for
 ↪ every smooth complex projective sixfold X and every ample Cartier divisor
 ↪ L . All existing problem statements, source guidance, checkpoints,
 ↪ fact/revocation stores, worker assignments and memories are preserved.
 ↪ Continue the existing fixed PROBLEM.md objective and coordinate the
 ↪ original seven Astra workers (four max, three xhigh) as the Astra ultra
 ↪ main. Recover from your latest checkpoint and perform your normal immediate
 ↪ scheduling and ongoing control beat. Your verifier is on the existing port
 ↪ 8407. Native workers and monitor are being restored by this same
 ↪ administrative recovery. This authorization is solely for Fujita; do not
 ↪ start or change any other project or operator. Preserve the original
 ↪ theorem, Goal, source boundaries and publication restrictions. This is
 ↪ research resumption, not a new project or a new Goal.

This resumed the same research thread after the interruption, preserving the seven-worker allocation, prior records, verifier, fixed theorem, and publication restrictions. It created no new mathematical objective.

9 September 2026, manuscript authorization. After the research completion record, the operator requested native manuscript production and a separate cold review. The three recorded utterances were:

[48 bytes of discussion have been omitted because they contain private information.]

[51 bytes of discussion have been omitted because they contain private information.]

[49 bytes of discussion have been omitted because they contain private information.]

The resulting authorization required the exact unconditional target, complete proofs from the existing sources, native parallel writing and verification, truthful provenance, and a checked final build. It kept mathematical source records read-only and did not authorize restarting the completed research workers or publishing externally. The subsequent cold-review instructions required the explicit multiplicity-eight and multiplicity-nine completion described above, the bibliography and parameter-wording corrections, complete raw proof extraction, and exclusion of historical comparators from the premises. These directions were incorporated in the outline, section assignments, and whole-paper checks.

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DEPARTMENT OF MATHEMATICS, PEKING UNIVERSITY, NO. 5 YIHEYUAN ROAD, HAIDIAN DISTRICT, BEIJING 100871, CHINA

BEIJING INTERNATIONAL CENTER FOR MATHEMATICAL RESEARCH, PEKING UNIVERSITY, NO. 5 YIHEYUAN ROAD, HAIDIAN DISTRICT, BEIJING 100871, CHINA

Email address: liujihao@math.pku.edu.cn